Part I

Markets, Instruments, and Infrastructure

The first nine chapters build the market before they build any model. They introduce the four instrument families (forwards, futures, swaps, and options) and then the infrastructure that decides what a derivative is actually worth: the risk-free-rate benchmarks that replaced LIBOR, the collateral agreement, and the valuation adjustments that follow from it. Nothing here needs stochastic calculus. A reader who takes only this part should still be able to read a term sheet, say who is on the other side of it, and know which discount curve applies.

1 **Introduction: Derivatives and Their Markets**
what the instruments are, who trades them, and how large the market is

2 **Futures Markets, Clearing, and Central Counterparties**
the exchange, the margin cycle, and the default waterfall

3 **Hedging Strategies Using Futures**
basis risk, the minimum-variance ratio, and when a hedge fails

4 **Interest Rates in the Risk-Free-Rate Era**
SOFR and its peers, and what replaced a term rate formed from quotes

5 **Determination of Forward and Futures Prices**
cost of carry, and why forwards and futures are not the same

6 **Interest Rate Futures**
the Treasury contract, the conversion factor, and cheapest-to-deliver

7 **Swaps and the Multi-Curve Framework**
the instrument, and why one curve became several

8 **Collateral, Margin, and Funding**
the CSA as a pricing input rather than paperwork

9 **XVA: Valuation Adjustments in Practice**
CVA, DVA, FVA, and what each one charges for

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Chapter 1
Introduction: Derivatives and Their Markets

A derivative is a contract whose value depends on something else. That is nearly the whole definition, and its emptiness is the point: the something else can be a commodity price, an interest rate, an exchange rate, a share price, a credit event, an index of volatility, a quantity of rainfall, or the price of another derivative. What unites the instruments in this book is not what they reference but what they do—they separate exposure to a risk from ownership of the thing that carries it.

That separation is the entire economic content of the subject. A farmer who wants the price of wheat fixed in September need not sell the wheat in March. A pension fund that wants thirty-year interest rate exposure need not buy thirty-year bonds. A bank that has made a loan it wants to keep, to a borrower whose credit it no longer wants, need not sell the loan. In each case a contract does what a transaction in the underlying asset would otherwise have done, usually with less capital, less disruption, and more precision—and, as the rest of this book documents at length, with risks of its own that are easy to underestimate precisely because so little cash changes hands at the outset.

This chapter is the map; every claim in it is developed properly later.

Sections 1.1 to 1.3 set out the four instruments, where they came from, and why they exist. Sections 1.4 to 1.7 describe the markets: exchange-traded against over-the-counter, the legal architecture of master agreements and netting, how large the market is once gross notional is read correctly, and what can serve as an underlying. Sections 1.8 to 1.10 are the participants: who uses derivatives and for what, why the labels hedger, speculator, and arbitrageur blur, and the leverage that makes every position larger than its cash. Sections 1.11 and 1.12 record what has gone wrong and how regulation responded after 2008, and Section 1.13 explains how the chapters fit together.

1.1 THE FOUR BUILDING BLOCKS

Almost every derivative in existence is a combination of four contracts. Learning the four well is more useful than memorizing the hundreds of named structures assembled from them.

A forward obliges one party to buy, and the other to sell, a specified asset on a specified future date at a price agreed now, the **forward price**. No money changes hands at inception; the forward price is set so that the contract is worth zero to both sides on day one. It is the simplest derivative and the subject of Chapter 5.

A futures contract is a forward that has been standardized, listed on an exchange, and settled daily in cash; the price at which it trades is the **futures price**. Daily settlement is more than a technicality: it converts a single payment at maturity into a stream of payments along the way, so that whichever side the price moves against must find cash now rather than at maturity, and Chapters 2 and 8 are largely about the consequences.

A swap exchanges two streams of payments (most commonly a fixed interest rate for a floating one) over a series of dates. A swap is economically a portfolio of forwards, one per payment date, bundled into a single contract with a single price. Chapter 7 develops them.

An option gives its holder the *right*, not the obligation, to buy (a call) or sell (a put) at a set price, the strike price. Because the holder will exercise only when it pays to do so, the option can never be worth less than zero to them, and they must therefore pay for it at the outset with a **premium**. Part II is devoted to options.

The essential distinction is between the first three and the last. Forwards, futures, and swaps are *symmetric*: both parties are obliged, gains and losses run in both directions, and the **payoff**, what the contract delivers at maturity, is a straight line. Options are *asymmetric*: one party has a choice, the payoff is kinked, and the premium is the price of the kink. Figure 1.1 shows the difference in terms of **profit**, which is the payoff net of any premium paid or received.

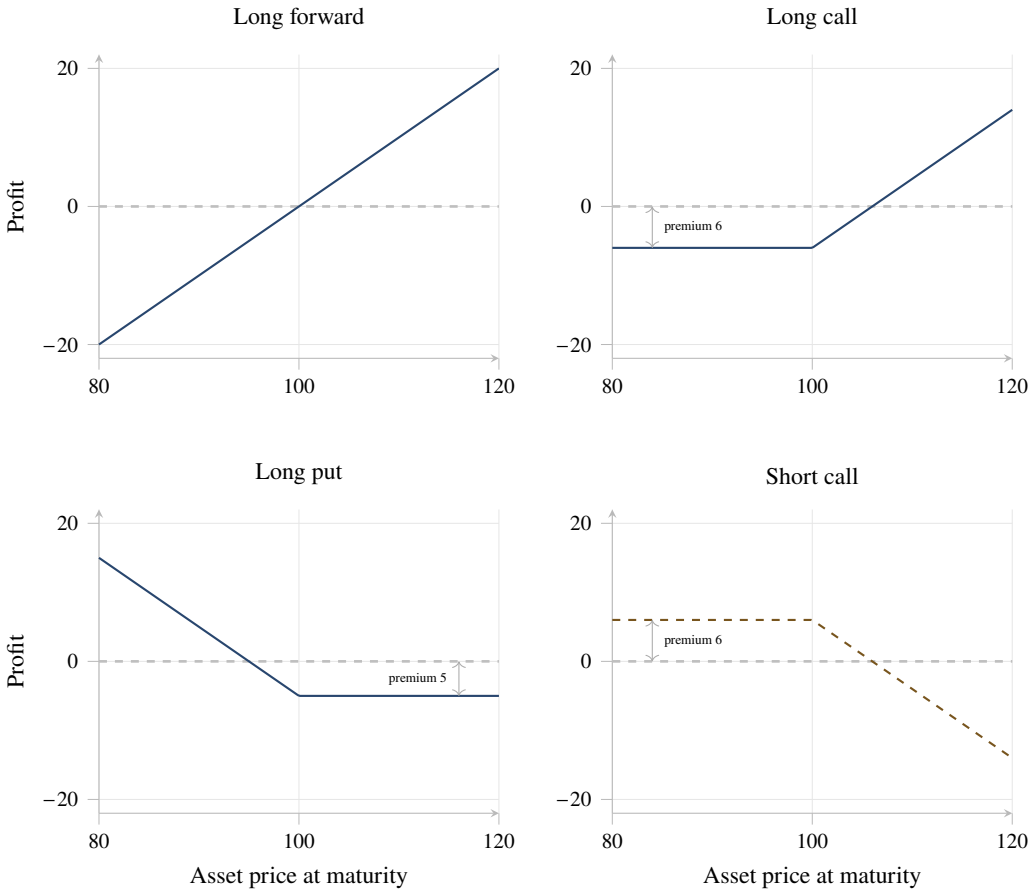


Figure 1.1: Profit at maturity, premium included. Strike and forward price 100; call premium 6, put premium 5.

Three features of Figure 1.1 are worth fixing now. The forward's line passes through zero at the forward price, so the contract starts life worth nothing and can end worth a great deal in either direction. The long call's worst case is bounded at the premium, however far the market falls. And the short call, in the fourth panel, has the mirror-image profile: a gain capped at the premium and an unbounded loss. That asymmetry is why option writing is a business for firms that can hedge and post collateral, and why it recurs in almost every chapter about things going

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wrong.

1.2 WHERE THESE MARKETS CAME FROM

Derivatives are often presented as a modern invention, and the impression is wrong in a way that obscures how the market works. Contracts for future delivery are ancient; what is modern is the standardization, the clearing, and the range of risks that can be referenced. More useful than a chronology is the pattern the chronology reveals: each major instrument appeared shortly after a risk became intolerable and a way of measuring it became available.

The Dojima rice market in Osaka, formally sanctioned in 1730, traded standardized claims on rice for future delivery, with a clearing house and margin: the essential apparatus of Chapter 2, two centuries early. The Chicago Board of Trade, founded in 1848, standardized grain contracts in the 1860s for the same reason: buyers and sellers separated by a season and a railroad needed a price agreed in advance.

The modern market dates from the breakdown of the Bretton Woods system. When fixed exchange rates ended in the early 1970s, currency risk became a problem every multinational suddenly had, and in May 1972 the Chicago Mercantile Exchange listed the first financial futures, on foreign currencies. Inflation and interest rate volatility followed, and with them the first interest rate futures in 1975–77 and the Eurodollar contract in 1981, on three-month US dollar deposits held at banks outside the United States—the first futures contract settled purely in cash because its underlying could not be delivered at all.

Options followed the same logic on a different track. They had traded over the counter for centuries, illiquidly and with unreliable pricing. The Chicago Board Options Exchange opened in April 1973 with listed calls on sixteen stocks; the Black–Scholes–Merton papers appeared the same year. It is tempting to say the model created the market. It is more accurate to say that standardization and a workable valuation method arrived together, and that neither would have sufficed alone, a point Chapter 15 returns to.

Swaps arrived from a different direction: not an exchange, but a single negotiated trade. The 1981 currency swap between IBM and the World Bank, arranged by Salomon Brothers, is conventionally cited as the first. Within a decade the swap market was larger than the exchange-traded market, entirely bilateral and documented under agreements that had to be invented as it grew (Section 1.5). Credit default swaps, which pay the buyer if a specified borrower defaults, emerged from the same bilateral tradition in the mid-1990s, developed by banks that wanted to keep lending to clients to whom they already had too much credit exposure.

The last two decades reverse the pattern. Their innovations have been structural, not new instruments: central clearing extended to over-the-counter (OTC) contracts after 2009, mandatory margining completed in 2022, and the replacement of the London Interbank Offered Rate (LIBOR) by overnight risk-free rates between 2021 and 2023—a change of foundations rather than of instruments. Newer underlyings continue to appear, and Section 1.7 surveys them, but the striking feature of the modern era is that the four building blocks of Section 1.1 have not changed. What changed is who stands between the two parties, and what they must post.

1.3 WHY DERIVATIVES EXIST

Why anyone bothers is a fair question, because the alternatives are real and sometimes better. A firm exposed to a price it dislikes has several choices that do not involve derivatives. It can hold physical inventory, which costs storage and capital. It can buy insurance, which works for events that are verifiable and idiosyncratic but not for market prices, since an insurer cannot diversify away a move that hits every policyholder at once. It can restructure operations (an airline can hedge fuel by owning a refinery, as Delta did in 2012), which works but ties up capital and management attention in a business the firm did not choose to be in. It can hold

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more equity, absorbing the risk rather than transferring it. Or it can simply accept the exposure, which for a diversified shareholder may be the correct answer and is discussed in Section 3.2.

Derivatives compete with these on three grounds:

Precision. A contract can be written on exactly the risk in question (one currency pair, one maturity, one delivery month) rather than on a bundle that happens to contain it.

Capital efficiency. Transferring $100 million of exposure requires posting margin measured in single-digit millions rather than committing $100 million. This is a genuine advantage and simultaneously the source of most of the disasters in Section 1.11; Section 1.10 treats it as one topic rather than two.

Reversibility. A hedge can be lifted when the exposure changes; the sale of a business cannot.

Against these sits the standing objection: because so little cash is required at inception, the size of a derivatives position is not disciplined by the size of the balance sheet behind it. Every episode in Section 1.11 is a variation on that sentence.

1.4 EXCHANGE-TRADED AND OVER-THE-COUNTER MARKETS

Derivatives trade in two structurally different places, and the difference used to be sharp. It is now more subtle, because regulation has pushed the two toward each other.

Exchange-traded contracts are standardized: the exchange fixes the size, the underlying, the delivery months, and the settlement terms, and participants choose only price and quantity. Every trade is novated to a central counterparty (CCP), which replaces it with two contracts, one between each party and the CCP, so the CCP becomes buyer to every seller and seller to every buyer, and no participant bears the credit risk of another. Standardization concentrates liquidity (any two participants wanting the same contract meet in the same order book) at the cost of fit: the contract may not match the exposure a hedger actually has, which is the basis risk of Chapter 3.

Over-the-counter contracts are negotiated bilaterally between two institutions and can specify anything the parties agree on: a non-standard maturity, an unusual underlying, a bespoke payoff. The fit is exact. Historically the price of that fit was credit risk: each party depended on the other's solvency for the life of the contract, which for a thirty-year swap is a long time. Figure 1.2 shows the two, and the hybrid that now dominates.

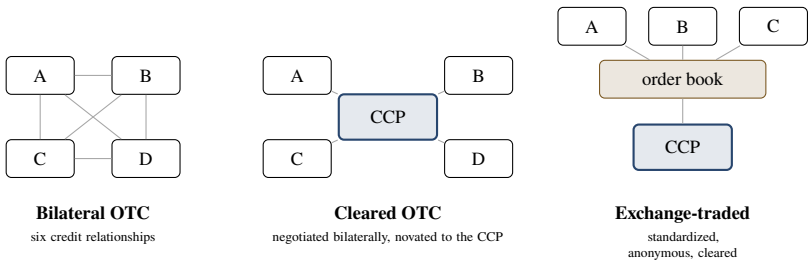


Figure 1.2: How a trade reaches its counterparty. Four participants shown.

The middle panel is where most of the market now sits. Following the G20's 2009 commitments (Section 1.12), standardized OTC contracts, above all interest rate swaps, must be cleared through a central counterparty. Such a trade is negotiated bilaterally, like an OTC contract, and then novated to a CCP, like a futures trade. It inherits the flexibility of one structure and the credit mechanics of the other, and that is why the old question "exchange-traded or OTC?" now has three answers rather than two. Chapter 2 works through what a CCP actually does with the risk it absorbs, which is not the same as making the risk disappear.

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1.5 THE LEGAL ARCHITECTURE: MASTER AGREEMENTS AND NETTING

A bilateral derivative is a promise extending years into the future, and what makes such a promise financeable is not the pricing model but the document. This is not a legal aside. The reduction from $23 trillion to $3.4 trillion in the next section is entirely an artifact of the arrangement described here, and Chapters 8 and 9 assume it throughout.

Two institutions did not exist when the swap market began and had to be invented as it grew. The first is the contract published by the International Swaps and Derivatives Association (ISDA), the **ISDA Master Agreement**, now the standard for essentially all bilateral derivatives. Its central provision is deceptively simple: all transactions between the two parties, however many and however different, form a **single agreement**. A party cannot default on one swap and perform on another.

That provision exists to enable **close-out netting**. If one party defaults, every outstanding transaction between them terminates at once, each is valued, and the values are summed to a single net amount owed one way or the other. Without it, an insolvency administrator could do what the market calls *cherry-picking*: enforce the contracts that favor the estate and disclaim the rest. Example 1.1 shows what that difference is worth.

Example 1.1 Why the single agreement clause exists

Bank A has five outstanding trades with counterparty B. Marked to market (valued at current market prices) from A's perspective:

$$+40, \quad -35, \quad +20, \quad -15, \quad -8 \qquad (\$ \text{ million})$$

The positives are amounts B owes A; the negatives are amounts A owes B. A's gross positive exposure is $60 million; the net is $2 million in A's favor.

With close-out netting, B's default terminates all five. They sum to $2 million, which A claims from the estate. A's loss is at most $2 million, and if collateral has been posted under the arrangement described below, close to nothing.

Without close-out netting, the administrator enforces the three contracts on which A owes money and disclaims the two on which A is owed. A must pay 35 + 15 + 8 = $58 million in full, and joins the queue of unsecured creditors for its $60 million of claims. At a 20% recovery rate (the share of its claims an unsecured creditor eventually collects), A collects $12 million:

$$\text{A's net payment} = 58 - 12 = \$46 \text{ million, in place of a } \$2 \text{ million claim}$$

The trades did not change. The document did. This is why the enforceability of close-out netting in a given jurisdiction is a precondition for a deep derivatives market there, and why regulators permit netted exposures to be used in capital calculations only where that enforceability has been established.

The second institution is the Credit Support Annex (CSA), an optional schedule to the Master Agreement under which the parties post collateral against the net amount as it moves. The CSA converts a credit exposure into a daily cash flow, which largely removes default risk and puts liquidity risk, the risk of having to find cash at short notice, in its place. That substitution is the subject of Chapter 8, because the second risk turned out to be the one that broke firms in 2022. Together the two documents explain the structure of the numbers in the next section: gross market value is what the contracts are worth, netting under the Master Agreement collapses it to a fraction, and collateral under the CSA covers most of what remains.

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1.6 HOW BIG IS THE MARKET?

Derivatives statistics are quoted in numbers so large that they are routinely misused. The Bank for International Settlements (BIS) puts the notional amount outstanding of over-the-counter derivatives at roughly $850 trillion in recent years ($845 trillion at the end of 2025), something like seven times world gross domestic product (GDP). The figure is correct and almost entirely uninformative.

Notional is the reference amount used to *calculate* payments. It is not an amount anyone owes, expects to receive, or can lose. On a $100 million ten-year interest rate swap, the $100 million is never paid by anyone: it is multiplied by a rate difference to determine a payment of a few hundred thousand dollars. Two more meaningful measures follow.

Gross market value is what the outstanding contracts are actually worth: the sum of what every position with a positive value would be paid if all contracts were settled today. Across the market this runs at roughly $23 trillion, under 3% of notional.

Gross credit exposure nets each pair of counterparties' offsetting positions against one another, as their master agreements permit them to do. That reduces the figure again, to roughly $3.4 trillion, under half a percent of the headline. Most of that remainder is collateralized, by the mechanisms of Chapter 8.

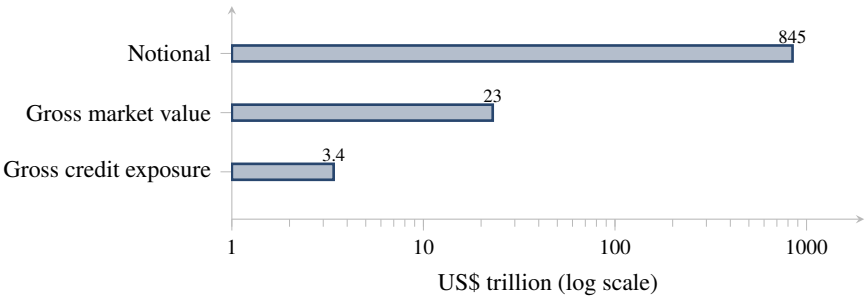


Figure 1.3: OTC derivatives at the end of 2025 (BIS). Log scale.

Figure 1.3 needs a log scale to show all three bars at once, which is itself the lesson: each step down is roughly an order of magnitude. A commentator who reports the first number and implies the third has overstated the risk by a factor of some two hundred and fifty. Example 1.2 traces the same three measures through a single swap.

Example 1.2 What $100 million of notional actually risks

A bank enters a five-year interest rate swap on $100 million notional, paying 4.00% fixed annually and receiving the floating rate.

The notional is never exchanged. In a year when the average floating rate sets at 4.60%, the bank receives

$$100{,}000{,}000 \times (0.0460 - 0.0400) = \$600{,}000$$

and in a year when floating sets at 3.50% it pays $500,000. The annual cash flows are measured in hundreds of thousands on a $100 million "position."

Now suppose rates move sharply and the swap's market value rises to $4 million in the bank's favor. That $4 million—not $100 million—is what the bank would lose if the counterparty failed and nothing had been collateralized. If the same counterparty also owes the bank $1.5 million on one swap and is owed $3 million on another, netting under the master agreement reduces the exposure again, to $2.5 million.

So the same trade supports three "sizes": $100 million of notional, $4 million of market value, $2.5 million of net exposure, which after collateral very likely falls to a few hundred thousand.

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Each is the right answer to a different question, and confusing them is the single most common error in commentary about this market.

Case Study 1.1 Why the headline number will not go away

Notional outstanding is a poor measure of risk, and everyone who works in the market knows it. It nevertheless remains the number that gets quoted, for two reasons worth understanding.

The first is that it is the only figure that can be reported consistently. Market value depends on the valuation date and on models; notional is written into the contract. The second is that notional is not *meaningless*: it scales operational load, and it is the base on which several regulatory measures are calculated. Two portfolios with the same market value but very different notionals are not equally easy to manage, close out, or transfer.

Whenever a derivatives figure is quoted, ask which of the three it is. If the source does not say, it is almost certainly notional, and almost certainly being used to suggest something the number does not support.

What the market is made of

The composition is as lopsided as the size is large. Figure 1.4 gives OTC notional by asset class at the end of 2025.

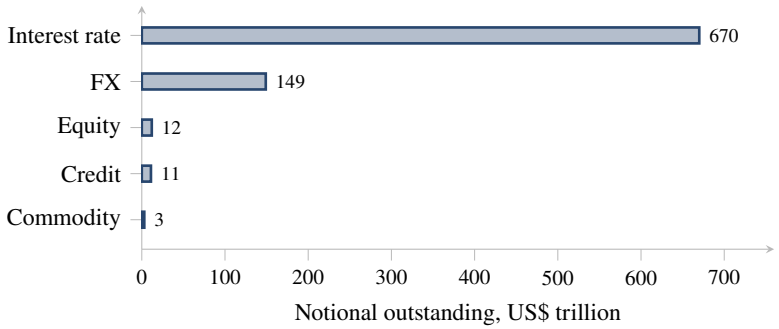


Figure 1.4: OTC notional by asset class, end of 2025 (BIS).

Interest rate contracts are roughly 80% of the total and foreign exchange most of the rest. Credit, equity, and commodity derivatives together are a rounding error on notional—which is precisely why notional misleads, since credit derivatives were at the center of a financial crisis while representing about 1% of it. This lopsidedness explains the shape of this book: Chapters 4 through 9 and 28 through 33 concern rates, because that is where the market is.

Exchange-traded volumes tell a different and more recent story. In 2024 the Futures Industry Association counted a record 206.5 billion futures and options contracts traded worldwide, and the largest single venue was India's National Stock Exchange. In 2025 the count fell 42% to 119.3 billion, most of the fall in equity contracts, though Asia–Pacific exchanges still accounted for well over half of the total. Contract counts are not comparable to notional (an Indian index option and a short-term interest rate futures contract on $1 million differ in size by orders of magnitude), and a swing of that size in a single year is a warning against reading a trend off them. Measured by *activity*, the center of gravity in listed derivatives has nonetheless moved toward Asia, and toward small-size, short-dated, retail-accessible contracts. Chapter 10 returns to this, including the growth of options expiring the same day they are traded, which now account for a large share of US index option volume.

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1.7 WHAT CAN BE AN UNDERLYING

The five asset classes of Figure 1.4 account for almost all the outstanding notional, but they do not exhaust what a derivative can reference, and the exceptions are instructive about why derivative markets succeed or fail.

Weather. The Chicago Mercantile Exchange (CME) lists futures and options on heating and cooling **degree days** for major cities, settling against temperature data published by a national meteorological service. The index is the number of degrees by which each day's average temperature falls below (heating) or rises above (cooling) 65°F, summed over a month or season. An energy utility's revenue falls in a mild winter; a weather contract pays when the winter is mild. The market is small but it works, because it meets the first condition for any derivative market: the settlement index is public, verifiable, and impossible for either party to influence.

Freight. Forward freight agreements, which fix the cost of carrying a cargo on a given route, settle against published shipping route indices. Their users, ship owners with capacity to fill and charterers with cargo to move, have genuinely opposite exposures, which is the second condition for a market to function.

Emissions. Futures on European Union allowances (permits to emit one metric ton of carbon dioxide under the EU's emissions trading system) are now among the more liquid commodity contracts in Europe. Here the underlying exists only because policy created it, which makes the contract's value dependent on the durability of the policy—a risk with no analog in a wheat contract.

Volatility. A derivative can reference the market's expectation of future variability rather than any price. The Cboe Volatility Index (VIX) measures the volatility that S&P 500 option prices imply for the next 30 days; VIX futures, listed in 2004, and options on them made volatility an asset class in its own right; Part IV is devoted to it.

Cryptoassets. CME listed bitcoin futures in December 2017 and ether futures subsequently. After US spot bitcoin exchange-traded funds (ETFs) were approved in January 2024, listed options on those funds followed later that year, adding options on a spot product to the options on bitcoin futures that CME had listed since 2020. The instruments are entirely conventional; what is novel is the underlying and the extreme volatility that pricing must accommodate.

Other derivatives. An option on a futures contract, or a swaption (an option to enter a swap), references a derivative rather than an asset. These are not curiosities: swaptions are among the largest option markets in the world, and Chapter 29 prices them.

What makes a derivative market possible. The list suggests three conditions, and they explain the failures as well as the successes. There must be an *observable, verifiable settlement reference* that neither party can manipulate. There must be participants with *genuinely opposite exposures*, so that trade is risk transfer rather than a wager between two people who both merely have opinions. And the exposure must be *material enough to be worth managing*, since a contract no one needs will not attract the liquidity that makes it usable.

Proposals for derivatives on macroeconomic aggregates (GDP growth, inflation-adjusted national income, house price indices for individual homeowners) have repeatedly satisfied the first condition and failed the second. Almost everyone is exposed to a recession in the same direction, so there is no natural party to take the other side, and the contracts have not sustained liquidity despite decades of advocacy. When a proposed contract seems obviously useful and does not exist, the missing party is usually the explanation.

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1.8 WHO USES DERIVATIVES, AND WHAT FOR

Textbooks traditionally divide participants into hedgers, speculators, and arbitrageurs. The taxonomy is useful and Section 1.9 takes it seriously, but on its own it tells a reader almost nothing about who is actually on the other side of a trade. Table 1.1 is more concrete: real institution types, the exposure each has, and the instrument each reaches for.

Table 1.1: Who trades what, and against which exposure.

Participant	Exposure	Typical instrument
Airlines, shippers	Fuel cost	Crude and refined product swaps, options
Exporters, importers	Receivables and payables in foreign currency	FX forwards, currency options
Corporate borrowers	Floating-rate debt service	Pay-fixed swaps, interest rate caps
Bond issuers	Locking a coupon before issue	Forward-starting swaps, swaptions
Pension funds, insurers	Long-dated liabilities discounted at market rates	Receive-fixed swaps, gilt and Treasury futures
Asset managers	Portfolio duration and equity beta	Bond and index futures
Banks (loan books)	Concentrated credit exposure	Credit default swaps, index tranches
Dealers	Inventory acquired from clients	Whatever offsets it, plus the futures strip
Hedge funds	Deliberately acquired relative-value positions	Everything in this book

Two observations about the table matter more than its contents. First, most of these participants are *removing* an exposure they acquired by doing something else: flying aircraft, writing insurance, lending money. Derivatives markets exist mainly to serve them, and the profits earned by the other side are, in the aggregate, the price those participants pay for the service. Example 1.3 follows one of them through a quarter.

Second, the dealer row is different in kind. A dealer does not begin with an exposure; it acquires one by accommodating clients, and then manages it. That is a business of intermediation, and understanding it explains a great deal that is otherwise puzzling about market prices: why a heavily demanded contract trades rich, why hedging costs rise when everyone hedges at once. This book returns to it repeatedly, and Chapters 8 and 9 are about what that intermediation costs.

Example 1.3 An airline hedging fuel

An airline expects to burn 12 million gallons of jet fuel next quarter. Fuel is roughly 25% of its operating cost, and a 30-cent move in the price per gallon changes quarterly fuel spend by $3.6 million, enough to swing a thin operating margin to a loss.

There is no liquid jet fuel futures contract, so the airline hedges with heating oil futures, whose underlying is a similar refined product. Each contract covers 42,000 gallons. Jet fuel has historically moved about 0.64 of a cent for each cent of heating oil (Example 3.3 estimates that ratio), so the airline buys

$$\frac{12{,}000{,}000}{42{,}000} \times 0.64 = 182.9 \rightarrow 183 \text{ contracts}$$

Suppose jet fuel then rises 30 cents a gallon and heating oil futures rise 34 cents. The airline pays $3.6 million more for fuel and gains

$$183 \times 42{,}000 \times 0.34 = \$2{,}613{,}240$$

on the hedge, leaving about $0.99 million of the increase uncovered. Jet fuel rose 30 cents for

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every 34 of heating oil, a ratio of 0.88 against the 0.64 the hedge was sized on, so in this quarter the hedge was too small. The ratio is set on the average relationship, and in any one quarter the realized ratio can sit well above or below it.

Three things in this example run through the whole book. The hedge is *approximate*, because the traded instrument is not the exposure (the cross-hedging problem of Section 3.5). The residual is not small relative to the margin being protected. And the gain on the futures leg arrives as cash, daily, whether or not the fuel has yet been bought—which is a benefit here and, in Chapter 3's failure cases, a catastrophe.

1.9 HEDGING, SPECULATION, AND ARBITRAGE—AND WHY THE LABELS BLUR

With the participants in view, the traditional triad can be stated properly. A **hedger** holds an existing exposure and enters a derivative that offsets it. The test is not the trade but the portfolio: the same short futures position is a hedge for a producer and a directional bet for someone with no production.

A **speculator** takes a position to profit from a view. Speculators are frequently described apologetically, as though merely tolerated, and that is a mistake. Hedgers rarely want to shed opposite exposures in the same amount at the same moment, and someone must take the imbalance; a market of hedgers alone would seldom clear. What speculation supplies is the willingness to hold an exposure no one else wants, and the price of that service is a real cost to hedgers, as Section 5.14 shows when it works out who pays whom.

An **arbitrageur** exploits a price inconsistency between related instruments, in principle without taking risk. This is the category that requires the most caution, and Section 5.15 is devoted to why: the textbook arbitrage assumes unlimited borrowing at a single risk-free rate, and the real trade requires a balance sheet, financing that may be withdrawn, and the ability to survive the position moving against you before it converges. Covered interest parity, the rule that a forward exchange rate equals the spot rate adjusted for the two currencies' interest rates, is violated persistently in the deepest currency markets in the world, and the violations exist because that assumption is false. Example 1.4 applies all three labels to a single trade.

Example 1.4 One trade, three labels

A trader buys 100 Treasury bond futures contracts. Which of the three is it?

If the trader runs a pension fund whose liabilities are longer-dated than its bonds, so that a fall in yields would raise the liabilities by more than the assets, it is a hedge: the futures offset an exposure the fund already carries.

If the trader believes long yields will fall and holds nothing else, it is speculation.

If the trader has computed that the futures are cheap after carry (the bond's coupon income less the cost of financing it) relative to the cheapest-to-deliver cash bond (the bond a futures seller would choose to deliver), and simultaneously sells that bond short, borrowing it through a reverse repo (lending cash against the bond as collateral; Section 4.1 describes the repo market), it is arbitrage: the basis trade of Section 6.5, run in reverse.

The three are the same instrument, the same direction, and the same quantity. Only the rest of the balance sheet distinguishes them, which is why regulators, risk managers, and accountants all find these categories so difficult to apply, and why a firm's own description of a position is weak evidence about what it is. Chapter 3 describes a firm whose "hedge" was economically sound and failed anyway, and Section 5.14 describes a fund whose "spread trade" carried more risk than an outright position would have.

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1.10 LEVERAGE

The feature that makes derivatives efficient is the same one that makes them dangerous, and it is worth isolating. To gain $10 million of exposure to the S&P 500 by buying shares requires $10 million. To gain the same exposure through index futures requires posting initial margin, a good-faith deposit sized to cover a few days of adverse moves, and returned when the position closes. If that margin is 5% of notional, the same exposure costs $500,000 to establish. Example 1.5 sets this route beside three others.

Table 1.2: $10 million of index exposure, four ways.

Route	Cash required	Loss on 10% fall	% of cash
Buy the shares	$10,000,000	$1,000,000	10%
Buy on 50% margin	$5,000,000	$1,000,000	20%
Index futures (5% IM)	$500,000	$1,000,000	200%
At-the-money call options	$600,000	$600,000	100%

Example 1.5 The same exposure, four ways

A fund wants $10 million of exposure to an equity index. Table 1.2 shows what each route requires and what a 10% market fall does to the capital committed.

The futures row is the one to study. A 10% fall produces a loss of twice the cash committed. The position is not merely wiped out—the fund owes $500,000 more, and will be asked for it in cash the following morning as variation margin, the daily settlement of gains and losses that Section 1.1 describes. Initial margin is not the maximum loss; it is a deposit, and Chapter 2 explains what happens when it is exhausted.

The option row (calls whose strike equals the current index level, which is what *at the money* means) differs in kind rather than degree. The premium is the maximum loss in every state of the world, which is why Section 12.10 argues that buying options is sometimes the right answer even when it looks expensive. The cost of that protection is the $600,000 premium, which is lost entirely if the index is unchanged.

Table 1.2 also explains a structural fact about the market: derivatives let a position be built that no lender would finance directly. A firm that wanted $10 billion of equity exposure would find it difficult to borrow $10 billion; it can obtain the same exposure through total return swaps with several dealers, none of whom sees the whole picture. Whether the resulting position is prudent is a separate question from whether anyone can observe it, and Archegos Capital Management showed in March 2021 what happens when the answer to both is no. It built concentrated positions in a few US and Chinese stocks almost entirely through total return swaps with several prime brokers, the banks that finance and execute trades for hedge funds and similar clients. The banks owned the shares, so Archegos filed no ownership disclosures, and each broker saw only its own exposure, with no way to know that competitors had written substantially the same trade against the same client. When the stocks fell, margin calls arrived at every broker at once, Archegos could not meet them, and the banks' simultaneous liquidations drove prices further down; the banks lost more than $10 billion between them. Section 27.14 works the numbers and Section 40.6 tells the episode.

1.11 WHAT GOES WRONG

Derivatives have a reputation for producing spectacular losses, and it is deserved. What is less well understood is that the losses follow a small number of repeated patterns. Table 1.3 lists the episodes this book returns to, and Chapter 40 takes them up together.

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Table 1.3: Episodes and the mechanism each illustrates.

Episode	Year	Mechanism
Metallgesellschaft	1993	A long-dated hedge sound in value but funded with short-dated futures; rolls and margin calls exhausted liquidity first (Chs. 3, 40)
Barings	1995	Unsupervised trader, unreconciled accounts; the loss was in plain futures and options (Ch. 40)
Long-Term Capital Management	1998	Convergence trades that were right eventually, financed in a way that could not survive being wrong first (Ch. 40)
Amaranth	2006	Calendar spreads with low margin and no exit; size itself was the risk (Chs. 5, 40)
Global financial crisis	2007–09	Credit derivatives on correlated collateral, valued with models calibrated on a benign sample (Chs. 24–26)
Negative WTI settlement	2020	Retail products whose design assumed prices could not go below zero (Chs. 5, 36)
Archegos	2021	Concentrated leverage invisible across dealers (Chs. 27, 40)
LME nickel	2022	A short-covering spike that broke the clearing house's own rules (Chs. 2, 40)
European energy	2022	Solvent hedgers unable to fund variation margin (Chs. 8, 40)
UK pension LDI	2022	Leveraged rate hedges, held in liability-driven investment mandates, liquidated into the market they were hedging against (Chs. 8, 40)

Read down the mechanism column and a pattern emerges that is not the expected one. Very few of these were valuation failures. The models mostly worked. What failed was *funding*: a position that was economically correct required cash at a moment when cash was unavailable, and had to be closed at the worst possible time.

This is the book's most persistent theme. A hedge that is right about price can still fail through liquidity. That sentence would have been an odd thing to build a derivatives textbook around in 2005. After 2008, 2020, and 2022, it is the central practical lesson of the subject, and it is why Chapter 8 (collateral, margin, and funding) exists as a chapter in its own right rather than as an appendix to the swaps chapter.

The second pattern is that scale and invisibility travel together. Barings, Amaranth, and Archegos each involved a position that had grown too large to exit, and in each case those who could have stopped it never saw its full size: senior management at Barings, the New York Mercantile Exchange, which saw only the part of Amaranth's position held there, and the dealers facing Archegos, each of which saw only its own slice.

1.12 REGULATION SINCE 2008

The market described in this book is substantially a creation of policy decisions taken after 2008, and a reader who learned the pre-crisis structure would find much of current practice inexplicable. At the Pittsburgh summit in September 2009, the G20 agreed four commitments for OTC derivatives:

- standardized contracts should be centrally cleared;
- they should trade on exchanges or electronic platforms where appropriate;
- all contracts should be reported to trade repositories;
- non-centrally-cleared contracts should attract higher capital.

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Implementation took over a decade and reshaped the market:

Clearing mandates moved the great majority of interest rate and index credit derivatives to CCPs, which is why Chapter 2 treats clearing houses as systemically important institutions rather than as infrastructure.

Trade reporting created the data underlying the statistics of Section 1.6, and, less happily, a compliance burden that consolidated dealing into fewer firms.

Uncleared margin rules, phased in from 2016 and completed in September 2022, require two-way initial margin on bilateral trades between large participants. This changed pricing, not merely operations: posting initial margin has a funding cost, which is the margin valuation adjustment (MVA) of Chapter 9.

Capital rules (the credit valuation adjustment charge, the standardized approach for counterparty credit risk, and the leverage ratio) made dealer balance sheets an explicit input into derivative prices. Section 5.15's argument that arbitrage requires a balance sheet is in large part an argument about these rules.

Who writes the capital rules is worth stating once, because every later chapter assumes it. The **Basel Committee on Banking Supervision**, a body of central banks and supervisors hosted at the Bank for International Settlements, agrees standards (Basel I in 1988, Basel II in 2004, and the post-crisis Basel III), which have no legal force until each jurisdiction writes them into its own rules, as the European Union, the United States, the United Kingdom, Canada, and China separately do. A bank must then hold capital, meaning common equity first and other loss-absorbing instruments after it, of at least 8% of its risk-weighted assets (each exposure multiplied by a weight meant to reflect its riskiness), plus buffers that take the requirement above 10%, and separately at least a few percent of its *unweighted* exposure under the leverage ratio. Everything Chapters 9, 26, 27, and 39 say about capital is an elaboration of those two ratios, Section 27.11 assembles them for a whole bank, and the divergences those chapters report (Europe exempting a charge the United States does not) come from the national step rather than from the accord.

Two developments are more recent and still unfolding. Benchmark reform retired LIBOR in favor of overnight risk-free rates, an event that changed the foundations of interest rate derivative pricing rather than its details; Chapter 4 is written from the new foundation rather than adapting the old one. And clearing continues to expand: the mandate from the Securities and Exchange Commission (SEC) to centrally clear US Treasury cash and repo transactions, adopted in December 2023 and phasing in over the following years, extends the post-crisis model to the largest government bond market in the world. Section 2.11 takes it up, with its compliance dates, and Section 6.5 its effect on the basis trade.

1.13 HOW THE CHAPTERS FIT TOGETHER

The book serves three audiences with different needs, and the shape it takes is the same for all three. Part I (Chapters 1–9) covers markets, forwards, futures, swaps, and the collateral and valuation machinery that modern practice requires. Part II develops options mechanics and strategies, and Part III their valuation, from the binomial tree through Black–Scholes–Merton, the Greeks (the sensitivities of an option's value to its inputs), and the volatility surface. Part IV treats volatility as an asset class. Parts V–IX cover credit, interest rate derivatives in depth, exotics and structured products, commodity and digital markets, and risk and regulation.

Figure 1.5 is the map. Within a part the chapters run in order and each assumes the one before it; the arrows are the principal cross-part dependencies, and they are what a reader assembling a path actually needs.

Three chapters are shaded as hubs, because the rest of the book stands on them. Chapter 4's risk-free-rate conventions are assumed by every rates chapter that follows. Chapter 13's

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binomial tree is assumed by everything in valuation, exotics, and computation. And Chapter 15 is the gate between those and the rest: later chapters open by naming it as a prerequisite almost as often as they name Chapter 13.

A hub is a chapter you must pass through, not one you keep turning back to, and the distinction is worth drawing because the two rankings disagree. By the number of later chapters that cite it at all, the busiest chapter in the book is Chapter 19, whose volatility surface is referred to from most of the chapters that follow it, but almost none of them need it as a prerequisite. Chapter 15 is the reverse: cited less often, and skippable by no one.

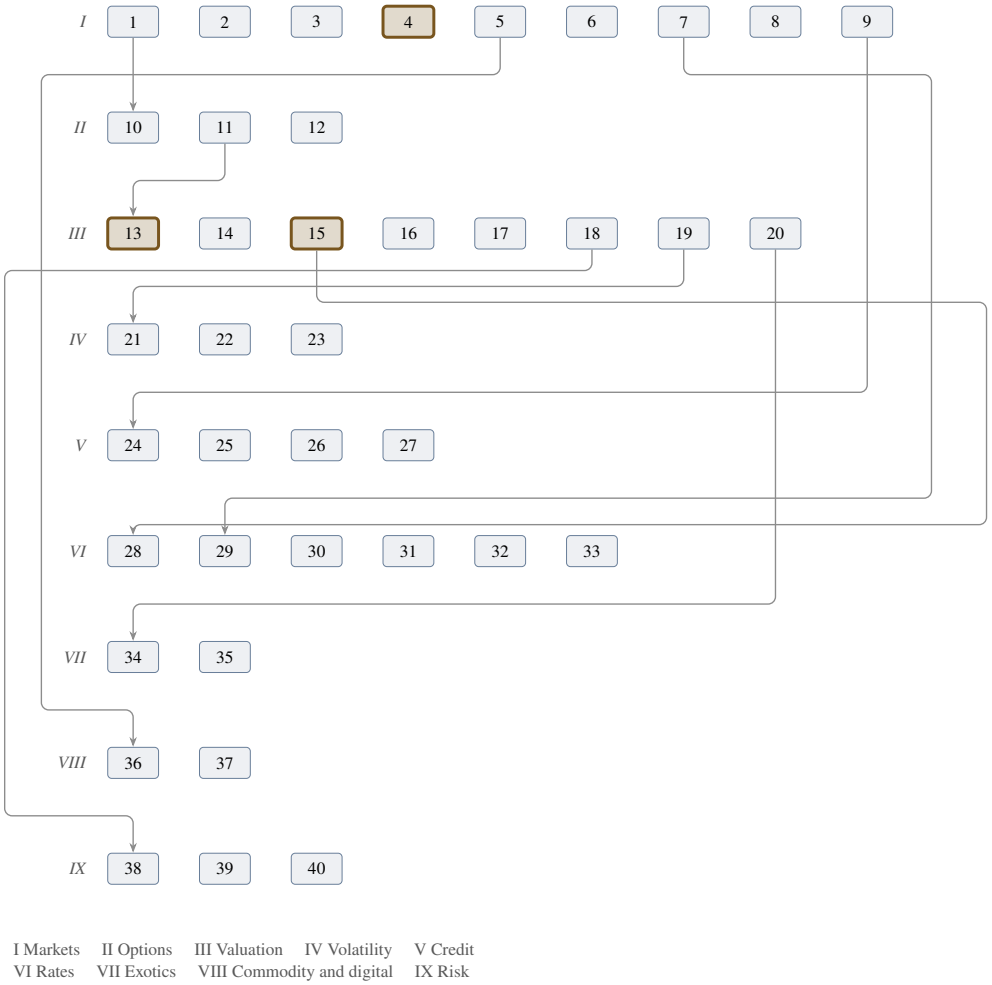


Figure 1.5: Chapters run in order within a part; arrows are the principal cross-part dependencies. Shaded chapters are the three hubs.

SUMMARY

A derivative is a contract whose value depends on something else, and its economic function is to separate exposure to a risk from ownership of the thing that carries it.

Four building blocks account for nearly everything: forwards, futures, swaps, and options. The first three are symmetric (both parties are obliged and the payoff is linear), while options are asymmetric, giving one party a choice and requiring a premium in exchange. The distinction

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organizes the whole subject.

Markets are exchange-traded, bilateral OTC, or—increasingly—bilaterally negotiated and centrally cleared. The third category is a post-2009 creation and now holds most of the outstanding interest rate risk in the world.

The market's size is routinely misstated. Notional outstanding is the figure usually quoted, but gross market value is a small fraction of it and exposure after netting a smaller fraction still, most of it collateralized. Interest rate contracts make up the large majority of the notional total, which is why this book weights them heavily.

Participants are better described by what they do than by the hedger–speculator–arbitrageur triad, though the triad remains useful once it is understood that the same trade falls into different categories depending on the rest of the balance sheet. Dealers are a distinct case: they do not begin with an exposure but acquire one by serving clients, and the cost of that intermediation shows up in prices.

Leverage is the source of both the efficiency and the danger. Initial margin is a deposit, not a maximum loss; a futures position can lose several times the cash committed, while a purchased option cannot lose more than its premium.

Most derivative disasters were not valuation failures. They were funding failures: positions that were economically correct but required cash when none was available. That theme recurs throughout the book and justifies the prominence given to collateral and margin.

FURTHER READING

The BIS publishes semiannual OTC derivatives statistics and quarterly exchange-traded data; they are the authoritative source for Section 1.6 and are worth consulting directly, since the definitions matter as much as the numbers. The Futures Industry Association's annual volume survey covers listed markets. For the post-crisis reform program, the Financial Stability Board's annual progress reports on OTC derivatives reform track what was promised against what was delivered. On Archegos, the Credit Suisse board's external investigation report sets out how the exposure went unmeasured.

PRACTICE QUESTIONS

1.1 Explain why a forward contract has zero value at inception but an option does not, and what this implies about when money changes hands in each.

1.2 A company enters a \$50 million notional five-year swap paying 3.8% fixed and receiving floating. In the first year floating averages 5.1%. Compute the net payment and state who pays whom. What fraction of notional is it?

1.3 Distinguish notional amount, gross market value, and gross credit exposure. A newspaper reports that a bank "has \$3 trillion of derivatives." What further question should a reader ask?

1.4 An investor obtains \$4 million of index exposure through futures requiring 6% initial margin. The index falls 12%. Compute the loss, express it as a multiple of the initial margin, and state what the investor owes.

1.5 Explain why a market consisting only of hedgers would not function, and identify what speculators are being paid for.

1.6 Classify each as hedging, speculation, or arbitrage, and state what additional information you would need in each case: (a) a copper producer selling copper futures; (b) a fund buying gold futures and selling gold forward at a higher price; (c) an importer buying dollars forward; (d) a bank buying protection on a borrower to whom it has lent.

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1.7 Using Figure 1.1, explain why the writer of a call is in a different business from the buyer of one, and what the writer must be able to do to be in it safely.

1.8 Explain what changes when the same economic trade is executed on an exchange rather than bilaterally. Identify which of those changes transfers risk to someone else and which merely changes when cash moves, and state one exposure that the exchange does not remove.

1.9 Using Example 1.1, explain what the single agreement clause prevents an administrator from doing in a bankruptcy, and use it to account for the fall from $23 trillion of gross market value to $3.4 trillion of gross credit exposure.

1.10 Archegos, as this chapter describes it, held a position no counterparty could see in full. Identify the instrument that made it invisible, state what each prime broker could observe and what it could not, and explain what would have had to change (in the instrument, the reporting, or both) for the aggregate position to be visible to any one of them.

FURTHER QUESTIONS

1.11 Table 1.3 attributes most of the listed episodes to funding rather than valuation failure. Choose two episodes and explain precisely what cash was required, when, and why it was not available. Then state what a risk report would have had to measure to give warning.

1.12 Central clearing was intended to reduce systemic risk by replacing a web of bilateral exposures with a hub. Construct an argument that it instead concentrates risk, and identify what would have to be true for the concentration to be worth accepting. Chapter 2 gives the full treatment; attempt it first.

1.13 In the Archegos episode, each dealer's individual exposure was defensible. Explain how a set of individually prudent positions produced a collectively imprudent one, and evaluate whether a disclosure rule attaching to economic exposure rather than share ownership would have prevented it.

1.14 An airline hedges fuel and the fuel price then falls, so the hedge loses money and the airline's unhedged competitor reports lower costs. Construct the argument that the hedging program should be continued, and the argument that it should not. Which turns on facts about the airline rather than about fuel?

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Chapter 2
Futures Markets, Clearing, and Central Counterparties

A forward contract is a promise. Its value depends entirely on the promisor being willing and able to perform, years after the promise was made, at a price that may by then look catastrophic. Futures markets exist to remove that dependence, and the machinery they use to do so—standardization, daily settlement, and a central counterparty standing between every buyer and seller—is the subject of this chapter.

That machinery works well enough that it is easy to take for granted. This chapter deliberately does not. The events of 2022 turned questions that had been footnotes in earlier textbooks into first-order concerns: how much margin a clearing house should demand, what happens when it demands too much too fast, and who absorbs the loss if a large member fails. Sections 2.8 through 2.10 take them up directly.

Sections 2.1 and 2.2 are the contract: what a futures specification fixes and why the futures price converges to spot. Sections 2.3 to 2.6 are the machinery: margin and daily settlement, the central counterparty, clearing members and their clients, and delivery. Section 2.7 covers regulation and manipulation; Sections 2.8 to 2.10 are the three questions above; Section 2.11 is the OTC markets, cleared and bilateral; Section 2.12 is who trades and how orders reach the market; and Section 2.13 closes by setting forwards against futures.

2.1 FUTURES CONTRACTS AND THEIR SPECIFICATION

A **futures contract** is an agreement to buy or sell an asset at a specified future time for a price agreed today. What distinguishes it from a forward is that it is standardized and exchange-traded, which permits a liquid secondary market: a position can be closed by taking the opposite position rather than by negotiating with the original counterparty.

Standardization means the exchange, not the parties, specifies everything except the price:

- **The asset.** For commodities, a grade and a set of permitted substitutions with price adjustments. This is not a detail, since a contract's deliverable grade determines what it actually tracks.
- **The contract size.** Too small and commissions dominate; too large and smaller hedgers are excluded. Exchanges sometimes list "mini" contracts for this reason.
- **Delivery arrangements.** Where delivery may be made, and for financial contracts whether delivery occurs at all or the contract is cash-settled against a published index.
- **Delivery months and the last trading day.**
- **Price quotation, tick size (the smallest permitted price change), and on some contracts daily price limits and position limits.**

Table 2.1 shows the specification of one such contract.

Two entries repay attention. The *deliverable grade* determines what the contract tracks: a producer whose output differs from the deliverable grade is exposed to the difference, which is

Table 2.1: The light sweet crude oil futures contract. Terms as of press date.

Exchange	New York Mercantile Exchange
Ticker	CL
Contract size	1,000 barrels
Deliverable grade	West Texas Intermediate light sweet crude, with listed substitutable grades
Delivery location	Cushing, Oklahoma
Delivery months	Every month, several years forward
Price quotation	US dollars and cents per barrel
Tick size	$0.01 per barrel, or $10 per contract
Last trading day	Third business day before the 25th of the preceding month
Position limits	Set by the exchange, tightening near delivery

the basis risk of Chapter 3. And the *delivery location* matters as much as the grade: a contract deliverable at an inland hub prices differently from one deliverable at a port, because the two are connected only by pipeline capacity that can bind. Table 2.2 shows that crude oil contract once it is trading.

Table 2.2: Quotes by delivery month. Illustrative, not market data.

Delivery	Settle ($/bbl)	Change	Volume	Open interest
December	78.29	+0.24	412,880	128,450
January	78.59	+0.21	198,340	265,910
February	78.88	+0.18	76,220	142,300
March	79.18	+0.16	41,150	98,660
June	80.07	+0.11	12,480	54,120
December (next)	81.90	+0.07	6,310	71,940

Five things can be read off it, and most are taken up again later:

- *The settle column is a forward curve*: the price of the same asset for delivery at each of six dates, generated here from Chapter 5's cost-of-carry relation at a spot price of $78.00 a barrel and a net carry rate of 4.5% a year, continuously compounded, with the six contracts delivering 1, 2, 3, 4, 7 and 13 months out. It is the object Section 2.12 says these markets exist to produce. It slopes upward here because storage and financing costs exceed any convenience yield (the benefit of holding the physical commodity rather than a claim on it), which is the storable-commodity case of Section 2.2.
- *Volume and open interest are different quantities and peak in different months*: December trades most because positions are being rolled out of it (closed in the expiring month and reopened in the next), while January holds the most open interest because that is where they are going.
- *Liquidity thins with maturity*, which is why the stack-and-roll hedge of Chapter 3 exists at all: a hedger wanting ten-year protection cannot simply buy it in the ten-year contract, so it holds the whole exposure in the liquid near contract and rolls it forward as each one expires.
- The December contract a year out carries more open interest than the nearer June, because annual hedging programs settle on December dates.
- The tick from Table 2.1 converts the change column into money: December's $0.24 is $240 a contract.

Price limits and position limits. A **daily price limit** halts trading if the price moves more than a specified amount from the previous settlement. The intention is to prevent large moves driven by disorderly trading. The effect is contested: a limit does not remove the information

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causing the move, and a market that is limit-locked is one in which no one can trade at all, which converts a price problem into a liquidity problem. **Position limits** cap the contracts a speculator may hold, to reduce the scope for cornering the market, that is, holding a position large enough that the shorts must buy back from the party that also controls the deliverable supply, which Section 2.7 takes up. They typically tighten as delivery approaches, when the deliverable supply is most easily squeezed.

Volume and open interest

Two quantities are reported for every contract and are constantly confused. **Volume** is the number of contracts traded during the day. **Open interest** is the number of contracts outstanding at its end, counted on one side only, since every long is matched by a short.

They differ because a trade may open positions, close them, or transfer one from one holder to another. Only the first and second change open interest.

Example 2.1 Volume and open interest

Trace a newly listed contract over four days.

Day	Trade	Volume	Change	Open interest
1	A opens long 10; B opens short 10	10	+10	10
2	C opens long 5; D opens short 5	5	+5	15
3	A closes 4; C opens 4	4	0	15
4	A closes 6; B closes 6	6	−6	9

On day 3 one participant left and another arrived, so the number of outstanding contracts was unchanged: the position was transferred, not created or extinguished. On day 4 both sides closed, so six contracts ceased to exist.

The distinction carries information. Rising prices on rising open interest suggest new money entering on the long side; rising prices on *falling* open interest suggest shorts closing out, which is a different phenomenon and typically less persistent.

2.2 CONVERGENCE TO THE SPOT PRICE

As delivery approaches, the futures price converges to the spot price. The argument is a simple arbitrage.

Suppose during the delivery period the futures price is above the spot price. A trader can sell the futures, buy the asset, and deliver immediately, earning the difference. This is riskless and will be done until the gap closes. If instead the futures price is below the spot price, those who wish to acquire the asset will buy futures and wait for delivery, pushing the futures price up.

Neither argument requires any view about where the price is going. It follows that at expiration the two prices must be essentially equal, and the futures contract can be treated as a claim on the spot asset at that date.

The futures price may approach the spot price from above (the normal pattern for a storable commodity, where the futures price includes storage and financing costs that shrink as delivery nears) or from below, which happens when holding the physical asset conveys a benefit that the futures contract does not. Chapter 5 develops what determines the gap *before* expiration; the point here is only that it must close.

What converges is the gap, not the prices. Spot and futures prices both move throughout, and neither approaches delivery along any predictable line. The gap between them (the spot price minus the futures price, called the basis) *narrows to zero anyway*, because the arbitrage that forces it does not care about the route.

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The convergence is what makes a futures contract useful as a hedge at all. A hedger who holds the physical asset and shorts futures relies on the two prices ending up together; if they did not, the hedge would leave a residual whose size was unknown in advance. Chapter 3 examines what happens when the hedge is lifted *before* delivery, when convergence is incomplete and the residual is exactly the basis risk. Figure 3.1 draws the convergence with that residual marked.

2.3 MARGIN AND DAILY SETTLEMENT

The exchange's central problem is that a futures contract can accumulate large losses for one side over its life, and the losing party may prefer to walk away. The solution is not to check creditworthiness but to make the question irrelevant: **losses are collected daily, in cash, before they can grow**.

The margin account. On opening a position, a trader deposits **initial margin** with the broker. At the end of each trading day the account is adjusted for that day's gain or loss, called **marking to market**, or **daily settlement**. Gains are credited to the account, and any balance above the initial margin may be withdrawn; losses are debited from it.

The balance need not stay at the initial margin. It may fall as far as a lower threshold, the **maintenance margin**, before anything happens; only if it falls below that does the trader receive a **margin call**, and must then restore the balance to the *initial* margin level. If the call is not paid, the broker closes the position. The daily gains and losses themselves, settled in cash, are the **variation margin**; older texts give that name to the deposit that answers a call, and Example 2.2 shows why the two are better kept apart.

What margin is posted in. Initial margin need not be cash. Clearing houses also accept Treasury bills and other government securities, valued after a haircut (a deduction from their market value; Section 4.1), and many hedgers post them because a bill keeps earning its yield while it sits in the account. Cash may earn interest too, but at a rate the broker or clearing house sets, usually below the overnight benchmark, so on a large position the forgone interest is a real cost of hedging. Variation margin is different. It is settled in cash every day, and the interest earned on gains or paid to finance losses is what can make a futures price differ from a forward price (Section 5.12).

Example 2.2 Operation of a margin account

An investor buys two gold futures contracts, each for 100 ounces, at a futures price of \$3,200 per ounce, a position of 200 ounces and \$640,000 of notional. The clearing house sets maintenance margin at \$15,000 per contract and initial margin ten percent above it at \$16,500, so the investor posts \$33,000 and must keep the account above \$30,000. The initial requirement is about 5% of notional, the order of magnitude a liquid futures contract attracts.

Day	Futures price	Daily gain	Cumulative gain	Balance	Margin call
0	3,200.00	—	—	33,000	—
1	3,188.00	−2,400	−2,400	30,600	—
2	3,176.00	−2,400	−4,800	28,200	4,800
3	3,168.00	−1,600	−6,400	31,400	—
4	3,190.00	+4,400	−2,000	35,800	—

On day 2 the balance of \$28,200 falls below the \$30,000 maintenance level, so the investor must deposit \$4,800 to restore the \$33,000 initial level. Day 3 therefore starts from \$33,000 and closes at \$31,400, still above maintenance, so no further call; on day 4 the price recovers to leave \$35,800.

Check the arithmetic overall: the price fell \$10 from 3,200 to 3,190, a cumulative loss of $10 \times 200 = \$2{,}000$. The investor deposited $33{,}000 + 4{,}800 = \$37{,}800$, and $37{,}800 - 2{,}000 =$

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$35,800.

The daily gain column is the variation margin. Each entry is the day's mark-to-market, debited from the account when the position loses and credited when it gains, and it is paid every day whether or not any call is made. The $4,800 on day 2 is not variation margin but a top-up of *initial* margin. The two meet in the balance column because a futures account runs them through one balance; Section 8.2 separates them. The separation matters most to a hedger: variation margin on the futures leg must be paid in cash every day, even while the position it hedges gains as much on paper, so a hedge that is working economically can still run its owner out of cash.

Two further features deserve attention. The investor was called for cash on day 2 even though the position was still open and might recover, as it did. And the $2,800 standing above the initial requirement on day 4 could be withdrawn immediately. Futures positions consume and release cash continuously, which is a different property from owning the underlying asset.

What the daily settlements add up to. The example's cross-check is an instance of a general fact. A position opened at F_1 and closed at F_2, settled every day in between, collects

$$F_2 - F_1 \text{ per unit if long}, \qquad F_1 - F_2 \text{ if short}, \tag{2.1}$$

whatever route the price took in between. Each intermediate price enters two consecutive settlements with opposite signs and cancels, so the sum telescopes to the difference between the first and last. Equation (2.1) is what allows a futures position to be treated as a single payoff at the date it is closed, which is how Chapter 3 reads a hedge and how Chapter 5 prices one. What does *not* cancel is the timing: the same total arrives as a stream rather than a lump, which is the subject of Section 2.9 and of Chapter 8, and the reason a futures price and a forward price are not quite the same number.

Why this matters beyond the mechanics. Example 2.2 looks like bookkeeping, and for a two-contract position it is. Scaled up, it is the mechanism behind several of the episodes in this book. A hedger whose futures leg loses money must find cash *today*, while the offsetting gain on the hedged item may not be realizable for years. The hedge is economically sound and the firm can still fail. Chapter 3 describes this happening to Metallgesellschaft; Section 2.9 describes it happening across an entire market in 2022.

And what it does to a speculator. Read the other way, the same table is the risk in a futures position taken for a view. Example 2.2 commits $33,000 against $640,000 of exposure, so gold need fall only 5.2% to consume the deposit, and past it the trader owes the rest of the move in cash the next morning; the option of Example 1.5, whose premium bounds the loss, differs in kind. The consequence is not that leverage is large but that it is *continuous*: a view expressed in futures must be solvent every day until it is vindicated, because a call unmet is a position closed by the broker at whatever price the market offers that morning. Section 40.4 is this arithmetic at institutional scale, and Chapter 38 explains why a risk measure built on the distribution of outcomes does not see it: the loss that matters is on the path, not at the horizon.

How initial margin is set

Example 2.2 took the margin requirement as given. How it is arrived at determines a great deal of what follows in this chapter.

Initial margin is intended to cover, at a high confidence level (typically 99% to 99.7%), the loss a position could suffer over the time it would take to close it out if the holder defaulted: one or two days for a liquid futures contract, five or more for a cleared swap portfolio. Clearing houses estimate the distribution of moves from historical data, often by simulating the portfolio's value under several years of observed daily moves and reading off the appropriate quantile. Strictly, this sets what a clearing member posts to the CCP. Exchange rules turn that figure

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into the customer's maintenance margin, with initial margin set above it (by ten percent in Example 2.2), and a broker may require more of its own clients.

These models have names, and a practitioner meets them before meeting any of the theory. The **Standard Portfolio Analysis of Risk** (SPAN) has been the long-standing framework for listed futures and options: it revalues the portfolio under a fixed grid of price and volatility scenarios and charges the worst outcome, which is why a margin requirement can be reproduced exactly from a published parameter file. Cleared swaps are margined instead by simulation across several years of historical moves, and the listed markets are migrating the same way. The simulation-based models are the historical simulation of Section 38.5 under another name, and the Standard Initial Margin Model (SIMM) of Chapter 8 is the bilateral cousin: the same problem, solved by an industry committee rather than by a clearing house.

The feature with the largest practical consequence is that margin is computed on the *portfolio*, not contract by contract. Positions that offset receive credit for it, as Example 2.3 shows.

Example 2.3 Portfolio margining and spread positions

A trader holds 100 long contracts in the March delivery month and 100 short in June. Taken separately, each leg might require \$2,000 per contract:

$$200 \text{ contracts} \times \$2{,}000 = \$400{,}000$$

But the two legs nearly offset: a parallel move in the underlying gains on one and loses on the other, and what remains is exposure to the *spread* between the two months, which is far less volatile. The clearing house might set the requirement for the paired position at \$200 per contract pair:

$$100 \text{ pairs} \times \$200 = \$20{,}000$$

a twentieth of the gross figure.

This is why calendar spreads, basis trades, and relative-value strategies generally can be run at leverage that would be impossible outright. Chapter 6 makes the point about the cash–futures basis trade, where a hedged position financed by repo (Section 4.1), with the lender withholding only a small haircut of the bond's value, runs at close to fifty times leverage.

The danger lies in the offset, which is granted on the assumption that the two legs move together. When they stop doing so (when the basis itself dislocates, as in March 2020), the margin credit is withdrawn at the same moment the position starts losing money. This is a more violent version of margin procyclicality, requirements rising as markets fall (Section 2.9).

2.4 CENTRAL COUNTERPARTIES

Daily settlement limits how large a loss can grow. It does not answer who bears the loss if a party fails to pay. That is the role of the **central counterparty**, or clearing house.

When a trade is cleared, the original contract between buyer and seller is replaced, or **novated**, by two contracts, each facing the CCP. The CCP becomes the buyer to every seller and the seller to every buyer. Neither original party has any further exposure to the other; both are exposed to the CCP.

Two things the rest of this chapter takes for granted follow from that substitution, and neither would exist without it. The first is **offset**. Section 2.6 observes that most positions are closed before delivery, and Example 2.1 watches open interest fall as traders do it, but closing out is only possible because the contract is with the CCP rather than with whoever sold it. A trader who is long sells to *anyone*, the CCP nets the two positions against each other, and the position is gone. Without novation the trader would hold two live contracts with two different parties, remain exposed to both, and be able to exit only by persuading the original counterparty to tear the trade up.

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The second is that the order book can be anonymous. Section 2.1 credits standardization with a liquid secondary market, and the reason is that any two participants wanting the same contract meet in the same place. That works only if neither has to price the other's credit before trading, and novation is what makes the contracts fungible: a March contract is one instrument whatever the identity of the seller. *The CCP is therefore not an adjunct to the futures market but a precondition for it*: liquidity, offset, and the daily settlement of Section 2.3 all rest on it.

The change in structure, from the web in the left panel of Figure 1.2 to the hub in the middle one, is not cosmetic. With n participants trading bilaterally there are up to $n(n-1)/2$ exposures (6 for the four participants drawn there, tens of thousands for a real market). Clearing replaces them with n. Three benefits follow.

Multilateral netting. A member long with one counterparty and short with another has no net position, and after novation no net exposure. Bilaterally it would have had two exposures.

Uniform, transparent margin. The CCP sets margin by rule rather than by bilateral negotiation, which removes the tendency for the strongest firms to demand collateral from the weakest while extending credit to each other.

Orderly default management. A CCP has a rehearsed procedure for closing out a failed member's positions, typically by auctioning the portfolio to survivors. In a bilateral market each of a failed firm's hundreds of counterparties scrambles simultaneously to replace its own hedge, which is what made September 2008 as disorderly as it was.

The netting benefit is the largest of the three and is worth computing rather than asserting, because its size is what justifies the concentration of risk that comes with it, and because the same arithmetic shows how easily it is given away.

Example 2.4 What multilateral netting is worth, and what fragmentation costs

Take five dealers, with every pair's obligations already netted bilaterally, so these ten figures are what a bilateral market leaves outstanding:

Owes		Owes	
A → B	120	C → A	80
A → D	45	C → D	90
B → C	60	D → B	110
B → E	25	D → E	70
E → A	30	E → C	55

Gross bilateral exposure is the sum of these, **685**, because each is owed to a different party and none offsets another. Novate all ten to one CCP and each dealer holds a single net position:

	A	B	C	D	E
Net position	−55	+145	−55	−45	+10

The five sum to zero, as they must. Total exposure is now the sum of the positive positions, **155**, 22.6% of the bilateral figure, a 77.4% reduction, and no netting agreement, collateral, or credit assessment produced it. It came from replacing ten counterparty relationships with five.

Now split the market. Suppose A, B, and C clear at one CCP, D and E at another, and the six trades that cross the boundary stay bilateral. Exposure rises to **485**, 70.8% of the bilateral figure, giving back 330 of the 530 that netting had saved. Fragmenting a cleared market across two CCPs destroys nearly two-thirds of the benefit of clearing it, and nothing either CCP does (margin, default fund, or waterfall) can win it back, because the lost offsets run between the two.

That arithmetic is the argument in every dispute about where a CCP should sit and about interoperability (links between CCPs that let members of one clear trades against members of another), and it explains why dealers resist being required to clear the same product in two places. The case for one CCP is quantitatively strong, and it is the same calculation that makes that CCP

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systemically important.

What clearing does not do. It does not eliminate risk; it relocates and concentrates it. Every participant now depends on one institution. That institution's resilience is therefore a matter of public interest, which is what the rest of this chapter is about.

2.5 CLEARING MEMBERS, CLIENTS, AND PORTING

Figure 1.2 implies that everyone faces the CCP directly. Almost no one does. Access is two-tier: a limited number of well-capitalized firms are **clearing members** of the CCP, and everyone else (asset managers, pension funds, corporates, hedge funds) clears *through* a member as a **client**. A client's positions reach the CCP only through its member, so if the member fails, what the client recovers turns entirely on how those positions and their collateral were held.

Segregation. Client collateral must be kept separate from the member's own. Under **individual segregation** each client's collateral sits in its own account; under **omnibus segregation** clients are pooled, which is cheaper and offers less protection, since a shortfall caused by one client in the pool can reach the others. Many clients do not know which they have.

What nets against what. A member holds more than one position at the CCP. Its own trades sit in a **house account**, where Example 2.4's netting applies in full; its clients' sit in separate accounts, and the two never net, which is what segregation means. Whether clients net against one another depends on how the account is margined: in a **gross** account the CCP computes initial margin client by client and adds the results; in a **net** omnibus account the pool offsets first, which is why it is cheaper. Variation margin nets in every regime: it settles a loss already suffered, so offsetting it gives nothing away. The choice governs initial margin alone.

Porting. If a clearing member fails, the CCP will attempt to **port** its clients (transfer their positions and collateral to a surviving member) rather than close them out. Porting preserves the client's hedge; closing out destroys it and leaves the client to rebuild in a market that is, by construction, disorderly. It is also what gross margining buys: collateral attributable to a client is collateral that can move with them, whereas a netted pool has a single margin figure out of which no one client can be lifted.

But porting depends on a surviving member being willing to accept the client, and that willingness is scarcest exactly when it is needed. A member contemplating a failed competitor's client book during a crisis is asked to take on unfamiliar positions and consume its own capital at the worst moment. Clients therefore hold a protection that is real but conditional—and its condition correlates with the event it protects against.

The two-tier structure also concentrates the market further than Figure 1.2 suggests: the number of firms willing to offer client clearing has fallen steadily, since the business consumes balance sheet under the leverage ratio (Section 1.12), while earning thin fees. A system designed to reduce concentration risk rests on a shrinking number of intermediaries.

2.6 DELIVERY

Most futures positions are closed before delivery (for financial contracts, the overwhelming majority). But the possibility of delivery is what ties the futures price to the spot price, so the mechanics matter even to those who never use them.

For a physically delivered contract the process runs from the short side. During the delivery month the short may issue a **notice of intention to deliver** to the exchange. The exchange assigns the notice to a long (typically the one with the oldest outstanding position), who must accept delivery and pay the invoice amount. The short chooses *when* within the delivery period to issue notice, and for some contracts *what* to deliver from a permitted set. Where that set spans

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grades or delivery points, the invoice amount is not the futures price alone: published **price adjustments** add a premium or discount for what was actually delivered, measured against the par grade and location, the ones the quoted futures price refers to. Without them the short would simply deliver the cheapest admissible item and keep the difference. Chapter 6 shows what those choices are worth in the Treasury market, where the adjustment is the conversion factor.

For a physical commodity, everything it costs to hold the asset falls on the short until delivery. Storing it, insuring it, and moving it to an approved delivery point are the short's expense, which is why those costs appear in the forward price of Section 5.8 rather than anywhere in the contract's terms. What happens at delivery then depends on whether the exchange licenses somewhere to put the asset. For metals and grains it does, and delivery is the transfer of a **warehouse receipt** (title to a specified lot already sitting in a licensed warehouse) rather than the movement of anything: storage comes with the title, and the long inherits the charges from that day. Getting the metal or grain physically out is a separate operation with a queue of its own, and Section 36.12 shows that the length of that queue moves the price.

Where no such warehouse stands behind the contract, the obligation is harder than it looks, and it belongs to the long. Crude oil is delivered into tankage at Cushing, Oklahoma, and a long that has not arranged capacity before expiration must find it in a market where everyone else needs it at the same moment. Section 5.9 describes the April 2020 expiration at which there was none to be had and the price settled below zero. This is the reason most participants close out rather than deliver, and the reason a retail product that mechanically holds the front contract is not the safe instrument it appears to be.

Financial contracts are usually **cash-settled** instead, against a published settlement index. There is no deliverable asset for a stock index or an overnight interest rate, and cash settlement avoids the artificiality of constructing one. It also removes the squeeze risk that delivery obligations create, at the cost of making the contract dependent on the integrity of the settlement index, which is its own vulnerability, as Chapter 4 shows for LIBOR.

2.7 REGULATION AND MARKET MANIPULATION

Futures markets are regulated to ensure prices are set by genuine supply and demand rather than by participants able to distort them. The characteristic abuse is the **corner** or **squeeze**: a trader accumulates a dominant long futures position *and* control of the deliverable supply, so that shorts approaching delivery must buy back from the same party that controls the physical market. The price rises not because the commodity has become scarce in any economically meaningful sense but because the shorts cannot obtain it. The defenses are position limits, tightening into the delivery period; surveillance of large positions; and the exchange's power to intervene by raising margin, imposing liquidation-only trading (positions may be closed but not opened), or in extremis altering delivery terms.

The nickel episode of Section 2.9 is instructive precisely because it was not a manipulation. The large short was a genuine producer hedge, and the price spike was driven by shorts scrambling to cover. Yet the market effect was much the same: prices dislocated from any plausible physical value, participants unable to transact. Regulation aimed at intent does not necessarily protect against outcomes that arise without it.

2.8 THE DEFAULT WATERFALL

A CCP that guarantees performance must be able to absorb the failure of a member. The resources it can call on are arranged in a strict sequence, the **default waterfall**, shown in Figure 2.1. Beyond the defaulter's own margin, the main resource is the **default fund**: a pool to which every clearing member contributes, held to cover losses larger than a defaulter's margin

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can absorb.

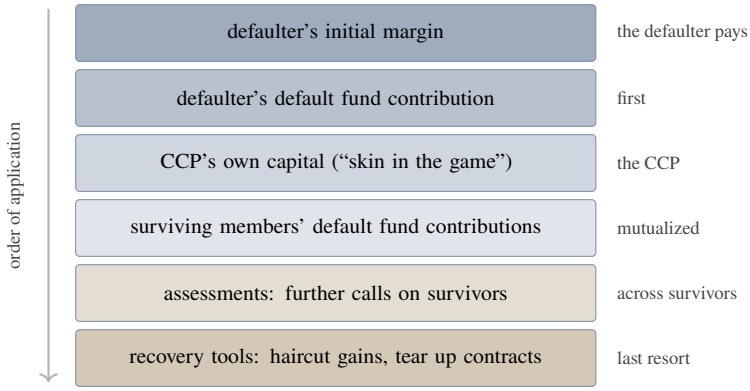


Figure 2.1: The sequence in which a CCP applies resources after a member default.

The ordering embodies a principle: *the defaulter pays first*. Only when the defaulter's own resources are exhausted does anyone else contribute, and the CCP's own capital (its *skin in the game*) sits ahead of surviving members' mutualized funds, so that a CCP which sets margin too low bears the first consequence itself. That layer is small relative to the fund below it, and how small it should be is one of the live disputes in the field. Example 2.5 takes one default through the layers.

Example 2.5 Applying the waterfall

A clearing member defaults. Closing out its portfolio in the market realizes a loss of $900 million. The CCP's resources are:

Layer	Amount ($m)
Defaulter's initial margin	500
Defaulter's default fund contribution	80
CCP skin in the game	50
Surviving members' default fund	2,000

Applying the waterfall in order: the defaulter's margin absorbs $500m, leaving $400m. Its default fund contribution absorbs $80m, leaving $320m. The CCP's own capital absorbs $50m, leaving $270m. That remainder is mutualized across the surviving members, consuming

$$\frac{270}{2{,}000} = 13.5\%$$

of their pooled contributions.

The survivors did nothing wrong, had no exposure to the defaulter bilaterally, and have lost 13.5% of the funds they posted, at a moment when a major member has just failed and markets are presumably under stress. Clearing did not make the loss disappear; it distributed it. That is what "mutualization" means, and members' willingness to accept it is what the system rests on.

2.9 MARGIN PROCYCLICALITY

Initial margin is calibrated to cover a large adverse move over the time it would take to close out a defaulted portfolio, conventionally a high confidence level over a few days. Since estimates of how far prices move are based on recent volatility, **margin requirements rise when volatility rises**. Because volatility is highest in a falling, stressed market, margin moves with the market cycle (lowest in calm, highest in a crisis), and a requirement that moves with the cycle reinforces

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it. That property is **procyclicality**.

This is prudent for the CCP taken alone and destabilizing for the system taken together. Consider what it means. Volatility spikes; every member's margin requirement increases at once; all of them must find cash simultaneously; and the natural way to raise cash is to sell assets into a market that is already falling. Margin models designed to protect the clearing house can amplify the very move they are protecting against. Example 2.6 puts a size on the jump.

Example 2.6 How fast margin can move

A fund holds a portfolio whose initial margin is set to cover a three-standard-deviation adverse move over three days, about 99.9% one-sided. Under normal conditions, with annualized volatility of 15%, the three-day standard deviation on a \$1 billion position is

$$1{,}000 \times 0.15 \times \sqrt{3/252} = \$16.4\text{m}$$

and at three standard deviations the requirement is about \$49m.

Volatility then triples to 45%, not extreme by the standards of 2022 in energy or rates. The same calculation gives a three-day standard deviation of \$49.1m and a requirement of about \$147m.

The fund must find an additional \$98 million in cash, on a position it has not changed, within a day. If it holds the cash, nothing happens. If it does not, it must sell—and so must everyone else in the same position at the same moment.

Case Study 2.1 LME nickel, March 2022

On March 7 and 8, 2022 the price of nickel on the London Metal Exchange (LME) more than tripled: from about \$29,000 a metric ton at the previous Friday's close it rose some 66% on the first day and then, in a few hours of the second morning, spiked above \$100,000.

Behind the move was a very large short position held by a producer that had hedged its output. As prices rose, the margin calls on that position grew to a size that threatened both the holder and the brokers standing between it and the exchange. Nickel is not an unusually large market, and the losses were plausibly beyond what the default waterfall could have absorbed without severe consequences for members.

The LME's response was extraordinary: it suspended trading in nickel and *canceled* several billion dollars of trades that had already executed that morning, resetting the market to the previous evening's close. Trading did not resume normally for over a week.

The cancellation protected the clearing system at the direct expense of those who had bought during the spike, who had transacted in good faith on a functioning exchange and had their profits annulled. Litigation followed. Whatever the legal merits, the episode established something uncomfortable that the textbook description of futures markets omits: when the alternative is the failure of the clearing house, the finality of a trade is not absolute.

Two further episodes the same year made the same point without any cancellation of trades. European energy utilities that had sold futures against future output faced margin calls in the tens of billions of euros when gas and power prices rose several-fold, and UK pension funds running liability-driven investment met collateral calls by selling gilts, which drove yields higher and triggered further calls. In each case a hedge correct in every economic sense was defeated by the timing of the cash it demanded; Sections 8.9 and 8.10 work both episodes (Example 8.9 for energy, the liability-driven investment case study and Example 8.11 for the pension funds), and Sections 40.8 and 40.12 tell them.

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What can be done. Regulators have pressed CCPs toward margin models that are less reactive: floors on the volatility estimate, buffers built in calm periods, and caps on how fast requirements may increase. Each reduces procyclicality at the cost of holding more margin permanently, which is a real cost borne in normal times to avoid a rare one. Where the balance should sit is unresolved, and reasonable people place it differently.

2.10 WHO ABSORBS A CCP'S LOSSES?

The waterfall of Section 2.8 is finite. If losses exhaust every layer, the CCP faces its own failure, and since clearing is now mandatory for large parts of the derivatives market, the failure of a major CCP would be a systemic event of the first order.

The tools available at the bottom of the waterfall are uncomfortable. **Assessment rights** allow the CCP to demand further contributions from surviving members, transmitting stress to firms that may already be struggling. **Variation margin gains haircutting** withholds part of the gains owed to members whose positions have profited, taking money from the winners to pay for the losers. **Tear-up** cancels contracts outright, leaving members who thought they were hedged suddenly unhedged.

Each of these is a way of allocating a loss that has already occurred. None creates resources. And each would be deployed in precisely the conditions in which it does most damage.

This has produced an unresolved policy debate with three broad positions. Some argue CCPs should hold much more of their own capital, so that they bear more of the consequence of setting margin too low. CCPs, the third party to the dispute, respond that their capital is small relative to the risks they manage and that heavier requirements would make clearing uneconomic. Others argue for an explicit public backstop, on the grounds that a systemically essential institution will be rescued in practice and that pretending otherwise mainly obscures who bears the risk. The objection is that an explicit backstop invites exactly the underpricing of risk it would then have to cover.

The question matters for this book because so much of what follows assumes cleared markets function, and that assumption is a judgment about institutional resilience, not a mathematical fact.

2.11 OTC MARKETS: CLEARED AND BILATERAL

Not everything trades on an exchange. The over-the-counter market, in which participants negotiate directly, remains larger than the exchange-traded market by notional and is where most of the instruments in this book are transacted. Post-crisis reform pushed it toward the futures model along two tracks.

Standardized OTC derivatives must be centrally cleared. Interest rate swaps in the major currencies and index credit default swaps are cleared through CCPs that operate essentially as described above. This is why Chapter 7 can treat cleared swaps as the standard case. Where they are cleared has become a policy question as well: since June 2025 the European Union's revised European Market Infrastructure Regulation (EMIR) has required EU firms above the clearing threshold in euro- and zloty-denominated interest rate derivatives and euro short-term interest rate derivatives to hold an active account at a CCP in the EU. Under technical standards in force since February 2026, they must also clear at least five trades on average through it in each of their most-used contract types, a rule aimed at reducing the EU's reliance on clearing houses in London.

Non-cleared derivatives face mandatory margin. Trades that cannot be cleared (because they are bespoke) are subject to uncleared margin rules requiring both variation margin and, above certain thresholds, initial margin posted to a segregated third party. The intent is to remove the incentive to avoid clearing simply to avoid posting collateral.

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The result is that the practical distinction between exchange-traded and OTC has narrowed considerably. What remains is the contract's flexibility: an OTC trade can be tailored to a hedging need exactly, while a futures contract cannot. That flexibility is why the OTC market persists, and the basis risk it eliminates is real, as Chapter 3 discusses.

And clearing has now been extended beyond derivatives altogether. In December 2023 the US Securities and Exchange Commission adopted rules requiring that eligible secondary-market transactions in US Treasury securities be cleared through a registered clearing agency: both outright purchases and sales, and the repurchase agreements that finance them. The requirement is being phased in: after a one-year extension in 2025, cash transactions must be cleared from December 31, 2026 and repo from June 30, 2027.

The reasoning belongs to this book rather than to securities-market infrastructure. A large share of Treasury repo was bilateral and uncleared, which meant leverage extended on privately negotiated haircuts, to positions no regulator could total. The cash–futures basis trade of Section 6.5 is financed that way, and March 2020 is what happens when the financing withdraws from a crowded one. The rule is a market-structure answer to a derivatives problem, which is why it appears in this chapter and not in a chapter on bonds.

The mechanism is the one already described. It is novation to a central counterparty, margin set by a published model rather than agreed across a telephone, and multilateral netting across members, applied to a market where the incumbent clearing agency, the Fixed Income Clearing Corporation, has long cleared a substantial share of interdealer activity while much client activity sat outside it. Access models matter as much as the mandate: a client can be sponsored into clearing by a member, and rules permitting execution with one dealer and clearing through another determine how much competition the change actually produces.

Section 2.8's default waterfall and Section 2.9's procyclicality now apply to the financing market for the asset that prices every other dollar asset. Concentrating that risk in a clearing house does not remove it; it makes it measurable, mutualizes it among members, and raises the stakes on the margin model. Whether that is a good trade is the same question Section 2.10 asks, on a larger underlying.

2.12 TRADERS, ORDERS, AND THE SHAPE OF THE MARKET

The three motives for trading (hedging, speculation, and arbitrage) are those of Section 1.9, which also makes the case that hedgers and speculators need each other. In a futures market the categories are less clean than they look (a dealer hedging inventory is simultaneously speculating on the basis), and what this section adds is how traders place their orders and what their trading produces.

Orders. A **market order** executes immediately at the best available price. A **limit order** executes only at a specified price or better, and may not execute at all; the trade-off between certainty of execution and certainty of price is the fundamental one, and no order type escapes it. A **stop order** becomes a market order once the price reaches a trigger, and is commonly used to cap losses, though in a fast market it executes at whatever price is available *after* the trigger, which may be far worse. A **stop-limit order** adds a limit to that price (a floor, for a stop to sell) at the risk of not executing at all, which in a falling market is precisely the outcome the stop was meant to prevent. The nickel episode is a reminder that every order type assumes a functioning market, and inherits the fragility of that assumption.

Price discovery

Hedging is not the only function these markets serve, and arguably not the most important one. A liquid futures market with contracts extending years forward produces something no individual participant could: a continuously updated, publicly visible **forward curve** for the

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underlying, the settlement prices of the successive delivery months plotted against those months, so that a single line shows what the market will pay for the asset at every date it trades. The process by which trading assembles many participants' scattered information into such a price is **price discovery**. Chapter 5 works out what determines its shape, and Figure 5.2 draws the characteristic ones.

That curve is used by firms that never trade a single contract. An oil refiner deciding how much crude to process next quarter, a farmer choosing what to plant, and a utility deciding whether to contract for supply all read prices off a curve established by others' trading. The information is a public good produced as a by-product of private speculation, which is the strongest argument for tolerating speculation that serves no hedging purpose of its own.

Two cautions attach. A futures price is not a forecast of the future spot price; it is a price that clears today's market for future delivery, and Chapter 5 shows it equals the expected spot price only when the asset carries no systematic risk, differing from it otherwise by a risk premium. And the signal degrades exactly when it matters most: a limit-locked or suspended market produces no price at all, and the LME's nickel cancellation produced prices that were subsequently declared not to have happened.

2.13 FORWARDS VERSUS FUTURES

Table 2.3 summarizes the differences.

Table 2.3: Forward and futures contracts compared.

Forward	Futures
Private contract between two parties	Traded on an exchange
Not standardized	Standardized
Usually one delivery date	Range of delivery dates
Settled at the end of the contract	Settled daily
Delivery or final cash settlement usual	Position usually closed before delivery
Some credit risk	Virtually no credit risk

The economically significant difference is daily settlement: because gains and losses are reinvested or financed at the short rate prevailing on the day they arrive, a futures price can differ from the forward price on the same terms, as Section 5.12 shows and Chapter 30 quantifies through the convexity adjustment.

SUMMARY

A futures contract standardizes everything except price, which permits a liquid secondary market in which positions are closed by offset rather than negotiation, and arbitrage forces convergence to spot at delivery. Volume counts contracts traded and open interest counts contracts outstanding, so the two series read together carry information a price series alone does not.

The exchange manages credit risk not by assessing it but by making it irrelevant: positions are marked to market daily and losses collected in cash before they accumulate, with a fall below the maintenance level triggering restoration to the initial level. Initial margin covers a high-confidence adverse move over the assumed close-out period and is computed on the portfolio, so a calendar spread can require a small fraction of what its two legs would need outright. That credit is what makes relative-value strategies viable at high leverage and what makes them fragile, since it is withdrawn precisely when the two legs stop moving together.

A central counterparty novates each trade into two, collapsing up to $n(n-1)/2$ bilateral exposures into n, permitting multilateral netting and an orderly default procedure. It concentrates

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risk rather than eliminating it. Access is two-tier, so most participants' protection depends on how their collateral is segregated and on whether they can be ported to a surviving member, which requires someone willing to take the book at the worst possible moment. Losses are absorbed in a fixed sequence on the principle that the defaulter pays first, and in the worked example the default reaches into the surviving members' pooled contributions: clearing distributed the loss rather than removing it. Beyond the waterfall, assessments, gain haircutting, and tear-ups allocate losses rather than creating resources.

Initial margin rises roughly in proportion to volatility, which is prudent individually and destabilizing collectively. The three episodes of 2022 (LME nickel, European energy, and UK liability-driven investment) shared that mechanism despite different causes: economically sound hedges were defeated by the timing of the cash they demanded. The nickel cancellation established that trade finality yields when the alternative is the clearing house failing. OTC markets have converged on this model, and what survives the convergence is flexibility: an OTC contract can match a hedging need exactly and a futures contract cannot.

FURTHER READING

The Committee on Payments and Market Infrastructures and the International Organization of Securities Commissions (CPMI–IOSCO) *Principles for Financial Market Infrastructures* (2012), together with the later guidance on CCP resilience and recovery, is the primary source for the clearing machinery of this chapter, and it is written for supervisors rather than for lawyers. Pirrong's ISDA discussion paper on the economics of central clearing is the skeptical counterweight to official-sector enthusiasm and is worth reading for exactly that reason. On the margin spiral of 2022, the BIS bulletin on margin dynamics in centrally cleared commodities markets and the Financial Stability Board's *Liquidity Preparedness for Margin and Collateral Calls* (2024) are the official-sector accounts, and Duffie's Jackson Hole paper on Treasury market resilience shows the same mechanism at work in the largest market there is.

PRACTICE QUESTIONS

2.1 Explain, using an arbitrage argument in both directions, why the futures price converges to the spot price at delivery.

2.2 An investor sells three silver futures contracts, each for 5,000 ounces, at \$28.00 per ounce. Initial margin is \$9,000 per contract and maintenance margin \$7,000 per contract. Construct the margin account over five days if the settlement prices are \$28.20, \$28.55, \$28.30, \$28.95, and \$28.70. On which days is there a margin call, for how much, and why is the account restored to the initial rather than the maintenance level?

2.3 Six firms trade with one another bilaterally. How many bilateral exposures exist, how many remain after novation to a CCP, and what is the general result for n firms?

2.4 A clearing member defaults, generating a close-out loss of \$1.4 billion. The defaulter's margin is \$600m, its default fund contribution \$120m, the CCP's skin in the game \$75m, and the surviving members' default fund \$2.5 billion. Work through the waterfall in order, state what percentage of the surviving members' fund is consumed, and explain why the CCP's own capital sits ahead of surviving members' contributions.

2.5 A \$500 million position has an initial margin requirement set at three standard deviations of a three-day move. Calculate the requirement at annualized volatilities of 12% and 30%, state how much additional cash must be found, and explain why a practice that is prudent for one clearing house may be destabilizing for the system.

2.6 Distinguish volume from open interest over three days: (i) two new participants open a

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20-contract position between them; (ii) one of them closes 8 to a new entrant; (iii) both original parties close 12. Give both figures for each day.

2.7 A trader is long 200 September contracts and short 200 December, with outright margin of $1,500 per contract and a spread rate of $250 per pair. Compare the portfolio and contract-by-contract requirements, and explain what happens to the position if the spread relationship breaks down.

2.8 Explain the difference between individual and omnibus segregation of client collateral, why a client might not know which it has, and why the chance of being ported is lowest exactly when it is most needed.

2.9 Explain what a corner or squeeze is and how position limits are meant to prevent it. Why did the nickel episode produce similar market effects without being a manipulation, and what does the cancellation of executed trades imply about finality?

2.10 Futures markets are defended on the grounds that they produce price discovery. Explain what this means, who benefits, why it is an argument for tolerating speculation that serves no hedging purpose, and under what conditions the argument fails.

FURTHER QUESTIONS

2.11 The 2022 episodes in energy, nickel, and UK pensions had quite different causes. Identify the mechanism common to all three, explain why being correctly hedged did not protect the firms involved, and state what a risk manager should have measured that a mark-to-market framework does not capture.

2.12 Set out the case that CCPs should hold substantially more of their own capital in the waterfall, then the case against. What evidence would move you?

2.13 Variation margin gains haircutting takes money owed to members whose positions have profited. Explain when it would be used, why it is preferred to simply failing, and what it does to the incentive to clear.

2.14 Central clearing concentrates exposures in a single institution. Construct the argument that this makes the system *more* fragile rather than less, and identify what would have to be true for the argument to be right.

2.15 A commodity producer hedges three years of output by selling futures. Prices triple. Show that the firm is economically better off than if it had not hedged, yet may fail. What financing arrangements would have prevented this, and what do they cost in normal times?

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Chapter 3
Hedging Strategies Using Futures

Most participants in futures markets are there to reduce a risk they already have, not to acquire one. This chapter is about how they do it and, more usefully, about the ways in which it does not quite work.

Futures are only one way to do it, and it is worth being explicit about why a firm chooses them. They are *liquid*, so a large hedge can be put on and taken off the same day. They are *cheap*, with spreads a fraction of the equivalent bank quote. They carry *no bilateral exposure to the other side of the trade*, because the clearing house of Chapter 2 stands between the two sides (exposure to the clearing house and to the clearing broker remains, as Chapter 2 explains). And they cost *nothing at inception*, which matters more to a treasurer defending a budget than it should. The alternatives each give one of these up: a forward matches the exposure exactly but is illiquid and carries the dealer's credit, a swap does the same across many dates, and an option keeps the upside but has to be paid for.

A hedge that worked perfectly would be uninteresting. The producer would lock in a price, the consumer would lock in a cost, and there would be nothing further to say. In practice four things intervene: the futures contract may not match the exposure, the hedge may be lifted before the contract expires, the exposure may last longer than any liquid contract, and (the theme Chapter 2 introduced and Chapter 8 develops) the hedge generates cash flows on a schedule the hedged item does not. Each is the shadow of one of the advantages just listed. Standardization is what makes futures liquid and what makes them fit imperfectly; a small number of listed maturities is what concentrates the liquidity and what forces both the early lift and the roll; daily settlement is what removes the bilateral credit risk and what demands the cash. A hedger does not get to take the advantages without their costs, and choosing futures is choosing that bundle.

Each of the four is a source of residual risk, and each has cost real firms large sums. Sections 3.3 through 3.9 take them in turn, and Section 3.10 examines what happens when they combine.

3.1 SHORT AND LONG HEDGES

A **short hedge** is appropriate when a firm already owns an asset, or expects to own one, and wants to protect against a fall in its price. The firm sells futures. If the price falls, the loss on the physical position is offset by a gain on the futures.

A **long hedge** is appropriate when a firm knows it will need to buy an asset in future and wants to protect against a rise in price. The firm buys futures.

Example 3.1 An oil producer's short hedge

On May 15 a producer contracts to sell 1 million barrels of crude on August 15 at whatever the spot price is on that day. Spot is currently \$80 and the September futures price is \$80.50. Each contract covers 1,000 barrels, so the producer sells 1,000 contracts.

On August 15 the spot price has fallen to \$72 and the September futures price is \$72.10.

Revenue from selling the oil	\$72,000,000
Gain on the futures position, $(80.50 - 72.10) \times 1{,}000{,}000$	\$8,400,000
Total	\$80,400,000

The effective price received is \$80.40 per barrel: close to, but not exactly, the \$80.50 futures price at which the hedge was placed. Note the discrepancy of 10 cents. It is not an error, and it is the subject of Section 3.3.

3.2 ARGUMENTS FOR AND AGAINST HEDGING

Before the mechanics comes a question that matters more than it is usually given credit for: should a firm hedge at all?

The case against. Shareholders can diversify. If an investor holds an oil producer alongside an airline, the oil price risk partly cancels within the portfolio, and the investor did not need either company to hedge. This is the Modigliani–Miller argument applied to risk: in frictionless markets, whatever a firm does to its risk its shareholders could do or undo on their own account, so hedging cannot add value. Hedging costs money and management attention, and a firm that hedges away the exposure its shareholders bought it for may be destroying the thing they wanted. Gold mining is the clearest case. Producers sold their output forward through the 1990s, and when the gold price rose through the decade that followed, shareholders who had bought the miners precisely to own gold discovered the exposure had been sold out from under them. The industry then spent years, and a great deal of money, buying those hedge books back.

There is also a competitive argument. If an entire industry faces the same input cost and passes it through, hedging does not reduce a firm's risk relative to its competitors—it *creates* it, because the hedged firm's costs now diverge from everyone else's. Airlines are the standing example: when fuel is expensive every carrier raises fares together, so the hedged carrier gains only until prices fall, at which point it is the one carrier that cannot cut.

The case for. The strongest arguments are about frictions the diversification story ignores. *Financial distress is costly*: a firm that runs out of cash loses customers, suppliers, and staff in ways that a shareholder's diversified portfolio does not compensate for, so reducing the probability of that state has value even if the expected outcome is unchanged. Lenders act on this directly: a reserve-based loan to an oil and gas producer, whose size (the **borrowing base**) is set by the value of the producer's reserves, commonly *requires* that a share of production be hedged as a condition of that borrowing base, because what the lender is buying is the reduced probability of distress and not a view on the oil price. Froot, Scharfstein, and Stein (1993) sharpen the point: when outside finance costs more than internal cash, a shortfall in cash flow forces a firm to pass up good investments, so a hedge that keeps internal funds available in bad states protects investment, not merely earnings. Taxes argue the same way, as Smith and Stulz (1985) show: where the tax schedule is convex, as progressive rates and limits on the use of losses make it, a smoother taxable income lowers the expected tax bill. *Managers know more than shareholders* about the firm's exposures, and can hedge more cheaply than shareholders can replicate. And *planning is easier* when input costs are known: a firm that can commit to a price can sign contracts it otherwise could not. An airline sells tickets months before it buys the fuel to fly them, and a utility files tariffs it cannot revisit for a year; neither can quote a price today without knowing a cost it will not pay until later.

Hedging is worth doing when it reduces the probability of genuinely bad states, and adds little when it merely smooths outcomes within a range the firm can absorb.

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Case Study 3.1 The hedger who is punished for hedging

A hedge that works generates a loss on one leg. That is what a hedge is.

The difficulty is that the two legs are usually not reported the same way. The futures loss is realized, visible, and appears in a specific line of the accounts with someone's name against it. The offsetting gain sits in an operating result, unlabeled, and looks like ordinary business performance. When prices move against the futures leg, the person who put the hedge on has a demonstrable loss to explain, and the person who did nothing has a good year.

Airlines have provided the clearest examples in both directions. Carriers that hedged fuel were criticized for the cost when prices fell, and praised when prices rose; carriers that did not hedge faced the reverse. The underlying decision, whether to accept a known cost or an unknown one, was the same in every year, but it was judged each time by an outcome that the decision was explicitly designed not to depend on.

The practical consequence is that hedging programs need to be justified and measured on the combined position, before the fact, with the mandate agreed in advance. A program evaluated leg by leg after the fact will be abandoned at the worst possible moment. That is what Section 3.10 describes happening to a real firm.

3.3 BASIS RISK

Example 3.1 did not lock in \$80.50. The hedge involved two prices (the spot price the producer received and the futures price at which the hedge was closed), and they need not move together.

The **basis** is defined as

$$\text{basis} = \text{spot price of the asset to be hedged} - \text{futures price of the contract used} \tag{3.1}$$

Chapter 2 showed the basis converges to zero at delivery. Before delivery it does not, and it varies.

Write S_1, F_1 for the spot and futures prices when the hedge is placed, S_2, F_2 for their values when it is lifted, and $b_1 = S_1 - F_1$, $b_2 = S_2 - F_2$ for the corresponding bases. For a short hedge the effective price received is

$$S_2 + (F_1 - F_2) = F_1 + b_2 \tag{3.2}$$

and the identical expression holds for a long hedge, as the effective price paid.

Equation (3.2) is the central result of this chapter. The hedger locks in F_1, which is known today, plus the basis at the time the hedge is closed, which is not. All the residual uncertainty is in b_2.

Figure 3.1 shows the two prices converging. The gap narrows to zero at delivery, but a hedger who lifts the hedge before then is exposed to wherever the gap happens to be at that moment.

The figure is also equation (3.2) drawn. The hedger sells the commodity at S_2 and closes the futures for $F_1 - F_2$, so S_2 plus the futures gain $F_1 - F_2$ reaches the dashed horizontal line above the two marked points, and that line sits above F_1 by exactly the gap b_2 between them. Both readings give the same height, which is what the identity says. Notice what has dropped out: the sinuous common movement of the two paths appears in S_2 and in F_2 and cancels between them, so the effective price does not depend on where prices went, only on the gap that remained.

Who gains from which move. An increase in the basis is called a **strengthening**; a decrease is a **weakening**. From (3.2), a short hedger receives $F_1 + b_2$ and therefore **benefits from**

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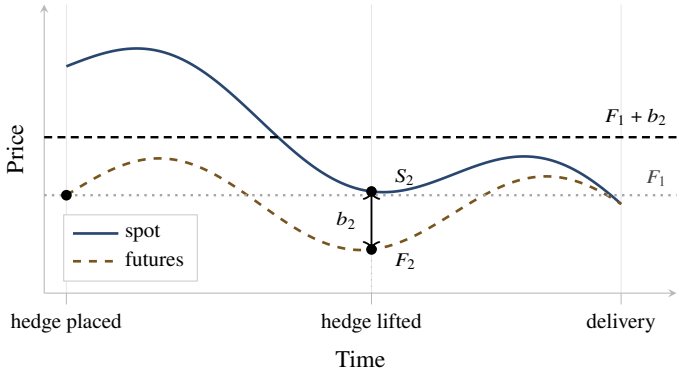


Figure 3.1: Spot and futures prices converging at delivery; b_2 is the basis at the lift date. Schematic.

an unexpected strengthening. A long hedger pays the same quantity and benefits from a weakening. Example 3.2 puts numbers on a weakening.

Example 3.2 Basis risk in numbers

A hedge is placed when the spot price is \$62 and the futures price is \$60, so $b_1 = +\$2$. It is lifted when the spot price is \$58 and the futures price is \$57.50, so $b_2 = +\$0.50$.

For a short hedger the effective price is

$$F_1 + b_2 = 60 + 0.50 = \$60.50$$

Had the basis held at \$2, the effective price would have been \$62. The basis weakened by \$1.50, and the short hedger lost exactly that amount per unit relative to expectation—despite the hedge having done its job in every other respect.

Compare the unhedged outcome. The spot price fell \$4 from the \$62 at which the hedge was placed. The hedge absorbed \$2.50 of the fall and left \$1.50, and that residual is basis risk.

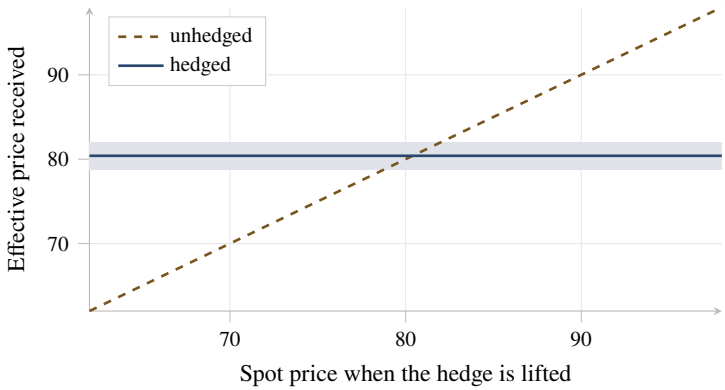


Figure 3.2: Effective price received against the spot price at the time the hedge is lifted. The band is basis risk.

Figure 3.2 makes the point visually. Unhedged, the effective price is whatever the spot price turns out to be (the diagonal). Hedged, it is $F_1 + b_2$, which is flat in the spot price: the hedge has removed the exposure to the level of the spot price. What remains is the vertical thickness of the band, which is the variability of b_2.

The figure also shows what the hedger has given up. Everything to the right of the crossing

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point is upside the hedger will not receive, and every hedging program has to survive a year in which that region is where the market went. This is the organizational problem of Case Study 3.1, drawn.

Where basis risk comes from. Basis risk has three sources, each corresponding to a mismatch:

- **Asset mismatch.** The asset hedged differs from the deliverable grade: jet fuel hedged with heating oil (since 2013 the New York Mercantile Exchange (NYMEX) contract of that name has specified ultra-low-sulfur diesel delivered in New York Harbor), a corporate bond portfolio hedged with Treasury futures. Chapter 2's remark that the deliverable grade determines what a contract tracks is the same point.
- **Location mismatch.** The same commodity at a different delivery point, connected only by transport capacity that can bind (the reason Chapter 2 flagged delivery location as economically significant).
- **Timing mismatch.** The hedge is lifted before the contract expires, so convergence is incomplete.

Choosing the contract. Two decisions reduce basis risk. Choose the asset whose price is most closely correlated with the exposure, and choose a delivery month *after* the hedge will be lifted, but not much after, since liquidity concentrates in near contracts. Hedging a June exposure with a July contract is usually better than with a June contract, since holding into the delivery month exposes the hedger to delivery mechanics and to erratic prices.

3.4 BASIS RISK BEYOND COMMODITIES

Basis risk is usually introduced with commodities, where the mismatches are physical and easy to picture. It is at least as important in financial hedges, where it is less visible and therefore more dangerous.

Credit. A portfolio of corporate bonds hedged with Treasury futures is protected against the level of interest rates and not at all against credit spreads. In calm markets the two move together enough that the hedge looks effective; in a credit sell-off, government yields *fall* while corporate yields rise, so both legs lose simultaneously. The hedge's measured R^2 over a quiet sample (the share of the exposure's variance the hedge removed, which Section 3.5 defines) says nothing about this, because the sample contains no such episode.

This is the general form of the problem, and it shapes how the next section should be read: a hedge ratio estimated over a period in which the basis behaved will fail in the period when the basis is what moves.

Currency. A firm with an exposure in a currency that has no liquid futures contract hedges with a correlated one: a Norwegian krone receivable hedged with euro futures, say. The hedge works while the two currencies track and fails when they do not, and they most reliably fail to track when something specific is happening to one of them, which is when the exposure matters.

Rates. Chapter 7's swap spread is a basis. A dealer hedging a swap book with Treasury futures, as Chapter 6 describes, is exposed to the spread between the swap curve and the government curve, and Chapter 7 shows that spread moving to a place no one expected it to go.

The common structure. In each case a hedger has substituted a liquid instrument for the exposure it actually holds and retained the difference. The difference is typically small, mean-reverting, and well behaved—until it is none of those things. A hedger should be able to state, before placing a hedge, what the basis is, what would make it move, and how much it could move; a hedger who can only state the historical correlation has not answered the question.

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3.5 CROSS HEDGING AND THE MINIMUM VARIANCE HEDGE RATIO

When the hedged asset differs from the futures asset (a **cross hedge**), a one-for-one position is generally not optimal. If the futures price is more volatile than the spot price, a smaller futures position gives a better hedge.

The **hedge ratio** h is the size of the futures position relative to the exposure. The value minimizing the variance of the hedged position, the **minimum-variance hedge ratio**, is

$$\boxed{h^* = \rho \frac{\sigma_S}{\sigma_F}} \tag{3.3}$$

where σ_S and σ_F are the standard deviations of changes in the spot and futures prices over the hedge's life, and ρ their correlation. [TN3] derives it, extends it to several futures contracts at once, and shows what the ratio costs when its three inputs are estimated rather than known. That cost decides whether a second hedging contract is worth using at all.

If the two prices move identically, $\rho = 1$ and $\sigma_S = \sigma_F$, giving $h^* = 1$: hedge one-for-one. Any imperfection moves the optimal ratio away from one. A correlation below one always moves it downward, relative to a perfectly correlated hedge with the same volatility ratio, because a futures position that does not reliably offset should be smaller. A volatility mismatch can move it either way: above one when the spot price is the more volatile, below one when the futures price is.

The connection to regression. Equation (3.3) is the slope coefficient from regressing changes in the spot price on changes in the futures price. This is not a coincidence: minimizing squared deviations is what both problems do. It is also how h^* is estimated in practice. The regression's R^2 then measures the proportion of variance the hedge removes.

The optimal number of contracts. With Q_A units of exposure and Q_F units per contract,

$$N^* = \frac{h^* Q_A}{Q_F} \tag{3.4}$$

Example 3.3 applies both formulas.

Example 3.3 Hedging jet fuel with heating oil

An airline expects to buy 2 million gallons of jet fuel in one month. There is no liquid jet fuel futures contract, so it uses heating oil futures, each covering 42,000 gallons, as a cross hedge.

From historical data on monthly changes: the standard deviation of the jet fuel price change is \$0.032 per gallon, that of the heating oil futures price change is \$0.040, and their correlation is 0.8. From (3.3),

$$h^* = 0.8 \times \frac{0.032}{0.040} = 0.64$$

and from (3.4),

$$N^* = \frac{0.64 \times 2{,}000{,}000}{42{,}000} \approx 30.5$$

so the airline buys 30 contracts.

Two observations follow. The hedge ratio is well below one (the airline's 30 contracts cover 1.26 million gallons against a 2 million gallon exposure) because heating oil futures are more volatile than jet fuel and imperfectly correlated with it. And the hedge is far from perfect: with $\rho = 0.8$, the regression R^2 is 0.64, so about 36% of the variance of the jet fuel cost remains after hedging. That the R^2 equals the hedge ratio here is a coincidence of the inputs: $R^2 = \rho^2$ while $h^* = \rho\,\sigma_S/\sigma_F$, and the two agree only because $\sigma_S/\sigma_F = 0.8$ happens to equal ρ.

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Estimating the hedge ratio in practice

Equation (3.3) requires three numbers that must be estimated, and the estimation choices matter more than the formula.

Use changes, not levels. Regressing the spot price on the futures price rather than their *changes* typically produces an R^2 near one, and a hedge ratio near one whenever the two levels share a trend of similar slope. Both series are nonstationary and trend together; the regression is measuring that shared trend, not the co-movement a hedge relies on. (Cointegration, a long-run tie between the two levels, is a separate question and does not change the point.) Regressing levels is among the most common errors in applied hedging, and it badly flatters the result.

Match the data frequency to the hedge horizon. A hedge held for a month should be estimated on monthly changes. Daily data offers more observations and a tighter standard error, but it measures a different thing: short-horizon co-movement is often weaker than long-horizon co-movement, because temporary dislocations between spot and futures wash out over time. Example 3.4 shows how large the difference can be.

Example 3.4 The hedge ratio depends on the horizon

The same spot and futures series are used to estimate (3.3) at two frequencies.

Data frequency	σ_S	σ_F	ρ	h^*
Daily changes	0.0070	0.0094	0.62	0.46
Monthly changes	0.0320	0.0400	0.80	0.64

The two estimates differ by nearly 40%. The daily figure is not wrong; it is the right hedge ratio for a hedge held one day. Applying it to a one-month hedge underhedges substantially, because it has been calibrated on a horizon over which spot and futures diverge more.

Most of the rise comes from the correlation, which climbs from 0.62 to 0.80; the rest comes from the volatility ratio σ_S/σ_F, which edges up from 0.745 to 0.800. The general rule is that correlation between related assets usually rises with the measurement horizon, so the hedge ratio does too. A hedger who has a monthly exposure and a daily dataset should aggregate the data to monthly changes, not apply the daily ratio.

The sample period is a judgment, not a technicality. A long sample gives precision; a short one gives relevance. The minimum-variance ratio is not a constant of nature: it shifts when market structure changes, when a new delivery point opens, when refining capacity changes the relationship between two grades. A ratio estimated over ten years spanning a structural break is precisely estimated and wrong.

Overlapping observations. Using monthly changes measured weekly, a common device for increasing sample size, creates overlapping windows and correlated residuals. The point estimate is usually acceptable; the standard errors are not, and a hedger who concludes the ratio is precisely determined on this basis is mistaken.

Tailing the hedge. Equation (3.4) works in units, and it slightly overhedges. Because futures are settled daily, a gain realized today earns interest until the hedge is lifted, so a smaller position delivers the same offset at the horizon. Discounting over the remaining hedge life $T - t$ at the interest rate r gives

$$N^* = \frac{h^* Q_A}{Q_F}\, e^{-r(T-t)} \tag{3.5}$$

a correction known as **tailing the hedge**; [TN3] derives it and tabulates the factor. The correction in (3.5) is small for short horizons and grows with maturity and with the level of interest rates: at $r = 4\%$ it removes 0.3% of the position over one month but 7.7% over two

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years, and the position is scaled back up as the horizon shortens. It comes from daily settlement, as the forward–futures price difference of Section 5.12 does, though unlike that difference it does not require interest rates to be uncertain.

Ratios estimated on returns. Futures hedge ratios are more often estimated on *proportional* changes than on absolute ones (returns on the spot against returns on the futures), which is the natural scale when the position is resized as prices move. Write $\hat{h}$ for a ratio estimated that way, and let S and F be the spot and futures prices when the hedge is placed (the S_1 and F_1 of Section 3.3, with the subscript dropped because only one date is in play here). Then $\hat{h}$ is not the h^* of (3.3). A proportional change becomes an absolute one by multiplying by the price, so $\sigma_{\Delta S/S} = \sigma_S/S$ and $\sigma_{\Delta F/F} = \sigma_F/F$, while ρ is unchanged because correlation does not depend on scale. Hence

$$\hat{h} = \rho \, \frac{\sigma_S/S}{\sigma_F/F} = h^* \, \frac{F}{S}.$$

A ratio estimated on returns is therefore applied to *values* rather than quantities,

$$N^* = \frac{\hat{h}\, V_A}{V_F} \tag{3.6}$$

where $V_A = SQ_A$ is the value of the position being hedged and $V_F = FQ_F$ the value of one contract. Substituting takes (3.6) straight back to (3.4):

$$\frac{\hat{h}\, V_A}{V_F} = h^* \, \frac{F}{S} \cdot \frac{SQ_A}{FQ_F} = \frac{h^* Q_A}{Q_F}.$$

The two are the same hedge written two ways, and either is correct on its own terms. What is not correct is to mix them: an h^* estimated on price changes and then applied to values gives

$$\frac{h^* V_A}{V_F} = \frac{h^* Q_A}{Q_F} \cdot \frac{S}{F},$$

rescaling the position by S/F for no reason at all, shrinking the hedge in contango (futures prices above spot and rising with maturity) and enlarging it in backwardation (below spot and falling with maturity).

3.6 HEDGING AN EQUITY PORTFOLIO

Stock index futures let a manager change a portfolio's market exposure without trading its constituents. This is the use Chapter 6 identifies for Treasury futures and duration.

The relevant hedge ratio is the portfolio's **beta**, its sensitivity to the index. To hedge fully,

$$N^* = \beta \, \frac{V_A}{V_F} \tag{3.7}$$

where V_A is the portfolio's value and V_F the value of one futures contract.

Example 3.5 Hedging a portfolio

A manager holds a \$5,050,000 portfolio with a beta of 1.5 against an index standing at 4,200. Futures contracts are worth \$50 per index point, so each contract is worth $4{,}200 \times 50 = \$210{,}000$. From (3.7),

$$N^* = 1.5 \times \frac{5{,}050{,}000}{210{,}000} = 36.07$$

so the manager shorts 36 contracts.

Suppose the index then falls 10% to 3,780. The gain on the futures is

$$36 \times (4{,}200 - 3{,}780) \times 50 = \$756{,}000$$

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while the portfolio, with a beta of 1.5, falls by roughly 15%, or $757,500. The two offset to within $1,500.

Removing the market also removes the market's return. Setting aside dividends and the gap between futures and spot prices, a fully hedged portfolio earns roughly the risk-free rate plus whatever the stocks earn beyond what their beta predicts (their alpha).

Changing beta rather than eliminating it. A manager who wants to reduce exposure without removing it shorts

$$N^* = (\beta - \beta^*)\,\frac{V_A}{V_F} \tag{3.8}$$

contracts, where β^* is the target. In Example 3.5, where $V_A/V_F = 24.05$, equation (3.8) moves beta from 1.5 to 0.75 by shorting $0.75 \times 24.05 = 18.04$, so 18 contracts; *raising* beta to 2.0 gives a negative count, and the manager goes long $0.5 \times 24.05 = 12.02$, so 12.

Why hedge rather than sell? If the manager wanted no market exposure, selling the portfolio would achieve it. Futures are used instead for three reasons: the manager may believe the specific stocks will outperform and want to keep that bet while removing the market, transaction costs on index futures are far lower than on the underlying holdings, and the position may be intended as temporary. The first of these is the most important: hedging with index futures is how a manager separates stock selection from market timing.

Hedging is not a one-off decision

Equations (3.4) and (3.7) give a position to put on today. The quantities they depend on then move.

The portfolio's value changes, so the hedge sized against V_A becomes wrong; the beta changes as holdings change and as the market's factor structure shifts; and for a commodity hedger the exposure itself changes as production forecasts are revised. A hedge placed once and left is correct only on the day it was placed.

How fast does it drift? For the equity hedge the answer is available in closed form. If the index moves by a proportion m, the portfolio moves by βm and one contract by m, so the position required by (3.7) changes by the factor

$$\frac{1 + \beta m}{1 + m}.$$

Two things follow. At $\beta = 1$ the factor is exactly one: market moves alone never call for rebalancing, because the exposure and the contract are revalued in step. What a manager is chasing is the amount by which β differs from one. And even away from $\beta = 1$ the drift is slow. In Example 3.5 the 10% fall leaves a portfolio of $4,292,500 and a contract worth $189,000, so the position now required is

$$1.5 \times \frac{4{,}292{,}500}{189{,}000} = 34.07$$

against the 36 contracts held, a two-contract adjustment after a 10% move in the market.

Rebalancing has a cost (each adjustment crosses a spread), so in practice firms use a schedule and a trigger together: monthly for an equity overlay (a futures position laid over a portfolio that is itself left untouched), on the planning cycle for a commodity hedger whose forecast exposure is the thing that moves. A band of a few percent of the required position catches large moves in between, and every roll forces an adjustment. The ratio itself is re-estimated far less often than the position is resized, quarterly at most, because re-estimating on each new observation feeds sampling noise straight into the position, and [TN3] puts a number on that

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cost. The choice of band is the same trade-off that recurs throughout this book: continuous adjustment is accurate and expensive, infrequent adjustment is cheap and approximate.

This is also the first appearance of an idea that dominates Part III. Chapter 18 shows that hedging an option requires exactly this kind of continuous rebalancing, because an option's sensitivity to the underlying changes as the underlying moves. The difference is one of degree: here the hedge ratio drifts slowly, and there it moves whenever the market does.

3.7 HOW MUCH TO HEDGE

Equation (3.3) answers a narrow question: what ratio minimizes the variance of a single exposure over a single horizon? Corporate hedging programs rarely answer only that question, and it is worth seeing what else enters.

Partial hedging is normal and often deliberate. Many firms hedge a fixed fraction of expected exposure (half the next year's production, a quarter of the year after) rather than the variance-minimizing amount. Three reasons follow, all defensible:

The exposure itself is uncertain. A producer hedging 100% of *expected* output is overhedged if output disappoints, and then holds a naked short futures position that loses if prices rise, which an industry-wide shortfall tends to make them do. Hedging a fraction leaves room for the quantity to come in low.

Shareholders may want some exposure. An investor buying a copper miner is buying copper exposure. A fully hedged miner is, in effect, a fixed-price extraction business, which is not what was bought.

Cash constraints bind. Section 3.10 is about this, and it is the reason a variance-minimizing ratio can be the wrong answer.

Layered hedging. A common structure hedges a declining fraction across a rolling horizon (perhaps 75% of the next six months, 50% of the following six, 25% of the year after), adding a new layer each quarter. This averages the entry price across many dates, which removes the risk of having placed the entire program on a single bad day, and it degrades gracefully if forecasts change.

The entry price is itself a risk. A firm that hedges its full three-year exposure in one transaction has replaced price risk with timing risk, and Case Study 3.1's organizational problem then attaches to a single decision made on a single day.

Or hedge with options instead. Futures are not the only instrument. A firm that buys an **option** (a producer a put, a consumer a call) removes the unfavorable tail and keeps the favorable one, in exchange for a premium. A purchased option whose premium is paid up front then owes no variation margin at all. That difference bears directly on the failures of Section 3.10, but options are not developed until Chapter 10, so the comparison is made in Section 12.10.

3.8 ACCOUNTING FOR HEDGES

Case Study 3.1 described the organizational difficulty of hedging. There is a formal version of it in the accounts, and it shapes which hedges firms actually choose.

Derivatives are carried at fair value with changes recognized in profit or loss. The hedged item usually is not: inventory sits at cost, future production is not on the balance sheet at all, and a forecast purchase does not exist as an accounting item. So a perfectly effective economic hedge produces *earnings volatility from the derivative alone*, with nothing offsetting it until the hedged transaction occurs.

Hedge accounting exists to fix this, in one of two ways depending on what is hedged. In a **fair value hedge**, of an asset or liability already on the balance sheet, the hedged item is remeasured for the hedged risk and the change goes to profit or loss beside the derivative's, so

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nothing is deferred; inventory hedged with futures, the case above, is of this kind. In a **cash flow hedge**, of a forecast transaction, there is no item to remeasure, so the effective portion of the derivative's gain or loss is held in other comprehensive income, a component of equity outside profit, and moved into profit when the hedged transaction affects earnings. Either way the two legs meet in the same period, which is what the economics always said.

The requirements are not trivial. Documentation must exist at inception, and the designation constrains what the firm may subsequently do with the position. The effectiveness test differs between the two main standards. International Financial Reporting Standard (IFRS) 9 requires an economic relationship between the hedged item and the hedging instrument, assessed prospectively, with any ineffectiveness recognized in profit or loss as it arises; it dropped the retrospective 80–125% test of its predecessor. US generally accepted accounting principles (US GAAP), in Accounting Standards Codification (ASC) Topic 815, still require the hedge to be highly effective, but after inception allow that to be assessed qualitatively.

The consequence for hedging choices. Because qualifying is onerous and failing is visible, firms sometimes prefer a hedge that qualifies for hedge accounting over one that is economically better. IFRS 9 eases one version of the problem by allowing a risk component to be designated as the hedged item, such as the crude oil component of a jet fuel purchase, so a crude hedge of jet fuel can qualify for that component. But a cross hedge whose link to the hedged item is real yet hard to evidence may still fail to qualify, or qualify with large recorded ineffectiveness, where a worse-matched but simpler instrument passes. A dynamic strategy that adjusts as exposures change may be harder to document than a static one that hedges less well.

The accounting treatment is a real input into hedging decisions, and it does not always point the same way as the economics. A reader who finds a firm's hedging program puzzling should check whether the puzzle dissolves once the accounting constraint is added.

And the tax rule is a third answer

Tax authorities also have a view, and it is not the accounting one. The specifics are jurisdictional and change, so what follows is the shape of the problem rather than any country's current rules; a reader acting on it should check the rules in force.

Three features recur across jurisdictions and matter to a hedger. *Timing*: gains on an exchange-traded futures position are commonly taxed as they accrue (marked to market at the year end whether or not the position was closed), while the hedged item is taxed when it is sold. *Character*: the derivative's gain may be a capital gain while the hedged item produces ordinary income, and capital losses often cannot be set against ordinary income. *Qualification*: some systems, notably the United States, provide a hedging exception that aligns both, taxing the derivative on the same timing and character as the item it hedges, but it must be claimed by identifying the hedge on the day it is entered into; others largely follow the accounting treatment.

The failure mode is the one this chapter keeps meeting. A hedge that is economically exact can be taxed asymmetrically: the loss leg deductible in one year at one rate, the gain leg taxable in another at another. The firm is then paying tax on a profit it did not make, in a year in which its economic position did not change—and it is paying cash, on the schedule of Section 3.10's margin calls rather than on the schedule of its production. Metallgesellschaft, whose collapse Section 3.10 describes, met the same asymmetry in its accounts rather than its tax bill: German accounting rules, which recognize unrealized losses but not unrealized gains, booked the losses on its futures while ignoring the offsetting gains on its customer contracts.

So a hedging decision faces three tests and they do not agree. The economics ask whether the exposure is offset, hedge accounting asks whether the offset can be documented and demonstrated, and the tax rule asks whether the hedge was identified in time and in the right

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form. *A hedge can pass any two and fail the third*, and the firm discovers which one it failed only afterward.

3.9 ROLLING THE HEDGE FORWARD

Section 3.3 assumed a contract exists spanning the exposure. Often none does. Liquidity in futures concentrates in the nearest few contracts, while a firm's exposure may extend years.

The response is to **stack and roll**: place the entire exposure in a near contract, and as it approaches expiration, close it and open the equivalent position in the next. A five-year exposure hedged with quarterly contracts requires nineteen rolls.

Each roll introduces basis risk of its own: the hedger closes at one contract's price and opens at another's, and the spread between them is not known in advance. Nineteen rolls means nineteen draws from that distribution. Example 3.6 works one sequence of rolls through, and Example 3.7 shows what the draws can cost when they run against the hedger.

Example 3.6 A rolling hedge, roll by roll

A producer will sell 1 million barrels in twelve months, but only three-month contracts are liquid. It shorts the near contract and rolls quarterly. The market is in mild contango throughout.

Month	Action	Price	Gain on leg
0	Short contract A	80.00	
3	Close A	78.00	+2.00
	Short contract B	78.60	
6	Close B	76.50	+2.10
	Short contract C	77.00	
9	Close C	75.00	+2.00
	Short contract D	75.40	
12	Close D	74.00	+1.40
	Sell the oil at spot	73.90	
	Total futures gain		**+7.50**

The effective price is 73.90 + 7.50 = \$81.40 per barrel—\$1.40 *better* than the \$80.00 at which the hedge was placed.

The improvement decomposes cleanly. Each roll closed a contract and opened a more expensive one: +0.60, +0.50, +0.40, totaling \$1.50 of roll gains. The final basis contributed 73.90 − 74.00 = −\$0.10. Together, 80.00 + 1.50 − 0.10 = \$81.40.

The producer gained because it was *short* in a market in contango. Reverse the direction (a consumer hedging with long futures in the same market) and the identical sequence of prices produces \$1.50 of roll *losses*.

Example 3.7 The cost of rolling

A producer must hedge 1 million barrels a month for three years, but only the front three contracts are liquid. It stacks the entire remaining exposure in the front contract and rolls monthly: 35 rolls, since the last month's barrels are delivered from the front contract without a further roll.

Suppose each roll costs 15 cents a barrel, because the market is in backwardation, which works against a short hedger. The position rolled shrinks from 35 million barrels to 1 million, an average of 18 million,

$$35 \text{ rolls} \times \$0.15 \times 18{,}000{,}000 = \$94.5 \text{ million}$$

of cumulative roll cost, against a hedge that was placed to protect a price.

Example 3.6 showed which direction the cost runs: a short hedger *gains* from rolling in contango and loses in backwardation, and a long hedger the reverse (Section 5.11 works the long

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investor's case). The producer above is short in backwardation and so pays. The mirror case, a firm hedging the other way with long futures, faces exactly this bill when the market moves into contango.

That is not a hypothetical. It is what happened to Metallgesellschaft.

3.10 WHEN HEDGING FAILS

The mechanisms above (basis risk, roll risk, and the cash flow timing of Chapters 2 and 8) combine badly. The clearest illustration remains a case from the early 1990s, told in Case Study 3.2.

Case Study 3.2 Metallgesellschaft

The US subsidiary of the German firm Metallgesellschaft sold long-term contracts to deliver heating oil and gasoline at fixed prices, some running as long as ten years. It had committed to sell roughly 160 million barrels at prices fixed in advance.

That is a short physical position, and it was hedged by going *long* futures. Since no liquid ten-year contract existed, the firm stacked the exposure in near-dated futures (about a third of the position) and short-dated over-the-counter swaps, and rolled them.

Three things then happened at once. Oil prices fell sharply through 1993, so the long futures position lost money—and because futures are settled daily, those losses had to be paid in cash immediately. The market shifted from backwardation into contango, so each roll now cost money rather than earning it. And the offsetting gains, sitting in customer contracts running out to a decade, produced no cash at all.

The margin calls ran to roughly $900 million by the end of 1993, and the losses eventually came to about $1.3 billion. The parent, MG AG, was itself in difficulty and was rescued by its banks in early 1994. Facing a cash demand of that size against gains it could not realize, it chose not to fund its subsidiary's margin: it liquidated the futures positions and unwound many of the customer contracts, crystallizing the losses while abandoning the position that would have offset them.

On the reading of Culp and Miller (1995) the hedge largely held in value, the futures losses being matched by gains on the supply contracts, and what it lacked was the cash to carry that offset through time. Mello and Parsons (1995) and Edwards and Canter (1995) disagree: a stack of short-dated contracts sized one for one against commitments a decade long was not a sound hedge in the first place. Whether the parent was right to liquidate rather than fund the position remains debated. What is not in dispute is the mechanism, and it is the one this book keeps meeting: a position hedged against price risk, defeated by the timing of the cash it demanded, and abandoned at the point of maximum loss because the combined position was judged one leg at a time.

The 2022 episodes of Chapters 2 and 8 are the same story with different commodities and a different resolution. European utilities that had sold output forward faced margin calls in the tens of billions against physical gains years away. The difference in outcome was not that those firms hedged better; it was that governments extended credit lines to them, while Metallgesellschaft's parent chose not to fund its subsidiary's margin.

What to take from this. Three practical conclusions follow, and they are the reason this chapter exists in the form it does.

Size the hedge to the cash you can produce, not only to the exposure. A hedge ratio derived from (3.3) minimizes price variance and says nothing about liquidity. A firm that cannot fund the margin on its variance-minimizing hedge should hedge less—and should know that in

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advance rather than discovering it. Often the cash problem is solved by a different instrument rather than a smaller hedge: the reserve-based borrowers of Section 3.2 usually hedge with swaps from their lending banks, secured by the same collateral as the loan and carrying no variation margin.

Stack and roll converts a long-dated exposure into a sequence of short-dated ones, and the sequence has its own risk. It is not a substitute for a matched-maturity hedge; it is a different strategy with different failure modes.

Judge the combined position. Every failure in this section involved someone looking at the futures leg alone.

SUMMARY

A short hedge protects a firm that owns or will own an asset; a long hedge protects one that must buy. Whether to hedge at all is a real question, since shareholders can diversify; the stronger arguments concern frictions, above all the cost of financial distress. Hedges are also hard to sustain, because the loss on the futures leg is visible while the offsetting gain is buried in operating results, so a program must be mandated in advance and judged on the combined position.

A hedge lifted before delivery locks in $F_1 + b_2$, so the risk that remains is the basis at the close, which arises from mismatches of asset, location, and timing. Basis risk is at least as important in financial hedges and less visible: a corporate bond portfolio hedged with Treasury futures keeps its credit spread risk, and a hedge in a correlated proxy currency fails exactly when the two currencies part company.

The variance-minimizing ratio $h^* = \rho\,\sigma_S/\sigma_F$ is a regression slope, and its R^2 says how much of the risk the hedge removes. It must be estimated on changes rather than levels and at the frequency of the hedge horizon. An equity portfolio is hedged with index futures in proportion to its beta, which separates stock selection from market exposure.

Firms rarely hedge the full variance-minimizing amount, and purchased options whose premium is paid up front, which then owe no variation margin, are the alternative that avoids the funding failures. Accounting and tax can make an economically sound hedge look bad: the derivative is marked to market while the hedged item often is not, hedge accounting realigns the two only for hedges that qualify, and a tax rule may treat the two legs differently. Exposures longer than the liquid contracts are stacked and rolled, which adds a fresh basis at every roll. Metallgesellschaft combined every failure mode, and the 2022 energy episodes repeated the mechanism; what separated the outcomes was whether someone would fund the margin, not better hedging.

FURTHER READING

Ederington (1979) is the origin of the minimum-variance hedge ratio in the form this chapter uses, and it is short. Froot, Scharfstein, and Stein (1993) make the modern case for why a firm hedges at all—the question behind every hedge ratio and the one a formula cannot answer. On Metallgesellschaft the literature is a genuine disagreement and should be read from both sides: Culp and Miller (1995) argue the strategy was sound and the liquidation was the error, while Edwards and Canter (1995) and Mello and Parsons (1995) take the opposing view. Smith and Stulz (1985) set out the tax and distress-cost arguments for hedging that Froot, Scharfstein, and Stein build on.

PRACTICE QUESTIONS

3.1 Set out the argument that a firm should not hedge because its shareholders can diversify, and give two reasons the argument may fail.

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3.2 Define the basis, show that a short hedger who lifts a hedge before delivery receives $F_1 + b_2$, and explain what this implies about where the residual risk lies.

3.3 A hedge is placed when the spot price is \$44 and the futures price is \$41.50, and lifted when the spot price is \$47 and the futures price is \$45.20. Compute the effective price for a short hedger, state whether the basis strengthened or weakened, and explain why a June exposure is usually better hedged with a July contract than a June one.

3.4 A company will buy 800,000 units of a commodity in three months. The standard deviation of the change in its price over three months is \$0.65, that of the futures price \$0.81, and the correlation 0.72. Each contract covers 50,000 units. Compute the minimum-variance hedge ratio and the number of contracts, state what proportion of the variance remains, and explain why that ratio is the slope of a regression.

3.5 A \$12 million portfolio has a beta of 0.9. Index futures are worth \$50 per point and the index stands at 4,000. How many contracts hedge it fully, and how many reduce beta to 0.4?

3.6 An analyst estimates a hedge ratio of 0.46 from daily data and 0.64 from monthly data for a hedge held one month. State which should be used, explain why the two differ in this direction, and explain what goes wrong if the regression is run on levels rather than changes.

3.7 Give three reasons a firm might deliberately hedge less than the variance-minimizing amount, and explain what layered hedging achieves that a single transaction does not.

3.8 A manager hedges an equity portfolio with index futures using the beta-adjusted contract count. Show that if the index moves by a proportion m, the position required changes by the factor $(1 + \beta m)/(1 + m)$. Explain why a portfolio with a beta of one never needs rebalancing on account of a market move alone, state the condition on β and m under which the hedge must be *increased*, and evaluate the factor for a portfolio with a beta of 0.8 after the index rises 12%.

3.9 A bank hedges a portfolio of corporate bonds with Treasury futures. Explain what basis risk remains, why a historical regression over a calm period would understate it, and what happens to both legs in a credit sell-off.

3.10 Using the roll table of Example 3.6, recompute the effective price if the market had been in backwardation, with each roll closing at a price 0.40 *above* the next contract, and state who gains in each case.

FURTHER QUESTIONS

3.11 A fund manager wishes to keep a portfolio's stock selection but remove its market exposure for three months. Explain why index futures are preferred to selling the portfolio, and identify what risk remains after the hedge is in place.

3.12 Explain what stack and roll is, why a firm would use it, and the two distinct risks it introduces that a matched-maturity hedge would not. Relate the direction of the roll cost to the shape of the forward curve.

3.13 Set out the Metallgesellschaft episode as a sequence of events, identifying at each stage which of this chapter's risks was operating. Was the strategy unsound, or was the liquidation the error? Defend a position.

3.14 A producer's variance-minimizing hedge would require it to post margin that could reach 40% of its annual revenue in a severe price move. Explain why the hedge ratio of

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equation (3.3) does not capture this, what the firm should do, and how you would set the hedge ratio instead.

3.15 A firm chooses a hedging instrument that matches its exposure less well than an available alternative, on the grounds that the alternative would not qualify for hedge accounting. Set out the argument for and against that decision, and say what you would need to know about the firm to judge it.

3.16 European utilities in 2022 and Metallgesellschaft in 1993 faced the same mechanism with different outcomes. Identify the mechanism precisely, explain what differed, and state what a treasurer should arrange in advance so that the difference does not matter.

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Chapter 4
Interest Rates in the Risk-Free-Rate Era

Interest rates are the raw material of derivatives pricing. Every valuation in this book discounts something, and most of what we discount is itself a function of rates. It is worth spending a chapter on how rates are defined, measured, and assembled into a curve before we start using them.

A reader who learned interest rates from an older textbook will find this chapter organized differently from the way the subject was taught for the preceding thirty years. That is deliberate, and it reflects a genuine change in the market rather than a change in taste.

Until recently, the reference rate at the center of the derivatives market was LIBOR: a rate with a term structure built into it, quoted for one month, three months, six months, and a year, and fixed at the *start* of each interest period. Everything about how we taught interest rate derivatives assumed a rate of that shape. Forward rate agreements, floating-rate notes, caps, floors, and swaps were all defined against a rate you knew in advance.

LIBOR is gone. The rates that replaced it are **overnight** rates. They have no term structure of their own, they are published one day at a time, and a three-month interest amount computed from them is not known until the three months have elapsed. Every convention in the market had to be rebuilt around that fact, and a great deal of the machinery in Chapters 6, 7, 29, and 30 rests on getting these mechanics right.

This chapter therefore builds from the overnight rate upward. Sections 4.1 through 4.6 cover the rates themselves: what they measure, why the old ones were replaced, and how an overnight rate is compounded into the term rate a contract actually needs. Sections 4.7 through 4.12 cover the traditional apparatus (compounding conventions, zero curves, forward rates, duration, and convexity), which the transition left essentially intact. Readers who have met that apparatus elsewhere can move through the later sections quickly, but should not skip Sections 4.4 and 4.5, which have no counterpart in older treatments.

4.1 TYPES OF INTEREST RATES

An interest rate in any situation defines the amount of money a borrower promises to pay a lender. The higher the credit risk of the borrower, the higher the rate. Rates are quoted with a compounding frequency and a day count convention attached; ignoring either is a common and expensive error, and we return to both in Section 4.7. A rate also has a **tenor** (the length of the period it applies to, such as overnight, one month, or three months), which is not the same thing as the maturity of the instrument that pays it. A ten-year swap whose floating rate resets every three months has a maturity of ten years and a tenor of three months. The distinction runs through the whole book: the tenor is what the rate measures, the maturity is how long the exposure lasts.

Treasury rates. These are the rates at which a government borrows in its own currency. Because a government can in principle raise taxes or print money to repay such debt, these

are conventionally treated as having negligible credit risk. Treasury rates are nevertheless *not* the rates derivatives markets discount at, for reasons that are partly technical and partly practical. Treasury yields differ from funding rates in either direction, depending on the supply of government debt, the balance-sheet cost of holding it, and the demand for it as collateral; Section 7.9 shows long dollar swap spreads negative since 2008, which puts Treasury yields above the swap curve. And a derivatives dealer cannot fund itself at the Treasury rate.

Repo rates. A repurchase agreement is a secured loan: one party sells securities and agrees to buy them back at a slightly higher price, the difference being the interest. The collateral need not be government securities. It can be almost any security the two sides agree on (agency mortgage-backed securities, corporate bonds, equities), and the rate depends on which it is. Better collateral commands a lower rate and a smaller **haircut** (the margin by which the collateral's market value exceeds the cash lent against it) because the lender of cash is less exposed if the borrower fails; repo against government bonds trades at the lowest rate, and repo against equities pays materially more. Credit risk is small but not zero, and it is a joint condition: the borrower must default *and* the collateral must have fallen in value before the lender loses anything.

Two further distinctions matter for what follows. A **general collateral** (GC) repo is one in which the lender of cash will take any security from an agreed class, so the trade is about the cash rather than any one security; a security in particular demand trades **special**, at a repo rate below general collateral and occasionally near zero, because the lender of cash is giving up interest for the privilege of borrowing *that* security. And repo trades at maturities from overnight out to several months. The dollar's reference rate is built from one narrow corner of all this (overnight, general collateral, US Treasury), and Section 4.2 explains why that corner and not another.

Overnight risk-free rates (RFRs). These are the successors to LIBOR and the subject of most of this chapter. Each is an average of actual overnight borrowing transactions in its currency, published once per business day by a central bank or exchange. The abbreviation **RFR** is used throughout this book, and in the market, for a rate of this kind.

Reference rates in derivatives. A **reference rate** is a published benchmark that contracts point at rather than negotiate individually. Its usefulness depends entirely on being both widely accepted and hard to manipulate. The story of the past decade is the market's discovery that its principal reference rate satisfied the first condition and not the second.

4.2 THE OVERNIGHT RISK-FREE RATES

Each major currency now has a designated overnight rate, selected by a working group of market participants and regulators: the Secured Overnight Financing Rate (SOFR) in dollars, the Sterling Overnight Index Average (SONIA), the Euro Short-Term Rate (€STR), the Tokyo Overnight Average Rate (TONA), and the Swiss Average Rate Overnight (SARON). Table 4.1 sets out how each is built.

Table 4.1: The principal overnight risk-free rates.

Currency	Rate	Basis	Administrator
USD	SOFR	Repo, US Treasuries	Federal Reserve Bank of New York
GBP	SONIA	Unsecured	Bank of England
EUR	€STR	Unsecured	European Central Bank
JPY	TONA	Unsecured	Bank of Japan
CHF	SARON	Repo, SNB GC basket	SIX Swiss Exchange

Three features are common to all of them, and each is a direct response to a defect in

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LIBOR.

They are transaction-based. Each rate is computed from actual trades, not from estimates. SOFR, SONIA, and €STR are published the next morning from the previous day's transactions; SARON is computed in real time and fixed at the close of the same day from trades and binding quotes; TONA has a same-day preliminary figure. SOFR is a volume-weighted median of overnight Treasury repurchase transactions and rests on well over $2 trillion of daily activity. LIBOR, at the end of its life, was in some tenors supported by a small number of transactions per day or none at all.

They are overnight. There is no one-month or three-month SOFR fixing in the way there was a one-month or three-month LIBOR. This is the feature with the largest practical consequences, and Section 4.4 is devoted to it.

They are close to risk-free. An overnight loan to a creditworthy counterparty carries almost no credit risk, and a secured overnight loan carries almost none even if the counterparty is weak. LIBOR, by contrast, was the rate at which *banks* could borrow unsecured for a term, and therefore embedded bank credit risk. That embedded credit component behaved unhelpfully: it widened when the financial system was under stress, which is exactly when the contracts referencing it mattered most.

The policy rate is not the benchmark

A reader who knows the US money market will object that the table omits an obvious overnight dollar rate: the **effective federal funds rate** (EFFR), at which banks lend reserves to one another, published by the same New York Fed that publishes SOFR. It is not the benchmark, for two reasons worth separating.

Volume. Trading in fed funds collapsed after 2008. Once reserves became abundant banks no longer needed to borrow them, and most of what remains is Federal Home Loan Banks (which cannot earn interest on reserve balances) lending to institutions that can. Daily volume runs on the order of $100 billion against SOFR's more than $2 trillion. A benchmark carrying hundreds of trillions of notional should not rest on the narrower market, which is the defect that ended LIBOR in a different form.

A policy rate makes a poor contract rate. EFFR is the rate the Federal Open Market Committee sets a target range for and steers toward, and it moves when policy decides it should. A benchmark ought to describe the market rather than be administered by an authority pursuing objectives of its own.

That second property is what keeps fed funds useful elsewhere. Being the policy rate makes it the rate in which expectations of policy are expressed (fed funds futures are how the market prices the next meeting), and the **fed funds–SOFR basis**, the spread between the two rates, trades actively through swaps that exchange one rate for the other, because a bank funds itself in one market and runs its derivative book in the other. The two rates can diverge with neither market under credit stress, for instance when Treasury collateral is scarce and repo rates fall relative to unsecured ones. Section 7.4 describes the October 2020 switch of dollar discounting from EFFR to SOFR, and the compensation it required.

Secured versus unsecured

Table 4.1 shows that the rates are not all built the same way, and the distinction matters. SOFR and SARON are *secured*: they measure the cost of borrowing against collateral. SONIA, €STR, and TONA are *unsecured*: they measure the cost of borrowing overnight without collateral, and so retain a very small credit component.

The two secured rates are not secured against the same thing. SOFR is Treasury repo and

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nothing else. SARON is computed from the Swiss repo market against the general collateral basket of the Swiss National Bank (SNB), which admits cantonal and foreign government bonds, bonds of supranational institutions (for example, the European Investment Bank and the World Bank), and covered bonds alongside Confederation debt. "Secured" describes the structure of the loan, not the identity of the collateral, and a benchmark's designers choose both.

Which repo rate? A repo market is not one rate, and turning it into a benchmark takes choices. SOFR is drawn from three segments of the overnight Treasury market (tri-party repo, the interdealer General Collateral Finance market, and cleared bilateral transactions) and is a volume-weighted *median* rather than a mean, so that a small number of extreme trades cannot move it. The cheapest quarter of the bilateral volume is discarded before the median is taken, and that is how specials are kept out: a trade struck at a low rate because the lender wanted one particular bond says something about that bond, not about the price of cash. The same data supports two further published rates, the Tri-party General Collateral Rate and the Broad General Collateral Rate, which are repo rates but are not benchmarks. Not every repo rate is a reference rate, and what separates them is these choices rather than anything in the underlying trades.

It follows that SOFR is a proxy for the risk-free rate of an arbitrage argument rather than the thing itself. What a given firm can actually finance a given security at depends on that security and on that firm, which is where Section 5.10 picks the question up.

Being secured is not the same as being immune to stress. A secured rate reflects conditions in the collateral market, and those conditions can move violently for reasons having nothing to do with credit.

Case Study 4.1 September 2019: when the risk-free rate spiked

On September 17, 2019, SOFR fixed at 5.25%, up from 2.43% the day before. Some overnight repo trades executed at rates approaching 10% intraday. Nothing had happened to the creditworthiness of any borrower; the collateral in question was US Treasury securities.

What had happened was a collision of cash flows. A large corporate tax payment date coincided with the settlement of a heavy Treasury auction, draining reserves from the banking system at the moment dealers needed to finance a large inventory of new securities. Demand for overnight cash overwhelmed supply, and the price of cash, the repo rate, went to whatever level cleared the market. The Federal Reserve intervened with repo operations within a day.

The episode carries two lessons for anyone using SOFR. First, a rate can be free of *credit* risk and still be exposed to *funding* and *liquidity* conditions; "risk-free" is a statement about default, not about volatility. Second, the effect was concentrated in a single day. A contract referencing an overnight rate compounded over three months would have absorbed the spike into an average and barely noticed it. That is a real, if incidental, argument for the compounded-in-arrears conventions of Section 4.4. Predictable calendar effects of the same kind, milder but recurring, appear at each quarter-end and year-end.

Negative rates

A feature of these rates that would have seemed impossible when LIBOR was designed is that they can be, and have been, negative. €STR was negative from its launch until 2022, after the European Central Bank took its deposit facility rate below zero in 2014. TONA was negative from 2016 until the Bank of Japan ended its negative rate policy in March 2024, and SARON was negative from 2015 until 2022 and has fixed slightly below zero again since the SNB cut

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its policy rate to zero in June 2025.

This is not a curiosity. A lognormal model of rates cannot represent a negative rate, so when European rates went below zero the market for rate options moved to **normal** (or Bachelier) and shifted lognormal volatilities, a change Chapter 29 takes up.

A second consequence is contractual. Many floating-rate loans carry a zero floor, so the borrower never receives interest. A floor is an option, and a loan with an embedded floor is a loan plus a short position in that option, a position many borrowers did not realize they had sold until it moved into the money. That short position showed in hedges. A borrower that had swapped a zero-floored loan to fixed with an unfloored swap found that, once the floating rate went negative, the loan's floating payment stopped at zero while the swap's floating leg kept falling, so the borrower paid the fixed rate plus the size of the negative rate. The hedge leaked, which is why such hedgers bought floors to close the gap.

4.3 WHY LIBOR ENDED

LIBOR was, for nearly four decades, the most important number in finance. At its peak it was referenced by contracts with a notional value in the hundreds of trillions of dollars: swaps, futures, floating-rate notes, syndicated loans, student loans, and adjustable-rate mortgages.

Its construction was simple and, as it turned out, fatally so. Each business day a panel of banks was asked a question: at what rate could you borrow funds, were you to do so by asking for and then accepting interbank offers in a reasonable market size? The highest and lowest quarter of the submissions were discarded and the rest averaged, and the result published as that day's fixing for each currency and tenor.

The question is hypothetical. It asks what a bank *could* borrow at, not what it *did* borrow at. During normal conditions the distinction was immaterial. Two developments made it fatal.

The submissions were manipulable, and were manipulated. Because a bank's own derivatives book was sensitive to the fixing, a submitter had a direct financial interest in the answer. Beginning in 2012, investigations across several jurisdictions established that traders at multiple panel banks had requested and obtained submissions favorable to their positions, and that some banks had understated their submissions during the financial crisis to avoid signaling funding difficulty. The resulting fines exceeded $9 billion and several individuals were prosecuted, though several of the convictions were later overturned on appeal (by the US Court of Appeals for the Second Circuit in 2022 and the UK Supreme Court in 2025).

The underlying market disappeared. More fundamentally, banks largely stopped funding themselves through term unsecured interbank borrowing after the crisis. By the late 2010s, some LIBOR tenors were supported by only a small number of transactions in a day. A benchmark that hundreds of trillions of dollars of contracts depend upon cannot rest on the expert judgment of panels of 11 to 16 banks per currency answering a hypothetical question about a market that no longer trades.

In July 2017 the UK Financial Conduct Authority (FCA) announced that it would not compel panel banks to make submissions beyond the end of 2021, which amounted to setting a date for LIBOR's death. Most settings ceased at the end of 2021. The remaining and most important ones (overnight, one-, three-, six-, and twelve-month US dollar LIBOR) ceased on June 30, 2023. A temporary "synthetic" version (no longer a bank submission but Term SOFR plus the fixed spread adjustment described below) was published for the one-, three-, and six-month US dollar settings only, for use in legacy contracts only, until September 30, 2024; the sterling and yen synthetic settings had ended earlier.

Not every interbank offered rate was retired. The Euro Interbank Offered Rate (EURIBOR), reformed in 2019 to a hybrid methodology that anchors submissions in transactions where they exist, remains the principal floating index for euro loans and swaps, and the Tokyo Interbank

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Offered Rate (TIBOR), Australia's bank bill swap rate (BBSW), the Hong Kong Interbank Offered Rate (HIBOR), and the Shanghai Interbank Offered Rate (Shibor) all continue. The Canadian Dollar Offered Rate (CDOR) ceased at the end of June 2024 in favor of the Canadian Overnight Repo Rate Average (CORRA). Where such a rate survives, it sits beside an RFR rather than in place of one, and the two-curve arithmetic of Section 4.10 applies to it directly.

Fallbacks: how existing contracts moved

Announcing that a rate will stop is straightforward. Moving the existing stock of contracts is not. A thirty-year swap written in 2015 against three-month LIBOR needs an answer to the question of what it references in 2030.

The answer for derivatives was the **ISDA 2020 IBOR Fallbacks Supplement and Protocol** (IBOR being the generic name for LIBOR-type interbank offered rates), which took effect in January 2021 and which the great majority of the market adhered to. It specifies that on cessation, a contract referencing LIBOR instead references

$$\text{compounded RFR in arrears} \;+\; \text{a fixed spread adjustment}$$

The spread adjustment is necessary because LIBOR contained bank credit risk and the RFR does not; simply substituting one for the other would transfer value from one side of every contract to the other. The adjustment was set as the five-year historical median of the spread between LIBOR and the compounded RFR, and, crucially, was fixed permanently on March 5, 2021, the date the FCA announced the cessation timetable. Table 4.2 gives the US dollar values.

Table 4.2: ISDA fallback spread adjustments for US dollar LIBOR, fixed March 5, 2021.

LIBOR tenor	Spread adjustment (bp)
1 month	11.448
3 months	26.161
6 months	42.826
12 months	71.513

Fixing the spread on a single date was the decisive design choice. It made every legacy contract's future economics knowable in advance, which is what allowed the transition to happen without a wave of disputes. It also means those numbers are now permanent features of the legacy book: a three-month dollar LIBOR swap that fell back in 2023 will reference SOFR plus 26.161 basis points for the rest of its life, regardless of what bank credit spreads do. Example 4.1 follows one such swap through a single payment.

Example 4.1 A legacy swap after fallback

A \$50 million interest rate swap written in 2015 pays three-month USD LIBOR quarterly. Following cessation, its floating leg references compounded SOFR in arrears plus the fixed adjustment of 26.161 basis points.

For an accrual period of 92 days over which compounded SOFR works out to 4.28%, the floating rate is

$$4.28\% + 0.26161\% = 4.54161\%$$

and, on an actual/360 basis, the interest amount is

$$\$50{,}000{,}000 \times 0.0454161 \times \frac{92}{360} = \$580{,}317$$

Had LIBOR survived and fixed at 4.55% for the same period, the payment would have been \$581,389, a difference of about \$1,072, roughly 0.2% of the coupon.

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That difference is deliberate. The spread adjustment was set to match the historical median gap between the two rates over five years; it was never intended to reproduce LIBOR period by period. Fixing the adjustment put its level beyond either party's influence, and made whatever value it transferred between the two sides known in advance and therefore priceable, the condition needed for the transition to proceed without renegotiating millions of contracts.

The protocol covered derivatives, whose counterparties could adhere to it. Cash products (loans, floating-rate notes, securitizations) carry their own fallback language, and many older US contracts had none that worked once LIBOR stopped. The Adjustable Interest Rate (LIBOR) Act of March 2022 and the Federal Reserve's implementing Regulation ZZ moved those contracts by statute, replacing US dollar LIBOR with a SOFR-based rate plus the same ISDA spread adjustment; for most cash contracts the replacement rate is CME Term SOFR, the forward-looking rate of Section 4.6.

4.4 COMPOUNDING AN OVERNIGHT RATE IN ARREARS

We now come to the mechanical heart of the chapter. A contract needs an interest amount for a period (a quarter, say). The reference rate supplies one number per business day. How do we get from one to the other?

The answer adopted almost universally is to **compound the daily rates over the interest period**. The economic logic is that of rolling an overnight deposit: lend for one day at today's rate, reinvest principal and interest tomorrow at tomorrow's rate, and continue to the end of the period. The compounded rate is whatever single rate would have produced the same terminal value.

Formally, for an interest period running from T_1 to T_2 and containing d_b business days and d_c calendar days, with day-count base B:

$$R \;=\; \left[\prod_{i=1}^{d_b}\left(1+\frac{r_i\,n_i}{B}\right)-1\right]\times\frac{B}{d_c} \tag{4.1}$$

where r_i is the overnight rate published for business day i and n_i is the number of calendar days that rate applies for. On most days $n_i = 1$; on a Friday $n_i = 3$, because Friday's rate accrues across the weekend; around holidays it can be larger. The base depends on the currency: SOFR and €STR use actual/360, so $B = 360$, while SONIA and TONA use actual/365, so $B = 365$.

Two things about (4.1) deserve emphasis. First, the product is taken over *business* days, each rate weighted by the calendar days it covers, but the final scaling is by *calendar* days; this is what makes weekends accrue correctly. Second, and more consequentially:

R is not known until the last business day of the interest period. A payment based on it cannot be calculated at the start of the period, only at the end. This is what "in arrears" means, and it is the single largest operational difference between the RFR world and the LIBOR world.

Figure 4.1 contrasts the two arrangements. Under LIBOR a single fixing at T_1 determined the entire coupon, so both parties knew the amount for the whole period. Under an RFR the coupon accumulates one day at a time and is complete only at T_2—which is why every convention in Section 4.5 exists.

Example 4.2 Compounding SOFR over one week

Consider a one-week interest period running from Monday the 6th to Monday the 13th: seven calendar days containing five business days, with Friday's rate applying for three days. SOFR

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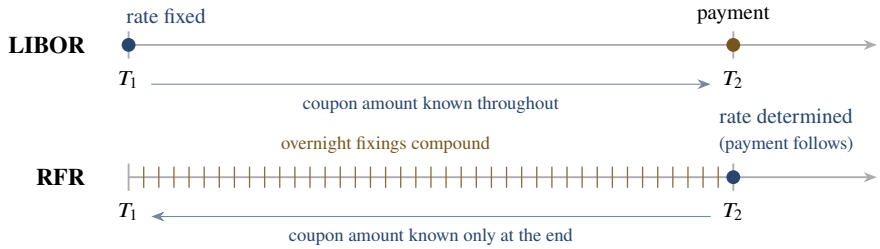


Figure 4.1: A rate set in advance against a rate determined in arrears.

accrues on actual/360, so $B = 360$.

Business day i	Date	r_i	n_i
1	Monday 6	4.31%	1
2	Tuesday 7	4.30%	1
3	Wednesday 8	4.32%	1
4	Thursday 9	4.33%	1
5	Friday 10	4.30%	3

The daily accrual factors are $1 + r_i n_i/360$:

$$1.000119722, \quad 1.000119444, \quad 1.000120000, \quad 1.000120278, \quad 1.000358333$$

Their product is 1.000838036, so from (4.1)

$$R = (1.000838036 - 1) \times \frac{360}{7} = 4.3099\%$$

For comparison, the simple day-weighted average of the same fixings is

$$\frac{4.31 + 4.30 + 4.32 + 4.33 + 3 \times 4.30}{7} = \frac{30.16}{7} = 4.3086\%$$

The compounded rate exceeds the simple average by 0.13 basis points over one week. The gap widens with the period: at a 4% rate level, compounding daily over a full year adds about 8 basis points relative to simple accrual. Over a quarter it is about 2 basis points: small, but not small enough to ignore in a market that quotes to a tenth of one. The gap grows roughly with the square of the rate level and in proportion to the term, so it is negligible when rates are near zero and becomes material quickly as they rise.

Published averages and indices

Because the arithmetic in Example 4.2 is tedious and error-prone, administrators publish it. The Federal Reserve Bank of New York publishes 30-, 90-, and 180-day compounded SOFR averages, and a **SOFR Index** that accumulates the daily compounding from a fixed start date. The index makes the compounded rate between any two dates a two-lookup calculation:

$$R = \left[\frac{\text{Index}(T_2)}{\text{Index}(T_1)} - 1 \right] \times \frac{B}{d_c} \tag{4.2}$$

where, as in (4.1), T_1 and T_2 are the start and end of the interest period, d_c is the number of calendar days between them, and $B = 360$ for SOFR. This is worth knowing not as a shortcut but because it eliminates a class of dispute: two parties reading the same two index values cannot disagree about the interest amount, whereas two parties each compounding sixty daily fixings can and do.

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4.5 CONVENTIONS THAT MAKE COMPOUNDING WORKABLE

The problem with a rate known only on the final day of a period is that payment systems need notice. A borrower with a coupon due on the last day of a quarter cannot wait until the close of business that day to learn the amount; by then it is too late to arrange the payment.

The market's response was a family of conventions, each trading a small amount of economic precision for a usable notice period. A practitioner needs to recognize all of them, because a contract's value depends on which one it uses.

Payment delay. The simplest fix is to compound over the full interest period and pay a few business days after it ends. It is economically the cleanest, since the observation window matches the accrual window exactly. Its cost is a small timing adjustment, treated in Chapter 30.

Lookback. The rate for each day of the accrual period is taken from a fixed number of business days earlier, typically five. The final observation therefore occurs five days before period end, giving the notice required. In a **lookback without observation shift**, each observed rate is weighted by the calendar days of the accrual date it is applied to; in a **lookback with observation shift**, it is weighted by the calendar days of the observation date. The distinction sounds pedantic and is not. Without the shift, each rate is weighted by days that belong to a different date, a weighting mismatch: the compounded rate can no longer be reproduced from discount factors or from the SOFR Index, so it is not exactly the rate the curve prices. With the shift the observation period is a genuine compounding period, which is why the shifted convention is generally preferred.

Lockout (rate cutoff). The rate is observed normally until a cutoff a few days before period end, after which the last observed rate is frozen and repeated for the remaining days. It is simple to operate, at the cost of ignoring genuine rate movements in the final days, an awkward property if the cutoff spans a central bank meeting.

Last reset. This abandons in-arrears determination entirely, using the *previous* period's compounded rate as the current period's coupon. The borrower knows the coupon at the start of the period, as under LIBOR. The cost is that the rate is paid one period after it is determined, a timing adjustment priced with the machinery of Section 30.2 (Chapter 30 discusses this case at the end of Section 30.4).

Table 4.3 sets them side by side, writing k for the lookback, lockout, or delay length in business days, typically two to five. The column that matters operationally is the third: how much notice the payer gets. The column that matters for valuation is the fourth.

Figure 4.2 shows the same information as windows on a timeline, which makes the family resemblance clearer than the table can: each convention is the accrual period with its observation window slid, truncated, or displaced. Payment delay observes exactly the accrual period and defers only the payment. A lookback slides the whole window earlier. A lockout truncates it and repeats the last observed rate across the pale segment. Last reset abandons the current period altogether and takes the coupon from the previous one.

The choice among these is not merely operational. Each convention implies a slightly different exposure, and a portfolio hedged on the assumption of one convention while transacted under another carries a small unhedged basis.

Cleared SOFR swaps use payment delay, usually two business days, and the interdealer market is uniform in this. Floating-rate notes use a lookback: five banking days with observation shift is the sterling standard for SONIA notes, while dollar notes have used both the shifted and unshifted forms, and the first agency SOFR notes of 2018 and 2019 paid a simple average of daily SOFR with a lockout before the market settled on compounding. Loans and securitizations, constrained by what a servicing system can compute, use lookbacks or fall back on last reset, or escape the question altogether by referencing the term rate of Section 4.6. The convention

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Table 4.3: Conventions for compounding an overnight rate in arrears.

Convention	Observation window	Notice before payment	Principal cost
Payment delay	Matches the accrual period exactly	k days	Timing adjustment (Section 30.2)
Lookback, no observation shift	Shifted back k days; weights from accrual dates	k days	Weighting mismatch; rate not reproducible from discount factors or the index
Lookback with observation shift	Shifted back k days; weights from observation dates	k days	Minimal; observation period length may differ slightly from accrual
Lockout	Accrual period, but the final rate is frozen for the last k days	k days	Ignores rate moves late in the period, awkward across a policy meeting
Last reset	The *previous* accrual period	Full period	Coupon reflects a stale rate; timing adjustment (Section 30.2)

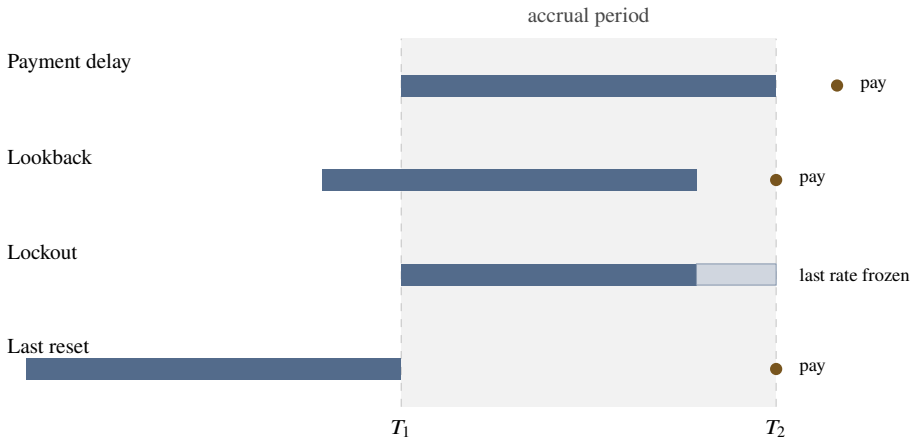


Figure 4.2: Observation windows (solid bars) against the accrual period (shaded).

the Alternative Reference Rates Committee (ARRC), the market group that steered the dollar transition, recommended for SOFR loans is **daily simple SOFR**, which accrues each day's rate without compounding and so lets a borrower repay or draw down on any day without recomputing the period's interest.

Futures draw a distinction of their own. Three-month SOFR futures settle against daily SOFR *compounded* over the reference quarter, while one-month contracts settle against a *simple arithmetic average* over the calendar month. The underlying rate is the same and the accumulation is not, which is part of what the basis between the two contracts prices.

The division between derivatives and loans is not accidental. The derivatives market could rebuild its infrastructure once and centrally, through its clearing houses, and so adopted the convention that is cleanest economically. The loan market took whatever its existing servicing systems could support, fragmented across lookbacks and last reset, and in the end reached for a forward-looking term rate instead—which is where the next section begins.

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4.6 TERM RATES AND CREDIT-SENSITIVE RATES

Two categories of rate sit alongside the compounded overnight rates, and both exist because some users genuinely wanted what LIBOR provided.

Term SOFR

A **term rate** is a forward-looking rate for a period, known at the start of that period (LIBOR's defining feature). Term SOFR is constructed not from submissions but from the SOFR derivatives market: CME publishes one-, three-, six-, and twelve-month term rates derived from SOFR futures prices, which embed the market's expectation of the average overnight rate over each period.

This is a sounder construction than LIBOR's, because it rests on traded prices. But it carries a structural vulnerability, circularity: Term SOFR is built from derivatives that reference overnight SOFR, so it is only as robust as the volume of those derivatives. If exposures migrated from overnight SOFR to the term rate, with hedges following them, the market the term rate is computed from would thin while the exposures depending on it grew, which is the failure mode that ended LIBOR.

For that reason the ARRC recommended Term SOFR for a deliberately restricted set of uses: business loans, and derivatives entered into to hedge those loans. It explicitly did *not* recommend it for the general derivatives market. The LIBOR Act added a third use by statute, as the replacement rate for legacy cash contracts (Section 4.3). The result is a two-tier structure, with loans referencing a forward-looking term rate and the derivatives market referencing compounded overnight rates, and the basis between them must itself be managed.

Credit-sensitive rates

A bank that funds itself unsecured and lends at a rate tied to SOFR has a genuine problem: in a crisis its funding costs rise while SOFR, being risk-free, does not. LIBOR's embedded credit component hedged that exposure automatically.

Several credit-sensitive benchmarks were launched to fill the gap, of which the most prominent was Bloomberg's Short-Term Bank Yield Index (BSBY). Regulators, including the SEC chair, questioned publicly whether it rested on sufficient transaction volume, and raised the concern that it reproduced LIBOR's central defect: a thin underlying market supporting enormous referencing volume, with the added feature that the thinness would be worst precisely during the stress the index was supposed to measure. Bloomberg announced its discontinuation in 2023 and ceased publication in November 2024. Smaller credit-sensitive indices such as Ameribor continue on a modest scale.

The episode is worth remembering when evaluating any proposed benchmark. The question is never only whether the index measures the right economic quantity. It is whether the market underneath it is deep enough to survive being referenced at scale, and whether it stays deep when conditions deteriorate.

4.7 MEASURING INTEREST RATES

We now turn to the traditional apparatus, beginning with a point that causes more confusion than any other: a quoted rate is meaningless without its compounding frequency. Suppose \$100 is invested for one year at 10%. With annual compounding the terminal value is \$110. With semiannual compounding it is $100 \times 1.05^2 = \$110.25$; with quarterly compounding $100 \times 1.025^4 = \$110.38$; with monthly, \$110.47. The same quoted number describes four different investments.

In general, an amount A invested for n years at rate R with compounding m times per year

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grows to

$$A\left(1+\frac{R}{m}\right)^{mn} \tag{4.3}$$

Letting $m \to \infty$ gives **continuous compounding**, under which the terminal value is

$$A\,e^{Rn} \tag{4.4}$$

Continuous compounding is used throughout the option-pricing chapters of this book, because it makes the algebra dramatically simpler: discount factors multiply as $e^{-R_1 t_1} e^{-R_2 t_2} = e^{-(R_1 t_1 + R_2 t_2)}$, with no adjustment terms. It is a mathematical convenience, not a market convention—no instrument actually pays interest continuously.

Converting between conventions is essential and straightforward. Equating (4.3) and (4.4), a rate R_m with compounding frequency m corresponds to a continuously compounded rate

$$R_c = m\ln\left(1+\frac{R_m}{m}\right) \tag{4.5}$$

and conversely

$$R_m = m\left(e^{R_c/m}-1\right) \tag{4.6}$$

Example 4.3 applies each in turn.

Example 4.3 Converting compounding frequencies

Consider a rate of 4% per annum with semiannual compounding. From (4.5), the equivalent continuously compounded rate is

$$R_c = 2\ln(1.02) = 2\times 0.019803 = 3.9605\%$$

Conversely, a rate of 4% with continuous compounding corresponds, from (4.6), to

$$R_2 = 2\left(e^{0.02}-1\right) = 4.0403\%$$

with semiannual compounding. The continuously compounded rate is always the lower of the two, since more frequent compounding achieves a given terminal value with a smaller quoted rate.

Day count conventions. A rate quotation is also incomplete without a day count, which specifies how a partial year is measured. The interest for a period is the notional times the rate times a year fraction, and the day count sets that fraction:

$$\text{interest} = N \times R \times \frac{\text{days in the period}}{\text{days in the year}},$$

with both counts defined by the convention. Four are in common use:

- **Actual/360**: actual days over 360. US dollar and euro money-market instruments and floating legs, the overnight rates SOFR and €STR, the annual fixed legs of swaps on SOFR and €STR, and China's Shibor;
- **Actual/365 (fixed)**: actual days over 365. Sterling (SONIA), the overnight rates TONA and CORRA, Japanese government bonds, and Chinese repo, swaps on the seven-day repo fixing FR007, and renminbi fixed legs;
- **Actual/actual**: actual days over the actual length of the year or coupon period. US Treasury coupons and most other government bonds;
- **30/360**: every month counts as 30 days and the year as 360. US corporate bonds, and the fixed legs of euro swaps and of the older US dollar LIBOR swaps.

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The difference is not trivial: an actual/360 rate of 4% earns interest on 365 days of a normal year, giving an actual/365-equivalent of $4\% \times 365/360 = 4.0556\%$. Mismatched day counts across the legs of a trade are a persistent source of small, systematic pricing errors.

4.8 ZERO RATES, BOND PRICING, AND BOOTSTRAPPING

The n-year **zero rate** (or spot rate) is the rate earned on an investment made today that pays nothing until year n. It is the term structure's fundamental object: every other rate is built from it.

A coupon-bearing bond's price is the sum of its cash flows discounted at the zero rate appropriate to each:

$$B = \sum_{i=1}^{n} c_i \, e^{-R_i t_i} \tag{4.7}$$

where c_i is the cash flow at time t_i and R_i the continuously compounded zero rate for maturity t_i. The bond's **yield** is the single rate y that, used for all maturities in (4.7), reproduces the observed price. The **par yield** is the coupon rate that makes the bond price equal to par.

Bootstrapping the OIS curve

Zero rates are not directly observable beyond the shortest maturities; they must be extracted from traded instruments. In the current market the natural instruments are **overnight indexed swaps** (OIS), swaps exchanging a fixed rate for the compounded overnight rate, because these are the most liquid and are what collateralized derivatives discount at. The extraction procedure, **bootstrapping**, works outward from the shortest maturity, using each newly determined discount factor to solve for the next.

Example 4.4 Bootstrapping a zero curve from OIS rates

Suppose annual-paying OIS rates are quoted as 4.00%, 4.20%, and 4.30% for one, two, and three years. Because a swap has zero value at inception, and its floating leg plus a final notional payment is worth par (Section 7.3 shows why), its fixed rate (the swap's **par rate**) is by construction a par yield: each quote says that a bond paying that coupon prices at par.

One year. A single payment of 1.04 at $t = 1$ must be worth 1, so

$$\mathrm{DF}_1 = \frac{1}{1.04} = 0.961538, \qquad z_1 = 4.0000\%$$

Two years. The two-year par bond pays 0.042 at $t = 1$ and 1.042 at $t = 2$:

$$0.042 \times 0.961538 + 1.042\,\mathrm{DF}_2 = 1 \implies \mathrm{DF}_2 = \frac{1 - 0.040385}{1.042} = 0.920936$$

so $(1 + z_2)^2 = 1/0.920936 = 1.085852$ and $z_2 = 4.2042\%$.

Three years.

$$0.043\,(0.961538 + 0.920936) + 1.043\,\mathrm{DF}_3 = 1 \implies \mathrm{DF}_3 = \frac{1 - 0.080946}{1.043} = 0.881164$$

so $(1 + z_3)^3 = 1.134863$ and $z_3 = 4.3072\%$.

At one year the zero and par rates coincide, since there is only one payment; beyond it each zero rate sits slightly above the corresponding par rate, which is the general pattern for an upward-sloping curve.

In practice the procedure is complicated by instruments that do not mature on convenient dates, by the need to interpolate between them, and by the requirement that the resulting curve be smooth enough that forward rates implied from it are not absurd. Curve construction is

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a specialty in its own right, and the choice of interpolation method is a genuine source of valuation differences between dealers.

4.9 FORWARD RATES AND FORWARD RATE AGREEMENTS

Forward rates are the future rates implied by today's zero curve. If investing for T_2 years must give the same result as investing for T_1 years and then reinvesting the proceeds at the forward rate until T_2, then the forward rate for the period $[T_1, T_2]$ satisfies

$$(1+z_1)^{T_1}(1+F_a)^{T_2-T_1} = (1+z_2)^{T_2}, \qquad 1+F\tau = \frac{\mathrm{DF}_1}{\mathrm{DF}_2} \tag{4.8}$$

where z_1 and z_2 are the annually compounded zero rates to T_1 and T_2, DF_1 and DF_2 the corresponding discount factors, and $\tau = T_2 - T_1$. The two forms state the same forward with different compounding: F_a is compounded annually, and F is a simple rate, the convention forward rate agreements use and the one meant below by "the forward rate." They coincide only for a one-year period, as in the applications below; otherwise $1 + F_a = (1+F\tau)^{1/\tau}$. In continuously compounded terms the relationship (4.9) is cleaner still:

$$e^{R_1 T_1} \times e^{F_c (T_2 - T_1)} = e^{R_2 T_2}, \qquad \text{so} \qquad F_c = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1} \tag{4.9}$$

where R_1 and R_2 are the continuously compounded zero rates and F_c the continuously compounded forward.

Applying (4.8) to Example 4.4 gives the one-year forward rate starting in one year, and the one-year rate starting in two years:

$$\frac{0.961538}{0.920936} - 1 = 4.4088\%, \qquad \frac{0.920936}{0.881164} - 1 = 4.5136\%.$$

Note how much more steeply the forward curve rises than the zero curve, as Figure 4.3 shows: zero rates are averages of forwards, so a modestly rising zero curve implies a considerably steeper forward curve. This amplification is a recurring feature and a frequent source of surprise.

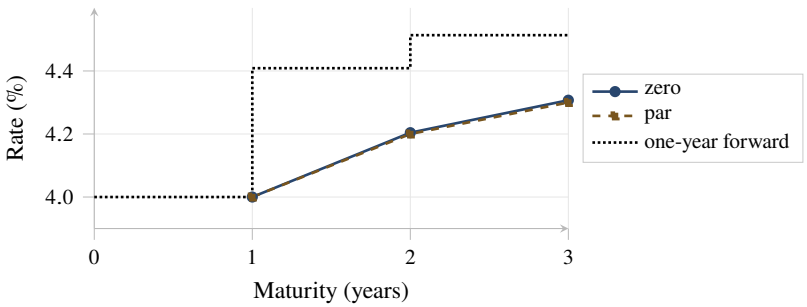


Figure 4.3: Par, zero and one-year forward rates from Example 4.4.

Forward rate agreements. A forward rate agreement (FRA) is an agreement to exchange a fixed rate for a floating rate on a notional principal for a future period. Its value follows from a single observation: an FRA is worth zero when the fixed rate equals the forward rate. More generally, an FRA in which the fixed rate R_K is received in exchange for the realized rate, with notional L over accrual τ, is worth

$$V = L\,\tau\,(R_K - F)\,\mathrm{DF}_2 \tag{4.10}$$

to the party receiving fixed, where F is the current forward rate. Equation (4.10) is the simplest instance of the general principle developed in Chapter 30: we value the contract by substituting

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the forward rate for the unknown future rate, an operation whose validity is not automatic and whose exceptions occupy that chapter.

An FRA traditionally settles at the start of the period, T_1, as soon as the rate has fixed, rather than at T_2 when the interest would be due. The payment is therefore the period's interest difference discounted back over the period at the realized rate R: the party paying fixed receives

$$L\,\tau\,\frac{R - R_K}{1 + R\tau}$$

at T_1, a negative amount meaning it pays.

FRAs were a major product in the LIBOR era, when a single fixing determined the settlement, and they still trade where a term rate survives: banks and corporate borrowers use EURIBOR FRAs to hedge the next few resets on euro loans. In the RFR currencies their role has narrowed considerably, since a short OIS achieves the same exposure with conventions that match the rest of the market. The short OIS market is where dealers, banks, and funds position for central-bank decisions, often through meeting-dated OIS whose accrual periods run from one policy meeting to the next.

4.10 THE MULTI-CURVE REALITY

One assumption made implicitly above needs flagging, because Chapter 7 dismantles it. We have spoken of "the" zero curve, as though a single term structure describes a currency. Before 2008 that was approximately true. It is not true now, for a simple reason: discounting a collateralized derivative correctly requires the rate paid on the collateral, which under a standard credit support annex is the overnight rate. Meanwhile the rate a contract's cash flows are *indexed* to may be something else: EURIBOR (the largest surviving interbank rate), Term SOFR, a foreign currency's RFR, or a legacy fallback rate.

The consequence is that valuation requires at least two curves: a discount curve built from OIS, and one or more forecast curves for the indices the contract references. Their difference is not noise. Cross-currency basis spreads (the additional spread required to swap one currency's RFR against another's) have at times exceeded 50 basis points and are themselves traded.

For the present chapter the single-curve treatment is adequate and keeps the mechanics visible. Chapter 7 develops the multi-curve framework properly, and treats it as the standard case rather than as a correction to a single-curve baseline.

4.11 DURATION AND CONVEXITY

We turn to the two standard measures of a bond's interest rate sensitivity, both of which recur throughout the book. Convexity in particular is the subject of Chapter 30.

Duration measures the sensitivity of a bond's price to a parallel shift in the zero curve. With continuous compounding, if the bond has price B and cash flows c_i at times t_i, its duration is the cash-flow-weighted average time to payment:

$$D = \frac{\sum_{i=1}^{n} t_i\, c_i e^{-y t_i}}{B} \tag{4.11}$$

from which

$$\Delta B \approx -B D\, \Delta y \tag{4.12}$$

When rates are quoted with compounding frequency m rather than continuously, the corresponding sensitivity uses **modified duration**, $D^* = D/(1 + y/m)$.

Equation (4.12) is a first-order approximation and inherits the usual limitation: it is a straight line fitted to a curve. The curvature is **convexity**,

$$C = \frac{1}{B}\frac{\partial^2 B}{\partial y^2} \tag{4.13}$$

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giving the improved approximation

$$\Delta B \approx -BD\,\Delta y + \frac{1}{2}B\,C\,(\Delta y)^2 \tag{4.14}$$

Since $C > 0$ for an ordinary bond, the convexity term is positive for a rate move in either direction: a bondholder gains more when rates fall than they lose when rates rise by the same amount. Convexity is therefore desirable, and, as Chapter 30 shows at length, it is priced.

Figure 4.4 shows why. The duration approximation is the tangent to the price–yield curve at the current yield, and because that curve is convex the tangent lies *below* it on both sides. Duration alone therefore understates the price whichever way rates move; the shaded region between the two lines is what the convexity term in (4.14) recovers.

The bond is the one of Example 4.5, and two features of the figure deserve attention. Plotting the percentage change rather than the price level makes the tangent's slope exactly minus the modified duration (here -2.7429 per percentage point of yield). And the yield range shown is deliberately far wider than anything a three-year bond would realistically see: over a plausible range the two lines are almost indistinguishable, which is precisely why duration alone suffices for small moves and fails for large ones.

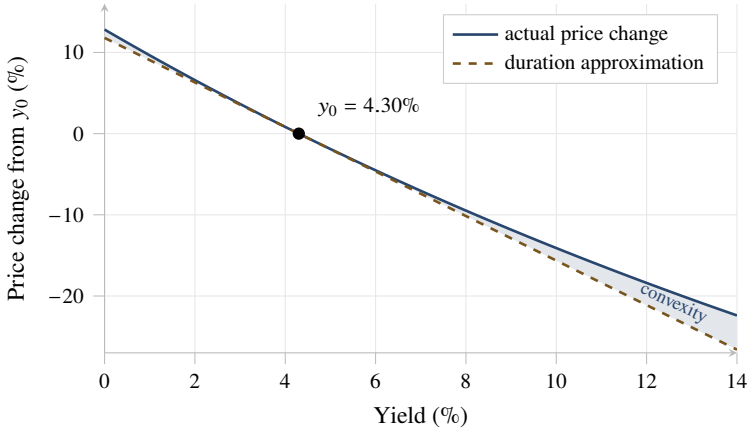


Figure 4.4: Percentage price change against yield for the three-year 5% bond of Example 4.5.

Example 4.5 Duration and convexity of a three-year bond

A three-year bond pays a 5% annual coupon and yields 4.30% with annual compounding. Discounting the cash flows of 5, 5, and 105 gives

$$B = 4.79386 + 4.59623 + 92.54147 = 101.93156$$

The duration from the discrete analog of (4.11), with each cash flow discounted by $(1+y)^{t_i}$ in place of e^{yt_i}, is

$$D = \frac{1\times 4.79386 + 2\times 4.59623 + 3\times 92.54147}{101.93156} = \frac{291.61071}{101.93156} = 2.8608 \text{ years}$$

and the modified duration is $D^* = 2.8608/1.043 = 2.7429$. The convexity, from the same discrete form of (4.13), is

$$C = \frac{1\times 2\times 4.79386 + 2\times 3\times 4.59623 + 3\times 4\times 92.54147}{1.043^2 \times 101.93156} = 10.3499$$

Now suppose the yield rises 50 basis points to 4.80%. The duration-only estimate is

$$\Delta B \approx -101.93156 \times 2.7429 \times 0.005 = -1.39794 \implies B \approx 100.5336$$

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Adding the convexity term from (4.14),

$$\frac{1}{2} \times 101.93156 \times 10.3499 \times 0.005^2 = +0.0132 \implies B \approx 100.5468$$

Repricing the bond exactly at 4.80% gives 100.5467. The duration estimate is off by 1.3 cents per 100 of face; adding convexity reduces the error to a hundredth of a cent. For a 50 basis point move on a three-year bond the refinement is marginal—but the convexity term grows with the square of the rate move and with the square of maturity, and on a thirty-year bond in a volatile market it is indispensable.

4.12 THEORIES OF THE TERM STRUCTURE

Why is the zero curve shaped the way it is? Three explanations are traditionally offered.

Expectations theory. Forward rates equal expected future spot rates. An upward-sloping curve then means the market expects rates to rise. This is clean, and Chapter 30 explains why it cannot be exactly right: forward rates are expectations only under a particular probability measure, and the adjustment between that measure and the real world is not zero.

Market segmentation theory. Short, medium, and long maturities are distinct markets with distinct participants (money funds at the short end, pension funds and insurers at the long end), and rates in each are set by local supply and demand. The theory explains persistent curve anomalies that expectations alone cannot, and it received strong support from the 2022 UK gilt episode, in which forced selling by liability-driven investors moved long gilt yields far more than the change in rate expectations.

Liquidity preference theory. Borrowers prefer to lock in long-term funding while lenders prefer to keep money short and available. Clearing that mismatch requires paying lenders a premium to extend, so forward rates exceed expected future spot rates and the curve slopes upward more often than not. This is the explanation most consistent with the historical record: yield curves have been upward-sloping most of the time, and under the pure expectations theory that would mean the market has persistently expected rising rates.

None of the three is complete, and modern practice treats the curve as an observable to be fitted rather than a prediction to be explained. But the intuitions matter when curves behave strangely, and knowing which force is operating is often the difference between recognizing an opportunity and taking a position the market is about to move against.

SUMMARY

The reference rate at the center of the derivatives market has changed, and with it the conventions that everything else is built on. LIBOR was a forward-looking term rate, fixed at the start of each interest period, derived from banks' estimates of their own unsecured funding costs, and therefore carrying an embedded bank credit component. It was manipulated, and the market underneath it had largely ceased to exist. It was replaced by overnight rates (SOFR, SONIA, €STR, TONA, SARON), each computed from actual transactions and each close to free of credit risk.

The defining consequence is that a term rate must now be built by compounding daily fixings across an interest period, which means the rate is not known until the period ends. The compounding formula weights each business day's rate by the calendar days it covers and scales the result to the full period, on a day-count base that differs by currency: actual/360 for SOFR and €STR, actual/365 for SONIA and TONA. Because in-arrears determination leaves no time to settle payments, the market developed conventions (payment delay, lookbacks with and without observation shift, lockouts, last-reset structures) that each trade a little economic precision for operational notice, and that each imply slightly different exposures.

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Legacy contracts moved to the new rates through the ISDA fallback protocol, which substitutes the compounded RFR plus a spread adjustment fixed permanently in March 2021. Forward-looking term rates exist, built from SOFR futures, but their recommended use is deliberately restricted to loans and their hedges, to avoid recreating LIBOR's vulnerability. Credit-sensitive benchmarks intended to restore banks' funding hedge have largely failed for the same reason, BSBY's discontinuation being the clearest case.

The traditional apparatus survives the transition intact. Rates must always be quoted with a compounding frequency and day count; continuous compounding simplifies the algebra of later chapters. Zero curves are bootstrapped from traded instruments, now principally OIS, and forward rates are implied from them, rising more steeply than the zero curve itself. Duration measures first-order rate sensitivity and convexity the curvature, the latter being both a desirable property of a bond position and, as Chapter 30 develops, a quantity the market prices explicitly.

One simplification made here is corrected in Chapter 7: there is no longer a single curve per currency. Discounting a collateralized trade uses the rate paid on its collateral, while its cash flows may be indexed to something else, so valuation requires a discount curve and one or more forecast curves.

FURTHER READING

Schrimpf and Sushko's "Beyond LIBOR: A Primer on the New Reference Rates" in the March 2019 *BIS Quarterly Review* is a short account of why the benchmarks changed and what replaced them. The Financial Stability Board's progress reports on *Reforming Major Interest Rate Benchmarks* (2014–2023) record the transition as it happened rather than in retrospect. For the mechanics, the Alternative Reference Rates Committee's fallback-language recommendations and its guidance on the scope of use for Term SOFR are the operative documents, and Henrard's 2019 note treats the fallback spread methodology in the detail a practitioner needs.

PRACTICE QUESTIONS

4.1 Explain the two distinct defects in LIBOR's construction that led to its replacement, and state which of the two would have justified replacement even if no manipulation had ever occurred.

4.2 An investment pays 12% per annum with quarterly compounding. Give the equivalent continuously compounded and annually compounded rates, and give the semiannual equivalent of a rate quoted as 5% continuously compounded.

4.3 Explain why SOFR can be described as risk-free even though it rose from 2.43% to 5.25% in a single day in September 2019, and state what kind of risk that episode illustrates.

4.4 An interest period runs Monday to Monday, seven calendar days. The overnight rates on Monday through Friday are 3.95%, 3.96%, 3.94%, 3.97%, and 3.96%. Using (4.1), calculate the compounded rate on an actual/360 basis and compare it with the day-weighted simple average.

4.5 Explain the difference between a lookback with an observation shift and one without, and why the market generally prefers the shifted convention.

4.6 Explain why it mattered that the ISDA fallback spread adjustments were fixed permanently on a single date rather than recalculated periodically.

4.7 Annual-paying OIS rates are 3.50%, 3.70%, and 3.85% for one, two, and three years. Bootstrap the zero rates and discount factors, calculate the one-year forward rates beginning in one and two years, and explain why an upward-sloping zero curve implies a forward curve that rises more steeply.

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4.8 A five-year bond pays a 6% annual coupon and yields 5%. Calculate its price, duration, and modified duration, estimate the price change for a 25 basis point increase in yield, and compare with the exact repricing.

4.9 A forward rate agreement fixes a rate on a notional for a future period. Explain how the forward rate is extracted from two zero rates, why the contract settles at the start of the period rather than at its end, and what that settlement convention requires of the discounting.

4.10 Explain why a single discount curve was adequate before 2008 and is not now. State which curve discounts a collateralized swap and which forecasts its floating payments, and explain why the two being different is a statement about collateral rather than about the credit of the counterparties.

4.11 State the theories of the term structure the chapter sets out, and give what each predicts about the shape of the curve when short rates are expected to stay flat. Explain which prediction the historical record supports, and what that implies for reading a forward rate as a forecast.

FURTHER QUESTIONS

4.12 A bank makes floating-rate loans indexed to Term SOFR and funds itself with unsecured deposits. Describe the exposure this creates, explain why LIBOR did not create it, and discuss why credit-sensitive benchmarks intended to solve the problem have largely failed.

4.13 Using the SOFR Index relationship (4.2), show that the compounded rate between any two dates can be obtained from two index values. Explain why this eliminates a class of settlement disputes, and identify what could still be disputed.

4.14 Term SOFR is derived from SOFR futures rather than from submissions. Explain why this makes it more robust than LIBOR, and why regulators nonetheless restricted its recommended scope of use. What would have to be true of the futures market for the restriction to be unnecessary?

4.15 A thirty-year swap written in 2015 referenced three-month USD LIBOR. Describe precisely what it references today, and explain how the value of the contract was preserved through the transition. Under what circumstances would a party have been economically disadvantaged by the fallback?

4.16 Consider a floating-rate note paying compounded SOFR in arrears with a five-day payment delay, and an otherwise identical note using a five-day lookback without observation shift. Explain qualitatively how their values differ, which adjustment from Chapter 30 applies to the first, and why the second cannot be valued exactly from the discount curve.

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Chapter 5
Determination of Forward and Futures Prices

Chapter 2 established that the futures price converges to the spot price at delivery. This chapter answers the question that leaves open: what determines it *before* delivery?

The answer, for a large class of assets, is remarkably tight. The forward price is not a forecast, not a consensus view, and not the result of supply and demand for the contract. It is pinned by arbitrage to a specific relationship with the spot price, and a trader who quotes anything else can be taken apart by someone with a bank account and no opinion whatsoever about the future direction of the underlying price.

That is the good news. The rest of the chapter is about where the argument weakens: assets that cannot be sold short, assets held for use rather than investment, and the fact that the arbitrage requires a balance sheet, which is not free (a theme of Chapters 6, 7, and 8).

What arbitrage means. An **arbitrage opportunity** is a trading strategy that costs nothing to set up, can never lose money, and makes money in at least one state of the world; equivalently, it pays money today and can never lose thereafter. The argument of this chapter, and the pricing in every chapter after it, rests on the **no-arbitrage assumption**: such opportunities cannot persist, because anyone who found one would trade it in unlimited size and prices would move until it was gone. It is an assumption about prices, not about people. It does not say that arbitrages never appear, only that the market does not leave them lying about, and Section 5.15 examines where it fails in practice.

Section 5.1 draws the distinction the chapter turns on, between assets held for investment and assets held for use, and sets out what short selling requires. Sections 5.2 to 5.5 are the arbitrage argument, its extensions to known income and known yield, and the value of a forward already on the books. Sections 5.6 to 5.8 apply it to the three markets where it is tested daily (index futures, currencies, and commodities), and Sections 5.9 to 5.11 collect the pieces into the cost of carry, the implied repo rate, and the roll yield of a position held across contracts. Sections 5.12 and 5.13 settle two questions that are often confused: whether forward and futures prices differ, and whether either forecasts the spot price. Section 5.14 shows how the shape of the curve is traded, and Section 5.15 is where the arbitrage argument stops.

5.1 INVESTMENT ASSETS, CONSUMPTION ASSETS, AND SHORT SELLING

The arbitrage arguments that follow depend on a distinction. An **investment asset** is held by a significant number of investors purely for investment (gold, silver, stocks, bonds, currencies). It need not be held *only* for investment; gold is also used industrially. What matters is that enough holders are indifferent between holding the physical asset and holding a claim on it, so that arbitrage can proceed by borrowing the asset from them.

A **consumption asset** is held primarily to be used: oil, copper, wheat, live cattle. No one holds crude oil as a store of value. Its holders fail the condition above: they are *not* indifferent

between the physical asset and a claim on it, because what they want from it is the use. A refinery needs barrels in the tank to keep the plant running, and a claim on barrels next month will not run it.

Two consequences follow, and the second matters more than the first. Such a holder will not lend the asset, so the short sale that the arbitrage of Section 5.2 depends on is simply unavailable: you cannot readily borrow oil from a refinery that intends to process it next week. But even a holder who owns the asset outright, and needs to borrow nothing from anyone, will not *sell it now and buy it back later*, however high today's price, because the whole point of holding it is to have it in the meantime. The trade that would keep a forward price from falling too far below what it costs to hold the asset until delivery is therefore not merely hard to finance: no one who has the asset wants to do it. That is what turns this chapter's equalities into one-sided inequalities for commodities, and it is the economic content of the convenience yield in Section 5.8.

Short selling. To sell short, an investor borrows the asset from a holder, sells it, and later buys it back to return. The mechanics matter for pricing. The short seller must pass through any income the asset generates (dividends, coupons) to the lender, and pays a **borrowing fee** for the loan. For liquid stocks that fee is a few basis points; for hard-to-borrow stocks it can be tens of percent, and the position may be recalled at any time.

Every result in this chapter assumes short selling is costless where it is possible. Where it is not possible at all, the arbitrage works in only one direction, and the results become inequalities rather than equalities, as Section 5.8 shows.

Notation and assumptions. Throughout, S_0 is today's spot price, S_T the spot price at delivery, F_0 the forward or futures price, T the time to delivery, and r the continuously compounded risk-free rate for maturity T (Section 4.7). The standing assumptions are that participants face no transaction costs, are subject to the same tax rate, can borrow and lend at r, and take arbitrage opportunities when they arise.

The third of these deserves skepticism from the outset: no one borrows at the risk-free rate. We keep the assumption because it makes the structure clear, and return to what it costs in Section 5.15.

5.2 THE BASIC ARBITRAGE

Consider an investment asset providing no income and costing nothing to store (a non-dividend-paying stock is the standard example). Then

$$\boxed{F_0 = S_0 e^{rT}} \tag{5.1}$$

The argument is worth setting out in both directions, because the pair of trades is the template for everything that follows. Each direction is an arbitrage opportunity in the sense defined at the start of the chapter: it costs nothing, cannot lose, and makes money if the forward price is out of line.

If $F_0 > S_0 e^{rT}$, borrow S_0 at rate r, buy the asset, and short a forward contract. At T, deliver the asset for F_0 and repay $S_0 e^{rT}$. The profit is $F_0 - S_0 e^{rT} > 0$, with no investment and no risk. This is the **cash-and-carry** trade: borrow cash, buy the asset, carry it to delivery. It is the trade the rest of the book refers to by that name.

If $F_0 < S_0 e^{rT}$, short the asset for S_0, invest the proceeds at r, and take a long forward. At T, receive $S_0 e^{rT}$ from the investment, pay F_0 for the asset under the forward, and return it to the lender. The profit is $S_0 e^{rT} - F_0 > 0$. This is the **reverse cash-and-carry**, and Section 5.1's question of who will part with the asset is the question of who can do this one.

Since neither trade can persist, equality must hold.

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Example 5.1 A forward on a non-dividend-paying stock

A stock trades at \$50, the six-month risk-free rate is 4%, and the stock pays no dividend. From (5.1) the six-month forward price is

$$F_0 = 50\, e^{0.04\times 0.5} = \$51.01$$

Suppose a dealer quotes \$52. Borrow \$50, buy the stock, short the forward. Six months later, deliver the stock for \$52 and repay \$51.01, keeping \$0.99 per share, with no capital committed and no exposure to where the stock went.

Suppose instead the quote is \$50. Short the stock, invest the \$50 to receive \$51.01, and buy the stock back at \$50 under the forward. Profit \$1.01.

No view on the stock is required, and no probability distribution enters. The forward price is determined by the spot price and the interest rate alone, which is why Chapter 2's remark that a futures price is not a forecast is more than a caution.

The constant rate is a convenience, not an assumption of the argument. If the purchase is financed by selling zero-coupon bonds maturing at T, rather than by borrowing at r, the same two trades show that

$$F_0 = \frac{S_0}{P(0,T)} \tag{5.2}$$

where $P(0,T)$ is today's price of a zero-coupon bond paying \$1 at T. This holds whatever rates do in between; only the bond's traded price enters, not a model for it. A constant rate makes $P(0,T) = e^{-rT}$, and (5.2) is (5.1) again. Section 28.7 starts from (5.2), in an economy where rates are stochastic by assumption.

When short selling is costly

Example 5.1 used both arbitrages, and the second borrows the stock. It is natural to expect that an expensive borrow degrades the equality into a band, and eventually it does, but not for the obvious reason, and not as readily.

The lower bound does not need a short sale at all. If $F_0 < S_0 e^{rT}$, an investor who *already owns* the asset can sell it, invest the proceeds at r, and take a long forward. At T that investor has the asset back, having received $S_0 e^{rT}$ and paid F_0, and is better off than by holding throughout. Nothing has been borrowed and no fee has been paid. This is exactly why an investment asset was defined in Section 5.1 by holders who are indifferent between the asset and a claim on it: their willingness to exchange one for the other is what defends the lower bound, and the cost of borrowing is nothing to them.

A borrowing fee therefore does not by itself widen the no-arbitrage range. What widens it is holders who will not or cannot make that exchange: an index fund mandated to hold the constituents, an insider under lockup, a taxable holder who would realize a gain by selling, or a holder already lending the shares and collecting the fee, for whom selling means surrendering that income. Where the sale route is closed and only borrowed stock is available, write b for the continuously compounded borrowing fee. The cash-and-carry trade is unaffected, so the upper bound stands; the reverse trade surrenders b to the lender, and

$$S_0\, e^{(r-b)T} \;\le\; F_0 \;\le\; S_0\, e^{rT} \tag{5.3}$$

If the fee accrues to the *marginal* holder (the one whose readiness to sell would close the gap), the asset is an investment asset paying income at rate b, and Section 5.4 gives the equality $F_0 = S_0 e^{(r-b)T}$, not a band. Equation (5.3) is the intermediate case: some holders earn the fee, but the holder who would otherwise arbitrage cannot act. Example 5.2 puts numbers on the band.

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Example 5.2 A hard-to-borrow stock

A stock trades at \$50 with a six-month rate of 4%. For an easy-to-borrow stock, Example 5.1 pins the forward at \$51.01.

Now suppose the stock is heavily shorted and the borrowing fee is 3% a year. From (5.3) the bounds are

$$50\,e^{(0.04-0.03)\times 0.5} = \$50.25 \qquad \text{and} \qquad 50\,e^{0.04\times 0.5} = \$51.01$$

Any forward price between \$50.25 and \$51.01 is free of arbitrage: a band 76 cents wide where previously there was a point. A forward quoted at \$50.40 is not mispriced to anyone who must borrow the stock to act on it.

The band is reached only if no holder will sell: a shareholder who owns the stock outright still closes the gap at \$50.40 by selling at \$50, investing to receive \$51.01, and buying the forward. Heavily shorted stocks tend to be markets where few holders will, which is why the two conditions arrive together: the borrow is expensive *because* the free float is already lent, and what remains is held by people who will not sell. So on a stock where fees exceed 20% a year, a persistently "cheap" forward is offering a profit only to whoever already owns the stock and will part with it, and the price is telling you how few of those there are.

5.3 KNOWN INCOME

If the asset provides a known cash income (coupons on a bond, a declared dividend), the holder of the physical asset receives it and the holder of a forward does not. The forward price must be lower by the present value of that income.

Writing I for the present value of the income received during the contract's life,

$$F_0 = (S_0 - I)\,e^{rT} \tag{5.4}$$

The arbitrage is the same as before, with one adjustment: the trader who buys the asset and shorts the forward receives the income along the way and can invest it. Example 5.3 applies it to a coupon-bearing bond.

Example 5.3 A forward on a coupon-bearing bond

A bond trades at \$900. It pays a \$40 coupon in four months, and a nine-month forward contract is being priced. The risk-free rate is 4% for all maturities.

The present value of the coupon is

$$I = 40\,e^{-0.04\times 4/12} = \$39.47$$

so from (5.4)

$$F_0 = (900 - 39.47)\,e^{0.04\times 0.75} = \$886.74$$

The forward price is below the spot price, which looks odd until one recalls that the forward buyer forgoes a \$40 coupon to save nine months of financing on \$900—and here the coupon is worth more.

5.4 KNOWN YIELD

Sometimes income is better expressed as a proportion of the asset's price than as a fixed amount: a stock index paying a dividend yield, or a foreign currency earning interest. Writing q for the continuously compounded yield,

$$F_0 = S_0\,e^{(r-q)T} \tag{5.5}$$

The intuition is direct: holding the asset earns q and costs r to finance, so the forward price grows at the net rate $r - q$. If the yield exceeds the financing rate, the forward price is *below* spot.

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Example 5.4 A stock index futures price

An equity index stands at 4,200. The dividend yield on the index is 1.8% and the three-month risk-free rate is 4.3%. From (5.5),

$$F_0 = 4200\, e^{(0.043-0.018)\times 0.25} = 4{,}226.3$$

Index arbitrage. When the futures price departs from (5.5), the trade is to buy the index basket and sell futures, or the reverse. It is executed by computer, since it requires simultaneous transactions in hundreds of stocks, and the profit per unit is small. It is the staple of index market makers and of bank *delta-one* desks (the desks that run exposures moving one-for-one with the underlying, as against the option books of Part II), for whom a few basis points is a business only because the volume is enormous. Example 5.5 shows how little of a quoted gap survives the costs.

This is an institutional trade, and only an institutional one. Simultaneity across hundreds of stocks requires automated basket execution and direct market access; the short direction requires standing stock-borrow relationships; and the financing has to be arranged at institutional rates. Equation (5.5) is enforced by the algorithmic desks of dealers, index funds, and quantitative managers. Read it as a description of how the price comes to be disciplined, not as an opportunity available on a retail platform.

The position is then held to settlement rather than traded out of. Index futures settle in cash against a published index level, so the basket must be liquidated *at that level*, which is why expiration days carry a concentration of orders to trade at the closing price and why volume in the constituents spikes at the moment the settlement level is set. A basket sold a few minutes early has stopped being a hedge and become a directional position in five hundred stocks.

The two directions are not equally available. Buying the basket against short futures needs cash and a broker. Selling the basket against long futures needs a borrow in every constituent: the constraint of Section 5.1 reappearing in an equity market, as borrowing arrangements for hundreds of stocks, at fees that differ from stock to stock, on loans that can be recalled without notice. The no-arbitrage band is therefore wider below theoretical value than above it: a futures price that is too cheap can persist, where one equally rich is arbitraged away more readily. When index futures nonetheless sit slightly rich (above theoretical value), the cause is not this asymmetry but the demand for exposure in futures form and the cost of the balance sheet needed to meet it, which Section 5.6 takes up.

Example 5.5 Index arbitrage with costs

The futures in Example 5.4 are quoted at 4,240, against a theoretical 4,226.3, a gap of 13.7 index points. The trade is to buy the basket and short the futures.

Round-trip transaction costs (commissions, bid–ask spreads, and market impact across the basket) come to perhaps 8 index points. The net is 5.7 points, and on a contract sized at \$50 per index point,

$$5.7 \times \$50 = \$285 \text{ per contract}$$

Two lessons follow. The arbitrage does not enforce equality but a *band*: no-arbitrage requires only that the futures price stay within transaction costs of the theoretical value. And the width of that band depends on execution capability, so the same mispricing is an opportunity for one participant and not for another.

When index arbitrage becomes impossible, as in a market where the underlying stocks cannot be traded or where trading halts break the simultaneity the trade requires, the band widens without limit, and futures and cash can decouple entirely. This happened in October 1987, and it is the reason the relationship is a mechanism rather than a law.

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5.5 VALUING AN EXISTING FORWARD CONTRACT

A forward contract is worth zero when entered. Later it is not, because the forward price has moved while the contract's delivery price K has not.

For a long forward position with delivery price K and current forward price F_0 for the same maturity, the value today, f, is

$$\boxed{f = (F_0 - K)\, e^{-rT}} \tag{5.6}$$

Derivation. Compare two portfolios. Portfolio A is the existing long forward at delivery price K; portfolio B is a newly entered long forward at the current forward price F_0, which by definition is worth zero.

At T both portfolios deliver the same asset, worth S_T. Portfolio A pays K for it and portfolio B pays F_0, so A's terminal value exceeds B's by exactly $F_0 - K$, whatever S_T turns out to be. Since this difference is certain, it can be discounted at the risk-free rate:

$$f_A - f_B = (F_0 - K)e^{-rT}$$

and since $f_B = 0$, equation (5.6) follows.

The step to notice is that the asset price cancels. The value of an existing forward does not depend on any model of the underlying, only on how far today's forward price has moved from the contract's own delivery price. This is why forward and swap books can be revalued from a curve alone. Example 5.6 values one such contract.

Example 5.6 Valuing a forward after the market moves

An investor is long a forward on a non-dividend-paying stock with a delivery price of \$48 and three months remaining. The stock now trades at \$52 and the risk-free rate is 4%.

The current three-month forward price is

$$F_0 = 52\, e^{0.04\times 0.25} = \$52.52$$

so from (5.6)

$$f = (52.52 - 48)\, e^{-0.04\times 0.25} = \$4.48$$

Equation (5.6) is the same structure as the swap valuation of Chapter 7: an off-market contract is worth the difference between its own rate and the current market rate, discounted.

5.6 STOCK INDEX FUTURES IN PRACTICE

Equation (5.5) treats the dividend yield as a known constant. For an index over a few months this is a reasonable approximation, since the constituent companies' dividends are largely announced in advance, but three practical points are worth making.

Dividends are lumpy, not continuous. The yield q is an approximation to a sequence of discrete payments concentrated in particular weeks. For contracts spanning a heavy dividend period the approximation matters, and desks price index futures off the actual projected dividend stream rather than a smoothed yield, returning in effect to equation (5.4).

Dividends are forecasts. Unlike a bond coupon, a dividend can be cut. In 2020 many companies suspended dividends within weeks, and index futures prices moved for that reason alone. The dividend assumption embedded in a futures price is a traded quantity in its own right: dividend futures and swaps exist precisely so it can be hedged.

Index futures are the cheapest way to hold the index. Because the contract requires only margin, a fund can obtain index exposure without buying the constituents. This is the same observation Chapter 6 makes about Treasury futures and duration, and it has the same consequence: steady demand for exposure in futures form tends to leave futures slightly rich to fair value. That richness is not an unclaimed riskless profit; it is what an arbitrageur is

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paid for committing the cash and the balance sheet to close the gap, which is why it survives (Section 5.15). The demand, not the borrow asymmetry of Section 5.4, decides which side the gap appears on: the asymmetry on its own would let a cheap futures price last longer than a rich one.

5.7 CURRENCY FORWARDS AND COVERED INTEREST PARITY

A foreign currency is an investment asset paying a known yield, the foreign risk-free rate r_f. Applying (5.5) with $q = r_f$ and S_0 quoted as units of domestic currency per unit of foreign currency,

$$F_0 = S_0\, e^{(r - r_f)T} \tag{5.7}$$

This is **covered interest parity**, and it says something economically strong: the forward exchange rate is determined entirely by the two interest rates, with no reference to anyone's view about where the currency is heading. A currency with a high interest rate trades at a forward *discount*, by exactly enough to offset the interest advantage.

How they are quoted. Equation (5.7) is stated in terms of "domestic" and "foreign," and the market does not use those words. A spot rate is quoted as units of the **quote** currency per one unit of the **base** currency: EUR/USD at 1.0850 is 1.0850 dollars per euro, and USD/JPY at 150.25 is 150.25 yen per dollar. Which currency is the base is pure convention: the euro, sterling, and the Australian and New Zealand dollars are quoted as base against the dollar, while the dollar is the base against most other currencies. Mapping this onto (5.7), *domestic is the quote currency and foreign is the base*, so for USD/JPY the domestic rate is the yen rate and the foreign rate is the dollar rate.

Forwards are not quoted outright. The interbank market quotes **forward points**: the difference between the forward and the spot rate, expressed in the last decimal place of the spot quote, which the client adds to spot to get the outright rate. Points are quoted rather than outright rates because they move far more slowly. Spot ticks continuously, while the interest differential driving the points does not, so a dealer can hold a points quote long after a spot quote has gone stale, and a foreign exchange (FX) swap, which is a spot leg against an offsetting forward leg, trades the points alone with no exposure to spot at all. The sign of the points carries the differential directly: negative points mean the base currency earns the higher interest rate.

Futures are quoted differently again, and for some pairs the other way up. Exchange-traded currency futures are quoted *outright*, in US dollars per unit of the other currency. For the euro that agrees with the interbank convention; for the yen it inverts it, so CME yen futures trade near 0.006650 dollars per yen where the interbank market quotes 150 yen per dollar. The two numbers are reciprocals of each other, and taking one for the other is a standard and expensive error.

Example 5.7 A currency forward

The spot rate is 150 yen per dollar. The one-year dollar rate is 4.3% and the one-year yen rate is 0.5%. The quote currency is the yen and the base currency is the dollar, so the yen is domestic and the dollar foreign, and

$$F_0 = 150\, e^{(0.005 - 0.043)\times 1} = 144.41 \text{ yen per dollar}$$

A dealer would not quote 144.41. It would quote forward points of −5.59 yen against a spot of 150.00, or −559 points in a market where a **pip** (one unit in the last decimal place of the quote) is 0.01 yen, leaving the client to add them. The negative sign says at once that the base currency, the dollar, carries the higher interest rate.

The dollar is worth fewer yen forward than spot: a forward discount of 3.7%, or 3.8%

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continuously compounded, which exactly offsets its 3.8% interest advantage. An investor who borrows yen, converts to dollars, invests at 4.3%, and sells the dollars forward earns, in yen, exactly the 0.5% it pays on the loan, and so makes nothing. Borrowing the low-rate currency to invest in the high-rate one is the **carry trade**, and equation (5.7) says there is no riskless profit in it once the currency risk is hedged.

An unhedged carry trade is a different matter entirely: it earns the interest differential and bears the currency risk, and whether that is attractive is an empirical question, not an arbitrage one.

Who uses them. Companies are the most familiar users. An American exporter invoicing in euros sells the euros forward to fix the dollar value of its receivables, and an importer buys its foreign currency forward to fix the cost of its payables; either way the forward removes the exchange rate risk between invoice and payment. The largest volume comes from asset managers, pension funds and insurers that hold foreign bonds and equities and hedge the currency back into their own, usually by rolling one- to three-month forwards or FX swaps, which removes the currency risk of the holding and leaves its local return. Banks use FX swaps to fund themselves: a bank whose deposits are in euros and whose loans are in dollars swaps the one into the other for a fixed term rather than borrowing dollars unsecured. When the demand to obtain dollars through FX swaps outruns the balance sheet dealers will commit to supplying them, the forward stops obeying equation (5.7), and that gap is the cross-currency basis that Section 7.8 prices.

And yet it fails. Chapter 7 shows that covered interest parity has been persistently violated since 2008, by tens of basis points, in the most liquid currency pairs in existence. That is a striking thing to say about an arbitrage relationship derived in three lines. Section 5.15 explains why, and the explanation is the most important qualification in this chapter.

Which foreign rate? Equation (5.7) contains r_f, and the choice is less obvious than it looks. It should be the rate at which the arbitrageur can actually invest the foreign currency for the contract's life, which for a bank executing the trade is not a government bond yield but a money market or repo rate, and may differ from the rate available to a different institution.

This is the same observation as Section 5.10: the relationship is enforced by a specific trade done by a specific participant, and the rate that belongs in the formula is the one that participant faces. Much of the apparent breakdown in covered interest parity dissolves, and the remainder becomes interpretable, once one stops asking what "the" risk-free rate is and starts asking what the marginal arbitrageur can actually borrow and lend at.

5.8 COMMODITIES, STORAGE, AND CONVENIENCE YIELD

Commodities cost money to store, which is negative income. Writing u for the storage cost expressed as a continuous proportional rate,

$$F_0 = S_0\, e^{(r+u)T} \tag{5.8}$$

for an investment commodity such as gold.

Storage is often charged as an amount rather than a proportion of value: so many dollars an ounce a year for a vault, so many cents a bushel a month at a grain elevator. It is then handled exactly as known income was in Section 5.3, with the sign reversed. Writing U for the present value of the storage costs incurred during the contract's life,

$$F_0 = (S_0 + U)\, e^{rT} \tag{5.9}$$

The arbitrage is the basic one of Section 5.2: a trader who buys the commodity and sells it forward must also pay to store it, so the forward price has to repay both the purchase and the storage, with interest. Example 5.8 prices the same contract both ways.

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Example 5.8 A gold forward

Gold trades at \$2,000 per ounce, the one-year rate is 4.3%, and storage and insurance cost 0.5% of value per year. From (5.8),

$$F_0 = 2000\, e^{(0.043+0.005)\times 1} = \$2{,}098.34$$

Suppose instead the vault charges a flat \$10 an ounce, payable at the end of the year: the same 0.5% of today's price, stated as an amount. Its present value is

$$U = 10\, e^{-0.043\times 1} = \$9.58$$

so from (5.9)

$$F_0 = (2000 + 9.58)\, e^{0.043\times 1} = \$2{,}097.88$$

Since the fee is paid at delivery, Ue^{rT} is the \$10 itself, and the forward price is simply the financed spot price, \$2,087.88, plus the fee. It sits slightly below the proportional figure because a proportional rate charges storage on a value that grows over the year, while a flat fee does not grow.

Consumption commodities and the convenience yield

For a consumption commodity the argument works in only one direction, and this is where the chapter's tidy equalities break down. If $F_0 > S_0 e^{(r+u)T}$, the usual arbitrage applies: buy the commodity, store it, sell forward. Anyone can do this, so this inequality cannot persist.

If $F_0 < S_0 e^{(r+u)T}$, the reverse trade requires *selling the commodity short* (borrowing oil from someone who holds it). But holders of consumption commodities hold them because they intend to use them. A refinery will not lend its crude, at any price, if lending it means shutting down. The arbitrage is therefore unavailable, and we can conclude only that

$$F_0 \le S_0\, e^{(r+u)T} \tag{5.10}$$

or, with storage stated as a present value, $F_0 \le (S_0 + U)e^{rT}$.

The gap that (5.10) leaves open is filled by defining the **convenience yield** y, the benefit of holding the physical commodity rather than a claim on it, so that equality is restored:

$$F_0 = S_0\, e^{(r+u-y)T} \tag{5.11}$$

With storage as a present value, the same definition reads $F_0\, e^{yT} = (S_0 + U)e^{rT}$.

The yield y is not observed and not modeled: it is *defined* as whatever makes the equation hold. That does not make it useless, because what it measures is real: the value of having the physical commodity available when you need it, which rises when supply is tight.

Example 5.9 Backing out a convenience yield

Crude oil trades at \$80 and the one-year futures price is \$76. The risk-free rate is 4.3% and storage costs 3% a year. From (5.11),

$$76 = 80\, e^{(0.043+0.03-y)} \implies \ln(0.95) = 0.073 - y$$

so

$$y = 0.073 + 0.051293 = 12.43\%$$

A convenience yield of over 12% says the market places a large premium on having oil now rather than in a year. That is a statement about physical tightness (low inventories, constrained logistics), and traders read the term structure of convenience yields as a supply indicator rather than as a pricing artifact.

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5.9 THE COST OF CARRY

All the results above are one formula. Define the **cost of carry** c as the cost of holding the asset: financing cost, plus storage, minus income. Then for an investment asset

$$F_0 = S_0\, e^{cT} \tag{5.12}$$

and for a consumption asset

$$F_0 = S_0\, e^{(c-y)T} \tag{5.13}$$

For a non-dividend stock $c = r$; for a stock index $c = r - q$; for a currency $c = r - r_f$; for a storable commodity $c = r + u$.

The cost of carry is a term structure, not a number. Writing c with no argument hides the fact that every ingredient varies with maturity: r is the zero rate to T, storage is contracted at different rates for different horizons, and the convenience yield moves most of all. What (5.12) needs at maturity T is $c(T)$, the *average* carry between now and T. It stands to the carry over each intervening month exactly as Chapter 4's zero rate stands to the forward rates underneath it (Figure 4.3), and inherits the same amplification: a mildly rising $c(T)$ implies a considerably steeper profile of carry between adjacent delivery months. This is why a real forward curve is not the single exponential of Figure 5.1, and why it can slope one way at the front and the other further out, which is what Figure 5.2 shows for four markets.

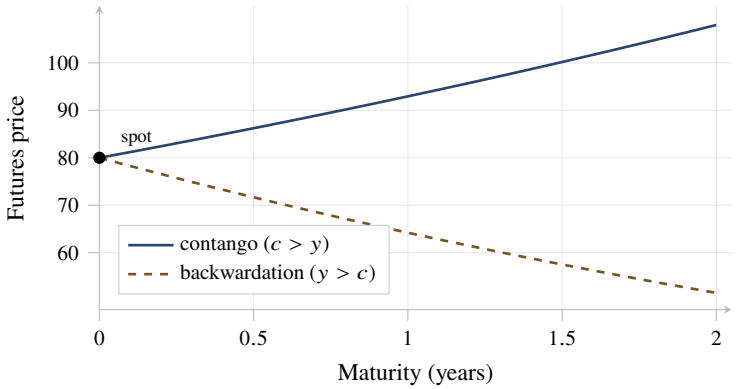


Figure 5.1: Forward curves when the cost of carry exceeds the convenience yield and when it does not. Constant carry, with the curvature exaggerated.

Figure 5.1 shows the two shapes this produces. When carry exceeds convenience yield the curve slopes upward (**contango**) and holding the physical asset is expensive relative to holding the futures. When convenience yield dominates the curve slopes downward (**backwardation**) and the market is paying a premium for immediate availability. Backwardation is the normal state for commodities in tight supply. For a financial asset, where there is no convenience in holding the physical, it arises only when the asset's income yield exceeds the financing rate, as for the dollar against the yen in Example 5.7.

What the curve looks like, market by market

Figure 5.1 draws the two shapes as though a market chooses between them. Real markets do not: each commodity has a characteristic curve, set by how cheaply it can be stored and how badly it is needed now. Table 5.1 summarizes, and Figure 5.2 draws four representative shapes.

Read that figure twice. The first reading is the shape. The second is the gap between the markers (the contracts that actually trade, at listed delivery months) and the dashed line, which is (5.13) with a single convenience yield applied to every maturity. A constant y fits none of

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them: gold's markers sit a shade below the curve and drift further as its small convenience yield accumulates, crude's pull away as the front-month scarcity decays, gas swings with the season, and corn runs below full carry because commercial storage caps the slope. The convenience yield is a term structure, not a number, and the only honest way to read y off a market is one maturity at a time, which is what Example 5.10 does at a single point, and what Section 5.15 says cannot be pushed too far.

Table 5.1: Characteristic forward curve shapes.

Market	Typical shape	What sets it
Gold, precious metals	Contango, almost always	Above-ground stock is decades of mine supply, so $y \approx 0$ and the curve is the financing curve
Base metals	Mild contango; sharp backwardation in squeezes	Storable at moderate cost; shape tracks exchange inventory
Crude oil	Alternates; backwardated more often than not	y swings with inventory, and storage capacity is finite
Natural gas	Seasonal; neither label applies	Winter above summer; storage is costly and capacity-constrained
Grains, oilseeds	Rises within a crop year, breaks at the new crop	Harvest cycle; commercial storage cost caps the slope
Livestock	No carry relation	Not storable; animals must be fed and change in value
Electricity	No carry relation	Not storable at all; the forward is expectation plus risk premium

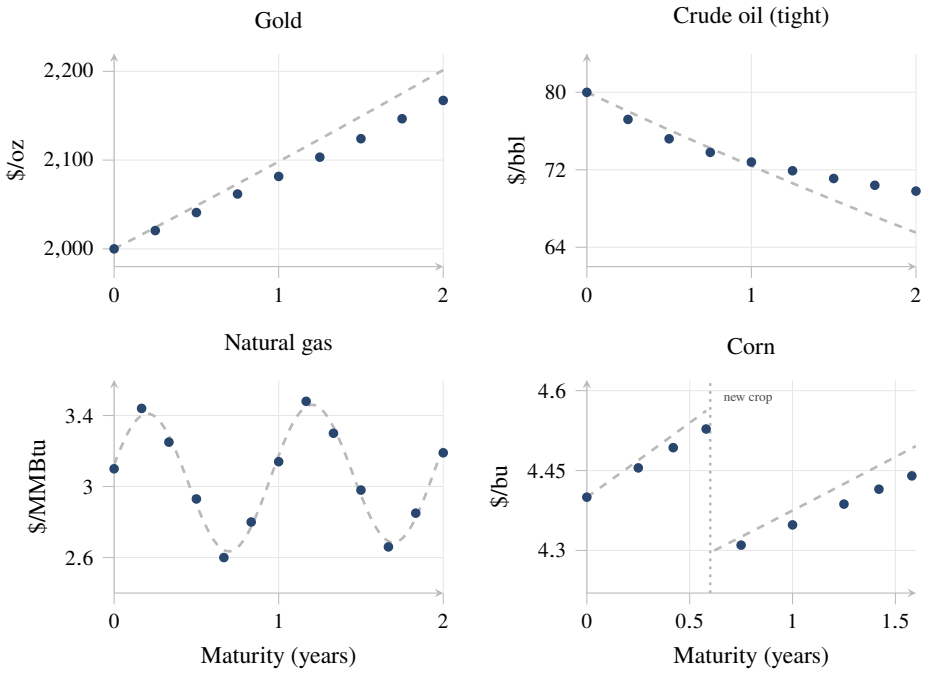


Figure 5.2: Four representative shapes. Markers are listed delivery months; the dashed line is what a constant convenience yield would give. Schematic.

Gold: the pure carry case. Gold is the cleanest illustration of equation (5.12) in the whole commodity complex, because it barely behaves like a commodity at all. Roughly sixty years of

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mine production sits above ground in vaults and jewelry, so no plausible shortage can develop; the convenience yield of holding physical gold instead of a claim on it is close to zero. Storage and insurance run a fraction of a percent a year. What remains is financing cost, and the gold forward curve is therefore essentially the interest rate curve drawn in dollars per ounce.

This is why gold is almost never backwardated. When it briefly is, the cause is not a shortage of gold but a dislocation in the market for *immediately deliverable* gold in a particular location (refineries closed, flights grounded, bars of the wrong specification), and it resolves in weeks.

Example 5.10 Backing out the gold lease rate

Example 5.8 computed the full-carry forward price of gold, assuming no convenience yield: at a spot of \$2,000, financing of 4.3%, and storage and insurance of 0.5%, it came to \$2,098.34.

Suppose the one-year forward actually trades at \$2,081.60. The gap is small but it is not noise. The market reads it as a lease rate, defined as the financing rate less the forward premium on metal:

$$l = 0.043 - \ln\left(\frac{2081.60}{2000}\right) = 0.043000 - 0.039990 = 0.003010 = 0.30\%$$

Those 30 basis points are the **gold lease rate**: what a holder earns by lending metal to a bullion bank, and equivalently what a borrower pays to obtain metal now rather than in a year. It is a quoted market rate, and this residual calculation is how it is quoted. Storage does not enter it: the convention compares the forward premium with the financing rate alone. Solving (5.11) for the convenience yield adds storage back, $y = l + u = 0.30\% + 0.50\% = 0.80\%$.

The magnitude carries the lesson. A lease rate of 30 basis points, and a convenience yield under one percent, is what "essentially zero convenience yield" looks like in a real market, and it should be set beside the 12.43% of Example 5.9. The same equation governs both, and one parameter separates them by a factor of fifteen.

Crude oil: the regime-switching case. Oil spends most of its time backwardated. The reason is structural: refiners cannot run without crude, inventories are small relative to consumption, and a genuine shortfall is expensive in a way no financial hedge repairs. That sustains a positive convenience yield most of the time. But when demand falls or production overshoots, inventories build, the convenience yield collapses, and the curve flips into contango. Then the arithmetic of storage takes over.

Suppose the front contract is \$42 and the twelve-month is \$54, a \$12 contango. A trader can buy physical crude, charter a tanker, and sell the deferred contract. A very large crude carrier holds about two million barrels; at \$60,000 a day the charter costs \$21.9 million a year, or about \$10.95 a barrel, and financing \$42 for a year at 2% adds \$0.84. Total carry is close to \$12, and the trade is marginal—which is the point. *Contango is capped near full carry*, because as soon as it exceeds the cost of storage, storage gets bought until it does not. This is not a hypothetical: floating storage on chartered tankers was a large and visible trade in 2009 and again in 2020.

The cap has a failure mode, and it is the one the next Case Study describes. The equilibrating mechanism works by *buying* storage. When there is no more storage to buy, nothing caps the contango, and prices do things that pricing formulas assuming a finite u cannot represent.

Natural gas: seasonal, so neither word applies. Gas demand is concentrated in winter heating and summer cooling, while production is roughly flat. Storage bridges the gap: gas is injected into underground facilities from April to October and withdrawn from November to March. The forward curve is therefore not monotone at all: it oscillates, with winter contracts above the summer contracts on either side, and calling it "in contango" or "in backwardation" is a category error.

The interesting point on such a curve is the March/April spread, which straddles the end of

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the withdrawal season. March is the last winter contract and April the first injection contract, so the spread is a direct bet on whether the winter ends with storage tight or comfortable. It is volatile enough to have earned the name *widow-maker*, and Section 5.14 returns to it: the largest hedge fund collapse of its era was a bet on this one spread.

Agriculture: carry within a crop year, a break across it. Grain arrives all at once at harvest and is consumed all year, so it must be stored, and within a crop year the curve normally rises at close to the cost of storing it. Grain markets have a name for the ceiling, **full carry** (interest plus commercial storage), and quote the spread as a percentage of it, as Example 5.11 does for corn.

What makes agricultural curves distinctive is that the carry relation *resets* at the new crop. A December corn contract and the following December contract are claims on different harvests, and no amount of storage links them, because grain stored from one harvest competes with grain that has not yet been planted. This is the discontinuity in the fourth panel of Figure 5.2, and it is why old-crop and new-crop contracts are quoted and traded as different markets.

Example 5.11 Reading a grain spread as a percentage of full carry

December corn trades at \$4.40 a bushel and March corn at \$4.55, both in the same crop year. Financing costs 5.0% and commercial elevator storage is 5 cents a bushel a month.

Over the three months from December to March, full carry is

$$\underbrace{4.40 \times 0.050 \times 0.25}_{\text{interest} = \$0.055} + \underbrace{3 \times 0.05}_{\text{storage} = \$0.150} = \$0.205$$

The market spread is $4.55 - 4.40 = \$0.15$, or

$$\frac{0.150}{0.205} = 73\% \text{ of full carry}$$

Read this as a signal about inventories. Near 100% of full carry, the market is paying anyone who stores grain their full cost, which it only needs to do when supplies are ample and someone must be induced to hold them. At 73% the market is paying most but not all of the cost: the 15-cent spread must first cover 5.5 cents of interest, so an elevator whose own storage cost for the three months is below the remaining 9.5 cents still profits by storing, one at full commercial rates does not, and grain moves out of the highest-cost storage first.

An *inverse* (March below December) reverses the message entirely. The market is then paying a premium to pull grain out of storage immediately, which is the agricultural version of backwardation and signals old-crop tightness. Storage cannot arbitrage it away, because the one thing carry cannot do is move grain backward in time.

Livestock and electricity: where the relation breaks. Two categories in Table 5.1 have no cost-of-carry relation at all, and they are worth naming because they mark the boundary of everything in this chapter.

Live cattle cannot be stored. An animal held for three months is not the same animal: it has been fed, at a cost, and has gained weight, changing what is delivered. Electricity is more extreme: outside limited battery and pumped-hydro capacity it cannot be stored for even an hour, so there is no trade that links today's price to next July's. In both markets the forward price is not spot compounded at anything. It is the market's expectation of the future spot price, plus a risk premium: the relationship of Section 5.13, standing alone because the arbitrage argument that usually dominates it is unavailable. This is why power forward curves show shapes no storable commodity ever displays, including negative prices in hours when the grid must dispose of generation.

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Case Study 5.1 When storage runs out

On April 20, 2020, the expiring May contract for West Texas Intermediate (WTI) crude settled at −$37.63 a barrel. Buyers were paid to take oil.

Equation (5.8) assumes storage is available at a cost u. In April 2020, with demand collapsed by the pandemic and production still running, the tanks at the delivery point at Cushing, Oklahoma were effectively full. Longs approaching expiration faced physical delivery of oil they had nowhere to put, and could not roll into the next month fast enough. Storage was not expensive; it did not exist. The carry relation itself cannot produce a negative futures price, since it makes F a positive multiple of the spot price. What went negative was the spot value of a barrel at Cushing: a barrel that had to be taken with nowhere to put it had become a disposal cost, and the expiring contract, converging to that spot value, settled below zero. Section 36.1 returns to the episode.

Two lessons follow. First, the parameters in these formulas are not always finite, and a model that treats u as a small constant will be wrong in precisely the situations that matter. Second, this is what a *consumption* asset looks like when the consumption stops: the convenience yield collapsed to something deeply negative, and holding the physical was a liability, not a benefit.

The episode also broke a great deal of software. Option pricing models assuming lognormal prices cannot represent a negative underlying, exactly the problem Section 4.2 describes for interest rates (and Chapter 29 prices), and exchanges had to switch commodity options to normal models within days.

It broke retail products too, and more painfully. The largest losses fell on holders of a Bank of China product called Crude Oil Treasure, which gave retail clients exposure to WTI futures. Its client trading window closed hours before the exchange struck the settlement price, so holders could neither see the collapse nor exit it. And because the product was unleveraged (clients funded it in full), it had no margin close-out mechanism at all, on the reasoning that an unleveraged position cannot lose more than its principal. That reasoning is sound only while prices stay positive. Roughly sixty thousand investors lost their principal and finished owing the bank money, and Chinese regulators subsequently barred banks from offering such products to individuals.

The general lesson is worth more than the episode. Every product and every model contains variables that someone has quietly assumed to be nonnegative. It is worth asking which ones, and what happens if they are not.

5.10 THE IMPLIED REPO RATE

Practitioners rarely compute a theoretical forward price and compare it with the market. They invert the question: given the two prices, what return would the cash-and-carry trade actually earn? That return is the **implied repo rate**.

Buy the asset today at S_0, finance it, and deliver it at T for the invoice amount F_0, the sum the futures buyer pays on delivery. Ignoring intermediate income, the annualized return is

$$r_{\text{implied}} = \left(\frac{F_0}{S_0} - 1 \right) \times \frac{\text{days per year}}{\text{days to delivery}} \tag{5.14}$$

Comparing this with the rate at which the position can genuinely be financed, the *actual* repo rate, answers the question directly. If the implied repo rate exceeds the actual repo rate, the cash-and-carry trade earns more than it costs and should be done. If it is lower, the trade loses, and the reverse is indicated for anyone able to borrow the asset. Example 5.12 makes the comparison for a deliverable bond.

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Example 5.12 Implied against actual repo

A bond has a cash price of 98.50. It can be delivered into a futures contract in 90 days for an invoice amount of 99.05. Financing in the repo market costs 4.30%.

From (5.14), using a 365-day year,

$$r_{\text{implied}} = \left(\frac{99.05}{98.50} - 1\right) \times \frac{365}{90} = 2.26\%$$

The trade returns 2.26% and costs 4.30% to finance: a loss of 2.04% annualized. Buying the bond and selling the futures is not profitable here: the futures are *cheap* relative to the cash bond, and the indicated trade is the reverse, subject to being able to borrow the bond.

The implied repo rate is the practitioner's version of everything in this chapter. It restates the cost-of-carry relationship as a comparison between two interest rates, which is how the trade is actually assessed. Chapter 6 uses the same quantity under a different name: the net basis of Section 6.4 is the gross basis less carry, and it is negative precisely when the implied repo rate exceeds the actual one.

Which financing rate? Equation (5.14) compares against the *repo* rate, not the risk-free rate, and the distinction is the practical content of Section 5.15. The relevant rate is what this particular asset can be financed at by this particular participant, which depends on the asset's quality as collateral, on whether it is trading "special" in the repo market, and on the borrower's own standing. Two firms looking at the same two prices can correctly reach opposite conclusions.

5.11 ROLLING A POSITION AND ROLL YIELD

Futures contracts expire; an investor wanting continuous exposure must roll, selling the expiring contract and buying the next. The shape of the curve in Figure 5.1 then has a direct and often underappreciated effect on returns, which Example 5.13 isolates by holding the spot price fixed.

Example 5.13 The cost of rolling in contango

An investor wants continuous exposure to a commodity trading at \$80 spot. The one-month futures contract trades at \$80.50, a market in mild contango.

The investor buys the one-month contract at \$80.50. Suppose the spot price does not move at all over the month. By Chapter 2's convergence argument, the contract must approach \$80 as it expires. The investor therefore loses \$0.50, sells at \$80, and buys the new one-month contract at \$80.50 to continue.

Repeating twelve times, the position loses

$$12 \times \$0.50 = \$6.00$$

or 7.5% of the initial \$80—on a commodity whose price never changed.

In backwardation the arithmetic reverses. With the one-month contract at \$79.50, the same investor gains \$0.50 each month and earns 7.5% a year from a flat spot price.

This is the **roll yield**, and for long-horizon commodity exposure it frequently dominates the spot price move entirely. It is the reason exchange-traded products tracking commodity futures can fall over years in which the underlying commodity rose, and why the offering documents of such products describe their roll methodology at length.

The roll yield is not a market inefficiency and there is no arbitrage in it. It is the cost of carry, collected month by month by whoever holds the physical and paid month by month by whoever wants exposure without holding it. What flows is the *net* carry $c - y$, so the direction can reverse. In Example 5.9 the convenience yield exceeds financing and storage: the holder of physical oil takes that yield in kind, and the futures price, set below spot, hands the futures

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investor a positive roll yield instead.

5.12 FORWARD VERSUS FUTURES PRICES

We have used "forward price" and "futures price" interchangeably. Chapter 2 explained why they are not the same instrument: futures are settled daily, forwards at maturity.

The consequence is precise. If interest rates were certain, the two prices would be identical: daily settlement would move cash around but the amounts would be known. When rates are stochastic the timing matters, and the direction depends on the correlation between the asset price and interest rates.

Consider an asset *positively* correlated with rates. A trader long futures makes money when the asset rises, and the gain arrives as variation margin at a moment when rates are high, so it is reinvested well. Losses arrive when rates are low and are financed cheaply. Being long futures is therefore better than being long a forward, and the futures price must be higher to compensate. For an asset negatively correlated with rates the argument reverses and the futures price is lower.

The magnitude is negligible for short-dated contracts on assets weakly correlated with rates, which is why the distinction is usually ignored. It is emphatically not negligible for interest rate contracts, where the asset *is* the rate and the correlation is one. Chapter 30 develops this into the convexity adjustment, which reaches tens of basis points, and Chapter 6 shows the consequences for curve construction. Section 28.7 states the same result in the language of probability measures.

5.13 FUTURES PRICES AND EXPECTED SPOT PRICES

We come finally to the question every newcomer asks: is the futures price the market's forecast of the future spot price? The answer is no, and the reason is instructive. Consider a speculator who takes a long futures position, holds it to delivery, and sets aside the present value of the futures price, $F_0 e^{-rT}$, in a risk-free investment. At T the investment pays F_0, which buys the asset under the contract, so the speculator ends with an asset worth S_T. The outlay today is certain and is discounted at r; the payoff is risky and is discounted at k, the rate appropriate to the asset's risk. Under standard asset pricing the present value of the whole position must be zero:

$$-F_0\, e^{-rT} + E[S_T]\, e^{-kT} = 0$$

so that

$$F_0 = E[S_T]\, e^{-(k-r)T} \tag{5.15}$$

If the asset has no **systematic risk** (risk that moves with the market as a whole and so cannot be diversified away), then $k = r$ and (5.15) gives $F_0 = E[S_T]$. If it has positive systematic risk (most assets), then $k > r$ and $F_0 < E[S_T]$: the futures price sits below the expected spot price, and the speculator earns the difference as compensation for bearing risk.

Figure 5.3 shows what this implies over time. If $F_0 < E[S_T]$, the futures price must drift *upward* on average as delivery approaches, since it must converge to S_T. Keynes called this **normal backwardation**, and explained it by hedging pressure: if producers are net short (as they typically are, hedging future output), speculators must be net long, and will require compensation. Where hedgers are net long instead, the argument reverses and $F_0 > E[S_T]$, which Keynes called contango.

A warning on terminology. The words are used in two different senses and the ambiguity causes real confusion. Section 5.9 used contango and backwardation to describe the *shape of the forward curve* today, which is observable. Keynes used them to describe the relationship between the futures price and the *expected* spot price, which is not observable, since no one

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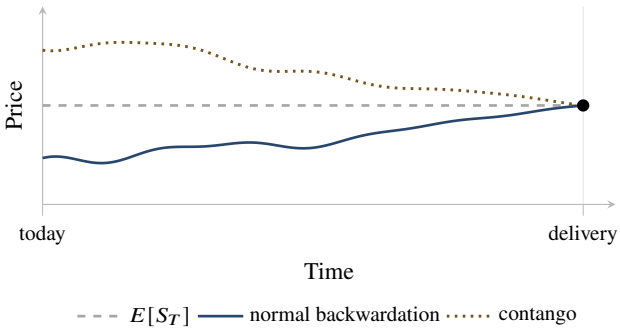


Figure 5.3: The futures price drifting toward the expected spot price as delivery approaches. Illustrative paths.

sees $E[S_T]$. Most market usage means the first. Statements that a market "in contango" implies traders expect prices to rise confuse the two, and are wrong: the shape of the curve is determined by carry and convenience, not by expectations.

5.14 TRADING THE SHAPE OF THE CURVE

Two results in this chapter describe money changing hands. Section 5.11 showed that the shape of the curve determines a roll yield, and Section 5.13 showed that a risk premium is compensation somebody is paying. Both are therefore things a speculator can try to earn, and three strategies are common.

Calendar spreads. Buying one delivery month and selling another is a direct bet on the shape of the curve, with no outright exposure to the level of the price. If the front contract is expected to richen relative to the second, because inventories are tightening, the trade is long the front and short the deferred. Section 2.3 showed why this is efficient: the two legs offset, so the margin requirement is a fraction of the outright, and Example 2.3's calendar spread required a twentieth of the gross amount.

Case Study 5.2 Amaranth and the widow-maker

In September 2006 Amaranth Advisors, a multi-strategy hedge fund managing about $9.2 billion, lost roughly $6.6 billion in a matter of weeks and closed. Almost all of it came from natural gas calendar spreads: principally long March against short April, the spread of Section 5.9 that straddles the end of the withdrawal season.

The position was not obviously foolish. A cold winter drains storage, March scarcity bids the last winter contract above the first injection contract, and the spread widens. It had done so before. But the 2005–06 winter had been mild, storage ended the season full, and the 2006 hurricane season passed without disrupting Gulf production. The spread collapsed instead of widening.

What turned a bad trade into a fatal one was size. Amaranth's positions were large enough that the Senate Permanent Subcommittee on Investigations later found it had at times held on the order of 40% of the open interest in NYMEX winter contracts. A position that large cannot be exited: every attempt to sell moved the spread further against the remaining position, because there was no one on the other side but the firm itself. The fund's positions were eventually transferred in bulk to JPMorgan and Citadel at a steep discount.

Three lessons follow. **Low margin is not low risk.** Spread positions receive the margin offsets of Section 2.3 precisely because the legs hedge each other against parallel

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moves, which is exactly what lets a trader accumulate a notional position that a directional book could never finance. **Liquidity is part of the position.** A trade you cannot exit is a different trade from the one you put on. **Margin arrives before the thesis resolves.** The variation margin mechanism of Section 2.3 does not wait to find out whether you were right about the winter.

The episode also had a regulatory afterlife. Amaranth had shifted positions from NYMEX to the then-unregulated Intercontinental Exchange (ICE) to escape NYMEX's position accountability levels (the position sizes above which the exchange can demand an explanation and order a position cut), and closing that gap was a direct motivation for extending Commodity Futures Trading Commission (CFTC) oversight to economically equivalent contracts traded off-exchange.

Carry harvesting. A systematic strategy holds long positions in backwardated markets and short positions in contangoed ones, collecting the roll yield of Section 5.11 across many commodities. This is the commodity version of a carry trade, and like all carry trades it earns a steady return punctuated by sharp losses when the curve inverts.

Taking the other side of hedging pressure. If Keynes was right that producers' hedging demand pushes F_0 below $E[S_T]$, a speculator who simply holds long positions in commodities where producers dominate should earn the difference.

Does the premium exist? The empirical answer is that it is real but neither large nor reliable. Studies such as Gorton and Rouwenhorst (2006) find returns consistent with a risk premium in some commodities and periods and not others, and the premium appears to move with the balance of hedging pressure, which is itself unstable, since whether producers or consumers dominate a market changes with industry structure.

This is a genuine risk premium and not an arbitrage. It is compensation for bearing a risk that someone else wanted to shed, it is earned on average and lost in specific episodes, and it disappears when hedging demand switches direction.

Why a hedger should care. This is not a digression from hedging; it is the same fact seen from the other side. The speculator's expected return is the hedger's expected cost. A producer hedging in a market where producers dominate, so that the futures price sits below the expected spot price, is systematically paying a premium to transfer risk, and that premium is the price of the insurance, not evidence that hedging is a mistake.

The same fact disposes of a common confusion. Contango is not "bad" and backwardation is not "good." Example 3.6 showed a short hedger gaining from rolling in contango, and Section 5.11 showed a long investor losing from it in the same market. Any statement about the shape of the curve that does not specify which side of the market one is on is empty. Chapter 36 returns to the curve in energy and commodity markets, where the term structure carries the most information.

5.15 THE LIMITS OF ARBITRAGE

Every result in this chapter rests on the assumption stated in Section 5.1: that participants can borrow and lend freely at the risk-free rate and will take arbitrage opportunities when they see them. It is time to examine what that assumption costs.

The arbitrage in Example 5.1 requires borrowing \$50 and holding a stock for six months. That is a balance sheet position. Post-crisis capital rules (the leverage ratio of Section 1.12 in particular) make holding it expensive, and the expense does not shrink because the position is hedged.

This is not an abstraction, and the next two chapters meet it three times:

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- **Covered interest parity** (Section 5.7) has been violated continuously since 2008. Chapter 7 shows the deviation is the price of the balance sheet required to close it.
- **Swap spreads** (Chapter 7) turned negative, offering an apparently riskless $811,100 on a $100 million ten-year position, which sat available for years because the trade requires financing a bond.
- **The Treasury cash–futures basis** of Chapter 6 is the arbitrage of Example 5.1 applied to government bonds. It persists, and is harvested by funds willing to run fifty times leverage—which is what the balance sheet constraint looks like when someone is prepared to bear it.

Case Study 5.3 Arbitrage requires a balance sheet

The three dislocations above are usually discussed separately (one in FX, one in rates, one in Treasuries) and are usually presented as puzzles. They are the same phenomenon.

In each case the textbook arbitrage exists and is riskless in the sense this chapter means. In each case executing it requires holding an asset and financing it. And in each case post-crisis regulation made that holding costly in a way that does not depend on the position's risk, so the cost cannot be hedged away.

The consequence is a genuine change in what "no-arbitrage" means. Before 2008 it was reasonable to treat these relationships as identities, because the intermediaries who enforced them faced negligible costs for doing so. They are now better understood as relationships that hold up to a spread, where the spread is the price of intermediation. That price is highest when dealers have the least balance-sheet capacity to spare, at reporting dates and in periods of stress, and those are the times when the dislocations are largest.

For a reader of this chapter, the practical implication is not to distrust the formulas, which remain the right starting point and the right way to think, but to be precise about what enforces them, and to expect the enforcement to weaken under stress rather than to strengthen.

One consequence deserves naming here, because this chapter has used r throughout without saying whose rate it is. For a *collateralized* derivative the question has a definite answer, and it is not a matter of judgment: the discount rate is the rate paid on the collateral, fixed by the collateral agreement rather than chosen by the valuer. Section 7.4 derives it. Where a position is collateralized in its own currency, this chapter's r is that rate; where it is not (an uncollateralized trade, or collateral posted in another currency), the difference becomes an explicit valuation adjustment in Chapter 9.

SUMMARY

For an investment asset, arbitrage in both directions forces $F_0 = S_0 e^{rT}$, and no view about the asset is needed. Income lowers the forward price, whether it is known in amount, $F_0 = (S_0 - I)e^{rT}$, or as a yield, $F_0 = S_0 e^{(r-q)T}$; storage raises it; a currency is an asset yielding the foreign rate, so a hedged carry trade earns nothing. An existing forward is worth $(F_0 - K)e^{-rT}$, with no model of the asset required.

The relation is an equality only where holders are willing to sell. A borrowing fee opens a band when they are not, and for a consumption commodity they never are, so only an upper bound survives and the convenience yield closes the gap. Everything reduces to the cost of carry, itself a term structure: carry above the convenience yield gives contango, below it backwardation. April 2020 showed what happens when storage is unavailable rather than merely costly.

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Practitioners run the relation in reverse: the implied repo rate says whether a cash-and-carry pays, and the cost of rolling in contango is the cost of carry, paid by whoever wants exposure without the physical. Neither the forward nor the futures price is a forecast; the futures price sits below the expected spot price when the asset carries systematic risk, so contango does not mean prices are expected to rise. Finally, arbitrage needs a balance sheet, which post-crisis regulation made expensive, so the formulas hold only up to a spread, and the spread is widest when dealers can least afford to close it.

FURTHER READING

Working (1949) is the origin of the theory of storage and states why a forward curve slopes the way it does; Keynes's *A Treatise on Money* (1930, vol. 2) is where normal backwardation comes from, and the distinction between the two ideas is worth getting straight. Cornell and Reinganum (1981) is the standard early test of forward against futures prices. On why the arbitrage this chapter relies on can fail, Shleifer and Vishny (1997) is the general argument and Du, Tepper, and Verdelhan (2018) the specific one, documenting deviations from covered interest parity that ought not to exist.

PRACTICE QUESTIONS

5.1 A non-dividend-paying stock trades at \$72 and the four-month risk-free rate is 3.6%. Compute the four-month forward price, and describe the arbitrage if the forward is quoted at \$74, giving the profit per share.

5.2 A stock will pay a \$1.50 dividend in two months and \$1.50 in five months. It trades at \$64 and the six-month rate is 4%, flat. Compute the six-month forward price.

5.3 An index stands at 3,800 with a dividend yield of 2.1% and a six-month rate of 4.0%. Compute the six-month futures price, state what it would be if the dividend yield exceeded the interest rate and why, and explain why that yield is a forecast rather than a known quantity.

5.4 An investor is short a forward on a non-dividend-paying stock with a delivery price of \$95 and five months remaining. The stock trades at \$88 and the rate is 3.5%. Value the position.

5.5 The spot rate is 1.08 dollars per euro, the one-year dollar rate 4.2% and the euro rate 2.6%. Compute the one-year forward rate, state which currency is at a forward discount and why, and explain why a hedged carry trade earns nothing while an unhedged one is a different proposition entirely.

5.6 Copper trades at \$4.20 per pound with a one-year futures price of \$4.05, a rate of 4.3%, and storage costs of 1.5%. Compute the implied convenience yield and interpret it. Repeat the carry calculation for silver at \$29 per ounce with storage of 0.4% and a rate of 4.3%, and explain why the two cases require different arguments.

5.7 Explain the difference between the two uses of the word "contango" in this chapter, and why it is wrong to say that a market in contango expects prices to rise.

5.8 A stock trades at \$120 with a three-month rate of 4% and a borrowing fee of 5% a year. Compute the no-arbitrage band for the three-month forward price, and explain why a quote inside the band is not a mispricing.

5.9 A bond has a cash price of 101.20 and can be delivered into a futures contract in 60 days for an invoice amount of 101.90. Compute the implied repo rate on a 365-day basis. If the bond can be financed at 4.10%, should the cash-and-carry trade be done?

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5.10 Derive the value of a long forward contract by comparing two portfolios. Explain why the underlying asset price does not appear in the result, and what this permits operationally.

5.11 A commodity trades at \$60 spot with the one-month futures price at \$60.75, and an investor holds continuous long exposure by rolling monthly. If the spot price is unchanged after a year, what is the return? Repeat for a one-month futures price of \$59.40, and identify who receives what the futures investor pays.

5.12 Silver trades at \$29 per ounce, and storage and insurance cost \$0.12 per ounce per quarter, payable at the start of each quarter. The interest rate is 4.3% for all maturities. Compute the nine-month futures price. If the nine-month futures contract trades at \$30.40, describe the arbitrage and give the profit per ounce.

FURTHER QUESTIONS

5.13 Explain why the arbitrage argument for a consumption commodity works in only one direction, and derive the resulting inequality. Is the convenience yield a model, a measurement, or a definition? Defend your answer.

5.14 In April 2020 the expiring WTI contract settled below zero. Explain which assumption in equation (5.8) failed, what it implies about the convenience yield in that month, and what pricing models had to be changed as a result.

5.15 Explain why forward and futures prices differ when interest rates are stochastic, deriving the direction of the difference from the correlation between the asset and rates. Why is the effect negligible for equity index futures and material for interest rate futures?

5.16 A trader observes that index futures are 20 points above their theoretical value. Set out the trade, identify every cost and risk it involves, and explain what would have to be true for the trade to be genuinely riskless. Under what market conditions would you decline to do it?

5.17 A stock is impossible to borrow at any price. Explain why this alone does not put the forward price below $S_0 e^{rT}$, state the condition on the shareholder register under which it does, and say why that condition and an expensive borrow tend to arrive together.

5.18 Explain why the no-arbitrage band around an index futures price is wider below theoretical value than above it. State what a participant needs to close the gap in each direction, and explain why, although index futures are more often rich than cheap, the asymmetry is not the reason.

5.19 A dealer quotes USD/JPY spot at 148.50 and one-year forward points of -480. Give the outright one-year forward rate, identify the base and quote currencies, say which currency carries the higher interest rate and how you know, and state roughly what a CME yen futures contract for the same date would be quoted at.

5.20 Equation (5.12) writes the cost of carry as a single number c. Identify which of its components vary with maturity, explain what $c(T)$ must therefore mean, and state the analogy with the zero and forward rates of Chapter 4. What does the analogy predict about the carry implied between two adjacent delivery months when $c(T)$ is rising gently?

5.21 Covered interest parity, negative swap spreads, and the Treasury cash–futures basis are usually discussed separately. Argue that they are one phenomenon. What does your answer imply about how the no-arbitrage results of this chapter should be taught and used?

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Chapter 6
Interest Rate Futures

Interest rate futures are the most heavily traded derivatives in existence, and they are where most of the hedging that the rest of this book describes actually gets done. A dealer who has warehoused the risk of a swap, a structured note, or a mortgage portfolio does not usually offload it into another bespoke trade; it hedges in the futures market, where the liquidity is.

Two families dominate. **Short-term interest rate futures** settle against an overnight rate averaged over a reference period, and are used to trade and hedge the front of the curve. **Treasury futures** require physical delivery of a government bond, and are used further out. The two are constructed on completely different principles, and each has a wrinkle that repays attention: the short-term contracts carry the convexity adjustment of Chapter 30, and the Treasury contracts embed a delivery option that decides which bond the contract tracks.

The chapter closes with a trade that arises directly from that second wrinkle: the cash–futures basis trade, which hedge funds have built to the high hundreds of billions of dollars, and whose unwinding in March 2020 was among the strains that roughly $1 trillion of Federal Reserve purchases were needed to arrest. It is a recurring subject of financial stability reports for that reason.

Section 6.1 sets out the quotation conventions without which none of the prices can be read. Section 6.2 covers the short-term contract, and Sections 6.3 and 6.4 the Treasury contract and the delivery option that decides which bond it tracks. Section 6.5 turns to the basis trade built on that option, and Sections 6.6 to 6.8 to the uses: hedging a swap book with the strip, duration-based hedging, and building the front of the curve from futures prices.

6.1 DAY COUNTS AND QUOTATION CONVENTIONS

Chapter 4 noted that a rate is meaningless without its day count. Futures markets add quotation conventions on top, and the combination is the subject's main source of confusion.

Day counts. US Treasury bonds accrue on actual/actual; corporate and municipal bonds on 30/360; money market instruments including SOFR on actual/360. The 30/360 convention treats every month as thirty days and gives no month a 31st, which produces the small anomaly that a bond accrues interest on February 29 and 30 and none on the 31st of a long month.

Price quotation for bonds. US Treasury bonds and notes are quoted in points and thirty-seconds of a point, per $100 of face value. A quote of `110-16` means $110 + 16/32 = 110.50$. Finer increments appear as a third digit: `110-165` means $110 + 16.5/32 = 110.5156$, and the shorter note futures are quoted in quarters of a thirty-second.

Clean and dirty price. A coupon is paid in full to whoever holds the bond on the payment date, although a seller may have held it for most of the period. The seller is compensated through **accrued interest**, the part of the next coupon earned since the last coupon date on the bond's day count:

$$\text{accrued interest} = \text{coupon} \times \frac{\text{days since the last coupon}}{\text{days in the coupon period}}$$

The buyer pays it to the seller on top of the price and receives the whole coupon back on the payment date. The quoted price is the **clean price**, which excludes accrued interest. The amount actually paid is the **dirty price**, or cash price:

$$\text{cash price} = \text{quoted price} + \text{accrued interest since the last coupon} \tag{6.1}$$

A 4% Treasury paying coupons on March 1 and September 1, bought at `101-08` for June 15 settlement, has accrued 106 of its 184-day coupon period on actual/actual, so per \$100 of face the buyer pays

$$101.25 + 2.00 \times \frac{106}{184} = 101.25 + 1.1522 = 102.4022$$

The convention exists so that a bond's quoted price does not sawtooth upward between coupons and drop on each payment date, which would obscure genuine price movement.

Price quotation for short-term rate futures. Short-term interest rate futures are quoted as

$$\text{quoted price} = 100 - R \tag{6.2}$$

where R is the rate in percent. A quote of 95.72 corresponds to a rate of 4.28%. The convention inverts the usual direction: a trader who expects rates to *rise sells* the futures contract. It makes the contract behave like a price, rising when the holder profits, as price-based margining systems require.

6.2 SOFR FUTURES

The Eurodollar futures contract, which settled against three-month LIBOR, was for decades the most actively traded futures contract in the world. Its replacement settles against SOFR, and the difference in construction has consequences.

Case Study 6.1 Retiring the largest futures contract ever traded

The Eurodollar contract launched in 1981 and became the benchmark for the front end of the dollar curve, with open interest at times above ten million contracts, over \$10 trillion of notional.

It could not simply be delisted: millions of open positions, some running for years, referenced a rate that would cease to exist. CME's solution, executed in April 2023, was to convert open Eurodollar positions into the corresponding SOFR futures, applying the same ISDA fallback spread adjustment that Chapter 4 described for swaps (26.161 basis points for the three-month tenor), so that the economics of each position carried over.

The conversion was, by the standards of such things, uneventful. The transition succeeded because the compensation mechanism was fixed in advance and applied uniformly, so that no participant had to negotiate. In cash markets, where fallback language was often ambiguous, disputes followed instead.

Contract construction

Two contracts cover the front end.

Three-month SOFR futures have a notional of \$1,000,000 and settle against the *compounded* daily SOFR over a three-month reference period running between consecutive **International Monetary Market (IMM) dates** (the third Wednesdays of March, June, September, and December). Since a basis point on \$1,000,000 for three months is

$$\$1{,}000{,}000 \times 0.0001 \times 0.25 = \$25$$

each basis point is worth \$25 per contract; Example 6.1 follows a position to its profit.

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One-month SOFR futures have a notional of \$5,000,000 and settle against the *arithmetic* average of daily SOFR over a calendar month. A basis point is worth

$$\$5{,}000{,}000 \times 0.0001 \times \frac{1}{12} = \$41.67$$

Example 6.1 Profit on a SOFR futures position

A trader expects the central bank to tighten more than the market has priced, and sells 200 three-month SOFR futures at 95.72, an implied rate of 4.28%.

Rates subsequently rise 12 basis points and the contract settles at 95.60, an implied rate of 4.40%. Because the trader is short the price, the gain is

$$200 \text{ contracts} \times 12 \text{ bp} \times \$25 = \$60{,}000$$

The trader profited from rates rising while being *short* the contract. The initial margin required would have been a small fraction of the \$200 million notional involved, which is what makes futures the instrument of choice for expressing rate views.

The averaging feature and delta decay

These contracts have a property with no counterpart in the Eurodollar world, one that Sections 29.4 and 30.3 return to, and it is worth seeing directly. A Eurodollar contract settled against a rate fixed on a single date. Until that date its sensitivity to rates was full; after it, the contract was over. A SOFR contract settles against an average accumulating across its reference period. Each day that passes locks in one more daily fixing, so the contract's remaining sensitivity falls, approximately linearly, from full at the start of the reference period to zero at its end. Rates desks call a first-order sensitivity of this kind a **delta**, which is why the effect is known as *delta decay*; the same word names an option's sensitivity to its underlying in Chapter 18, and the two are the same idea applied to different variables. Figure 6.1 contrasts the two contracts.

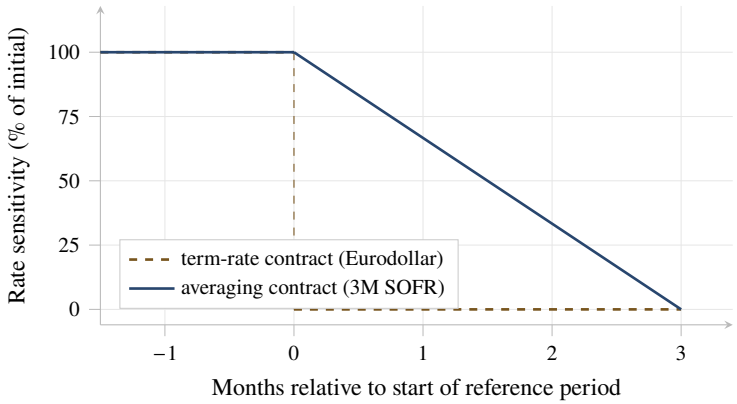


Figure 6.1: Rate sensitivity through the reference period.

The practical consequence is that a hedge built from SOFR futures must be rebalanced *through* the reference period rather than rolled at a single date. A position that is correctly hedged on the first day of the quarter is under-hedged by the last, because the futures leg has lost sensitivity that the hedged item has not.

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Who uses them, and for what

The contracts are rarely traded one at a time. A **strip** is a set of contracts of the same type spanning a run of consecutive dates (or, in the option markets of Chapter 22, a run of strikes) bought or sold together and priced as a single position. The SOFR strip is the sequence of quarterly contracts stretching out along the curve, and most of what follows is about the strip rather than about any one contract in it. Four groups trade the SOFR strip, and their separate motives explain the contract's liquidity.

Borrowers and lenders hedging near-term funding. A corporate treasurer expecting to roll commercial paper (short-term unsecured company borrowing) in six months, or a bank funding a loan book at short rates, locks in the cost by selling futures for the relevant quarter. This is the textbook hedge and the contract's original purpose.

Dealers hedging swap books. This group is by far the largest source of volume. A dealer that has received fixed on a five-year swap holds front-end delta it does not want, and offloads it by selling a strip of SOFR futures covering the swap's early accrual periods. Chapter 7 describes the residual risks this leaves; the futures market is where the first-order risk goes.

Traders positioning on monetary policy. The strip of consecutive contracts is a direct reading of the market's expected path for the policy rate, so a view that the central bank will move faster or slower than priced is expressed by buying one contract and selling another. Trades of this kind (the **steepener**, which profits if later rates rise relative to earlier ones, and the **flattener**, which profits if they fall) are among the most common macro trades.

Curve builders. The futures strip supplies the market's most liquid information about the forward rates of Section 4.9 between roughly three months and five years, and is a primary input to the curve construction of Chapter 7, after the convexity adjustment described next.

Futures rates are not forward rates

The final property, developed at length in Section 30.3, must be flagged here because it governs how these contracts are used to build a curve. Daily settlement favors the short side of the contract, which gains when rates rise: its gains arrive when rates are high and can be reinvested at those rates, and its losses arrive when rates are low and are cheap to finance. A forward rate agreement, settled once, offers neither side that advantage, so the futures rate must sit *above* the forward rate to compensate the long. The gap is the **convexity adjustment**. How large it is depends on the model used for the short rate; the simplest form, and the one this chapter quotes, comes from the Ho–Lee model of Chapter 31, in which the short rate moves by normally distributed amounts with constant volatility:

$$\text{Forward rate} = \text{Futures rate} - \frac{1}{2}\sigma^2 T_1 T_2 \tag{6.3}$$

where T_1 is the time until the contract's rate is determined, T_2 the time until the end of the period that rate covers, and σ the standard deviation of the absolute change in the short rate over a year (a normal volatility, which is the market convention for this parameter). Strictly, (6.3) adjusts the continuously compounded rate; Section 30.3 adds the small term for the simply compounded rate the contract settles on. At $\sigma = 1.00\%$ the adjustment is under a basis point on the one-year contract and about 51 basis points on the ten-year, growing with the *square* of maturity—though Ho–Lee has no mean reversion, no pull of the rate back toward a long-run level, and Section 30.3 shows that this overstates the long end once mean reversion is allowed for. Curve construction that treats quoted futures rates as forward rates will be visibly wrong beyond the first two years or so.

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6.3 TREASURY FUTURES

Treasury futures rest on a different principle: physical delivery of a government bond. Two features of that delivery shape what follows, and both are the short's choice. *Delivery is not a date but a month.* Any business day of the contract's delivery month will serve, and it is the **short** who declares which one, by giving notice; the long takes delivery when told. *And the short chooses the bond as well*, from a basket set by the exchange. Neither choice is a technicality; together they are worth a material amount, priced in Section 6.4.

Physical delivery raises a further difficulty: no single bond stays suitable. A contract referencing one specific ten-year note would become a contract on a nine-year note within a year, and would be vulnerable to a squeeze if someone accumulated the issue. The exchanges' solution is to define a **deliverable basket**: the short may deliver any bond satisfying a maturity range on the delivery date. Table 6.1 lists the principal US contracts.

Table 6.1: Principal US Treasury futures contracts.

Contract	Approximate deliverable maturity	Face value
2-year note	1¾–2 years	$200,000
5-year note	4¼–5¼ years	$100,000
10-year note	6½–10 years	$100,000
Treasury bond	15–25 years	$100,000
Ultra bond	25 years and over	$100,000

Conversion factors

Allowing any bond in the basket to be delivered creates an obvious problem: a high-coupon bond is worth more than a low-coupon one, so the short would always deliver the lowest-coupon bond, and the contract would track that one bond alone. The remedy is the **conversion factor**. Each deliverable bond is assigned a factor equal to its price per $1 of face value if it were priced to yield a notional standard rate (6% in the US contracts), so that a bond paying more than 6% is scaled up and one paying less is scaled down. The amount the long pays on delivery, writing CF for the conversion factor, is

$$\text{invoice} = (\text{settlement price} \times \text{CF} + \text{accrued interest}) \times \frac{\text{face}}{100} \tag{6.4}$$

Example 6.2 computes it for one delivery.

Example 6.2 The invoice amount on delivery

A ten-year note futures contract settles at 110-16, that is 110.50. The short delivers a bond with conversion factor 0.8456 on which accrued interest is $1.20 per $100 of face; the contract covers $100,000.

From (6.4),

$$(110.50 \times 0.8456 + 1.20) \times 1{,}000 = (93.4388 + 1.20) \times 1{,}000 = \$94{,}638.80$$

The factor scales the payment down because the delivered bond's coupon is below the 6% standard.

Who uses them, and for what

Treasury futures are the principal tool for moving duration around, and the identity of the users matters because it explains a persistent feature of the price.

Asset managers adjusting duration. A bond fund that wants to shorten duration ahead of an expected sell-off can sell futures rather than sell bonds: faster, cheaper in transaction costs,

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and without disturbing a carefully constructed credit portfolio. A fund that wants *more* duration than its cash allows buys futures, obtaining the exposure for margin rather than the full purchase price. This second use is important: it means a large and fairly price-insensitive stream of buying interest sits in the futures market, which tends to leave futures *rich* (expensive) relative to the cash bond. That richness is precisely what the basis trade of Section 6.5 harvests.

Dealers hedging inventory. A dealer holding bonds bought from customers hedges the price risk with futures until the position is distributed. Because the hedge is against a basket rather than the specific bond, what remains is basis risk, the subject of Section 6.4.

Mortgage portfolios. Holders of mortgage-backed securities face duration that *shortens* when rates fall, as borrowers prepay, and lengthens when rates rise, the property called **negative convexity**, since the price gains less in a rally than it loses in a sell-off. Keeping such a portfolio hedged requires constant rebalancing in the same direction as the market is already moving, and Treasury futures are the usual instrument. This is a well-known amplifier of rate moves.

Curve and spread traders. Buying one contract and selling another expresses a view on the shape of the curve (long two-year against short ten-year futures is a steepener) without an outright duration position. Trading futures against the bonds expresses a view on the basis itself.

Liability-driven investors. Pension funds and insurers with long-dated liabilities use long-maturity contracts to lengthen asset duration toward liability duration. Section 8.9 works through what happened in 2022 when this exposure was held with leverage.

How conversion factors are calculated

The definition is operational, and worth carrying out, because the two conventions buried in it are what make conversion factors imperfect.

Step 1: round the maturity. Measure the bond's remaining life from the first day of the delivery month and round it *down* to the nearest three months. A bond with 8 years and 5 months remaining is treated as having 8 years and 3 months; one with 8 years and 2 months is treated as having exactly 8 years. This rounding is what allows a single factor to be published for the whole delivery month rather than recomputed daily.

Step 2: price the bond at 6%. Discount the bond's coupons and principal at exactly 6% per annum, semiannually compounded, using the rounded maturity and \$1 of face; the result is the conversion factor.

For a bond whose rounded maturity is a whole number of half-years, this is an ordinary bond price calculation.

Example 6.3 Conversion factor for a bond with 8 years remaining

A bond pays a 5% annual coupon semiannually and has exactly 8 years remaining after rounding: $n = 16$ half-years, a coupon of \$2.50 per \$100 of face, and a discount rate of 3% per half-year.

The annuity factor is

$$a = \frac{1 - 1.03^{-16}}{0.03} = 12.5611$$

so the price per \$100 of face is

$$2.50 \times 12.5611 + \frac{100}{1.03^{16}} = 31.4027 + 62.3167 = 93.7194$$

The conversion factor is this divided by 100:

$$\text{CF} = 0.9372$$

It is below 1 because the bond's 5% coupon is below the 6% standard.

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Now take an odd quarter. Suppose the same bond has 8 years and 3 months remaining after rounding. Three months from now it will have exactly 8 years to run and will just have paid a coupon, so at that date it is worth the \$93.7194 above plus the \$2.50 coupon:

$$93.7194 + 2.50 = \$96.2194$$

Discount that back one quarter at 3% per half-year. Today the bond is already three months into a six-month coupon period, and the interest accrued since the last (notional) coupon date is paid separately by the buyer, as accrued interest, so it must not be charged for twice; deduct it:

$$\frac{96.2194}{1.03^{1/2}} - 2.50 \times \frac{3}{6} = 94.8078 - 1.25 = 93.5578 \qquad \Longrightarrow \qquad \text{CF} = 0.9356$$

The factor is slightly below the 0.9372 of the whole-half-year case, because the extra quarter carries a below-coupon bond a little further from par. The general formula below is built for this case.

The general formula. When the rounded maturity ends on an odd quarter, the bond is valued at the next coupon date, three months away, discounted back, and reduced by the interest accrued since. The exchange publishes a single formula covering both cases. Writing c for the annual coupon rate, n for the number of whole years, and z for the remaining months ($z \in \{0, 3, 6, 9\}$), set $v = z$ if $z < 7$ and $v = z - 6$ otherwise. Then

$$\text{CF} = \frac{1}{1.03^{\,v/6}} \left[\frac{c}{2} + \kappa + \frac{c}{0.06}\,(1 - \kappa) \right] - \frac{c}{2} \cdot \frac{6 - v}{6} \tag{6.5}$$

where $\kappa = 1.03^{-2n}$ if $z < 7$ and $\kappa = 1.03^{-(2n+1)}$ otherwise. The final term deducts the interest accrued over the $6 - v$ months since the last coupon; the leading factor discounts back over the odd quarter. Example 6.4 checks the formula against Example 6.3.

Example 6.4 Applying the formula

Check against Example 6.3. Here $c = 0.05$, $n = 8$, $z = 0$, so $v = 0$ and $\kappa = 1.03^{-16} = 0.623167$:

$$\text{CF} = 1 \times \left[0.025 + 0.623167 + \frac{0.05}{0.06} \times 0.376833 \right] - 0.025 = 0.962194 - 0.025 = 0.9372$$

which reproduces the direct calculation exactly.

Now raise the coupon. With $c = 0.07$ and the same maturity,

$$\text{CF} = \left[0.035 + 0.623167 + \frac{0.07}{0.06} \times 0.376833 \right] - 0.035 = 1.0628$$

The factor is above 1, as it must be for a coupon above 6%.

Finally, take an odd quarter. With $c = 0.05$, $n = 8$, $z = 3$, so $v = 3$ and κ is unchanged at 0.623167:

$$\text{CF} = \frac{1}{1.03^{0.5}} \left[0.962194 \right] - 0.025 \times \frac{3}{6} = 0.9356$$

reproducing the direct construction at the end of Example 6.3. The bracket is that example's \$96.2194 per \$100, the leading factor is its discount back over the odd quarter, and the final term is its accrued deduction. The formula is that construction written out once.

Coupon	Rounded maturity	Conversion factor
5%	8 years	0.9372
7%	8 years	1.0628
5%	8 years 3 months	0.9356

Why the factors do not equalize perfectly. Two approximations are now visible. The factors are computed at a *fixed* 6% yield, so they correctly equalize the basket only when the curve is flat at 6%; at any other level the scaling is systematically off, and increasingly so the further the

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curve sits from 6%. And the maturity rounding discards up to three months of a bond's life, which distorts factors slightly and in a direction that varies across the basket. Both errors are small but not zero, and the residue decides which bond is cheapest to deliver.

6.4 THE CHEAPEST-TO-DELIVER BOND

Because conversion factors equalize the basket only approximately, some bonds end up cheaper to deliver than others. The short, who chooses, will deliver whichever is cheapest—and since the contract is then priced off that bond, identifying it comes first.

For each candidate the short compares the cost of buying the bond against what it receives on delivery. Ignoring carry, the relevant quantity is the **gross basis**:

$$\text{gross basis} = \text{quoted bond price} - \text{futures price} \times \text{conversion factor} \tag{6.6}$$

and the **cheapest-to-deliver** (CTD) bond is the one that minimizes it (Example 6.5).

Example 6.5 Identifying the cheapest-to-deliver bond

The futures price is 110.50 and three bonds are deliverable:

Bond	Quoted price	Conversion factor	Gross basis
A	98.50	0.8912	0.02240
B	104.25	0.9430	0.04850
C	112.75	1.0195	0.09525

For bond A, $98.50 - 110.50 \times 0.8912 = 98.50 - 98.4776 = 0.0224$; the others follow the same way. Bond A is cheapest to deliver, at a cost 2.24 cents per \$100 of face above the futures invoice.

The differences look trivial (seven hundredths of a point separates A from C), but on the \$100 million position of Example 6.8 the gap between choosing correctly and incorrectly is \$72,850.

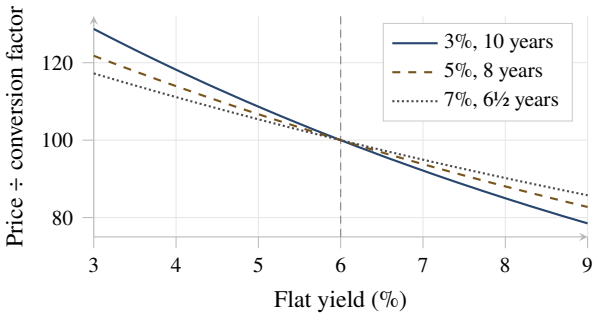


Figure 6.2: Price divided by conversion factor for three deliverable bonds of the ten-year basket, at a flat yield.

Rules of thumb. Two regularities recur. When yields are above the 6% notional coupon, low-coupon long-maturity bonds tend to be cheapest to deliver; when yields are below 6%, high-coupon short-maturity bonds tend to be. The reason is duration. A factor fixes each bond's value as if it yielded 6%; when yields are higher, every bond is worth less than that, and the long-duration bonds fall furthest below their factor, while when yields are lower every bond is worth more, and the short-duration bonds rise least above it. Since yields have been below 6% for most of the contract's recent history, the CTD has usually sat at the short end of the basket. Figure 6.2 draws this for three bonds of the ten-year basket, the middle one being the 5% bond of Example 6.3. Divided by its conversion factor, each bond's price is exactly 100 at a 6% yield, so the three lines cross there, and the cheapest to deliver, the lowest line, switches

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from the short high-coupon bond to the long low-coupon one as yields rise through 6%. Near 6% the lines sit close together and a small move can change which is lowest, which is why the quality option described below is worth most when yields are near the notional coupon.

Carry and the net basis

The gross basis of (6.6) ignores what happens between now and delivery. Meanwhile the bond's holder earns its coupon and pays repo to finance it; the difference is **carry**:

$$\text{carry} = \text{coupon income over the holding period} - \text{repo financing cost} \tag{6.7}$$

and the quantity that actually measures the cost of the delivery option is the **net basis**:

$$\text{net basis} = \text{gross basis} - \text{carry} \tag{6.8}$$

Example 6.6 computes both for one deliverable bond.

Example 6.6 Carry and net basis

A deliverable bond carries a 4.5% coupon, is quoted at 101.20, and can be financed at a repo rate of 4.30% for the 90 days to delivery. Its gross basis is 0.0500 per \$100 of face.

Coupon income accrues on an actual/actual basis over 90 days:

$$4.50 \times \frac{90}{365} = 1.1096 \text{ per \$100 face}$$

Repo interest is charged at actual/360, here on the quoted price for simplicity (strictly it is charged on the cash price, accrued interest included):

$$101.20 \times 0.0430 \times \frac{90}{360} = 1.0879$$

so carry is $1.1096 - 1.0879 = +0.0217$, and from (6.8)

$$\text{net basis} = 0.0500 - 0.0217 = 0.0283$$

That 0.0283 is not a residual to be explained away: it is what the short's delivery options cost. A trader who buys this bond and sells futures against it acquires those options and pays 0.0283 per \$100 of face for them (0.028% of the cash price, or about 11 basis points annualized over the 90 days). The gross basis of 0.0500 confuses that price with the carry; the net basis separates them, and is the number a basis trader watches.

The sign of carry depends on whether the coupon exceeds the financing rate. In the low-rate decade carry was reliably positive and the distinction was a refinement; once short rates rose above the coupons on older bonds, carry turned negative for much of the deliverable basket, and net basis exceeded gross basis. Any statement about basis-trade profitability that ignores carry is incomplete, including the deliberately simplified Example 6.8 below.

The same ingredients give a theoretical futures price. Set the delivery options aside and treat the contract as a forward on the cheapest-to-deliver bond, which the short could buy today, finance in repo, and deliver. The bond's forward price is its cash price (quoted price plus accrued interest) carried to delivery at the repo rate, less the value of any coupon paid in between, as for an asset with known income (Section 5.3). Deducting the accrued interest the long will pay separately at delivery gives a quoted forward price, and dividing by the conversion factor turns that into a futures price, since the invoice of (6.4) pays the futures price times the factor. Example 6.7 carries it out.

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Example 6.7 The futures price from the cheapest-to-deliver bond

Take the bond of Example 6.6 as the cheapest to deliver, with a conversion factor of 0.9180. Its last coupon was 60 days ago, none falls in the 90 days to delivery, and accrued interest is counted as there.

Today's cash price is the quoted price plus 60 days of accrued interest:

$$101.20 + 4.50 \times \frac{60}{365} = 101.20 + 0.7397 = 101.9397$$

Carried to delivery at the 4.30% repo rate (actual/360), it becomes the bond's forward cash price:

$$101.9397 \times \left(1 + 0.0430 \times \frac{90}{360}\right) = 103.0356$$

At delivery the long pays 150 days of accrued interest separately, so the quoted forward price is

$$103.0356 - 4.50 \times \frac{150}{365} = 103.0356 - 1.8493 = 101.1863$$

and the theoretical futures price is

$$\frac{101.1863}{0.9180} = 110.2247$$

The gross basis of 0.0500 in Example 6.6 implies a market futures price of $(101.20 - 0.0500)/0.9180 = 110.1852$, which is 0.0395 lower. Multiplied by the conversion factor, the gap is 0.0363 per \$100 of face: the net basis of 0.0283, plus 0.0080 of repo interest on the accrued interest, which that example left out for simplicity. The futures contract trades below its forward-based value by what the short's delivery options are worth.

The delivery options

The short's right to choose which bond to deliver is a genuine option, and it is not the only one. Four are conventionally distinguished, all belonging to the short and all therefore depressing the futures price below what a no-arbitrage calculation on today's CTD alone would give.

The quality option is the one just described: the right to deliver any bond in the basket. Its value comes from the possibility that the CTD *changes* before delivery. Since the identity of the CTD depends on the level and shape of the curve, a large enough move switches it, and the short benefits from switching to whichever bond has become cheapest. The option is therefore worth more when the basket is wide and yields sit near the 6% notional coupon, where small moves can flip the ranking, and worth less when one bond is cheapest by a wide margin.

The timing option is the right to choose which business day of the delivery month to deliver. When carry is positive the short prefers to wait, earning more coupon than financing costs; when carry is negative it prefers to deliver early. Since the sign of carry is known, this option is worth little in practice—but it is not zero, because carry can change.

The wild card option arises from a mismatch in trading hours. The futures settlement price is struck at 2:00 p.m. Chicago time, but the short's notice of intention to deliver is not due until 6:00 p.m., while the cash market's New York session runs to 5:00 p.m. New York time (4:00 p.m. in Chicago). So the invoice amount is fixed for four hours, of which the first two have a liquid cash market to act in. If bond prices fall in that window, the short buys the deliverable cheaply and delivers it against the higher invoice already fixed at 2:00. It recurs every day of the delivery month, so a small daily edge adds up.

The end-of-month option extends the same idea: futures trading ends some days before the last delivery date, so for those days the invoice price is fixed while the cash market moves.

None of these is large individually. Collectively they are the reason a Treasury futures contract cannot be priced as a simple forward on the current cheapest-to-deliver bond, and the reason that basis traders speak of being short an option rather than holding an arbitrage.

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6.5 THE CASH–FUTURES BASIS TRADE

The relationship between a Treasury futures contract and its cheapest-to-deliver bond gives rise to a trade that has grown from an arbitrage niche into a staple of financial stability reports: buy the cash bond, sell futures against it, finance the bond in repo, and hold to delivery. The profit is worth stating carefully, because the sign is easy to get wrong. The trade pays away the gross basis (you bought at the cash price and will receive the futures invoice) and earns the carry along the way. Its profit and loss (P&L) to delivery is therefore

$$\text{profit} = \text{carry} - \text{gross basis} = -\,\text{net basis} \tag{6.9}$$

using the net basis of Section 6.4. The trade is profitable precisely when the *net basis is negative*, which happens when the futures are rich relative to the cash bond, the condition that the asset-manager duration demand of Section 6.3 tends to create. Equivalently, and as Section 5.10 puts it, it is profitable when the implied repo rate exceeds the actual repo rate.

A *positive* net basis means the reverse: the cash-and-carry loses, and what it loses is the value of the short's delivery options from Section 6.4. That is the textbook state of affairs, and it is why the trade is not simply a riskless profit waiting in the market.

Two forces therefore set the sign, and they push opposite ways. The delivery options belong to the short and *depress* the futures price, driving the net basis positive. Demand for duration in futures form (from asset managers, and from liability-driven investors who want exposure without balance sheet) makes futures *rich*, driving it negative. Which dominates is an empirical matter, and since the 2010s the demand side has usually won in the liquid front contracts. That is what turned this from an occasional arbitrage into a standing business large enough to be a financial-stability question, and it is also why the trade is not riskless: the sign it depends on is set by a balance of flows, not by an identity. Figure 6.3 sets out the three legs: the bond bought in the cash market, the futures sold against it, and the repo that finances the position until delivery closes all three.

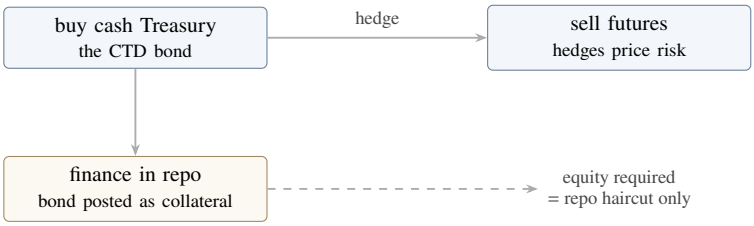


Figure 6.3: Structure of the cash–futures basis trade.

Example 6.8 Leverage in the basis trade

A fund buys \$100 million face of a cheapest-to-deliver bond at 98.50 and sells the corresponding futures. Asset-manager demand has left the futures rich, so the net basis is −0.0180 per \$100 of face. From (6.9) the position earns, held to delivery,

$$\$100{,}000{,}000 \times \frac{0.0180}{100} = \$18{,}000$$

That is 1.8 basis points of the position. What transforms it is financing: the repo lender advances the bond's value less a haircut (Section 4.1), which the borrower funds itself. Treasury repo haircuts are small (take 2%), so the fund posts

$$0.02 \times \$98{,}500{,}000 = \$1{,}970{,}000$$

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of its own money and borrows the rest. The return on equity is

$$\frac{18{,}000}{1{,}970{,}000} = 0.91\%$$

over the life of the trade. If that is three months, the return on equity is roughly 3.7% a year above the repo rate the equity itself earns (carry was charged on the whole price), from a position whose market risk is hedged.

The leverage is close to fifty times. At a 1% haircut it would be a hundred times.

Why it matters beyond the fund

Example 6.8 explains both why the trade is attractive and why regulators watch it. The economics work only at scale and with high leverage, and a Federal Reserve estimate puts hedge funds' aggregate position at about $830 billion in September 2025 (Monin 2026). Three features make that uncomfortable:

- **The leverage is procyclical.** Repo haircuts widen under stress. A move from 2% to 4% halves the position a given equity can support, forcing sales when markets are illiquid.
- **Margin on the futures leg moves the other way.** A sharp rally raises variation margin calls on the short futures position, and that margin must be met in cash even though the offsetting gain on the bond is unrealized. This is the mechanism of Chapter 8, appearing here.
- **Everyone is in the same trade.** The positions are concentrated in a small number of funds running similar strategies, so unwinding is correlated.

Case Study 6.2 March 2020

In the second week of March 2020, the US Treasury market (the world's deepest securities market, and the benchmark risk-free asset) stopped functioning properly. Bid–ask spreads widened by an order of magnitude, and on some days prices for on-the-run (most recently issued) and off-the-run (older) issues diverged without precedent.

The causes were several: a global dash for cash, foreign official selling, and dealer balance sheets already at their limits. A widely cited contributor was the forced unwinding of leveraged basis positions. As volatility spiked, haircuts rose and futures margin calls landed, and funds sold cash Treasuries into a market that could not absorb them, which pushed cash prices further from futures, widened the basis, and triggered more selling.

The Federal Reserve ended the episode by buying Treasuries on a scale not previously contemplated, roughly $1 trillion within weeks. The lesson is that a trade genuinely hedged against *price* risk can still fail catastrophically through *funding* risk, and leverage that looks prudent at normal haircuts does not survive stressed ones.

The clearing response

The principal regulatory response has been to push Treasury cash and repo activity into central clearing, under the SEC rules that Section 2.11 sets out with their compliance dates. Their intended effect is the multilateral netting of Example 2.4, together with uniform margin standards in place of bilaterally negotiated haircuts.

For this trade the rules bite through the financing leg. Once the repo that funds the position is cleared, its haircut comes from the clearing house's published margin model rather than from a negotiation with one dealer, and that puts a ceiling on the leverage in Example 6.8: a haircut that a dealer might set at 1% for a good client becomes a number the clearing house sets for everyone, and the example's fifty-to-a-hundred-times leverage needs a small haircut. The offsetting benefit is netting (a fund whose futures and repo positions are margined together

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posts against the net exposure, not the gross), and the practical question, unresolved as this is written, is which of the two effects dominates for a trade whose economics are a few basis points.

6.6 HEDGING A SWAP BOOK WITH THE STRIP

Section 6.2 named dealers hedging swap books as the largest source of SOFR futures volume. Chapter 7 treats swaps; only their outline is needed here. An interest rate swap exchanges a fixed rate for a floating one on a notional principal, period by period, and only the difference between the two payments changes hands. In each period the party that receives fixed is paid the fixed rate and pays the floating rate set for that period, here SOFR compounded over the quarter, so it gains when rates fall and loses when they rise. The party that pays fixed holds the opposite position.

For a dealer that has received fixed, the exposure is not a single number: each accrual period of the swap carries its own forward rate risk, and the futures strip lets those be hedged period by period. Example 6.9 hedges the first year of a two-year swap this way.

Example 6.9 Hedging the front of a swap

A dealer has received fixed on a \$500 million two-year swap with quarterly floating periods. Consider the first four accrual periods, each with $\tau = 0.25$.

A one basis point rise in any one period's forward rate costs the dealer about

$$\$500{,}000{,}000 \times 0.0001 \times 0.25 = \$12{,}500$$

before discounting. At \$25 per basis point per contract, the dealer sells

$$\frac{\$12{,}500}{\$25} = 500 \text{ contracts}$$

against each accrual period, 2,000 across the first year.

That match needs the dates to line up, and usually they do not. It assumes the swap's accrual periods coincide with the futures' reference periods, which holds only if the swap starts on one of the quarterly IMM dates, on which those reference periods begin and end. Suppose instead the first period runs from May 10 to August 10: 92 days, of which 39 fall before the third Wednesday of June and 53 after. The March contract covers the first stretch and the June contract the second, so the period's \$12,500 of sensitivity is split between them in proportion to days,

$$\frac{39}{92} \times \$12{,}500 = \$5{,}299 \to 212 \text{ March}, \qquad \frac{53}{92} \times \$12{,}500 = \$7{,}201 \to 288 \text{ June}$$

still 500 contracts, but spread across two delivery months. The residual is real: the two contracts do not move identically, so the hedge retains a small basis between the two reference periods, and a dealer whose book is mostly off-IMM swaps carries it on every one.

Three qualifications follow. Discounting reduces the required position slightly for later periods, since the \$12,500 arrives at the end of the accrual period rather than today. The futures hedge the futures rate, which by Section 30.3 exceeds the forward rate, so the hedge inherits convexity risk that must be managed separately. And as Figure 6.1 showed, each contract's sensitivity decays through its reference quarter while the swap's does not, so the ratio must be revisited within the period rather than only at the roll.

Beyond two years or so the strip thins out and dealers switch to hedging with other swaps or with Treasury futures, accepting the basis risk that comes with hedging a swap exposure using a government bond instrument. That basis, the swap spread of Section 7.9, is itself a traded risk.

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6.7 DURATION-BASED HEDGING

The most common use of Treasury futures is to adjust the interest rate exposure of a bond portfolio without trading the bonds themselves. Chapter 4 gave the first-order relationship $\Delta B \approx -BD\Delta y$. Applying it to both the portfolio and the futures contract, and requiring the changes to offset, gives the number of contracts to short:

$$N^{*} = \frac{P\, D_{P}}{F\, D_{F}} \tag{6.10}$$

where P is the value of the portfolio, D_P its duration at the hedge horizon, F the contract value of one futures contract, and D_F the duration of the cheapest-to-deliver bond at contract maturity.

Example 6.10 Hedging a bond portfolio

A fund holds \$10 million of bonds with a duration of 6.8 years and wishes to hedge with ten-year note futures. The futures price is `110-16`, so with \$100,000 face per contract,

$$F = \frac{110.50}{100} \times \$100{,}000 = \$110{,}500$$

The cheapest-to-deliver bond has a duration of 7.6 years at contract maturity. From (6.10),

$$N^{*} = \frac{10{,}000{,}000 \times 6.8}{110{,}500 \times 7.6} = \frac{68{,}000{,}000}{839{,}800} = 80.97$$

so the fund shorts 81 contracts.

Check: a one basis point rise in yields costs the portfolio about $10{,}000{,}000 \times 6.8 \times 0.0001 = \$6{,}800$, while 81 contracts gain about $81 \times 110{,}500 \times 7.6 \times 0.0001 = \$6{,}802$.

What this hedge does not do. Equation (6.10) immunizes against a *parallel* shift only. A portfolio hedged this way remains exposed to changes in the shape of the curve (the bucketed exposures of Section 29.10) and to the possibility that the CTD bond changes, which alters D_F discontinuously. It also inherits the limitation of duration itself: it is a first-order approximation, and Chapter 4's convexity term reappears for large moves.

Hedging to a target duration

Managers more often want to *change* duration than to eliminate it. If the target duration is D_T rather than zero, the required position becomes

$$N^{*} = \frac{P\,(D_{P} - D_{T})}{F\, D_{F}} \tag{6.11}$$

which is positive when duration must be reduced (short futures) and negative when it must be raised (long futures); Example 6.11 works the second case.

Example 6.11 Raising portfolio duration

The fund of Example 6.10 decides its \$10 million portfolio should carry a duration of 9.0 years rather than 6.8, because it expects yields to fall. From (6.11),

$$N^{*} = \frac{10{,}000{,}000\,(6.8 - 9.0)}{110{,}500 \times 7.6} = -26.2$$

The negative sign means buying: the fund goes long 26 contracts.

The alternative, selling short-dated bonds and buying long ones, achieves the same duration but requires trading a large part of the \$10 million portfolio, several days of execution in less liquid issues, and realized gains or losses on every position touched. The futures route requires margin on 26 contracts and takes seconds.

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Futures contracts expire while the exposures they hedge often do not, so a five-year liability hedged with three-month contracts is stacked and rolled, as Section 3.9 describes: each roll crosses a bid–ask spread and is made at whatever spread between the two contracts prevails on the day, which is not known in advance. Section 3.10 tells the Metallgesellschaft episode, in which such a hedge failed through roll costs and margin calls while the hedged position performed as designed; the same funding and roll risk recurs in Section 6.5.

6.8 BUILDING THE FRONT OF THE CURVE

Section 4.8 bootstrapped a zero curve from OIS quotes. In practice the region from roughly three months to five years is pinned down instead by the futures strip, because that is where the liquidity concentrates. The procedure has one wrinkle, and it is the subject of Chapter 30.

Each contract gives a futures rate for a specific three-month period. Converting the strip into forward rates requires subtracting the convexity adjustment of (6.3); only then can the forwards be chained into discount factors. Example 6.12 does so for three contracts.

Example 6.12 From futures quotes to forward rates

Three consecutive contracts settle at 95.72, 95.55, and 95.41, implying futures rates of 4.28%, 4.45%, and 4.59% for periods beginning in 1.00, 1.25, and 1.50 years. Take a short-rate volatility of 1.00% per year and apply the Ho–Lee adjustment $\frac{1}{2}\sigma^2 T_1 T_2$ from Section 30.3:

T_1	T_2	Futures rate	Adjustment	Forward rate
1.00	1.25	4.280%	0.63 bp	4.2738%
1.25	1.50	4.450%	0.94 bp	4.4406%
1.50	1.75	4.590%	1.31 bp	4.5769%

For the first contract, (6.3) gives $\frac{1}{2} \times 0.0001 \times 1.00 \times 1.25 = 0.0000625$, or 0.63 basis points; the others follow.

At this point on the curve the corrections are about a basis point and a curve builder might be tempted to skip them. The temptation should be resisted, because the adjustment grows with the square of maturity: by the eight-year contract it reaches 33 basis points.

Beyond the liquid futures region, the curve is completed with swap quotes, and the two must be made consistent; that is the simultaneous calibration that Section 7.6 describes.

SUMMARY

Interest rate futures come in two families: short-term contracts settle in cash against an overnight rate averaged over a reference period, and Treasury futures require delivery of a bond from a defined basket. Short-term contracts are quoted as $100 - R$, so a trader expecting higher rates sells; because the settlement rate accrues through the reference period, a contract's sensitivity decays toward zero across it, and hedges must be rebalanced within the quarter rather than rolled on one date. Bonds are quoted clean, with accrued interest added to reach the cash price.

Each contract's users explain much of its behavior. The SOFR strip serves borrowers hedging funding, dealers hedging the front of their swap books, and macro traders positioning on the policy path. Treasury futures serve asset managers and liability-driven investors moving duration, dealers hedging inventory, and mortgage holders rebalancing negative convexity, and the steady demand for duration in futures form tends to leave futures rich to the cash bond.

Conversion factors set at a 6% notional coupon make the basket only approximately equivalent, so one bond is cheapest to deliver and the contract tracks it, and which bond that is switches as yields cross the notional coupon. The futures price is that bond's forward price divided by its conversion factor, less the value of the short's options to choose the bond and the

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delivery day; the net basis, the gross basis less carry, is what those options cost.

The basis trade buys the cheapest bond, sells futures against it and finances it in repo. It earns a convergence of a few basis points, and thin repo haircuts make the leverage high, so it runs at scale and is fragile under stress. March 2020 showed that a trade hedged against price can still fail through funding, and central clearing of Treasury cash and repo is the main regulatory response.

Duration-based hedging sells $N^* = PD_P/(FD_F)$ contracts against a parallel shift and leaves curve reshaping, cheapest-to-deliver switches and convexity unhedged; a dealer instead matches the strip to a swap book period by period. Hedges that outlive their contracts must be rolled, which exposes the hedger to the roll spread every time. Finally, the strip anchors the front of the curve, once the convexity adjustment, negligible for near contracts and material for distant ones, is subtracted from the futures rates.

FURTHER READING

Burghardt and Belton's *The Treasury Bond Basis* is the standard practitioner treatment of conversion factors, the cheapest-to-deliver, and the delivery options, and it is the book to own if this material is your job. The Committee on the Global Financial System's *Repo Market Functioning* (2017) covers the financing leg that the basis trade depends on. On what happens when that financing withdraws, Duffie's Brookings paper on redesigning the Treasury market after March 2020 and the Financial Stability Board's thematic reviews of leverage in non-bank financial intermediation (2023–2024) are the two accounts to read together.

PRACTICE QUESTIONS

6.1 A Treasury bond is quoted at `104-20` with accrued interest of \$0.85 per \$100 of face. Express the quoted price as a decimal and give the cash price.

6.2 A trader buys 150 three-month SOFR futures at 96.10 and the contract settles at 96.34. Calculate the profit, state what happened to the implied rate, and explain why the contract is quoted as $100 - R$ in the first place.

6.3 Explain why the rate sensitivity of a three-month SOFR futures contract declines through its reference period, why this did not happen with Eurodollar futures, and what it implies for a hedger.

6.4 A bond pays a 6% coupon semiannually and has exactly 10 years to maturity after rounding. Calculate its conversion factor at the 6% notional coupon rate and explain the result. Then use equation (6.5) to calculate the conversion factor of a 4% coupon bond with 7 years and 3 months remaining after rounding.

6.5 The futures price is 108.75 and three bonds are deliverable, quoted at 96.20, 101.85, and 118.40, with conversion factors 0.8825, 0.9350, and 1.0880. Identify the cheapest-to-deliver bond.

6.6 Distinguish gross basis from net basis. A deliverable bond has a 3.0% coupon, is quoted at 94.60, can be financed at 4.40% for the 60 days to delivery, and has a gross basis of 0.0410. Calculate carry and the net basis, and comment on the sign.

6.7 A portfolio worth \$25 million has a duration of 5.4 years, ten-year note futures are priced at `109-00`, and the CTD has a duration of 8.1 years at contract maturity. How many contracts should be shorted to immunize it? What position instead moves a \$40 million portfolio of duration 4.2 years to a target of 7.5 years, with futures at `111-00` and a CTD duration of 8.0?

6.8 Describe the cash–futures basis trade, and state where the profit comes from and where

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the leverage comes from.

6.9 A ten-year note futures contract settles at `112-08`. The delivered bond has a conversion factor of 0.9125 and accrued interest of \$0.60 per \$100. Calculate the invoice amount for one \$100,000 contract.

6.10 Explain why conversion factors do not make all deliverable bonds exactly equivalent, and state which bonds tend to be cheapest to deliver when yields are below 6% and why.

6.11 Explain why steady demand from asset managers for duration in futures form tends to leave Treasury futures rich relative to the cash bond, and connect this to Section 6.5.

6.12 A dealer has received fixed on a \$300 million swap with quarterly floating periods. State how many three-month SOFR futures it should sell against each accrual period, ignoring discounting, and give two reasons the true hedge ratio differs.

6.13 Drawing on Sections 3.9 and 3.10, explain what a stack-and-roll hedge is and the two costs it incurs, and why a hedge of this kind can fail even when the underlying exposure behaves exactly as expected.

FURTHER QUESTIONS

6.14 A fund runs a \$500 million basis position at a net basis of -0.031 per \$100 with a repo haircut of 1.5%. Calculate the profit on full convergence, the equity required, the return on equity, and the leverage ratio. If the haircut widens to 4%, how much additional equity must the fund find, and what are its options if it cannot?

6.15 The basis trade is hedged against the price of Treasuries, yet it failed in March 2020. Explain precisely which risk was unhedged, identify the two distinct channels through which it materialized, and explain why both worsened simultaneously.

6.16 Explain why the short's delivery options depress the Treasury futures price. Would you expect the effect to be larger when the yield curve is flat or steep, and when yields are near 6% or far from it?

6.17 A dealer hedges a swaps book with SOFR futures. Explain three distinct reasons the hedge will not be exact, drawing on this chapter, Chapter 7, and Chapter 30.

6.18 Central clearing of Treasury cash and repo is intended to reduce systemic risk. Set out the mechanisms by which it should do so, and then argue the opposite case as strongly as you can. What would you want to know about the clearing house before deciding?

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Chapter 7
Swaps and the Multi-Curve Framework

The interest rate swap is the largest derivatives market in the world by notional outstanding, and it is the instrument through which almost every other rate derivative is hedged. It is also, for our purposes, the place where a simplification made in Chapter 4 has to be paid for.

That chapter spoke of "the" zero curve for a currency, as though a single term structure described it. The assumption was flagged as provisional, and this chapter retires it. A modern swap valuation requires *at least two* curves: one that says what cash flows are worth, and one that says how large the floating cash flows will be. They are built from different instruments, they move independently, and the gap between them is traded in large volume.

It would be easy to present this as a technical correction added to a simpler theory—which is how it entered the market, in the years after 2008, and how most textbooks still frame it. That framing is misleading. The single-curve world was not a simpler truth later refined; it was a coincidence. Two things happened to line up: the rate at which banks lent to each other for three months, and the rate at which a derivative's collateral earned interest. Once those separated, they never rejoined, and there is no reason they should have. This chapter therefore develops the two-curve structure as the standard case and treats the single-curve world as the special case it always was.

Sections 7.1 to 7.3 are the contract, why it exists, and how it is valued. Sections 7.4 to 7.6 are the two-curve structure: why the collateral agreement determines the discount curve, why one curve is not enough, and how the curves are built. Sections 7.7 to 7.9 are the instruments and prices that structure produces (basis swaps, cross-currency swaps and their basis, and swap spreads as the price of balance sheet), and Sections 7.10 to 7.12 close with credit, netting and clearing, what the transition did to the swap book, and the other swaps the market trades.

7.1 MECHANICS OF AN INTEREST RATE SWAP

A **swap** is an agreement to exchange cash flows on a schedule. In the standard interest rate swap, the **plain vanilla** swap, two parties agree to exchange, on a notional principal L, a stream of fixed interest payments for a stream of floating payments. The notional is never exchanged; it exists only to scale the payments, which is why an enormous notional outstanding corresponds to a much smaller amount of money at risk.

Since the transition described in Chapter 4, the floating leg of a standard swap in dollars, sterling, Swiss francs, yen and Canadian dollars references a compounded risk-free rate in arrears. Such a contract is the overnight indexed swap of Section 4.8: a **SOFR OIS** in dollars, a SONIA OIS in sterling, and so on for each currency's designated rate. In those currencies it is now the market standard rather than a specialist product. The euro is the large exception: EURIBOR survived the transition (Section 4.3), and most euro swaps still pay an annual fixed rate against six-month EURIBOR, with €STR OIS trading beside them; Australian, Hong Kong and Chinese swaps likewise still reference term rates. In dollars the name OIS once

meant a different contract. Before October 2020 the OIS curve was built from federal funds swaps, not SOFR, and Section 7.4 describes the switch; where this chapter writes OIS without qualification, the currency's current rate is meant. On each payment date the floating payer owes

$$L\,\tau_i\,R_i \tag{7.1}$$

where τ_i is the length of accrual period i as a fraction of a year, R_i is the compounded RFR over that period computed as in Section 4.4, and the fixed payer owes $L\,\tau_i\,K$ for the agreed fixed rate K. Only the difference changes hands. Figure 7.1 shows the arrangement.



Figure 7.1: Cash flows in a plain vanilla interest rate swap. The notional L is not exchanged.

Example 7.1 A swap's cash flows over its life

Company B enters a three-year annual swap on a notional of \$100 million, paying 4.20% fixed and receiving the compounded RFR. Suppose the compounded rate turns out to be 4.00% over the first year, 4.35% over the second, and 4.60% over the third. All figures below are in millions of dollars.

Payment date	Compounded RFR	Floating received	Fixed paid	Net to B
End of year 1	4.00%	+4.00	−4.20	−0.20
End of year 2	4.35%	+4.35	−4.20	+0.15
End of year 3	4.60%	+4.60	−4.20	+0.40
Total				**+0.35**

Three features of the table deserve attention. Only the net column changes hands: the gross \$4 million legs are never exchanged, and neither is the \$100 million notional.

The rate in the second column is not known at the start of the period it applies to. Under the conventions of Chapter 4 it accumulates from daily fixings across the year and is complete only at the end of the period, a few days at most before it is paid. Under the LIBOR conventions this table would have replaced (where each rate was fixed twelve months in advance), B would have known each payment a year ahead. This is the single largest operational change the transition brought.

B did well here only because rates rose. Had they fallen to 3.5%, B would have paid \$700,000 in that year. That is not a loss in any meaningful sense: B entered the swap to *fix* its cost, and Example 7.2 shows the combined position doing exactly that.

Conventions. A swap confirmation specifies far more than the fixed rate: the payment frequency on each leg (usually annual on both legs for an OIS, while swaps on the surviving term rates mix frequencies: annual fixed against semiannual EURIBOR in euros, quarterly against quarterly BBSW in Australian dollars), the day count on each leg, the business day convention, the calendar, and, critically after Chapter 4, the exact compounding convention and payment lag on the floating leg. Two swaps quoted at the same rate under different conventions are not the same trade, and the difference is not always small. Table 7.1 sets out a confirmation for a five-year dollar swap, and later sections return to it: Section 7.3 values it, and Section 7.4 shows that its last line, the collateral, is what fixes the curve used to do so.

The fixed rate is 4.31% rather than the 4.295% midpoint of the dealer's quote because the customer is paying fixed and therefore deals at the higher of the dealer's two rates, as the next subsection explains. Every other line is a convention, and every one of them changes the cash flows.

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Table 7.1: Terms of a five-year dollar swap. Illustrative, not a market quote.

Notional	\$100,000,000
Effective date	March 18, 2026
Maturity date	March 18, 2031
Fixed rate	4.31%, paid by the customer
Fixed leg	Annual, actual/360
Floating index	SOFR, compounded daily in arrears
Floating leg	Annual, actual/360
Payment lag	Two business days
Business day convention	Modified following
Calendar	New York
Clearing	Cleared at a CCP
Collateral	Variation margin in USD cash, interest at SOFR

How swaps are quoted

A dealer quotes a swap by quoting the *fixed* rate; the floating leg is understood to be the index flat, with no spread. A quote of "4.28 / 4.31" on the five-year means the dealer will receive fixed at 4.31% and pay fixed at 4.28%, and the three-basis-point difference is the bid–ask spread. The customer pays a slightly higher fixed rate than the mid when paying fixed and receives a slightly lower one when receiving. This is the usual arrangement, with the notable feature that the spread is quoted in rate rather than in price. Three basis points sounds negligible and is not: on the notional of Table 7.1 it is worth about \$134,000 over the life of the trade, by the calculation Section 7.3 sets out.

Swaps are often traded on the IMM dates of Section 6.2, the standardized quarterly dates used by futures markets, so that a swap position can be hedged against futures without a date mismatch. A particular class of standardized contracts, **market agreed coupon** swaps, fixes the coupon at a round number in increments of 25 basis points and settles the difference from the par rate as an upfront payment. These are one product among many rather than the general case: the swap of Table 7.1 is an ordinary par swap, struck at whatever rate makes it worth zero, and most of the market trades that way. What the round coupon buys is *fungibility*. Two par swaps struck a day apart carry different fixed rates and net against each other only in value; two market agreed coupon swaps of the same coupon and dates are the same instrument, so they compress and clear more cheaply. That matters most to dealers and to hedge funds and other active traders, who open and close positions often and want them to net; a company hedging one particular loan has no use for fungibility and trades a par swap matched to the loan. What it costs is cash on day one. Written that way, Table 7.1's swap would carry a 4.25% coupon (the nearest quarter-point to its 4.295% par rate), and the 4.5 basis point difference, multiplied by the swap's annuity of 4.48 (the present value of the year fraction accrued in each of its five years, about 1.014 under actual/360, discounted at the par rate; Section 7.3), gives

$$\$100{,}000{,}000 \times 0.00045 \times 4.48 = \$201{,}600$$

half as much again as the entire bid–ask. Because the coupon is the nearest quarter-point and not the one below, the payment runs in either direction: here the fixed payer makes it, having been given a coupon below the market rate, but at a par rate of 4.205% the same coupon would sit above par and the fixed payer would receive it. Half an increment also bounds it at inception (about \$560,000 on this trade), though a standing coupon the market has since moved away from can exceed that.

A dealer running a swap book is not taking a view. Having received fixed from one customer, it hedges by paying fixed elsewhere: to another customer, in the interdealer market, or with futures. What remains after hedging is a residual exposure to the shape of the curve, to the

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basis spreads of Section 7.5, and to the counterparty credit of Section 7.10. Managing those residuals, rather than predicting rates, is the business. The residual is measured bucket by bucket, as the dollar value of one basis point (DV01) of each quoted maturity separately: the book's change in value for a one-basis-point move in that maturity alone. This bucketed or key-rate DV01 shows exposure to the shape of the curve that a single parallel-shift number nets away. The short buckets are typically hedged with the futures of Chapter 6, the longer ones with swaps.

7.2 WHY SWAPS EXIST

A swap changes the interest rate character of an exposure without disturbing the contract that created it. That single sentence covers most of the market, but the four standard uses are worth working through, because the cash flow accounting is what makes the idea concrete.

Transforming a liability

This is the most common use. A company has borrowed at a floating rate and wants certainty about its interest cost. Figure 7.2 shows the three flows and where the floating rate cancels.

Example 7.2 Converting floating debt to fixed

Company B has a three-year loan of $50 million on which it pays the RFR plus 80 basis points. It would rather know its cost. It enters a swap on the same notional, paying 4.20% fixed and receiving the RFR.

Its three cash flows are now:

Pays on the loan	−(RFR + 0.80%)
Pays on the swap	−4.20%
Receives on the swap	+RFR
Net cost	−5.00% fixed

The RFR appears twice with opposite signs and cancels. B has converted a floating liability into a fixed one at 5.00%, or $2.5 million a year, without renegotiating the loan or informing the lender.



Figure 7.2: Company B pays its lender on the left and swaps on the right.

Note what has *not* changed. B still owes the loan, still bears the 80 basis point credit spread, and, as Section 7.10 discusses, has added a counterparty to its balance sheet. What it has removed is one specific risk: the level of the reference rate.

Transforming an asset

The same operation applies to the other side of the balance sheet, and its diagram is the mirror image of Figure 7.2, with every arrow reversed. An investor holding a floating-rate asset who wants a fixed return receives fixed and pays floating.

Example 7.3 Converting a floating-rate asset to fixed

Company A holds $50 million of floating-rate notes yielding the RFR plus 25 basis points, and wants a fixed return to match a fixed obligation. It enters a swap receiving 4.20% fixed and paying the RFR:

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Receives on the notes	+(RFR + 0.25%)
Receives on the swap	+4.20%
Pays on the swap	−RFR
Net return	+4.45% fixed

A has synthesized a 4.45% fixed-rate asset out of a floating one.

These two examples are mirror images, and in principle A and B could face each other directly. In practice they do not, for the obvious reason that they do not know each other and would not want each other's credit.

The role of the financial intermediary

A dealer stands between them, as in Figure 7.3 and Example 7.4, entering offsetting swaps with each and quoting a bid–ask spread on the fixed rate, say 4.18 / 4.22, read as in Section 7.1: the dealer pays fixed at 4.18% and receives fixed at 4.22%.

Example 7.4 The dealer in the middle

The dealer pays A 4.18% fixed and receives the RFR; it receives 4.22% fixed from B and pays the RFR. The floating legs offset exactly, so the dealer is left with

$$4.22\% - 4.18\% = 4 \text{ basis points}$$

on \$50 million, or \$20,000 a year, for as long as both swaps run.

The all-in outcomes shift slightly. A now earns 4.18% + 0.25% = 4.43% rather than 4.45%, and B pays 4.22% + 0.80% = 5.02% rather than 5.00%. Between them they have paid the dealer four basis points for finding the other side and standing between them.

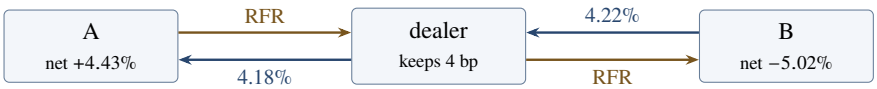


Figure 7.3: The dealer faces both customers. Its two floating legs offset, leaving the four basis point spread.

The dealer is not merely a matchmaker. It takes B's credit risk and A's credit risk, and it need not have found A and B simultaneously: in practice it warehouses one side, hedges the rate risk in the futures market of Chapter 6, and waits for offsetting business. What it earns is compensation for that warehousing and for the credit it has taken, not a broker's fee.

The comparative advantage argument, and what is wrong with it

The fourth traditional motivation is comparative advantage, and it deserves both a fair statement and a critique. Suppose Company A and Company B can each borrow for five years in either market, on these terms:

	Fixed	Floating
Company A (stronger credit)	5.0%	RFR + 0.3%
Company B (weaker credit)	6.4%	RFR + 1.0%
Difference	1.4%	0.7%

A is cheaper in both markets, but its advantage is larger in the fixed market (1.4%) than in the floating market (0.7%). Suppose further that A *wants* floating and B wants fixed, which is the inconvenient case, since each wants the market in which it is relatively worse off.

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The apparent total gain is 1.4% − 0.7% = 0.7%. Splitting it evenly, each should end up 35 basis points better off than by borrowing directly. Example 7.5 shows how.

Example 7.5 Splitting the comparative advantage gain

A borrows fixed at 5.0%, its stronger market; B borrows floating at RFR + 1.0%. They then swap, with A receiving 5.0% fixed and paying RFR − 0.05%.

Company A:

$$\underbrace{-5.0\%}_{\text{loan}} + \underbrace{5.0\%}_{\text{swap received}} - \underbrace{(\text{RFR} - 0.05\%)}_{\text{swap paid}} = -(\text{RFR} - 0.05\%)$$

A borrows floating at RFR − 0.05%, against RFR + 0.30% directly, better by 35 basis points.

Company B:

$$\underbrace{-(\text{RFR} + 1.0\%)}_{\text{loan}} - \underbrace{5.0\%}_{\text{swap paid}} + \underbrace{(\text{RFR} - 0.05\%)}_{\text{swap received}} = -6.05\%$$

B borrows fixed at 6.05%, against 6.4% directly, also better by 35 basis points.

The two gains sum to the 0.7% predicted, and the RFR cancels for both parties.

The arithmetic is correct. The economics is not, and it is worth seeing why, because the flaw illustrates something general about apparent riskless profits.

The two rate pairs are not comparable commitments. A five-year fixed loan is a five-year commitment by the lender. A floating-rate loan is typically rolled at intervals, and the spread over the index is reset (or the facility withdrawn) if the borrower's credit deteriorates. The floating lender therefore has an option the fixed lender does not, and the weak borrower B pays a smaller *quoted* spread in the floating market precisely because the lender retains the right to walk away.

When B enters a swap to convert its floating borrowing into synthetic fixed, it does not thereby acquire a five-year fixed rate. It acquires a five-year swap plus a rolling floating loan that may be repriced against it. The "gain" is compensation for retaining exactly the risk that made the floating quote look cheap. If B's credit deteriorates, the swap keeps paying but the loan spread widens, and the synthetic fixed rate rises.

Comparative advantage is therefore a poor explanation of why the swap market is the size it is. The better explanation is the mundane one contained in Examples 7.2 and 7.3: swaps let institutions separate the interest rate character of an exposure from the funding decision that created it, and they do so with far less balance sheet than transacting in the underlying bonds would require.

Other users

Beyond the four classical uses, four groups account for much of the market's size.

Banks managing the gap. A bank whose deposits reprice overnight while its loan book or securities portfolio is fixed is structurally exposed to rates rising, and closes that mismatch with swaps rather than by changing its lending. This is one of the largest sources of end-user demand. Silicon Valley Bank (SVB) in March 2023 is the case of a bank that left the mismatch largely open: it held long fixed-rate securities against deposits that could leave overnight, the 2022 rise in rates cut the securities' value well below cost, and the deposits left when depositors saw the loss.

Mortgage hedgers. A mortgage servicer or a holder of mortgage-backed securities loses duration when rates fall, because borrowers refinance, and gains it when rates rise. To stay hedged it must pay fixed in swaps as rates rise and receive as they fall, which is buying high and

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selling low on a large scale; in a sharp move that flow pushes swap rates further in the direction they were already going.

Pension funds and insurers matching liabilities. Long-dated liabilities discounted at market rates create enormous duration exposure. Receiving fixed in long-dated swaps lengthens asset duration toward liability duration without buying bonds, which is why, as Section 7.9 notes, this demand compresses swap spreads, and why the 2022 crisis in UK pension funds' liability-driven investment (LDI) strategies, which Chapter 8 examines, was driven by collateral calls on swap and repo positions rather than by losses on bonds held outright.

Issuers arbitraging their own funding. A company that can issue a fixed-rate bond cheaply but prefers floating exposure issues the bond and swaps it. This is what remains of the comparative advantage idea: the gain is real, but it comes from the issuer's access to a particular investor base, not from a mispricing between two markets.

What makes all of this practical for a nonfinancial company is hedge accounting. A swap is a derivative and would otherwise be marked to market through earnings each quarter while the debt it hedges sits at cost, so the hedged position would show volatility it does not have. Under IFRS 9 and its US counterpart, ASC 815 (Section 3.8), a swap documented as a hedge of specific debt has its gains and losses recognized alongside the hedged item, and reported earnings show the fixed or floating cost the company chose.

7.3 VALUING A SWAP

There are two equivalent ways to value a swap, and it is worth having both.

As an exchange of bonds. A swap is a long position in one bond and a short position in another. The fixed payer is short a bond paying coupon K and long a floating-rate note.

As a portfolio of forward contracts. Each exchange of payments is an FRA of the kind valued in Section 4.9. The swap is their sum. This is the more useful view, because it extends directly to the multi-curve case.

Take the second route, and write DF_i for the discount factor to payment date t_i and F_i for the forward rate for accrual period i. The ith exchange alone is an FRA: the fixed payer owes $L\tau_i K$ and is owed $L\tau_i R_i$, and valuing it means substituting the forward rate F_i for the unknown R_i and discounting, so it is worth

$$f_i = L\,\tau_i\,(F_i - K)\,\mathrm{DF}_i \tag{7.2}$$

which is the FRA value of Section 4.9 seen from the other side, there written for the party receiving fixed, here for the party paying it. Summing (7.2) over the payment dates gives the swap:

$$V_{\text{payer}} = L\sum_{i=1}^{n}\tau_i\,F_i\,\mathrm{DF}_i \;-\; L\,K\sum_{i=1}^{n}\tau_i\,\mathrm{DF}_i \tag{7.3}$$

The second sum is the **annuity**

$$A = \sum_{i=1}^{n}\tau_i\,\mathrm{DF}_i \tag{7.4}$$

the present value of one unit of fixed rate received on the swap's schedule; multiplied by $0.0001L$ it is the present value of one basis point of fixed rate on the notional, which is why it is also called the PV01 leg. It will be used repeatedly: it is what converts a difference between rates into an amount of money, it is the swap's sensitivity to its own fixed rate, and Chapter 29 will take it as a numeraire (the unit in which other prices are measured) in its own right. A five-year annual swap discounted at 4%, with each year fraction taken as one, has $A \approx 4.45$, so

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a one-basis-point move in the par rate changes the value of $100 million of notional by about $44,500.

Setting $V = 0$ and solving gives the **par swap rate**, the fixed rate at which a new swap has zero value:

$$S = \frac{\sum_{i=1}^{n} \tau_i \, F_i \, \mathrm{DF}_i}{\sum_{i=1}^{n} \tau_i \, \mathrm{DF}_i} \tag{7.5}$$

so the par rate is a weighted average of the forward rates, with the discount factors as weights. From (7.3) and (7.5) the value of an existing swap has the compact form

$$V_{\text{payer}} = L \, (S - K) \, A \tag{7.6}$$

which says that a swap is worth the difference between the current par rate and its own fixed rate, annuitized. Most swap risk management reduces to this expression.

A telescoping identity—and a warning

Something convenient happens if the floating index and the discount curve come from the *same* curve. In that case the forward rate satisfies $\tau_i F_i = \mathrm{DF}_{i-1}/\mathrm{DF}_i - 1$ by definition, with $\mathrm{DF}_0 = 1$ for today, so

$$\sum_{i=1}^{n} \tau_i F_i \, \mathrm{DF}_i = \sum_{i=1}^{n} (\mathrm{DF}_{i-1} - \mathrm{DF}_i) = 1 - \mathrm{DF}_n \tag{7.7}$$

The whole floating leg collapses to $L(1 - \mathrm{DF}_n)$: the forward rates cancel, and only the first and last discount factors survive. This is the classic result that a floating-rate note prices at par on its reset dates.

Remember this identity, and remember equally that it is *false in general*. It holds only when the rate being forecast is the same rate being used to discount. That was approximately true before 2008 and is not true now. Section 7.5 shows what replaces it.

Example 7.6 Valuing an existing swap

A three-year annual swap on a notional of $100 million pays fixed at 4.00% and receives the compounded RFR. Using the OIS curve bootstrapped in Chapter 4, the discount factors are 0.961538, 0.920936, and 0.881164, and the corresponding forward rates are 4.0000%, 4.4088%, and 4.5136%.

Floating leg. Discounting each projected payment,

$$100\big[0.040000 \times 0.961538 + 0.044088 \times 0.920936 + 0.045136 \times 0.881164\big] = \$11{,}883{,}596$$

The telescoping identity (7.7) gives the same answer in one step, to the rounding in the forward rates, since index and discount curve agree here:

$$100 \, (1 - 0.881164) = \$11{,}883{,}600$$

Fixed leg. The annuity is $A = 0.961538 + 0.920936 + 0.881164 = 2.763638$, so the fixed leg is worth

$$100 \times 0.04 \times 2.763638 = \$11{,}054{,}552$$

Value. To the fixed payer the swap is worth 11,883,600 − 11,054,552 = $829,048.

Checking with (7.6): the par rate is $(1 - 0.881164)/2.763638 = 4.30\%$, so the swap is worth $100 \times (0.0430 - 0.0400) \times 2.763638 = \$829{,}091$, agreeing to rounding. The payer is receiving a floating stream now worth 4.30% while paying only 4.00%.

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7.4 THE COLLATERAL AGREEMENT DETERMINES THE DISCOUNT CURVE

We now come to the question the previous section left open: discount at *what*? The answer is not a matter of preference. Consider a swap that is fully collateralized under a CSA, the standard arrangement described in Chapter 8. Each day, the party for whom the swap has negative value posts collateral equal to that value, and receives interest on the posted collateral at a rate specified in the CSA, almost always the overnight rate of the collateral currency.

Now think about what a dealer holding such a position actually owns. If the swap's value rises by $1, the dealer receives $1 of collateral, on which it owes the poster the overnight rate until the position reverses. The dealer may use that cash, so the gain on the swap is financed by a liability that accrues at the collateral rate: the position is, in cash flow terms, self-financing at the overnight rate. Replicating it therefore costs the overnight rate, and by the arbitrage arguments of Chapter 5, the correct discount rate for a fully collateralized derivative is **the rate paid on its collateral**.

This has a consequence that surprises people the first time they meet it: the same contract, with the same cash flows, has a different value under different CSAs. A swap collateralized in dollars earning SOFR and an otherwise identical swap collateralized in euros earning €STR are not worth the same. The CSA is part of the instrument.

A cleared swap, such as the one in Table 7.1, works the same way with the clearing house's rulebook in place of a CSA: variation margin is paid in cash and earns price alignment interest at the overnight rate, so cleared dollar swaps are discounted on the SOFR curve. Bilateral CSAs change more slowly, and some older dollar CSAs still pay the effective federal funds rate; trades under them are discounted on a federal funds curve, a live case of the same contract carrying two values.

Case Study 7.1 The day discounting changed

For years after the crisis, cleared swap markets discounted at the effective federal funds rate in dollars and the euro overnight index average (EONIA) in euros, because those were the rates paid on cleared collateral. When the market moved to SOFR and €STR, the collateral rate changed—and therefore so did the value of every cleared swap.

The clearing houses did not simply re-mark and move on. LCH (originally the London Clearing House) and CME executed the dollar switch over a weekend in October 2020, changing the discounting rate on their entire cleared portfolios at once. Because the change transferred value between members (some portfolios gained, others lost), each house paid cash compensation to make members whole, and additionally distributed basis swaps to offset the change in risk profile that the discounting switch produced. The euro switch, in July 2020, needed only the cash: EONIA had been redefined in 2019 as €STR plus a fixed 8.5 basis points, so the spread between the old and new discount rates was known for every future date and left no basis risk to hand back.

Two lessons follow. First, the choice of discount curve is not an accounting convention; it is worth enough money that changing it required a coordinated, compensated, market-wide operation. Second, discounting risk is a real risk: a portfolio can be flat to rates and still gain or lose from a change in the rate its collateral earns.

Uncollateralized trades. If a swap is not collateralized, as many corporate trades are not, the replication argument above does not apply, because the dealer must fund the position from its own resources. The discount rate then involves the dealer's funding cost, which introduces the funding valuation adjustment of Chapter 9. It also means the trade's value depends on *who is*

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holding it, which is uncomfortable but true.

7.5 WHY ONE CURVE IS NOT ENOUGH

We now have two distinct roles for a curve, and no reason for one curve to fill both.

- The **discount curve** answers: what is a cash flow at time t worth today? Section 7.4 settled this: it is determined by the collateral rate in the CSA, and is built from OIS.
- A **forecast curve** answers: how large will the floating payment be? This is determined by whatever index the contract references, which may be the same overnight rate, or Term SOFR, or a foreign currency's RFR, or a legacy fallback rate.

Figure 7.4 sets out the division of labor. When the index and the collateral rate coincide, the two curves coincide and everything collapses to the single-curve formulas, including the telescoping identity (7.7). When they do not, the curves must be built and used separately, and Example 7.7 measures the error of letting one curve do both jobs.

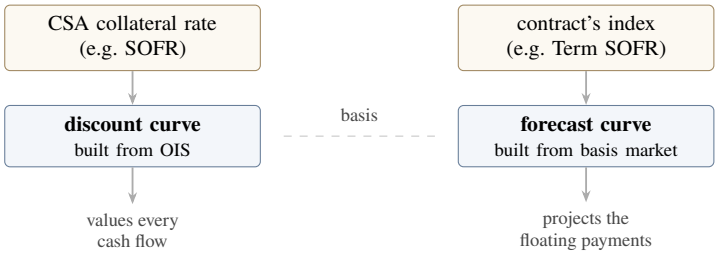


Figure 7.4: The two roles a curve plays. Before 2008 the same curve filled both.

The basis is not noise

The gap between the two curves is the **curve basis**, and it is a traded quantity with its own market. Historically the largest example was the spread between LIBOR and OIS, which measured the credit and liquidity premium in interbank lending. Before 2007 it sat at a small number of basis points and was widely ignored; in late 2008 it exceeded 350 basis points. A dealer valuing a swap book on the assumption that the two were the same would have been wrong by an amount that dwarfed every other modeling choice in the portfolio.

That particular basis has gone with LIBOR. Others remain and are structural rather than crisis-driven:

- **Term versus compounded.** Term SOFR is a forward-looking rate derived from futures; compounded SOFR is backward-looking. They are not equal, and the difference is traded, though narrowly: under the ARRC's scope-of-use rules, Term SOFR derivatives are confined to end users hedging loans and other cash products that reference it, and the dealers who face them lay the exposure off in Term SOFR against SOFR basis swaps.
- **Tenor basis.** Where a currency retains term indices of different lengths, a swap referencing the shorter index against one referencing the longer trades at a spread. The euro is the main case: three-month against six-month EURIBOR is quoted as a basis swap (the "3s6s" basis), and euro swap systems carry a separate forecast curve for each tenor. The economics is the same as the old LIBOR tenor basis: rolling a short index carries different risk than fixing a long one.
- **Cross-currency basis.** The subject of Section 7.8.

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Example 7.7 Where the single-curve identity breaks

Take the swap of Example 7.6, but suppose the floating leg references an index whose forward rates sit 10 basis points above the OIS forwards (a term rate, say). The forecast forwards become 4.1000%, 4.5088%, and 4.6136%, while the discount factors, being set by the CSA, are unchanged.

The floating leg is now

$$100\big[0.041000 \times 0.961538 + 0.045088 \times 0.920936 + 0.046136 \times 0.881164\big] = \$12{,}159{,}960$$

against \$11,883,596 from the same calculation in Example 7.6. The difference of \$276,364 is exactly the basis times the annuity:

$$100 \times 0.0010 \times 2.763638 = \$276{,}364$$

The telescoping identity (7.7) would have given \$11,883,600 and been wrong by more than a quarter of a million dollars—because the rate being forecast is no longer the rate being used to discount. The par swap rate rises from 4.30% to 4.40%: the 10 basis point basis passes through one-for-one into the par rate.

7.6 BUILDING THE CURVES

Curve construction in a multi-curve world proceeds in a definite order, because the pieces depend on one another.

Step 1: the discount curve. Bootstrap from OIS quotes, exactly as in Section 4.8. These instruments have a single index, and that index is the collateral rate, so the construction is self-contained.

Step 2: the forecast curves. For each index the book references, build a curve that reprices the instruments referencing that index, while discounting them on the curve from Step 1. This is the step that makes the procedure genuinely multi-curve: the forecast curve is defined by what it must reprice, given a discounting choice already fixed.

Step 3: check consistency. Basis swaps quoted between two indices must be repriced by the pair of forecast curves. In practice this is not a sequential bootstrap but a simultaneous calibration: the curves are solved jointly so that every quoted instrument (OIS, index swaps, basis swaps, futures with their convexity adjustments from Chapter 30) is repriced at once. Example 7.8 carries the three steps through by hand.

Example 7.8 Two curves from two sets of quotes

Payments are annual and $\tau_i = 1$ throughout, so the construction can be followed by hand.

Step 1: the discount curve. OIS quotes are 4.00%, 4.20%, and 4.30% for one, two, and three years. An OIS forecasts the same rate it discounts at, so its floating leg telescopes and the par condition is simply $S \sum_i \mathrm{DF}_i = 1 - \mathrm{DF}_n$. Solving in order,

$$\mathrm{DF}_1 = \frac{1}{1.04} = 0.961538, \qquad \mathrm{DF}_2 = \frac{1 - 0.042 \times 0.961538}{1.042} = 0.920936$$

$$\mathrm{DF}_3 = \frac{1 - 0.043 \times (0.961538 + 0.920936)}{1.043} = 0.881164$$

which are the discount factors of Example 7.6. The forward rates this curve implies are 4.0000%, 4.4088%, and 4.5136%.

Step 2: the forecast curve. The book also holds swaps referencing a twelve-month term index, quoted at 4.10% for one year and 4.32% for two. These are not OIS: the index is not the collateral rate, so the floating leg does not telescope and the forwards have to be solved for, while discounting on the Step 1 curve, which is already fixed. The one-year swap gives $F_1 = 4.1000\%$ directly. The

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two-year swap then gives F_2 from

$$0.0410 \times 0.961538 + F_2 \times 0.920936 = 0.0432 \times 1.882475$$

where $1.882475 = \mathrm{DF}_1 + \mathrm{DF}_2$ is the two-year annuity factor, so $F_2 = 4.5497\%$. The term index is forecast 10.00 basis points above the overnight rate in the first year and 14.09 in the second. The basis is not flat, and no single curve can carry both sets of forwards.

Step 3: the consistency check. A two-year basis swap pays the term index and receives the overnight rate plus a spread b. Setting its value to zero,

$$b = \frac{(0.0410 - 0.0400) \times 0.961538 + (0.045497 - 0.044088) \times 0.920936}{1.882475}$$

which comes to 0.001200, or 12.00 basis points. That is what the market quotes, so the pair of curves is consistent.

Notice what that check did *not* establish. With one schedule and one discount curve, b is forced to the difference between the two par rates, $4.32\% - 4.20\% = 12$ basis points, and the basis swap has told the calibration nothing the outright quotes had not. Basis swaps are needed where that equality breaks (legs on different payment frequencies, or maturities and indices at which no outright swap trades), which is the usual reason a curve is pinned by basis quotes at the long end and by outright quotes at the short end.

Two practical warnings carry over from Chapter 4 and become sharper here. Interpolation choices matter more, because errors in one curve propagate into every other through the basis instruments. And the resulting object is not unique: two desks calibrating to identical quotes with different interpolation and different instrument selection will produce curves that agree at the quoted points and disagree between them. That disagreement is a genuine source of valuation difference, not a rounding artifact.

7.7 BASIS SWAPS

A **basis swap** exchanges two floating legs. Neither leg is fixed; the contract trades the spread between two indices, quoted as a number of basis points added to one leg to make the swap worth zero at inception.

Basis swaps serve two purposes. Commercially, they let an institution move exposure from one index to another without taking a directional rate position: a bank whose assets reference one index and whose liabilities reference another closes the mismatch with a basis swap, not with an ordinary swap. A euro borrower whose loan resets on three-month EURIBOR while its existing hedge pays six-month uses one to align the two, and a dealer that has written Term SOFR swaps to loan borrowers lays off the term-against-overnight exposure the same way. Technically, they pin down the forecast curves of Steps 2 and 3 above wherever no outright swap on an index trades, and there the relationship between the curves would otherwise be unobservable. Within a single currency the euro 3s6s basis is now the main example; Section 33.9 returns to basis swaps as a product and to what their spreads price.

The spread is not arbitrary. It reflects the genuine economic difference between the two indices: differences in credit content, in the term over which the rate is fixed, and in the supply and demand for exposure to each. When the two indices are close substitutes the spread is small and stable; when they differ in credit character it widens under stress, exactly when it matters.

7.8 CROSS-CURRENCY SWAPS AND THE CROSS-CURRENCY BASIS

A **cross-currency swap** exchanges payments in two different currencies. Unlike a single-currency swap, the notional principals *are* exchanged, at the start and again at maturity, because

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the two legs are denominated in different money and there is nothing to net. Figure 7.5 shows the structure.



Figure 7.5: A EUR/USD cross-currency basis swap, from the perspective of a party raising dollars against euros. Notional is exchanged at both ends.

The economic content is a collateralized loan of one currency against another. A European bank needing dollars can borrow euros at home and swap them for dollars for five years, returning the dollars and reclaiming the euros at maturity. It is, in substance, secured dollar borrowing.

Three kinds of user account for most of the market. Banks outside the United States use it, as here, to fund dollar lending and dollar securities. Companies and supranational issuers borrow in whichever market is cheapest for them, often by issuing dollar or euro bonds, and swap the proceeds and the coupons back into the currency their revenues are in. And investors holding foreign bonds, such as a Japanese insurer holding US corporate debt, swap the foreign coupons and principal into their own currency so that the bond's return is not swamped by the exchange rate. The cash flows are easiest to see in the fixed-for-fixed version, where neither leg floats, as in Example 7.9.

Example 7.9 Cash flows on a fixed-for-fixed currency swap

A European company raises dollars for five years. At a spot rate of 1.10 dollars per euro it pays €100 million and receives $110 million at the start, pays 4.2% on the dollars and receives 2.5% on the euros annually, and reverses the principal exchange at maturity. Amounts are in millions.

Year	USD cash flow	EUR cash flow
0	+110.00	−100.00
1	−4.62	+2.50
2	−4.62	+2.50
3	−4.62	+2.50
4	−4.62	+2.50
5	−114.62	+102.50

The interest amounts are $110 \times 4.2\% = 4.62$ and $100 \times 2.5\% = 2.50$; year 5 adds the principal back. Compare this with Example 7.1. Three differences stand out.

The principal is exchanged, twice. In a single-currency swap the two notionals are identical and cancel, so exchanging them would be pointless. Here they are in different money and cannot be netted, so they must actually change hands—and because they do, the contract carries exchange rate risk that an interest rate swap does not.

Nothing nets. Both legs are paid gross, in their own currency, on every date.

It is a bond issued and a bond bought. Read the dollar column alone and it is exactly the cash flow of issuing a $110 million 4.2% bond; the euro column is exactly that of buying a €100 million 2.5% bond. That is the whole instrument: borrow in one currency, lend in the other.

The exchange rate risk sits overwhelmingly in the final row. The annual interest amounts are small and partly offsetting, but the $114.62 million repayment against €102.50 million received is where a move in spot is felt, which is what Example 7.11 quantifies.

What actually trades in volume is the floating-for-floating version of Figure 7.5, where each

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leg pays its own currency's overnight rate and the whole price of the transaction is compressed into a single spread: the basis. Two distinctions matter in practice. Between dealers the standard contract is the **mark-to-market** (resettable) cross-currency swap, in which the dollar notional is reset to the current spot equivalent at each payment date and the difference is paid in cash, so the exchange rate exposure that Example 7.11 measures is settled period by period instead of accumulating to maturity; Section 33.9 prices the difference. The constant-notional swap of the examples here is what a company hedging a fixed-size bond issue or foreign investment uses. And a cross-currency swap is not an **FX swap**, which is a spot exchange of currencies reversed by a forward exchange, with no interest payments in between: the interest differential is built into the forward points of Section 5.7. FX swaps are by far the larger market, concentrated at maturities under a year, and are how banks and investors fund and hedge foreign-currency positions from day to day; cross-currency swaps carry the same funding trade out to many years.

The cross-currency basis

Textbook theory says the two floating legs should exchange flat: covered interest parity holds, and any spread would be arbitrageable by borrowing in one currency, swapping, and lending in the other. Since 2008 this has been persistently false. A spread, the **cross-currency basis**, is quoted on one leg, and for the euro, yen, Swiss franc and sterling against the dollar it has been negative for most of the period since, meaning the party *providing dollars* is paid a premium. (The Australian and New Zealand dollars, whose banks and companies borrow abroad and swap the proceeds home, have often run a positive basis: there the scarce side is the local currency.) The basis widened dramatically during the 2008 crisis and again during the 2011–12 European sovereign crisis, and although it narrowed afterward it did not return to zero. It also spikes predictably at quarter-ends and year-ends.

Case Study 7.2 The end of covered interest parity

Covered interest parity is not a theory about expectations or risk preferences. It is an arbitrage relationship: borrow dollars, convert to euros at spot, invest in euros, and sell the euro proceeds forward for dollars. If the resulting dollar return differs from the dollar interest rate, someone has been handed a riskless profit.

For decades the relationship held to within a basis point or two. Since 2008 it has failed continuously—not by a basis point but by tens of basis points, for years at a time, in the most liquid currency pairs in the world.

Traders did not stop noticing. The explanation has two phases. In 2008 and again in 2011–12 the basis measured stress: banks outside the United States needed dollars, their credit was in doubt, and dollar lenders would supply them only at a premium. Dollar swap lines between the Federal Reserve and other central banks, opened in 2008 and widened again in March 2020, are what capped those spikes. The persistence afterward has a different cause: the arbitrage requires balance sheet. Post-crisis regulation (the leverage ratio of Section 1.12 in particular, phased in from 2015) made it expensive for banks to expand their balance sheets to capture small spreads. Against that limited supply sits steady demand: Japanese and European banks, insurers and pension funds holding dollar assets fund or hedge them through FX and cross-currency swaps, and they pay the premium. The trade is still there; no one wants to hold it. This is also why the basis widens at reporting dates, when balance sheet is scarcest. The episode is an instance of the limits of arbitrage set out in Section 5.15.

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Example 7.10 The cost of a negative basis

A European company swaps €100 million into dollars for five years at a cross-currency basis of −20 basis points. It has lent euros and borrowed dollars, so it pays SOFR flat on the dollar leg and *receives* €STR less 20 basis points on the euro leg. Receiving 20 basis points less than the euro overnight rate is precisely how it pays for the dollars.

With euro discount factors of 0.975610, 0.951814, 0.928599, 0.905951, and 0.883854, the euro annuity is

$$A_{\text{EUR}} = 4.645828$$

The present value of the 20 basis point spread is

$$100{,}000{,}000 \times 0.0020 \times 4.645828 = €929{,}166$$

So the negative basis is worth just under a million euros on this trade, paid by the party that wanted dollars, earned by the party that supplied them. It is precisely the quantity that a single-curve valuation would set to zero.

Valuation

A cross-currency basis swap is most easily valued as a pair of floating-rate notes: long a note in the currency lent, short a note in the currency borrowed, with the two converted at the current spot rate. At a reset date each note prices at par *provided* its spread equals the current market basis; when the market basis has moved, the off-market spread is worth its difference times the annuity, exactly as in Section 7.3.

Example 7.11 Valuing a cross-currency swap after the market moves

The company of Example 7.10 entered its five-year swap three years ago: it lent €100 million and borrowed $110 million at a spot rate of 1.10 dollars per euro, receiving €STR less 20 basis points. Two years remain. The euro discount factors for those two years are 0.975610 and 0.951814, so the remaining annuity is $A_{\text{EUR}} = 1.927424$.

Case 1: only the exchange rate has moved. Spot is now 1.05, and the market basis is still −20 basis points. Because the contract's spread equals the market spread, the euro note prices at par and so does the dollar note:

$$V = €100{,}000{,}000 - \frac{\$110{,}000{,}000}{1.05} = 100{,}000{,}000 - 104{,}761{,}905 = -€4{,}761{,}905$$

The company borrowed dollars and the dollar has strengthened, so the liability has grown in euro terms.

Case 2: the basis has also widened. Suppose in addition the market basis has moved to −45 basis points. The contract pays €STR less 20, where the market now requires €STR less 45, so the euro note pays 25 basis points *more* than the market demands and is worth more than par. The company receives that leg, so the move is in its favor:

$$100{,}000{,}000\,(1 + 0.0025 \times 1.927424) = €100{,}481{,}856$$

and the swap is worth

$$100{,}481{,}856 - 104{,}761{,}905 = -€4{,}280{,}049$$

The widening basis has offset €481,856 of the exchange rate loss, because the company locked in dollar funding when dollars were cheaper than they are now.

Two lessons follow. The exchange rate move is an order of magnitude larger than the basis move (€4.76 million against €0.48 million), which is why cross-currency swap books are managed first as FX exposures and only then as basis exposures. And the basis is a genuine mark-to-market risk in its own right: a position that is fully hedged in both currencies' rates and in spot FX still moves when the basis does.

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Valuation also requires a decision about the collateral currency, because Section 7.4 applies here too. A trade collateralized in dollars and a trade collateralized in euros have different values even with identical cash flows. Many CSAs permit the poster to choose among several eligible currencies, which makes the collateral a rolling option to deliver whichever currency is cheapest, an option that must itself be valued and that Section 8.5 takes up alongside the rest of collateral practice.

7.9 SWAP SPREADS AND THE PRICE OF BALANCE SHEET

The **swap spread** is the difference between the swap rate and the yield on a government bond of the same maturity. For most of the swap market's history it was positive, and the explanation was obvious: the swap curve carried bank credit risk through its LIBOR index, while the government curve did not, so the swap rate stood above the government yield by roughly the market's assessment of that credit difference.

Dollar swap spreads have now been negative for years: at thirty years since late 2008, at ten years since the middle of the 2010s, and, since the switch to SOFR, across most of the curve. The switch itself lowered the series, because a SOFR swap carries no bank credit at all, so spreads measured before and after 2021 are not directly comparable. Read literally, this says the market accepts a lower fixed rate from a bank than from the sovereign (that banks are better credits than the government), which would be an extraordinary claim, and is not what it means.

The resolution is the argument of the previous section, transplanted. Example 7.12 sets out the trade a negative swap spread appears to offer.

Example 7.12 The arbitrage no one wants

The ten-year swap rate is 4.00% and the ten-year government bond yields 4.10%, so the swap spread is -10 basis points. The apparent trade is:

1. Buy \$100 million of the ten-year bond, yielding 4.10%, funding it in the repo market at approximately the overnight rate.
2. Pay fixed at 4.00% in a ten-year swap, receiving the compounded overnight rate.

The floating leg received on the swap approximately offsets the repo cost of funding the bond. What remains is a fixed 4.10% received against a fixed 4.00% paid: 10 basis points, apparently for no market risk at all. With a ten-year annuity of 8.111 at 4%, the present value is

$$100{,}000{,}000 \times 0.0010 \times 8.111 = \$811{,}100$$

The trade is real, and it sat available for years. What it requires is a balance sheet: the bond must be held and financed. Under the leverage ratio (Section 1.12), holding \$100 million of government bonds consumes capital whether the position is hedged or not. A dealer weighing \$811,100 against the capital that position ties up for a decade may reasonably decline.

So a negative swap spread is not a statement about relative creditworthiness. It is the price of the balance sheet needed to close it—the same phenomenon, measured a different way, as the cross-currency basis of Section 7.8. Heavy government bond supply pushes yields up; regulation makes the offsetting arbitrage expensive; and demand from pension funds and insurers to receive fixed in swaps, which requires almost no balance sheet, pushes swap rates down. The spread settles wherever those forces balance, and there is no reason that point should be positive.

Two recent regulatory changes bear directly on it. In November 2025 the US banking agencies recalibrated the enhanced supplementary leverage ratio for the largest banks, in force since April 1, 2026, with the stated aim of making it a backstop rather than a binding constraint; the more balance sheet that releases for holding Treasuries, the less negative swap spreads

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need be. And the SEC's Treasury clearing mandate of Section 2.11, which applies to cash trades from December 31, 2026 and to repo from June 30, 2027, lets dealers net cleared Treasury and repo positions against one another, which also reduces the balance sheet the trade consumes. Relative-value funds that do put the trade on, financed in repo, carry the two risks a dealer avoids: that the spread moves further against them before it converges, and that their repo funding is withdrawn. In March 2020 both happened at once. Swap spreads, like the cross-currency basis, are the limits of arbitrage of Section 5.15 at work, and they can stay dislocated for years without anyone behaving irrationally.

7.10 CREDIT RISK, NETTING, AND CLEARING

A swap creates credit exposure, but of an unusual shape. Unlike a loan, where one party is always the creditor, a swap can have positive value to either side depending on where rates go. Exposure is therefore *two-sided* and changes sign over the contract's life.

It is also far smaller than the notional suggests. The notional is never exchanged; what is at risk is the swap's replacement cost (what it would cost to replace the contract on current terms if the counterparty defaulted), which by (7.6) is $L(S-K)A$. A \$100 million ten-year swap that has moved 50 basis points in your favor exposes you to roughly \$4 million, not \$100 million. This is why aggregate notional statistics (hundreds of trillions) are a poor guide to systemic exposure. Three mechanisms reduce what remains.

Netting. The ISDA Master Agreement provides for close-out netting, which Section 1.5 and Example 1.1 set out, cherry-picking included. Netting enforceability is a legal question decided jurisdiction by jurisdiction, which is why China's Futures and Derivatives Law, in force since August 2022, mattered so much to the dealers trading with Chinese counterparties: it was the first Chinese statute to recognize close-out netting. Until a jurisdiction settles the question, the netted figure a dealer reports is not necessarily the figure that would survive its counterparty's bankruptcy court.

Collateral. A credit support annex requires the out-of-the-money party to post collateral against current exposure, reducing it toward zero between margin calls. Section 7.4 showed the valuation consequence; Chapter 8 examines what posting collateral does to liquidity, which is a different and sometimes larger problem.

Central clearing. Post-crisis reforms require standardized swaps to be cleared, which novates each one to a central counterparty by the mechanism of Section 2.4. Bilateral credit risk is replaced by exposure to the CCP, and Chapter 2 examines what that concentration achieves and what it costs.

The same reforms changed how swaps are traded and recorded. In the United States, standard swaps that the regulator has declared "made available to trade" must be executed on a swap execution facility (SEF), an electronic venue where requests for quotes go to several dealers at once, and the European Union imposes a matching trading obligation. Every swap, cleared or not, is reported to a trade repository, which is what makes the market's size measurable at all. Dealers periodically take part in portfolio compression, tearing up offsetting trades among many participants to cut gross notional without changing anyone's net risk (Section 33.12). And swaps that are not cleared carry mandatory initial margin under the uncleared margin rules of Section 8.4, which is deliberately more expensive than clearing and is one of the main reasons standard trades clear.

What survives all three is the residual counterparty risk priced as a credit valuation adjustment (CVA) in Chapter 9 and capitalized under the framework of Chapter 27. It is smaller than it once was, but it is not zero, and the Archegos episode of 2021, discussed in Chapter 27, shows how quickly it returns when a portfolio is concentrated and margining is lax.

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7.11 WHAT THE TRANSITION DID TO THE SWAP BOOK

Chapter 4 described the ISDA fallback mechanism in the abstract. It is worth seeing what it meant for an actual portfolio, because the swap book was where the transition would succeed or fail.

A dealer entering 2021 held millions of swaps referencing LIBOR, with maturities running decades out. Three things had to happen. New trades had to move to RFR references, which market convention accomplished by coordinated "big bang" dates on which interdealer trading switched. Cleared legacy trades were converted by the clearing houses before each LIBOR setting ended (sterling, yen and Swiss franc in December 2021, dollars in April and May 2023): each LIBOR swap was replaced by a standard OIS, with the fallback spread adjustment folded into the fixed rate and cash paid for the remaining differences in convention. Bilateral legacy trades fell back under the protocol.

The two routes left different things behind. A converted cleared swap is an ordinary OIS, indistinguishable from one traded new. A bilateral LIBOR swap that fell back references compounded SOFR plus the fixed spread of Table 4.2 for the rest of its life, under the protocol's own conventions (Example 4.1), so those spreads remain a structural feature of the legacy book. Because the spread is a constant over compounded SOFR, such a trade needs no curve of its own; what it needs is a booking and confirmation system that records it correctly, and the stock of these trades shrinks only as they mature or are restructured.

The broader observation is the one from Chapter 4, now visible at portfolio scale: the transition did not eliminate the multi-curve structure. It changed which curves the structure contains.

7.12 OTHER SWAPS

The vanilla structure generalizes in many directions, of which a few deserve naming here; Chapter 33 treats them in full.

Amortizing and accreting swaps vary the notional over time, typically to match an underlying loan (Section 33.1). **Forward-starting swaps** begin at a future date (Section 33.2); a company planning to issue a bond in six months pays fixed on one today to lock in the rate, and they are the underlying for the swaptions of Chapter 29. **Constant-maturity swaps** (CMS) reference a long-tenor swap rate on each reset; Section 33.5 describes them and Section 30.5 derives the convexity adjustment they need. **Inflation swaps** exchange a fixed rate for the realized change in a consumer price index and are how pension funds hedge inflation-linked liabilities (Section 33.10).

Equity swaps exchange the total return on an equity or index for a floating rate. **Total return swaps** extend the same exchange to any reference asset (a bond, a loan portfolio, a single stock), and they transfer the full economics of an asset without transferring the asset. Hedge funds use them to hold exposure they would otherwise have to finance and disclose, and banks provide them as a form of secured lending, which is why they are the instrument through which leverage accumulates quietly. Section 33.11 treats both, and Chapter 27 examines what total return swaps made possible for Archegos in 2021.

Commodity swaps exchange a fixed price for a floating commodity index (Section 33.8), and are how airlines fix their fuel cost and producers their selling price for years ahead; **variance swaps** exchange a fixed variance for realized variance and are treated in Chapter 22. **Credit default swaps** (CDS), despite the name, are structurally quite different (one leg is a periodic premium, the other pays only if a specified issuer defaults) and belong to Chapter 25.

What unites them is the mechanism of this chapter: identify the cash flows, project the floating ones on the appropriate forecast curve, and discount all of them on the curve the collateral agreement dictates.

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SUMMARY

An interest rate swap exchanges fixed for floating payments on a notional that is never exchanged; in most major currencies the standard contract is now an overnight indexed swap referencing a compounded risk-free rate in arrears, while euro swaps still reference EURIBOR. Valued as a portfolio of forward rate agreements, it is worth $L(S - K)A$, the gap between the current par rate and its own fixed rate, annuitized. The comparative advantage argument traditionally used to motivate swaps does not survive inspection, because the fixed and floating borrowing rates it compares are not comparable commitments: the floating lender retains the right to reprice or withdraw, and the apparent gain is compensation for that. Swaps are better explained by their ability to separate the interest rate character of an exposure from the funding decision that created it.

The discount curve is not a modeling choice. For a collateralized derivative the correct discount rate is the rate paid on its collateral, because that is what replicating the position costs; the same cash flows are therefore worth different amounts under different credit support annexes, which is why the move to SOFR discounting in October 2020 required coordinated compensation across cleared markets. Valuation needs at least two curves: a discount curve determined by the CSA and one or more forecast curves determined by the indices the contracts reference. Where the two coincide the floating leg telescopes to $L(1 - \mathrm{DF}_n)$ and the familiar single-curve results follow; where they do not, that identity fails and the error equals the basis times the annuity. Curves are built discount first, and in practice by simultaneous calibration.

Basis swaps exchange two floating legs, hedge the mismatch between indices, and are the instruments that pin down the forecast curves. Cross-currency swaps exchange payments and notionals in two currencies and are economically secured borrowing across currencies; their basis has been persistently nonzero since 2008, first as a measure of dollar funding stress and then because the arbitrage that would close it consumes balance sheet that post-crisis regulation made expensive, which is also why it widens predictably at reporting dates. Negative long-dated dollar swap spreads are the same phenomenon measured a second way: closing that gap means financing a bond, and the leverage ratio charges capital for it regardless of the position's risk.

Swaps create two-sided credit exposure far smaller than the notional, being the replacement cost $L(S - K)A$ rather than the principal. Close-out netting under the ISDA Master Agreement collapses a defaulted counterparty's trades to a single net amount and prevents cherry-picking; collateral under a CSA reduces the residual toward zero between margin calls; and central clearing replaces bilateral exposure with exposure to a CCP that nets multilaterally. What remains is priced as CVA (Chapter 9) and capitalized under the framework of Chapter 27.

FURTHER READING

Bianchetti's "Two Curves, One Price" (*Risk*, August 2010) is an early statement of the multi-curve framework, and Piterbarg's "Funding Beyond Discounting" (*Risk*, February 2010) derives the collateral-rate discounting result that the framework rests on. Henrard's *Interest Rate Modelling in the Multi-Curve Framework* is the book-length treatment. Du, Tepper, and Verdelhan (2018) is the standard reference on the cross-currency basis, and it is the empirical counterpart to this chapter's claim that the basis is a price rather than an error.

PRACTICE QUESTIONS

7.1 State the comparative advantage argument for swaps, explain the flaw in it, and give the better explanation for the size of the swap market.

7.2 A two-year annual swap on a notional of \$50 million pays fixed at 3.50% and receives floating. Discount factors are 0.9709 and 0.9426; forward rates are 3.00% and 3.30%.

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Value the swap to the fixed payer and find the par swap rate.

7.3 Show that when the floating index and the discount curve are the same, the floating leg is worth $L(1 - \mathrm{DF}_n)$, and state precisely the condition under which this fails.

7.4 Explain why the correct discount rate for a fully collateralized derivative is the rate paid on its collateral, and state what follows about two otherwise identical swaps collateralized in different currencies.

7.5 Repeat Question 7.2 assuming the floating index has forward rates 15 basis points above those given, with discount factors unchanged, and verify that the change in value equals the basis times the annuity.

7.6 Describe the order in which curves must be built in a multi-curve framework, explain why the discount curve must come first, and explain why a single currency needs more than one forecast curve.

7.7 Explain what a basis swap is, give two distinct reasons the market needs them, and explain why notional principals are exchanged in a cross-currency swap but not in a single-currency one.

7.8 The eight-year swap rate is 3.60% and the eight-year government bond yields 3.72%. Compute the swap spread, describe the trade it appears to offer, calculate its present value on a \$250 million position using an eight-year annuity of 6.85, and explain both what prevents the trade from being done and why a negative spread does *not* imply that banks are better credits than the government.

7.9 Explain why a swap creates two-sided credit exposure and why the exposure is far smaller than the notional. For the swap of Question 7.2, state the exposure of each party.

7.10 Explain what close-out netting achieves, what would happen without it, and why its legal enforceability is a jurisdiction-by-jurisdiction question.

FURTHER QUESTIONS

7.11 A five-year cross-currency swap on a notional of €200 million carries a basis of −35 basis points. Using the euro discount factors of Example 7.10, find the present value of the basis. Who pays it, and what are they paying for?

7.12 Covered interest parity is an arbitrage relationship, not a behavioral theory. Explain carefully why it has failed persistently since 2008, and why the failure is largest at quarter-ends and year-ends.

7.13 When clearing houses switched dollar discounting from the federal funds rate to SOFR in October 2020, they paid cash compensation to members and also distributed basis swaps. Explain what each of the two was compensating for, and why cash alone would not have been sufficient.

7.14 A dealer's swap book is flat to a parallel shift in rates but holds large positions in different tenor buckets and in two currencies. Identify at least three distinct risks the book still carries that Chapter 4's single-curve framework could not have represented.

7.15 A CSA permits the collateral poster to deliver any of three currencies. Explain why this makes the collateral an option, why that option has value, and how you would begin to quantify it. Which chapter's machinery would you need?

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Chapter 8
Collateral, Margin, and Funding

Several chapters have now deferred a question to this one. Chapter 2 showed margin calls transmitting stress across a market; Chapter 6 showed them threatening a hedged basis position; Chapter 7 showed the collateral agreement determining a derivative's discount curve. Each treated collateral as a means to some other end. This chapter treats it as the subject. The reason is a proposition that would have seemed eccentric twenty years ago and is now the organizing fact of derivatives risk management:

> A position can be perfectly hedged against price risk and still destroy the firm holding it, because the hedge demands cash on a schedule the hedged item does not supply.

This is not a hypothetical. In 2022 it happened three times in one year (in European energy, in LME nickel, and in UK pensions), to firms in unrelated businesses, none of which had misjudged the direction of any market. Sections 40.7, 40.8, and 40.12 tell all three episodes. The mechanism is always the same, and it lives entirely in the collateral arrangements that earlier chapters passed over.

Sections 8.1 to 8.3 cover the collateral agreement itself (the CSA, variation and initial margin, haircuts), Sections 8.4 and 8.5 the rules for uncleared trades and the optionality a CSA can carry, and Sections 8.6 to 8.8 what happens to collateral once posted (transformation, rehypothecation, segregation, disputes). Sections 8.9 and 8.10 work the proposition above through the 2022 energy and pension episodes and show how the resulting liquidity risk is measured and managed. Section 8.11 asks whether there is enough collateral in the system, and Section 8.12 says where all this leaves valuation, the starting point of Chapter 9.

8.1 THE CREDIT SUPPORT ANNEX

Derivatives between two parties are documented under an ISDA Master Agreement, which Section 1.5 described as providing close-out netting. Collateral is governed by the CSA, a separate document attached to it.

The CSA is where the economics of a trade are quietly decided. Its principal terms are:

- **Eligible collateral.** Which assets may be posted (cash in which currencies, which government bonds, occasionally corporate bonds or equities), and the haircut applied to each, the percentage by which its market value is marked down when it is counted (Section 8.3).
- *The interest rate paid on collateral.* Usually the overnight rate of the currency posted. Section 7.4 showed that this single line determines the discount curve for the whole trade.
- **Threshold.** An amount of exposure below which no collateral is required, in effect an unsecured credit line. The uncleared margin rules of Section 8.4 require a zero threshold on variation margin between the firms they cover, so a threshold now survives only on initial margin (the allowance of Section 8.4) and on legacy or uncovered CSAs.

- **Independent amount.** Collateral one party posts on top of whatever the mark-to-market requires, fixed in advance or set by formula. Before the uncleared margin rules it was how dealers took initial margin from hedge funds.
- **Rating triggers.** Credit-dependent terms: a threshold that falls, or an independent amount that rises, when a party is downgraded, so that a downgrade becomes a collateral call. Such clauses turned the 2008 downgrade of American International Group (AIG) into calls it could not meet (Section 40.9).
- **Minimum transfer amount.** A floor on the size of any single call, to avoid transferring trivial sums daily.
- *Frequency and timing.* How often the portfolio is valued and calls are made, how quickly a call must be met, and how disputes are handled.
- *Direction.* Whether both parties post (a two-way CSA) or only one (a one-way CSA).

These terms combine into a single number each day. The **credit support amount** is the collateral the receiver is entitled to hold:

$$\text{credit support amount} = \text{exposure} + \text{independent amount} - \text{threshold},$$

floored at zero, and the call is the difference between it and the collateral already held, valued after haircuts. Example 8.1 works one through.

Example 8.1 Computing a margin call

A dealer faces a corporate client under a CSA outside the uncleared margin rules. The client posts an independent amount of \$5 million and has a threshold of \$10 million; the minimum transfer amount is \$500,000, and calls are rounded up to the next \$100,000. The dealer holds collateral worth \$12.0 million after haircuts.

At today's valuation the portfolio is worth \$23.47 million to the dealer. The credit support amount is

$$23.47 + 5 - 10 = \$18.47\text{m},$$

so the shortfall is $18.47 - 12.0 = \$6.47$ million. That exceeds the minimum transfer amount, and rounding makes the call \$6.5 million. The dealer issues it on the valuation date and the client delivers within a business day or two, the settlement period the CSA sets. The dealer then holds \$18.5 million.

The next day the portfolio is worth \$23.2 million. The credit support amount falls to \$18.2 million and the dealer holds \$0.3 million more than it is entitled to, but that is below the minimum transfer amount, so nothing moves.

Even when every call is met, the dealer is collateralized only to within the threshold net of the independent amount: on the first day it held \$18.5 million against \$23.47 million of exposure. That gap of nearly \$5 million is the unsecured credit line Chapter 9 prices.

Two-way and one-way. In a one-way CSA the party that posts is traditionally the dealer, when the client is a sovereign, supranational, or large corporate with the bargaining power to insist. Since the post-crisis CVA charge and funding costs made such arrangements expensive for dealers, a number of sovereign debt offices have moved to two-way CSAs to obtain better pricing, but one-way arrangements remain common for sovereigns and supranationals, which the uncleared margin rules exempt.

One-way CSAs look like a commercial concession and are in fact a funding commitment. The dealer hedges the client trade with an offsetting trade that is itself collateralized. When the client trade moves in the dealer's favor, the hedge moves against it, so the dealer posts collateral on the hedge and receives nothing from the client; it therefore funds that collateral for the life of the trade, out of its own resources, at its own cost of funds. This is the origin of the

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funding valuation adjustment of Chapter 9, and it is why a trade that appears to be priced at the mid-market level can be substantially loss-making once the CSA is read.

Title transfer and security interest. Two legal forms exist. Under **title transfer**, collateral becomes the recipient's property outright, to be returned in equivalent form later; the recipient may use it freely, including by passing it on to a third party. Under a **security interest**, the poster retains ownership and the recipient holds a charge, a legal claim on the assets that it can enforce only if the poster defaults. The English-law CSA is a title transfer; the New York-law CSA creates a security interest, but its Paragraph 6(c) gives the recipient a right to use the collateral, including to rehypothecate it (Section 8.7), unless the parties disapply that right. The distinction is invisible until a default, at which point it determines whether the poster is an owner reclaiming property or an unsecured creditor in a queue.

8.2 VARIATION MARGIN AND INITIAL MARGIN

Two quite different things are both called margin, and conflating them is the most common error in this area.

Variation margin (VM) covers exposure that has *already arisen*. Each day the portfolio is revalued and the party for whom it has moved unfavorably transfers that amount. VM is a transfer of value that has already been lost; it does not make anyone worse off in economic terms, since the payer has suffered the loss whether or not any collateral changes hands.

Initial margin (IM) covers exposure that *might arise in the future*: specifically, the further move that could occur between a counterparty's last VM payment and the completion of close-out. It is posted at inception, recalculated daily by a model, and held, in a changing amount, until the trade matures. The distinction matters enormously for funding.

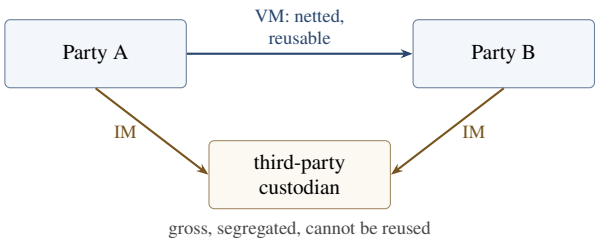


Figure 8.1: Variation margin passes between the parties; initial margin is posted gross to a custodian.

As Figure 8.1 shows, VM flows between the parties and is netted: A pays B or B pays A, never both. The recipient may use it, so it is not withdrawn from the system. IM is different: *both* parties post, *gross*, to a segregated account with a third-party custodian, and neither may touch it. The figure shows the bilateral arrangement under the uncleared margin rules; at a clearing house only the member posts initial margin. IM is therefore genuinely encumbered for the life of the trade, and it must be funded.

Example 8.2 Variation and initial margin on the same portfolio

A fund and a dealer hold a swap portfolio. At inception it is worth zero, and each posts initial margin of $25 million to a custodian. Neither can use the other's.

Rates then move and the portfolio is worth −$40 million to the fund. The fund pays $40 million of variation margin to the dealer, who may use it.

The two obligations are not comparable:

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	Variation margin	Initial margin
Covers	loss already incurred	possible future move
Direction	one way, netted	both ways, gross
Amount	\$40m from fund	\$25m each
Reusable by recipient	yes	no
Economic cost to poster	none (value already lost)	funding spread, life of the trade

Posted IM is not idle in every sense: cash earns interest and bonds keep their yield. What it costs is the poster's funding spread over that return. At the 60 basis point spread Chapter 9 uses for this cost, the fund's IM costs it

$$\$25\text{m} \times 0.0060 = \$150{,}000 \text{ a year},$$

and the dealer bears the same on its own posting: a real cost of doing the trade that appears nowhere in the swap's valuation.

8.3 HAIRCUTS

Collateral is not counted at its full market value. A haircut (Section 4.1) discounts it, so that the collateral remains sufficient even if its own price falls before it can be liquidated. If the haircut is h, then covering an exposure E requires collateral with market value

$$M = \frac{E}{1 - h} \tag{8.1}$$

Haircuts reflect the collateral's price volatility, its liquidity, and the time it would take to sell. Cash in the CSA's own currency carries none. Short-dated government bonds carry half a percent to two percent; long-dated government bonds more, because their prices move more; corporate bonds and equities considerably more. For firms that do not model their own, the uncleared margin rules of Section 8.4 prescribe a standardized schedule:

- cash, none;
- high-quality government debt, 0.5% up to one year to maturity, 2% from one to five years, and 4% beyond five;
- either of these in a currency other than the one the trade settles in, a further 8% currency add-on.

The currency add-on bears on the currency choice of Section 8.5. It is waived for cash variation margin in a major currency, where most of that option lives, but it applies to bonds and to initial margin posted in another currency. Example 8.3 shows what happens when a haircut widens.

Example 8.3 Haircuts and how they behave under stress

A fund must cover a £100 million exposure with gilts (UK government bonds) subject to a 4% haircut. From (8.1) it must deliver

$$\frac{100}{0.96} = £104.2 \text{ million of gilts}$$

Markets then become volatile and the haircut is raised to 10%. Covering the same £100 million now requires

$$\frac{100}{0.90} = £111.1 \text{ million}$$

an additional £6.9 million of gilts—on an exposure that has not changed at all.

The example holds the exposure fixed; a real stress does not. When markets turn volatile enough to raise the haircut, three things happen together, and they compound. The exposure itself has probably grown, because the volatility that widened the haircut also moved the portfolio. The

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gilts have probably fallen in price, so the fund holds fewer pounds of collateral than it did. And the haircut has widened. The fund must find more assets, from a smaller pool, to cover a larger exposure. This is the same procyclicality Section 2.9 identified in initial margin models, operating through a second channel. A firm that stress-tests its margin requirements using *current* haircuts has tested the easy case.

8.4 UNCLEARED MARGIN RULES AND SIMM

Before the financial crisis, initial margin on bilateral derivatives was rare between dealers. Dealers did collect it from hedge funds, as independent amounts, but held it commingled with their own assets, which the failure of Lehman Brothers exposed. The post-crisis reforms of Section 2.11 pushed standardized products into clearing; for what remained bilateral, regulators imposed the uncleared margin rules (UMR), phased in over several years by size of counterparty and now applying broadly.

UMR requires variation margin on uncleared trades between covered firms, and initial margin as well once both groups hold more than €8 billion of uncleared derivatives notional ($8 billion in the United States), averaged over a prescribed window. Even then the parties may agree on a threshold of up to €50 million per counterparty group and post only the excess, to a segregated third-party custodian. The intent is that staying bilateral should not be a way to avoid posting collateral. Initial margin reached the largest dealers first and, in the final phase in September 2022, firms on the investing side such as asset managers and pension funds above the notional threshold. Some funds deliberately keep their uncleared notional below it so as never to post initial margin.

Calculating IM. Firms may use a model (which some regulators must approve) or a standardized schedule. Almost all use the model, because the schedule is punitive, and almost all use the same one: **ISDA SIMM**, an industry-standard methodology. The industry converged on one model deliberately, for a reason Section 8.8 explains. Adopting it is not automatic: in the European Union, the European Market Infrastructure Regulation as revised in 2024 brings in supervisory authorization of initial margin models, starting with the largest groups, and gives the European Banking Authority a central role in validating industry-wide models such as SIMM.

SIMM works from sensitivities rather than simulation. Each risk factor sensitivity is multiplied by a prescribed risk weight, and the weighted sensitivities are aggregated using prescribed correlations.

Example 8.4 A simplified SIMM calculation

A portfolio has interest rate sensitivity in two tenor buckets, the maturity ranges into which SIMM groups rate sensitivities:

Bucket	Sensitivity per bp	Risk weight (bp)	Weighted sensitivity
2-year	$80,000	60	$4.8m
10-year	$120,000	50	$6.0m

With a prescribed correlation of $\rho = 0.7$ between the two buckets, the margin is

$$\text{IM} = \sqrt{4.8^2 + 6.0^2 + 2 \times 0.7 \times 4.8 \times 6.0} = \sqrt{99.36} = \$9.97\text{m}$$

Compare this with the two figures that bracket it. With zero correlation the two charges would combine as $\sqrt{4.8^2 + 6.0^2} = \$7.68\text{m}$. Simple addition of the weighted sensitivities gives $10.8m, what would be required with no diversification recognized.

The diversification benefit here is modest, only 8% below the undiversified sum, because rate buckets are highly correlated. The benefit is larger where correlations are lower, as between interest

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rates and FX, which SIMM treats as separate risk classes within one product class. Across its four product classes (rates and FX, credit, equity, commodity) it recognizes none: their margins are simply added. Netting, too, stops at the edge of a relationship, since each counterparty margins only its own portfolio. This is why firms concentrate trades with fewer counterparties: splitting an offsetting book across two dealers means posting IM twice, on two portfolios each less diversified than the whole.

Cleared against bilateral

Because IM is model-driven, the same trade can require quite different amounts depending on where it is booked. Two features drive the gap.

The **margin period of risk** (the assumed time to close out a defaulted portfolio) is conventionally five days for cleared trades and ten for bilateral ones, on the reasoning that a CCP has a rehearsed auction process and a bilateral counterparty does not. Since margin scales roughly with the square root of the holding period, that alone makes bilateral IM about $\sqrt{10/5} = 1.41$ times larger.

Netting then compounds it. At a CCP a member's trades net across all its counterparties; bilaterally, each relationship nets only within itself. Example 8.5 puts numbers on both features.

Example 8.5 The same trade, cleared and bilateral

A fund executes a swap whose five-day 99% risk (the loss over five days that is exceeded with only 1% probability) is $12 million.

Cleared. IM is $12 million, posted once to the CCP. The dealer on the other side faces the CCP as well, but its leg offsets against the rest of its large cleared book and adds little to its own IM, so the trade's own cost is the fund's $12 million.

Bilateral. The ten-day margin period raises it to roughly $12\sqrt{2} = \$17$ million. Under UMR both parties post, gross and segregated. Other trades in the relationship have already used up its €50 million threshold, so each posts the full $17 million and $34 million is immobilized against the same economic risk.

Immobilized collateral is therefore nearly three times larger in the bilateral case. Without that offset (a dealer whose cleared book did not absorb the leg), the comparison would be $24 million against $34 million, and the gap would be the margin period's factor of 1.41 alone. Add the diversification point of Example 8.4 (the fund's other trades net at the CCP but not across separate bilateral relationships) and the difference widens further.

The rules were designed to make clearing cheaper than not clearing. The corollary is that genuinely bespoke hedges, which cannot be cleared, now carry a collateral cost that may exceed the value of the tailoring—which is why some hedges that would reduce a firm's risk are no longer worth doing.

Some firms were spared the clearing mandate altogether. European pension schemes were exempt from it, because clearing would have obliged them to hold cash for variation margin. The EU exemption expired in June 2023; the United Kingdom made its own permanent in 2025, so UK schemes may still keep their swaps bilateral.

The wider consequence of UMR is that initial margin has become a large, permanent, system-wide demand for high-quality assets, locked away with custodians where neither party may use it (Figure 8.1). Section 8.11 takes up that demand.

8.5 CSA OPTIONALITY

Section 7.8 noted in passing that a CSA permitting several collateral currencies gives the poster an option. It deserves explicit treatment as a real position that must be valued and hedged.

Suppose a CSA permits cash in dollars, euros, or sterling, each earning its own overnight

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rate. The poster chooses which to deliver, and may switch. It will deliver whichever is cheapest to fund. Since the currencies' overnight rates and the cross-currency basis of Section 7.8 both move, the cheapest currency changes over time.

Because the collateral rate determines the discount curve, this is an option over the discount curve itself. The tools for valuing it come from two places: it is an option on the best of several rates, of the kind Section 34.10 treats, and comparing a foreign overnight rate with a domestic one requires the changes of measure of Sections 30.1 and 30.6. Neither alone is enough, which is part of why the position is so often left unpriced. Example 8.6 gives a first estimate of its size.

Example 8.6 The value of collateral choice

A dealer owes $1 billion on a portfolio under a CSA permitting dollars, euros, or sterling, so the dealer is the poster. Dollar collateral earns SOFR. To post euros instead, the dealer swaps dollars into euros: it lends dollars and receives SOFR, borrows euros and pays €STR plus the basis b (the quoting convention of Example 7.10), and earns €STR on the euros it posts. Its net return, in dollar terms, is

$$\text{SOFR} - (€STR + b) + €STR = \text{SOFR} - b$$

The euro overnight rate cancels, so what a currency's collateral earns relative to dollar collateral is its basis against the dollar with the sign reversed. With the basis at -15 basis points for euros and -10 for sterling, euro collateral earns the equivalent of SOFR plus 15 basis points, sterling SOFR plus 10, and dollars SOFR flat. The dealer posts euros, the cheapest currency to fund. Discounting at the euro collateral rate converted into dollars, rather than at SOFR, raises the effective discount rate by 15 basis points.

Take a portfolio annuity of 8, the annuity of Section 7.3: the present value of one unit a year over the portfolio's life. Posting euros rather than dollars then lowers the dealer's liability by roughly

$$\$1{,}000\text{m} \times 0.0015 \times 8 = \$12\text{m}$$

Using the annuity assumes the collateral balance stays at $1 billion over the portfolio's life, which overstates the value as a swap book amortizes.

This is only a first approximation: it prices the *current* cheapest currency, not the option to switch as relative rates move, which is worth more. Valuing it properly requires modeling the joint dynamics of the overnight rates and the cross-currency basis, a genuine multi-currency term structure problem. A dealer that ignores this and discounts on the domestic curve alone is systematically mispricing every trade under a multi-currency CSA. Two firms quoting the same trade under the same CSA can differ by millions purely on this treatment.

Similar optionality arises wherever the CSA offers a choice: eligible bonds as well as cash, a range of maturities, or substitution rights. In every case the poster will deliver the cheapest, and the receiver should price on that assumption. How much of this optionality survives depends on the agreement's vintage. The ISDA Standard CSA of 2013 was designed to remove it, and the variation margin CSAs signed since the uncleared margin rules are mostly cash-only, often in few currencies, so the option now lives mainly in legacy CSAs.

8.6 COLLATERAL TRANSFORMATION

The margin requirements of Sections 8.2 and 8.4 together create a problem for a particular kind of institution. Variation margin on cleared trades, and increasingly on bilateral ones, must be posted in *cash*. But the institutions with the largest derivatives hedges (pension funds, insurers, asset managers) hold almost no cash. They hold bonds, because holding cash would defeat their purpose. When margin is called they have the assets but not the form.

Collateral transformation is the business of converting bonds into cash: the fund repos its bonds, receiving cash to post as margin, and pays the repo rate while continuing to earn the

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bond's yield. Figure 8.2 shows the transaction, which leaves the fund with a financing leg and a rollover it did not have before. Insurers also use securities lending for the same purpose: in a collateral upgrade trade they lend corporate bonds or equities that a clearing house would not accept, and receive government bonds or cash against them.

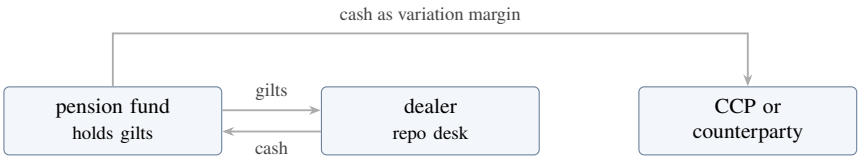


Figure 8.2: A fund repos bonds for cash to meet a margin call.

Example 8.7 The cost of transformation

A pension fund must post £500 million of cash variation margin. It holds gilts and repos them with a 2% haircut. Take illustrative rates: the repo costs 4.25%, and the cash, once posted, earns SONIA at 4.00% (the CSA rate, or price alignment interest at a clearing house).

To raise £500 million it must deliver

$$\frac{500}{0.98} = £510.2 \text{ million of gilts}$$

The gilts keep earning their yield whether or not they are repoed, so the yield drops out of the cost. The fund pays the repo rate on the cash and receives SONIA on the same cash as posted margin, a drag of 25 basis points, or

$$£500\text{m} \times 0.0025 = £1.25 \text{ million a year}$$

The haircut adds no running cost, since the fund owns the extra £10.2 million of gilts outright, but those gilts are encumbered and cannot raise cash anywhere else.

Two features make this fragile rather than merely expensive. The haircut means the fund needs *more* assets than the cash it raises, and haircuts widen under stress exactly as Example 8.3 showed: if the haircut is raised to 8% when the repo is next rolled, the same £500 million needs $500/0.92 = £543.5$ million of gilts, £33.3 million more. And the transformation depends on a dealer being willing to roll the repo, a service that consumes balance sheet under the leverage ratio of Section 1.12, and that dealers withdraw when balance sheet is scarce, which is precisely when the fund needs it.

Collateral transformation therefore converts a market risk the fund has hedged into a dependence on the continued functioning of the repo market. That is a different risk, not a smaller one.

8.7 REHYPOTHECATION, SEGREGATION, AND WHAT HAPPENS ON DEFAULT

Section 8.1 distinguished title transfer from security interest. The distinction becomes concrete when the recipient of collateral fails. **Rehypothecation** is the reuse of collateral received. A dealer that receives bonds as variation margin may post them onward to meet its own obligations, and the same security can support several transactions in a chain. Both legal forms allow it: a title transfer outright, and the New York-law security interest through the right of use that Section 8.1 described. This is efficient: it means the system needs less collateral in total to support a given volume of activity. It is also fragile, because a chain of claims on one security unwinds messily when a link fails, and because the original poster may find that its property has traveled somewhere it cannot follow.

The protections differ sharply by margin type, and the arrangement in Figure 8.1 is what produces the difference:

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- *Variation margin* is transferred outright, or under a New York-law CSA held with a right of use, and may be reused. Either way the poster has parted with control of it: if the recipient fails, the poster is an unsecured creditor for the return of equivalent assets, ranking with everyone else.
- *Initial margin* under UMR must be held by a third-party custodian and may *not* be rehypothecated. The poster remains the owner; if the recipient fails, the IM is returned rather than joining the estate.

That asymmetry is deliberate and was learned expensively. Client assets held by failed intermediaries, Lehman Brothers in 2008 and MF Global in 2011, proved far harder to recover than clients had assumed, and in several cases had been reused in ways the clients did not know about and could not have discovered. The segregation requirement for initial margin exists because the alternative was tested and failed.

Wrong-way risk in collateral. A separate hazard is taking collateral whose value falls precisely when the counterparty defaults. The extreme case, accepting a counterparty's own securities, is obvious and prohibited, but weaker versions are easy to miss: taking the bonds of a bank's sovereign as collateral against exposure to that bank, or taking energy-sector securities against exposure to an energy trading firm. In each case the collateral and the exposure share a driver, so the protection evaporates in the state where it is needed. Chapter 27 develops this as wrong-way risk in general.

8.8 DISPUTES AND RECONCILIATION

A margin call is a demand for money based on one party's valuation of a portfolio. The other party has its own valuation, and the two will differ: over the marks on illiquid trades, over which trades are in the portfolio at all, over the discount curve of Section 7.4, and over the terms of the CSA itself.

Disputes are therefore routine rather than exceptional, and the procedure for handling them is part of the arrangement. Small differences never become calls at all, because no collateral moves below the CSA's minimum transfer amount (Section 8.1). The uncleared margin rules cap it at €500,000 across initial and variation margin combined ($500,000 in the United States), which is small enough that it removes only the noise. Above it, the undisputed amount is transferred while the difference is investigated, and the CSA specifies a resolution procedure: recalculation, and failing that, polling reference dealers for quotations. Firms also reconcile portfolios on a schedule set by regulation (daily for the largest books) to catch discrepancies before they become material, and a dispute large enough and old enough must be reported to the supervisor. Three observations make this more than an operational footnote.

First, a dispute is a *funding* event. If a firm calls for $80 million and receives $50 million pending resolution, it is short $30 million while the disagreement lasts, and must fund that gap.

Second, disputes cluster exactly when they are most dangerous. Illiquid positions are hardest to mark when markets are disorderly, and both parties have an incentive to mark in their own favor. A firm relying on incoming collateral to meet its own outgoing calls has built a chain whose weakest link is a valuation disagreement, at the moment when everyone is disagreeing.

Third, and least obviously, a dispute is expensive long after it is settled. A **netting set** (the trades with one counterparty that a single enforceable netting agreement covers) that produces repeated disputes lasting longer than the margin period of risk has that period *doubled* for the following two quarters in the dealer's counterparty capital calculation. The add-on for potential future exposure measured there (a high quantile of how large the exposure could grow, which Section 9.1 defines) scales with the square root of the period, through the maturity factor of Section 27.3, as margin does in Section 8.4. Disputing is therefore not free even for the party whose valuation is right; Example 8.8 prices the funding cost and compares the two.

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Example 8.8 What a dispute costs, and when

A dealer calls a counterparty for $80 million of variation margin. The counterparty accepts $50 million and pays that, disputing the rest. The $30 million gap is outstanding for twelve days before the two sides reconcile, and the dealer funds it at 5.0%:

$$\$30\text{m} \times 0.050 \times \frac{12}{360} = \$50{,}000$$

That is the visible cost, and it is trivial. The cost that can matter more is in capital: if this is the third dispute on the netting set in two quarters to outlast the margin period of risk, the dealer's counterparty capital calculation doubles that period for the next two quarters, raising the capital held against the netting set whether or not the dealer's valuation was the correct one (Section 27.3 works through the rule).

The lesson is not that firms should concede disputes. It is that the cost of a valuation disagreement is mostly not in the disputed amount, so the investment that pays is in not disagreeing: reconciled portfolios, agreed curves, and a common margin model.

The cost of disputes also explains why the industry converged on SIMM, a choice that would otherwise look like bureaucratic uniformity. Initial margin is model-driven, and two counterparties running their own proprietary models would disagree about the number on every single call: not occasionally, but always, and by amounts far larger than the marks. A common model does not make either party's risk estimate better. It makes the two estimates the *same*, which is what converts initial margin from a standing dispute into an operational transfer. SIMM is best understood as dispute-avoidance infrastructure that happens to also measure risk.

8.9 MARGIN CALLS AS LIQUIDITY RISK

The sections above describe how collateral works. This section states the chapter's thesis: why any of it matters.

A hedge and the item it hedges generate cash on different schedules. The hedge is marked daily and settles daily. The hedged item (physical output to be delivered over years, a pension liability payable over decades, a bond held to maturity) does not. When the hedge loses money, the loss is a cash outflow *today*, while the offsetting gain on the hedged item generally brings in no cash at all. This asymmetry is invisible in a mark-to-market framework, which nets the two and reports approximately zero. It is the entire problem.

Example 8.9 The liquidity cost of a correct hedge

A power producer expects to generate 10 TWh over the next three years and hedges by selling forward at €80 per MWh. Its revenue is locked in at

$$10{,}000{,}000\ \text{MWh} \times €80 = €800\ \text{million}$$

Prices then rise to €250 per MWh. The producer's short forward position has lost

$$10{,}000{,}000 \times (€250 - €80) = €1.7\ \text{billion}$$

and, because the position is margined, it must pay that €1.7 billion in cash, now.

Against it stands a gain of exactly €1.7 billion on the electricity it will generate. But that electricity does not exist yet. It will be produced and sold over three years, and the cash will arrive then.

The firm is better off than if it had not hedged: it has locked in €800 million of revenue in a market that might equally have fallen to €30. It is also, quite possibly, unable to pay what it owes on the day: illiquid, though not insolvent.

Set the two sides of Example 8.9 against time and they sum to approximately zero: €1.7

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billion out today, about €567 million a year in over the next three years. Figure 8.3 plots the cumulative cash: the net position opens €1.7 billion short and climbs back to zero only as the last of the power is sold. The difficulty is entirely in the timing. The hedge remains economically sound right up until the firm fails. Solvency and liquidity are different questions, and derivatives collateral is the mechanism that most reliably separates them.

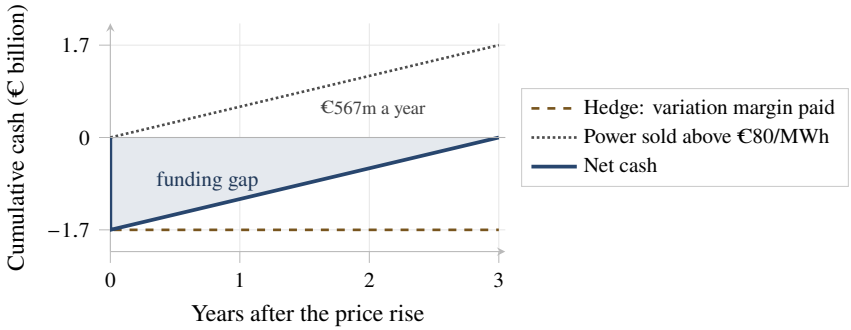


Figure 8.3: Cumulative cash in Example 8.9 (schematic: even sales, prices held at €250, no discounting).

This is why several European governments extended emergency credit lines to energy companies in 2022: the utilities in difficulty were the ones that had sold their output forward and faced Example 8.9's arithmetic at national scale, while those that had not hedged faced no margin calls and took windfall profits. The European Union's regulatory answer, in late 2022, was to widen what its clearing houses could accept: as a temporary measure, bank guarantees not backed by collateral became eligible from non-financial firms that clear directly, and public guarantees from financial and non-financial clearing members alike. The 2024 revision of the European Market Infrastructure Regulation made the relief permanent and extended it to non-financial firms that clear through a member. Section 40.12 tells the episode, and what it says about a system that prices risk correctly and liquidity not at all.

Case Study 8.1 Liability-driven investment, 2022

UK pension funds face a structural problem: liabilities extending fifty years, and not enough long-dated gilts in existence to match them. The standard solution, liability-driven investment, uses leverage (gilt repo and interest rate swaps) to obtain the required duration from a smaller pool of assets.

The leverage was collateralized, and the collateral was gilts. When the September 2022 fiscal announcement pushed gilt yields sharply higher, the funds' swap and repo positions moved against them and collateral was called. To meet the calls they sold gilts. Selling gilts pushed yields higher. Higher yields generated further calls.

The loop was not a panic: the mechanism was structural. The assets pledged as collateral were the same assets whose price the forced selling was driving down, so the more collateral was called, the less the remaining collateral was worth.

The Bank of England intervened by buying gilts on a temporary basis, explicitly to break the loop rather than to set a yield. The subsequent regulatory response set collateral buffers: the Bank's Financial Policy Committee (March 2023) and the Pensions Regulator (April 2023) required funds to withstand a rise of about 250 basis points in gilt yields, which caps leverage indirectly by fixing the collateral held against it. The target was liquidity rather than solvency. In January 2025 the Bank opened a contingent repo

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facility, activated at its discretion, through which it can lend cash against gilts to eligible insurers, defined benefit pension schemes, and liability-driven investment funds.

8.10 MEASURING AND MANAGING MARGIN LIQUIDITY

The obvious response to the previous section is to hold enough cash. The difficulty is knowing how much, since the requirement depends on a market move that has not happened. Firms therefore run **margin liquidity stress tests**, which differ from ordinary market risk measurement in three ways.

They measure gross outflow, not net loss. A value-at-risk (VaR) framework reports the net change in portfolio value. The relevant quantity here is the *cash that must leave the firm*, and the offsetting gain does not count, because it produces no cash.

They model the initial margin response, not only variation margin. A large move generates a VM call equal to the loss, *and* raises the IM requirement because volatility has risen (the procyclicality of Section 2.9), at a clearing house within days and bilaterally at the next recalibration. At a clearing house both land at once.

They assume the transformation channel narrows. A stress test that assumes normal repo access (Example 8.7) has assumed away the problem.

Example 8.10 Sizing a liquidity buffer

A fund holds a swap portfolio of $2 billion notional with a DV01 of $1.6 million, so a one basis point move in rates changes its value by $1.6 million.

It stresses a 100 basis point adverse move over one week, which is severe but well within the range observed in 2022:

$$\text{VM outflow} = 100 \times \$1.6\text{m} = \$160 \text{ million}$$

Volatility rises with the move, and the initial margin model responds. The portfolio is cleared, and a clearing house's margin model reacts to a volatile week within days; SIMM, by contrast, is recalibrated only periodically. If the clearing house's initial margin on the book rises 50%, from $60 million to $90 million, that is a further $30 million call.

The total requirement is $190 million of cash within days, against a portfolio whose *value* may be unchanged once the hedged item is included. If the fund holds $100 million of cash and expects to repo bonds for the rest, it should ask what haircut it will face at that moment, and whether dealers will still lend against its bonds at all.

The tools. Firms manage this with cash buffers, committed repo or credit lines, CSAs that permit non-cash collateral where possible, and leverage limits that keep the plausible call within reach. Each has a cost, paid continuously, to avoid an event that may not arrive. That trade-off is the same one Section 2.9 identified for CCP margin models, and it is resolved no more cleanly here. Sizing these tools requires anticipating the call, which is why the Basel Committee, CPMI, and IOSCO's final reports of January 2025 ask clearing houses to give members margin simulators that show requirements under stress scenarios, and to disclose a standardized measure of how strongly their initial margin responds to market volatility.

Why the problem is worse than each firm's own numbers suggest

Three effects make the aggregate picture worse than the sum of single-firm calculations like Example 8.10.

Calls are correlated. A large market move calls margin from everyone positioned the same way at the same moment. Each firm's stress test may be individually adequate while the

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system's total demand for cash exceeds what can be raised, which is precisely what turns a price move into a liquidity event.

The supply of cash contracts as demand for it rises. Repo capacity depends on dealer balance sheet (Example 8.7), so demand and supply move in opposite directions at once.

Selling to raise cash moves the price that generated the call. Where the asset sold is the asset driving the exposure (gilts sold by gilt-collateralized funds), the loop closes on itself, as the 2022 pension episode of Case Study 8.1 showed.

A single firm can estimate the second, and its own contribution to the third, as Example 8.11 does; the first, and the rest of the third, depend on what everyone else is doing.

Example 8.11 The loop, quantified

A pension fund holds £1.2 billion of gilts with a duration of 12 and closes its duration gap with £3 billion notional of twenty-year receive-fixed swaps. The twenty-year annuity at 4% is 13.590, so the swap DV01 is

$$£3\text{bn} \times 13.590 \times 0.0001 = £4.077 \text{ million per basis point}$$

Rates rise 100 basis points. On a funding-ratio basis (assets divided by liabilities, the measure a pension fund is managed to) the fund is fine: its liabilities fall by more than its assets, which is what the hedge was for. On a cash basis, assuming the swaps are cleared so that their initial margin rises with volatility (swaps margined bilaterally for variation only would call the first line alone):

Variation margin on the swaps	£407.7m
Initial margin increase as volatility doubles	£120.0m
Cash required within days	**£527.7m**

At the same instant the gilt pool has fallen 12% to £1,056m, and the repo haircut on it has widened from 2% to 8%:

	Before	After
Gilt pool	£1,200m	£1,056m
Cash raisable against it	£1,176m	£972m

Capacity fell 17.4% in the same move that raised the requirement from nothing to £527.7m. That requirement is now 54.3% of what remains. Neither number appears in a net-loss stress test, which reports approximately zero.

And the loop has not started yet. Raising £527.7m at an 8% haircut would tie up £573.6m of gilts, and on a day like this the repo may not be there. Suppose the fund sells £527.7m of gilts instead, and that selling on that scale moves twenty-year yields a further 15 basis points (modest against what was actually observed). That is another £61.2m of variation margin, met by selling another £61.2m of gilts, which moves yields 1.74 basis points more, and so on; the further fall in the gilt pool from these later rounds is ignored:

Round	Yield move	Extra VM
1	15.00 bp	£61.2m
2	1.74 bp	£ 7.1m
3	0.20 bp	£ 0.8m

The series converges to £596.9m, a 13.1% amplification of the fund's own requirement, and it converges only because one fund's selling has a bounded price impact. The series diverges when enough funds sell at once, which is the precise sense in which this is a system-level property rather than a risk management failure: the same calculation performed by every fund individually gives the right answer for each of them and the wrong answer for all of them.

No firm-level stress test, however conservative, captures this, which is why the supervisory

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response, including the Financial Stability Board's work on margin preparedness, has come through financial stability channels rather than individual firm requirements.

No one behaved badly. In the energy and pension episodes of 2022 the utilities hedged as textbooks instruct, the pension funds matched their liabilities as their mandates required, and the clearing houses raised margin as their models and their regulators demanded. Every participant did what it was supposed to do, and the aggregate outcome was a liquidity crisis. Frameworks that locate such failures in imprudence will not anticipate the next one.

8.11 IS THERE ENOUGH COLLATERAL?

Clearing mandates, uncleared margin rules, and higher haircuts have each increased the high-quality assets immobilized against derivatives exposures. Is the supply adequate?

The pessimistic case is straightforward. Initial margin is segregated and cannot be reused (Section 8.2), so every dollar of IM is a dollar of government bonds or cash withdrawn from productive use for the life of a trade. Bank regulation has simultaneously increased demand for the same assets as liquidity buffers. And collateral transformation, which lets bond-holders meet cash calls, depends on dealer balance sheet that regulation has made expensive.

The optimistic case is equally straightforward. Government debt outstanding has grown enormously over the same period, so eligible collateral has expanded alongside demand. Netting and compression (tearing up offsetting trades among many participants at once) have reduced gross exposures substantially. And the pre-crisis alternative, in which large exposures went uncollateralized, was demonstrably worse.

The aggregate quantity has not been the binding constraint; the *distribution* and *timing* have. The 2022 episodes were caused not by a shortage of eligible assets but by particular firms needing cash on particular days and unable to convert what they held quickly or cheaply enough. A system can hold ample collateral in aggregate and still fail at the moment of transformation, which is why Section 8.10's stress tests focus on the channel rather than the aggregate.

8.12 WHERE THIS LEAVES VALUATION

Collateral changes what a derivative is worth, in three distinct ways that later chapters develop.

The *discount curve* is set by the CSA's collateral rate, as Section 7.4 established, and by the cheapest-to-deliver option of Section 8.5 where the CSA permits choice.

Funding enters when collateral does not fully cover the exposure. A one-way CSA, an uncollateralized trade, or a CSA with a large threshold all leave the dealer funding an exposure from its own resources. Chapter 9 prices this as a funding valuation adjustment (FVA).

Initial margin must itself be funded for the life of the trade, at a cost that no discounting convention captures. Chapter 9 prices this as MVA; Example 8.2 gave its size.

The phrase "the value of the swap" is therefore incomplete: a swap's value depends on who holds it, under what CSA, funded how, and Chapter 9 makes that dependence explicit. What these three do *not* capture is the proposition this chapter opened with. All of them price the *cost* of collateral: what it is worth to discount on one curve rather than another, to fund an uncollateralized exposure, to leave initial margin encumbered for ten years. None of them prices the risk of being unable to deliver collateral on the day it is called. The producer of Example 8.9 owed €1.7 billion in cash against a hedge that was working exactly as designed; no valuation adjustment would have shown a problem, because in valuation terms there was not one. That exposure is a liquidity risk, measured by the stress tests of Section 8.10 and not by any adjustment to a price. Collateral changed what derivatives are worth; it also changed what can bring a firm down, and the two are not the same list.

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SUMMARY

The credit support annex sets what may be posted, at what haircut, above what threshold, and at what rate. These terms are not administrative: the collateral rate sets the discount curve, a one-way CSA is a funding commitment, and a choice of collateral currency gives the poster an option over the discount curve itself.

Variation margin settles a loss already incurred. It is netted, may be reused, and costs the poster nothing beyond the loss. Initial margin covers a possible future move. It is posted gross by both parties, segregated and barred from reuse so that it survives the failure of whoever holds it, and funded for the life of the trade, so it is a running cost. A bilateral trade can immobilize far more collateral than a cleared one by design, which makes genuinely bespoke hedges expensive to hold.

Haircuts widen under stress, so a firm must find more assets, from a pool that has itself shrunk, to cover an exposure that has grown: the procyclicality of margin models, through a second channel. Disputes are funding events: the party that made the call funds the disputed amount until it is resolved, and disputes multiply when markets are disorderly. A firm holding bonds rather than cash depends on the repo market to transform them, and that market narrows in the same stress.

The chapter's central proposition is that a hedge and the hedged item generate cash on different schedules. A producer that sells its output forward owes margin in cash today against revenue that arrives later, so it can be better off for having hedged and still unable to pay. That is why European governments lent to energy firms in 2022, and why the UK pension episode was a collateral loop rather than a solvency problem.

Managing this requires stress tests of gross cash outflow that assume the repo channel narrows, and even those understate the problem, because calls are correlated across firms and forced sales move the prices that caused them. No participant in the energy and pension episodes of 2022 behaved imprudently, and the result was still a liquidity crisis. The system holds enough collateral in aggregate; what fails is converting it into cash on the day it is called. The valuation adjustments of Chapter 9 price the cost of collateral, not the risk of being unable to deliver it.

FURTHER READING

Gregory's *The xVA Challenge* is the standard practitioner reference for collateral, margin, and the funding that follows from them. The ISDA SIMM methodology document is the operative specification for initial margin and changes often enough that the current version is the one to use. On margin procyclicality, the Basel Committee on Banking Supervision and CPMI–IOSCO's *Review of Margining Practices* (2022) is the official record. Two final reports of January 2025 set out the policy that followed from it: the same bodies' *Transparency and Responsiveness of Initial Margin in Centrally Cleared Markets: Review and Policy Proposals* for clearing houses, and the Basel Committee and IOSCO's *Streamlining Variation Margin Processes and Initial Margin Responsiveness of Margin Models in Non-centrally Cleared Markets* for bilateral trades. The Financial Stability Board's *Liquidity Preparedness for Margin and Collateral Calls* (2024) is the supervisory response. The Bank of England's December 2022 *Financial Stability Report* is the place to start on liability-driven investment and gilt market functioning in the 2022 episode.

PRACTICE QUESTIONS

8.1 Explain the difference between a one-way and a two-way CSA, why a one-way CSA makes an apparently mid-market trade loss-making for the dealer, and why the rate paid on collateral is the most economically important term in the document.

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8.2 A fund posts $40 million of variation margin and $25 million of initial margin, and its funding costs 60 basis points more than posted margin earns. Give the annual economic cost of each, and explain why they differ along the dimensions of what each covers, direction, and reusability.

8.3 Explain why initial margin is posted gross by both parties to a third-party custodian rather than netted between them.

8.4 A portfolio has sensitivities of $50,000 per basis point in a bucket with a risk weight of 55 bp and $90,000 per basis point in a bucket with a risk weight of 45 bp, with a prescribed correlation of 0.6. Calculate the initial margin, compare it with the undiversified sum, and explain why the result leads firms to concentrate their trades with one counterparty.

8.5 Explain why a CSA permitting several collateral currencies gives the poster an option, and why that option is an option over the discount curve.

8.6 A pension fund must post £300 million of cash margin, which earns SONIA at 4.0% once posted. It holds bonds yielding 3.8% and can repo them at 4.3% with a 3% haircut. How many bonds must it deliver, and what is the annual cost of the transformation?

8.7 An airline expects to buy 500 million gallons of jet fuel evenly over the next two years and hedges by paying a fixed $2.80 a gallon in swaps that are margined daily. Jet fuel then falls to $2.10 a gallon. Calculate the variation margin the airline must pay and the offsetting saving on its fuel bill, and state when each becomes cash. Explain why the airline can be perfectly hedged against price risk and still fail.

8.8 A fund must cover a $250 million exposure with bonds subject to a 5% haircut. How much collateral must it deliver? Recalculate at a 12% haircut, and explain why the two other things happening at the same moment make the position worse than the comparison suggests. Now suppose the $250 million is initial margin under the uncleared margin rules, posted in euro-denominated government bonds with more than five years to maturity against trades that settle in dollars. Using the standardized schedule (4% on the bonds plus the 8% currency add-on), how much must the fund deliver, and how much of that is due to the currency?

8.9 A swap has a five-day 99% risk of $20 million. Compare the collateral immobilized if it is cleared with the amount immobilized bilaterally under UMR, stating your assumptions. Why were the rules calibrated this way, and what is the cost of that design?

8.10 Explain what rehypothecation is, why it is economically efficient, and why initial margin may not be rehypothecated while variation margin may.

8.11 Give two examples of wrong-way risk in the choice of collateral, and explain the common feature that makes them dangerous.

8.12 A dealer calls a counterparty for $60 million of variation margin. The counterparty pays $35 million and disputes the rest, and the gap is outstanding for 20 days, which the dealer funds at 4.8%. Calculate the funding cost of the dispute. The dispute is the third on the netting set in two quarters to outlast the margin period of risk, so the period doubles from ten days to twenty for the next two quarters. If the netting set's potential future exposure add-on is $150 million and capital costs 40 basis points a year on the exposure, what does the doubling cost? Explain why a margin dispute is a funding event that clusters at the worst possible time.

8.13 A CSA gives a counterparty a threshold of $5 million and requires it to post an independent amount of $2 million. The minimum transfer amount is $500,000, and calls are rounded up to the next $100,000. The dealer's exposure is $14.23 million and it holds collateral

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worth $8.0 million after haircuts. Calculate the margin call. The next day the exposure is $14.6 million: is there a further call? How much of the exposure is then uncollateralized, and why?

8.14 A dealer posts $600 million of cash collateral under a CSA permitting dollars, euros, or sterling. Dollar collateral earns SOFR, the cross-currency basis against the dollar is −20 basis points for euros and −8 for sterling, and the portfolio annuity is 6. Which currency does the dealer post, and what is the approximate value of that choice relative to posting dollars? In what respect does this estimate understate the value of the CSA's optionality, and in what respect does it overstate it?

8.15 A dealer's portfolio with one counterparty holds an interest rate swap whose SIMM margin on its own is $8 million and an equity swap whose margin on its own is $5 million, and the two positions have tended to move in opposite directions. What initial margin does SIMM require on the two together? Replace the equity swap with an FX forward whose margin on its own is $5 million. SIMM prescribes a correlation of about 0.3 between the interest rate and FX margins, whatever the positions' signs: what is the margin now? Explain the difference.

FURTHER QUESTIONS

8.16 A gas producer hedges 25 TWh of output at €60 per MWh, and prices then rise to €200. What financing arrangement would have allowed the firm to meet its margin calls and survive, and what would that arrangement have cost in a year when prices did not move?

8.17 Explain the liability-driven investment loop of 2022 step by step. Identify the feature that made it self-reinforcing rather than merely costly, and explain why the regulatory response limited leverage through a minimum collateral buffer rather than by capping it directly.

8.18 A margin liquidity stress test differs from value at risk in three ways. Set them out, and explain why a firm passing a VaR-based test could still fail the liquidity one.

8.19 A $5 billion portfolio has a DV01 of $3.2 million. Stress a 150 basis point move over a week and assume the clearing house's initial margin rises 60%, from a base of $120 million. Calculate the total cash requirement, and discuss what the firm should assume about repo haircuts and dealer willingness at that moment.

8.20 Collateral was introduced to reduce credit risk, and it does. Argue that it has substituted a liquidity risk that is harder to measure and more correlated across firms than the credit risk it replaced. Then argue the other side. Which view do the 2022 episodes support?

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Chapter 9
XVA: Valuation Adjustments in Practice

Chapters 7 and 8 arrived, by different routes, at the same uncomfortable conclusion. A swap's value depends on the curve used to discount it, and the curve depends on the collateral agreement. Where collateral does not fully cover the exposure (under a one-way CSA, a threshold, or no CSA at all), the dealer funds the position itself; where initial margin is required, that too must be funded for the life of the trade. Behind all of it stands the possibility that the counterparty defaults. The phrase "the value of the swap" therefore does not identify a number: the same cash flows are worth different amounts to different holders, under different agreements, funded differently. The adjustments that convert a textbook valuation into a price someone will actually trade at are collectively called **XVA**, and this chapter is about what they are, who charges them, and where they overlap.

Default needs a precise meaning, because every adjustment prices it. A counterparty **defaults** when it fails to meet its obligations on any of its contracts, or becomes bankrupt or insolvent. Default belongs to the counterparty, not to a trade, and it reaches all of its liabilities at once: cross-default and acceleration clauses, where a contract carries them, put its other obligations into default when one is missed, and bankruptcy stays them all together. Creditors differ only in how much each recovers, which depends on seniority and collateral. A swap that is performing perfectly is therefore caught when the counterparty defaults on a bond, a loan or another dealer. For derivatives, the ISDA Master Agreement sets the consequences. An **event of default** (among other things, a failure to pay or deliver under the agreement that is not cured within a short grace period, bankruptcy or insolvency, or, where the parties have agreed it, a default on other borrowing above a threshold amount) entitles the other party to terminate every trade under the agreement and replace them with a single net close-out amount (Section 1.5). That amount, less the collateral held, is what the counterparty may never pay. Default is a legal event, not a market price: a downgrade or a wider credit spread is not a default, although a rating trigger in the CSA can turn one into a collateral call (Section 8.1). It also differs from two neighboring definitions. Bank capital rules treat an obligor as defaulted when it is more than 90 days past due on a material obligation or is judged unlikely to pay, and a credit default swap pays on its own list of credit events (Chapter 25).

Section 9.1 defines exposure, the quantity every adjustment is built on. Sections 9.2 to 9.7 take the adjustments one at a time (credit, debit, funding, margin, collateral, and capital), and Section 9.8 adds them up on one swap. Section 9.9 treats wrong-way risk and Section 9.10 the desk that manages all of it. Section 9.11 confronts the fact that the same trade now carries three defensible values, and Section 9.12 asks what that does to the idea of a price, where the framework falls short, and where it came from.

Two warnings are needed at the outset. First, XVA is not a refinement: on a long-dated uncollateralized trade the adjustments can exceed the bid–ask spread by more than an order of magnitude and dominate the economics of the trade. Second, this is the least settled material

in the book: the components are not agreed, some may double-count, and one of them, the subject of Section 9.4, has been argued to have no business existing at all. The disputes are set out rather than resolved.

9.1 EXPOSURE

An **exposure** is the amount a party would stand to lose if its counterparty failed at a given moment, before any recovery. It is not the notional, which in a swap is never at risk, nor the portfolio's value, which can be negative. It varies with the date, and collateral held against it reduces it; the definition below leaves collateral out until equation (9.2) subtracts it. If a counterparty defaults at time t, the loss depends on what the portfolio is worth at that moment. If the portfolio has *negative* value (the dealer owes money), there is no loss; the obligation survives the counterparty's default and must still be paid. If the portfolio has *positive* value, the dealer joins the queue of unsecured creditors and recovers a fraction.

The exposure is therefore

$$E(t) = \max\left(V(t),\, 0\right) \tag{9.1}$$

where $V(t)$ is the value at time t of the trades held with that counterparty, taken together and signed from the dealer's point of view, so that $V(t) > 0$ means the counterparty owes the dealer. Which trades may be taken together is a legal question, taken up below. Equation (9.1) is the payoff of a call option on that value struck at zero: *counterparty risk is an option*, and everything that makes options hard (volatility, correlation, the need to model a distribution rather than a point) applies to it.

Three summaries of the exposure distribution are used:

- **Expected exposure** (EE), the average over scenarios at each future date: $\mathrm{EE}(t)$ is the average of $\max(V(t), 0)$.
- **Expected positive exposure** (EPE), the average of $\mathrm{EE}(t)$ over the life of the trade, a single number summarizing the profile. Banks using the internal model method capitalize a variant, effective EPE (Section 27.5).
- **Potential future exposure** (PFE), a high quantile of the exposure distribution, used for credit limits rather than valuation.

How exposure is computed

Obtaining $\mathrm{EE}(t)$ means obtaining the distribution of the portfolio's value at every future date, by a Monte Carlo simulation in three stages.

1. *Evolve the risk factors.* Simulate paths of the underlying risk factors (interest rate curves, FX rates, credit spreads, commodity prices) from today to T on a chosen time grid.
2. *Revalue the portfolio.* On every path, at every grid date, revalue every trade in the netting set. This is the expensive step: N paths $\times$ m dates $\times$ K trades valuations.
3. *Aggregate.* On each path and date, net across trades, subtract collateral, take the positive part, and average across paths.

   Formally, for path j and date t_i,

$$E_j(t_i) = \max\left(\sum_{k=1}^{K} V_k^{(j)}(t_i) - C^{(j)}(t_i),\; 0\right), \qquad \mathrm{EE}(t_i) = \frac{1}{N}\sum_{j=1}^{N} E_j(t_i) \tag{9.2}$$

where $C^{(j)}(t_i)$ is the collateral held on that path and date.

The maximum in (9.2) is taken *after* netting across trades and subtracting collateral. Taking it trade by trade and summing overstates exposure, by convexity (the netting inequality (9.4) below), in some portfolios by a large multiple.

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Computing cost is what limits the calculation, which is among the largest a dealer runs: 5,000 paths and 100 dates on a netting set of 2,000 trades is a billion trade valuations for one counterparty, repeated across thousands of counterparties nightly. Two techniques make it feasible. **American Monte Carlo** (the least-squares regression of Section 20.11, borrowed for its regression step rather than for the early-exercise decision) replaces exact revaluation with a fitted function of the state variables. **Adjoint algorithmic differentiation** (AAD, Section 20.10) produces every sensitivity in a small multiple of one valuation's cost, and the desk of Section 9.10 hedges the charge's sensitivity to every market input, not only its level.

The shape of an exposure profile

For an interest rate swap the exposure profile has a characteristic hump. Two forces work against each other. *Diffusion*: the longer rates have had to move, the wider the range of possible values, which pushes exposure up roughly with $\sqrt{t}$. *Amortization*: as payment dates pass, the swap's sensitivity to rates falls, which pulls exposure down roughly linearly. The product peaks at about one third of the way through the trade.

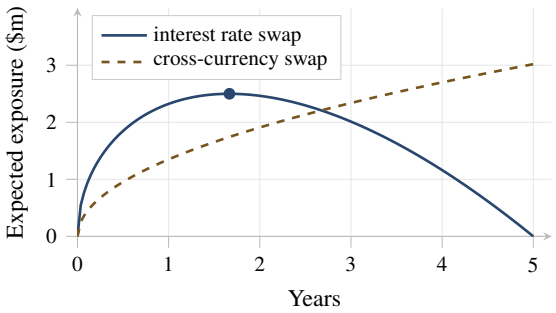


Figure 9.1: Expected exposure for a five-year interest rate swap and for a cross-currency swap (the latter schematic).

Figure 9.1 contrasts this with a cross-currency swap. There the notional is exchanged at maturity (Chapter 7), so the largest cash flow is the last one and nothing amortizes: diffusion operates unopposed and exposure rises to maturity. A cross-currency swap therefore carries more counterparty risk than an interest rate swap of the same notional and tenor, and carries it later, when the counterparty has had longest to default. The curve drawn is schematic and understates the difference (on Example 9.2's inputs the two curves give CVAs only about 9% apart), because the final exchange of notional is exposed to the whole move in the exchange rate, which at ordinary currency volatilities puts a real profile well above the one drawn. Between dealers the effect is usually contained by a mark-to-market resetting cross-currency swap, which adjusts one notional to the current exchange rate at each reset and so returns the currency move to the counterparty period by period (Section 7.8; Chapter 33 prices it); the corporate swaps that carry most cross-currency CVA rarely reset.

Other products have other shapes. A purchased option has exposure equal to its full value throughout, since that value can never be negative, while a sold option has none once the premium is received, so two trades that look symmetric carry radically asymmetric counterparty risk. An FX forward rises with $\sqrt{t}$ to maturity like the cross-currency swap, because nothing amortizes. A repo or fully collateralized financing is exposed only to the extent that the collateral can lose more than the haircut over the margin period of risk, which is why Chapter 8's haircuts belong to credit as well as to liquidity. In general, exposure grows with the uncertainty remaining and shrinks with the obligations discharged.

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Expected exposure and potential future exposure

Expected exposure and potential future exposure serve different purposes, and confusing them produces numbers a factor of four apart (Example 9.1). Suppose that at the peak of the profile the portfolio value is normally distributed about zero with standard deviation σ. Then the expected exposure has a closed form,

$$\mathrm{EE} = \int_0^\infty v \, \frac{1}{\sigma\sqrt{2\pi}} \, e^{-\frac{v^2}{2\sigma^2}} \, dv = \frac{\sigma}{\sqrt{2\pi}} = 0.3989\,\sigma \tag{9.3}$$

while the potential future exposure at the 95% level is the corresponding quantile of the positive part, $1.645\,\sigma$.

Example 9.1 Expected against potential future exposure

The swap of Figure 9.1 peaks at an expected exposure of \$2.5 million. From (9.3) the implied standard deviation of the portfolio's value at that date is

$$\sigma = \frac{2.5}{0.3989} = \$6.27 \text{ million}$$

so the 95% potential future exposure is

$$\mathrm{PFE}_{95} = 1.645 \times 6.27 = \$10.3 \text{ million}$$

roughly four times the expected exposure.

Both numbers describe the same trade, and both are used, for different things. *Pricing* uses expected exposure, because the hedge of Section 9.2 is sized to the average and its cost is what the client is charged. *Credit limits* use potential future exposure, because a limit is meant to cap a bad case rather than an average one. In practice the two are usually simulated differently as well: exposure for pricing from risk factors calibrated to market prices such as implied volatilities, exposure for limits from dynamics estimated on history.

A credit officer refusing a trade on a \$10.3 million PFE and a trader pricing it off a \$2.5 million EE are not disagreeing; they are answering different questions.

Netting matters, and it matters at the right level. Exposure is computed on the netting set (the group of trades covered by one master agreement), not trade by trade:

$$E(t) = \max\left(\sum_i V_i(t),\, 0\right) \;\le\; \sum_i \max\left(V_i(t),\, 0\right) \tag{9.4}$$

The inequality is strict whenever some trades have positive values and others negative ones, so a new trade's contribution to exposure can be *negative*: a trade that offsets the existing portfolio reduces the netting set's exposure and with it the CVA charge. In the limiting case, a client that already pays fixed on a swap and now asks to receive fixed on identical terms closes the netting set's exposure to zero, and the new trade's incremental CVA is minus the whole CVA of the existing one. A dealer that quotes a better price for a trade that hedges its existing exposure to a client than for one that adds to it is not giving a discount; it is pricing correctly.

9.2 CVA

The **credit valuation adjustment** is the market value of the counterparty's option to default: the amount by which a trade is worth less than the same cash flows promised by a counterparty that cannot fail. Everything in this book is priced the same way: the value of a position is what it costs to hedge it, a principle Chapter 13 makes exact. Here it applies directly, because the hedge for this option already trades. A credit default swap on the counterparty (introduced in Chapter 1, named in Chapter 7, and analyzed in Chapter 25) pays its buyer the loss on a stated

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amount if that counterparty defaults, in exchange for a running premium, so the premium rate, the **CDS spread**, is the market's price for insuring one dollar owed by that counterparty for one year. CVA is the cost of insuring the exposure. No probability is estimated; the spread is a price, and it is used as one. The formula is worth building rather than quoting, because everything questionable about CVA enters as an assumption along the way.

What a default costs. Suppose the counterparty defaults at some time τ before the final maturity T. Write $V(t)$ for the portfolio's value assuming no default, and R for the **recovery rate**, the fraction of its claim that a creditor eventually collects from the estate, so that a default costs it $1 - R$, the fraction the credit chapters call the loss given default. Recovery is treated here as a known constant, which it is not: it varies with seniority and with the state of the economy, and Chapter 25 describes how the market fixes it by auction. As Section 9.1 showed, the loss is one-sided, which is what makes it an option:

$$L = (1 - R)\max\left(V(\tau),\, 0\right) \quad \text{if } \tau \le T, \qquad \text{and zero otherwise.} \tag{9.5}$$

What insuring it costs. A CDS with notional N pays $(1 - R)N$ on default, the same $1 - R$, because the protection seller makes good the loss on a bond of that face amount. A dealer that knew the exposure path in advance could buy protection on an amount equal to $E(t)$ for each future date, and the payout would match the loss in (9.5) exactly. The premium is the spread s times the amount covered, per year, paid only while the counterparty is still alive. The cost of the hedge is therefore a stream of premiums, $s \times E(t)$ a year, running until maturity or default, whichever comes first, and CVA is the present value of that stream. Two assumptions turn this into something a desk can compute.

Assume the amount to insure is the average exposure. The exposure is a distribution at each date, not a known path, and the dealer insures its average, the expected exposure $\mathrm{EE}(t)$. That is the right amount only if the size of the exposure when the counterparty fails is unrelated to the failure itself; where the exposure is largest in the states in which the counterparty is likeliest to fail, the average understates what needs cover (Section 9.9).

Assume discounting separates from exposure. This is exact under fixed rates and otherwise a good approximation, the link between the discount factor and the exposure being second order.

Discretizing on a grid, $0 = t_0 < t_1 < \cdots < t_n = T$, and writing $c(t_{i-1}, t_i)$ for the cost of covering one dollar of exposure over the interval (the premium for that interval, scaled by the fraction of scenarios in which the counterparty is still alive to be insured), and $\mathrm{DF}(t_i)$ for the discount factor to t_i:

$$\boxed{\mathrm{CVA} = \sum_{i=1}^{n} \mathrm{EE}(t_i)\; c(t_{i-1}, t_i)\; \mathrm{DF}(t_i)} \tag{9.6}$$

The spread is the *counterparty's*: CVA prices the risk that it fails, and Section 9.3 turns the same calculation on the dealer's own. It comes from market spreads (the counterparty's own CDS or bonds where they trade, otherwise a proxy curve for its rating, sector, and region) rather than from historical default rates.

Chapter 24 arrives at the same cost from the other side. A spread of s with a loss given default of $1 - R$ means the market is charging as though the counterparty fails at a rate of about $\lambda = s/(1 - R)$ a year, a relation Chapter 24 calls the credit triangle; with that rate, the cost of cover for an interval is the loss on default times the chance of a default falling in it,

$$c(t_{i-1}, t_i) = (1 - R)\, q(t_{i-1}, t_i), \qquad q(t_{i-1}, t_i) = e^{-\lambda t_{i-1}} - e^{-\lambda t_i}.$$

The two readings must agree, because a CDS is priced so that its premiums and its payouts balance, and Example 9.2 checks that they do.

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Example 9.2 Computing CVA on a swap

A dealer has an uncollateralized five-year swap with a corporate client on a notional of $100 million. The expected exposure profile is the interest rate swap curve of Figure 9.1, and equation (9.3) produces it. By date t the swap's value has moved by roughly the change in the swap rate times the notional L times the annuity of the payments still to come, about $T - t$ years of them; the rate change has a standard deviation of $\sigma_r \sqrt{t}$, so the value is roughly normal about zero with standard deviation $L\,\sigma_r \sqrt{t}\,(T - t)$, and

$$\mathrm{EE}(t) = 0.3989 \times L\,\sigma_r \sqrt{t}\,(T - t)$$

With L = $100 million, T = 5 years and a rate volatility σ_r of 1.456% a year, the profile rises with $\sqrt{t}$ as the rate diffuses and falls as the swap amortizes, and at mid-year points it gives the expected exposure column of the table below. Section 20.6 confirms it by simulation, at the same volatility. The client's CDS trades at 120 basis points a year; recovery is assumed to be 40%, so the loss given default is 60%, and the credit triangle puts the market's default rate for this issuer at $0.012/0.60 = 2\%$ a year. Discounting is at 4% continuously compounded.

Equation (9.6) is evaluated on an annual grid, using mid-year exposures and discount factors. The cost of cover in each year is the 120 basis point premium scaled by the fraction of scenarios in which the client is still alive to be insured, $e^{-0.02\times0.5} = 0.990$ in the first year and falling by 2% a year after that:

Year	EE ($m)	Cost of cover	Discount factor	Contribution ($m)
1	1.848	1.188%	0.9802	0.02152
2	2.490	1.165%	0.9418	0.02731
3	2.296	1.141%	0.9048	0.02371
4	1.630	1.119%	0.8694	0.01585
5	0.616	1.097%	0.8353	0.00564
			CVA	0.09404

$$\mathrm{CVA} = \$94{,}000$$

That is 9.4 basis points of notional, charged up front. It is not a reserve the dealer sets aside from prudence; it is the market price of an option the dealer has written (the client's option to walk away), and if the dealer does not charge it, the trade is mispriced from inception.

The cost-of-cover column is also what Chapter 24's reading gives. In the first year the chance of a default is $1 - e^{-0.02} = 1.980\%$, and $0.60 \times 1.980\% = 1.188\%$, the figure in the table; the same holds in every row to the third decimal. Insuring the exposure and bearing the expected loss on it cost the same, as they must when the insurance is fairly priced.

Years 2 and 3 dominate, because that is where the exposure hump sits. A trade whose exposure peaked later, like the cross-currency swap of Figure 9.1, would carry a larger CVA for the same notional and the same counterparty: about $102,000 with that curve as drawn, some 9% more, and more again at realistic currency volatility.

Why the spread, and not the record. The cost of cover in the table is about 1.2% a year. Borrowers that trade at that kind of spread have, historically, cost their creditors much less, by a factor of several for an issuer of this quality and far more for the safest credits (Section 24.9). Concluding that the CDS market overprices, and charging the client the historical figure, would be a mistake. The spread is not a forecast of how often the counterparty will fail; it is what a seller of protection demands for a risk that pays off in exactly the bad times, when the seller least wants to pay, and so it includes compensation for that timing, for the seller's own funding and capital, and for the possibility that this counterparty turns out worse than its history. CVA uses the spread because CVA is what the hedge costs, not what the loss is expected to be. A dealer that charged historical CVA would undercharge by that same several-fold factor, and

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would find every hedge it bought costing more than the revenue meant to pay for it. Chapter 28 shows that the same is true of every price in this book: derivatives are priced by what they cost to replicate, and the probabilities a hedging cost implies are not the probabilities a statistician would estimate.

9.3 DVA

If the counterparty's option to default has value, so does the dealer's. The **debit valuation adjustment** (DVA) is the mirror image of CVA: the value to the dealer of its own option to default, computed on the *negative* exposure (the states in which the dealer owes money and would not pay it) at the dealer's own CDS spread, exactly as the counterparty would compute CVA on the dealer.

DVA has an impeccable derivation and an absurd implication. It is required for symmetry: if the client charges the dealer CVA for the dealer's credit risk, and the dealer charges the client CVA for the client's, then each party's CVA is the other's DVA, and the two agree on the trade's value. The implication is that a deterioration in the dealer's own credit is a gain. As the dealer's spread widens, its own option to default becomes more valuable, DVA rises, and the dealer books a profit.

Case Study 9.1 Profiting from your own decline

In 2011, as bank credit spreads widened sharply in the euro area crisis, several large banks on both sides of the Atlantic reported substantial DVA gains. The accounting was correct and the logic unimpeachable: their liabilities, including their derivative liabilities, had become less likely to be paid in full and were therefore worth less, and a fall in the value of a liability is a gain.

The presentational problem was obvious (institutions reporting profits derived from the market's declining confidence in them), and the prudential problem was worse. Capital exists to absorb losses, and capital that increases as the firm's creditworthiness falls does the opposite: a firm approaching failure would report its largest DVA gains just as its capital was most needed.

Regulators resolved this by filtering DVA out of regulatory capital: the gain may be recognized in accounting profit but is deducted from common equity. The effect is a permanent gap between the accounting value of a derivatives book and its regulatory value.

Much of the 2011 gain was not on derivatives at all but on the banks' own bonds carried at fair value, and the accounting standards have since moved that part out of profit and into other comprehensive income (IFRS 9 and, in the United States, Accounting Standards Update (ASU) 2016-01, both required from 2018). The DVA on derivative liabilities still runs through profit.

Most dealers treat DVA as real for valuation and hard to use for management. It cannot be hedged directly, since that would mean selling protection on oneself, which no one will buy in the size required; banks have hedged it approximately by selling protection on peer banks whose spreads move with their own, taking on those banks' credit. And it is realized, if at all, mostly by defaulting, or in part through the funding benefit of Section 9.4, where the two adjustments overlap. It therefore appears in the accounts and carries little weight in pricing.

Two refinements matter for a precise bilateral calculation. Each party's charge should be conditioned on the other surviving (the first-to-default effect), which makes bilateral CVA and DVA each slightly smaller than their one-sided versions. And the amount claimed on default depends on the close-out convention: if the surviving party may claim the cost of replacing

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the trade with a new counterparty, that replacement value includes the surviving party's own DVA, and the loss on default is smaller than a risk-free close-out implies. The 2002 ISDA Master Agreement frames its close-out amount as a replacement cost, which points toward the replacement reading, though how far in practice is not settled.

9.4 FVA

The **funding valuation adjustment** is the cost of funding an uncollateralized derivative position, and it is the most contested item in this chapter. Section 8.1 gave the mechanism: the dealer hedges the uncollateralized client trade with a margined one, so when the client trade moves in the dealer's favor the hedge calls variation margin in cash that nothing from the client offsets, and the difference has to be borrowed. If the trade is against the dealer, the dealer keeps cash that a CSA would have required it to hand over, while the hedge returns margin. Either way the trade leaves a funding position (a cost in the first case, a benefit in the second), carried at the dealer's own cost of funds, which is above the risk-free rate by its funding spread. Under the one-way CSA of Section 8.1, which sovereigns, supranationals and central banks typically insist on, the dealer must pass to the client the margin its hedge returns, so the second case disappears and FVA is all cost.

Example 9.3 Computing FVA

Take the swap of Example 9.2 and suppose the dealer's funding spread over the risk-free rate is 60 basis points. The discounted expected exposure uses the exposure and discount columns of Example 9.2 with the cost of cover left out, each year weighted by its length $\Delta t_i = 1$:

$$\sum_i \mathrm{EE}(t_i)\,\mathrm{DF}(t_i)\,\Delta t_i = 1.848 \times 0.9802 + 2.490 \times 0.9418 + 2.296 \times 0.9048$$
$$+ 1.630 \times 0.8694 + 0.616 \times 0.8353 = \$8.17 \text{ million-years}$$

The funding cost is

$$\mathrm{FVA} = 0.0060 \times \$8.17\text{m} = \$49{,}000$$

So on this trade FVA is about half of CVA. This is the cost side alone, the funding of positive exposure; the benefit on negative exposure is taken up below. The ratio is not general. It is roughly the funding spread against the counterparty's credit spread, here 60 basis points against 120, and it comes out a little above one-half because the cost of cover decays with survival while the funding cost, as computed here, does not. But the example makes the point that funding is the same order of magnitude as credit, not a rounding item.

Case Study 9.2 The FVA debate

In 2012 Hull and White argued, in a widely read article, that FVA should not exist.

Their argument applies a principle this book has used throughout: the value of an asset is determined by its risk, not by how the buyer chose to finance it. A firm with a high funding cost does not get to charge more for the same cash flows; that the cheapest-funded firm would win every trade is a statement about competition, not about value. Charging FVA, on this view, confuses the dealer's funding problem with the trade's worth, and amounts to passing the dealer's own credit risk to the client twice, since the client is already compensated through DVA.

The practitioner response is that this is correct in a frictionless world and irrelevant in this one. A dealer that ignores FVA and prices at theoretical value will win exactly the trades that are expensive for it to fund, and will discover the cost in its funding line rather than in its trading book. Whatever the theory says, the cash outflow is real.

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The debate was settled, in the way such debates usually are, by accounting. Firms began recognizing FVA in their published results (most visibly when JPMorgan took a $1.5 billion FVA charge in the fourth quarter of 2013), and once one major dealer had recognized it, the others followed rather than explain the discrepancy.

The theoretical question was answered later, and the answer gives each side part of what it claimed. Andersen, Duffie, and Song (2019) showed that funding an uncollateralized position transfers value from shareholders to the bank's creditors, so FVA is a real cost to the shareholders who decide whether to trade, and belongs in the price the desk quotes; but it is not a property of the cash flows, and the trade's value to all of the bank's stakeholders together does not contain it. It is worth understanding both positions, because the same argument recurs whenever a cost that is real to a firm is not a property of the instrument it trades.

FVA and DVA overlap. A genuine technical problem underlies the dispute. The dealer's funding spread reflects its own credit risk, which DVA also prices, so charging both counts the same thing twice, at least in part. Practice varies: some firms compute FVA using only the liquidity component of their funding spread; others restrict FVA to a funding cost adjustment on positive exposure and set the corresponding funding benefit adjustment on negative exposure against DVA. No treatment is universally agreed, and two dealers can quote materially different prices on identical trades for this reason alone.

9.5 MVA

The **margin valuation adjustment** is the cost of funding initial margin over the life of a trade. It exists because of the rules Chapter 8 described: initial margin is posted gross, segregated, and cannot be reused, so it must be funded from inception to maturity.

Example 9.4 Computing MVA

Suppose the swap of Example 9.2 is instead cleared. The clearing house sizes initial margin to cover the swap's value change over a five-day close-out at a high confidence level, 99% or more; at the rate volatility behind Figure 9.1 the five-day standard deviation of the value is about $1 million, so the margin starts at about $2.5 million and declines roughly linearly to zero as the trade amortizes, an average of $1.25 million. MVA is charged on the margin the trade adds to the dealer's cleared portfolio, and this dealer's cleared book is assumed not to offset the trade, so the increment is the whole stand-alone amount; Example 8.5's dealer, whose cleared leg nets against its existing book, would pay far less. The dealer funds the margin at the same 60 basis point spread.

A charge accruing at a constant rate over the trade's life is worth that rate times the trade's discounted life, which at the 4% used throughout this chapter is

$$\int_0^5 e^{-0.04t}\,dt = \frac{1 - e^{-0.2}}{0.04} = 4.53 \text{ years}$$

Using that discount-weighted duration,

$$\text{MVA} = 0.0060 \times \$1.25\text{m} \times 4.53 = \$34{,}000$$

Compare this with the figures already computed: CVA of $94,000 and FVA of $49,000 (Examples 9.2 and 9.3). On this trade the cost of funding initial margin is about a third of the credit charge it replaces.

The proportion is not fixed. MVA scales with the margin model's output and with the funding spread, and both rise in exactly the conditions that raise everything else: a stressed lookback period in the margin model, a ten-day close-out for an uncleared trade instead of five, a longer maturity, or a dealer whose own funding has become expensive. Practice Question 8 doubles the margin and raises the spread by half, and MVA overtakes CVA. Clearing and the uncleared margin rules

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> replaced part of the credit risk they removed with a funding requirement, and a firm that dropped its CVA desk after mandatory clearing, on the grounds that counterparty risk had gone, would have been charging nothing for a cost that had changed form rather than disappeared.

MVA is also awkward to compute, because it requires forecasting a *margin model's output* on future dates and scenarios: a clearing house's own initial margin model for a cleared trade, SIMM for an uncleared one under the rules of Section 8.4. Under the uncleared rules the cost is also marginal in a way the cleared one is not: two firms exchange no initial margin until the requirement between them passes the €50 million threshold, so a trade that fits under it carries no MVA, and once the threshold is used up every further trade carries its full margin.

On the same grid as (9.6),

$$\mathrm{MVA} = \sum_{i=1}^{n} s_F(t_i)\, \overline{\mathrm{IM}}(t_i)\, \mathrm{DF}(t_i)\, \Delta t_i \tag{9.7}$$

where $s_F(t)$ is the dealer's funding spread over the rate paid on posted margin (60 basis points in Example 9.4), $\overline{\mathrm{IM}}(t_i)$ is the initial margin outstanding at t_i, averaged across scenarios, and $\Delta t_i = t_i - t_{i-1}$. Like the other adjustments, this is written as a positive cost; Section 9.8 carries the signs.

The difficulty in (9.7) is entirely in $\overline{\mathrm{IM}}(t_i)$: computing it exactly means running the margin model *inside* the Monte Carlo, evaluating sensitivities on every path and date and aggregating them through the margin formulas, which multiplies an already expensive calculation by the cost of a full sensitivity run. Three approximations are used.

A deterministic profile scales today's initial margin by the remaining risk (for a swap, roughly by remaining duration). That is what Example 9.4 does with its linear decline, and it is adequate while the portfolio's composition is stable.

Regression fits IM as a function of the simulated state variables on a subset of paths and evaluates the fitted function everywhere: the regression Section 9.1 borrowed from American Monte Carlo for exposure, turned on the margin model rather than on value.

Sensitivity scaling applies the model's risk weights to the sensitivities simulated on each path, but replaces the step that combines them through the model's correlation formulas, which depends on the whole portfolio on that path, with a simpler fixed rule.

All three approximate a quantity that is itself an approximation, a standardized margin model being no more than a proxy for a risk measure.

9.6 COLVA AND THE TERMS OF THE CSA

Chapter 8 showed that a CSA is rarely the idealized arrangement in which collateral exactly tracks the mark-to-market in cash, daily, in one currency. Real CSAs have thresholds, minimum transfer amounts, non-cash eligible collateral, and currency choice, and each departure has a value. Strictly, the **collateral valuation adjustment** (ColVA) is the value of the terms on which collateral is remunerated and chosen: a collateral rate (or, for non-cash collateral, a return) other than the overnight rate the curve is built on, a floor on that rate, and the right to choose what to post, such as the cheapest of several currencies, an option over the discount curve that Example 8.6 values. Thresholds, minimum transfer amounts, and the margin period of risk (the gap between the last collateral movement and close-out) act differently. They leave exposure uncovered, a $5 million threshold being a $5 million unsecured credit line, so their cost is CVA and FVA computed on the exposure that remains; adding it again under ColVA would count it twice. Desks discuss the whole set together, as the terms of the CSA, and so does this section. Benchmark reform moved most CSAs' cash rate from EFFR or EONIA to SOFR or €STR, and since 2016 the uncleared margin rules have required daily variation margin between the firms

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they cover, so the standard CSA has drifted toward the idealized one and the residual terms matter most on legacy agreements.

Thresholds, minimum transfer amounts, and the margin period of risk enter the simulation through one term, the $C^{(j)}(t_i)$ of (9.2), and the modeling has a single essential feature: collateral reflects the portfolio's value at an earlier date. Writing δ for the margin period of risk, H for the threshold and M for the minimum transfer amount, and setting to zero the independent amount that Chapter 8's credit support amount also includes,

$$C(t) = \max\left(V(t-\delta) - H,\ 0\right) \quad \text{subject to changes of at least } M \tag{9.8}$$

Equation (9.8) describes collateral the dealer *holds*. Under a two-way CSA the dealer also posts when V is negative, symmetrically, and posted collateral matters in two ways the equation omits: it is an exposure of its own if the counterparty does not segregate it, and it must be funded, which is where FVA enters a collateralized trade.

The consequence is that a perfect CSA does not give zero exposure. Even at $H = M = 0$, equation (9.8) leaves $C(t) = \max(V(t-\delta), 0)$: the collateral held reflects the value one margin period ago, so the collateralized exposure is at most the portfolio's move over that window, and exactly that whenever the dealer was holding collateral,

$$E(t) = \max\left(V(t) - V(t-\delta),\ 0\right)$$

At Chapter 8's ten days bilateral and five days cleared, this residual is why a fully collateralized trade still carries a small CVA rather than none. Initial margin is sized to cover exactly this move, which is what it is for, and why it costs what Section 9.5 says it does.

Example 9.5 The cost of a threshold

The swap of Example 9.2 is with a corporate client outside the uncleared margin rules, which would otherwise require a zero threshold (Chapter 8). The client insists on a \$5 million threshold in its CSA, below which no collateral moves. The dealer agrees, believing the CSA to be otherwise standard.

For exposures up to \$5 million the trade is effectively uncollateralized. In most scenarios the exposure stays below that level, so on the normal model of Section 9.1 the expected exposure capped at \$5 million averages about \$1.4 million over the mid-year dates, against \$1.78 million uncapped. Insuring that amount at the client's 120 basis point spread for the trade's discounted life carries

$$\text{CVA on the threshold} \approx 0.012 \times \$1.4\text{m} \times 4.53 = \$76{,}000$$

plus funding on the same amount at 60 basis points,

$$0.0060 \times \$1.4\text{m} \times 4.53 = \$38{,}000$$

A concession that sounded operational ("we would rather not exchange small amounts") costs about \$114,000 on a single trade, most of the way to the uncollateralized CVA and FVA. (Weighting the premiums by survival, as Example 9.2 does, would take a few thousand dollars off; at this precision it does not matter.) Thresholds are priced, and clients are frequently surprised by how much.

9.7 KVA

The **capital valuation adjustment** (KVA) is the cost of the regulatory capital a trade requires over its life: the equity a bank must hold against the position because the rules say so, rather than because it chose to. How much capital a trade requires is Chapter 27's calculation (counterparty exposure under the standardized approach for counterparty credit risk (SA-CCR) in Section 27.3, the charge against CVA's own volatility in Section 27.6), and Example 9.6 takes the result as given.

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Capital is not free. Shareholders require a return above the risk-free rate (a hurdle of perhaps 8 to 12% over it), and a trade that consumes capital for ten years consumes that return for ten years. If the trade does not earn it, the shareholders are worse off even though the trading book shows a profit.

Example 9.6 Computing KVA, and its size

Example 27.3 computes the counterparty credit capital for the uncollateralized swap of Example 9.2 at $0.33 million at inception, against a corporate at a 100% risk weight. Counterparty capital rises in the first years as replacement cost builds, then falls as the swap runs off. The example takes the case in which the client, a nonfinancial corporate, is exempt from the CVA capital charge, as it is under the EU rules (Section 27.7), so the capital is counterparty credit capital alone; suppose it averages $0.4 million over the swap's life, and that the firm's hurdle rate over the risk-free rate is 10%. Then

$$\text{KVA} = 0.10 \times \$0.4\text{m} \times 4.53 = \$181{,}000$$

This is about twice the CVA and nearly four times the FVA. Where the CVA capital charge applies, as it does to the same trade in the United States and Canada (Section 27.7), it adds roughly as much capital again and KVA is materially higher. On long-dated uncollateralized trades KVA is often the largest adjustment of all, and it is the reason such trades became expensive and less common after the post-crisis capital rules took effect. Practitioners compute it on a projected capital profile, trade by trade within the portfolio, and include the leverage ratio where it binds, since a bank must hold capital against whichever rule is the stricter.

CVA the price and CVA the capital charge

The same three letters denote two different things, a persistent source of confusion.

CVA as used in Section 9.2 is a *price*: the market value of the counterparty's default option, deducted from the value of the trade.

The **CVA capital charge** is a *regulatory requirement*: capital held against the risk that CVA itself *moves*. Its rationale is empirical: in the financial crisis, by the Basel Committee's count, roughly two-thirds of losses on counterparty exposures came not from counterparties actually defaulting but from the mark-to-market deterioration of CVA as credit spreads widened. Firms held capital against default and were hit by volatility.

The charge is computed under the Basel framework of Section 1.12, either by the basic approach (BA-CVA), a formula on exposures and supervisory risk weights, or, with supervisory approval, by the standardized approach (SA-CVA), which works from CVA's sensitivities to credit spreads and to the market factors driving the exposure. The Basel standard took effect in 2023, in the European Union in 2025 and in the United Kingdom in 2027; the US agencies re-proposed theirs in March 2026, and they were not final at the time of writing. Section 27.6 sets out the approaches and Section 27.7 the exemptions. The charge feeds into KVA, since it is part of the capital the trade consumes.

The distinction matters because the two respond to different actions. Single-name CDS protection on a counterparty hedges the price and reduces the charge (under BA-CVA by at most three-quarters), and novating the trade to a clearing house eliminates both. But for most counterparties no single-name CDS trades (Section 9.10), and the index or proxy protection a desk can actually buy hedges the *price* while earning only partial relief from the *capital charge*. Tightening a CSA lowers both by different amounts: the price by what the firm's own exposure model says, the charge by what the standardized formulas allow.

KVA is the most conceptually troubled of the adjustments. Capital requirements are set by regulators and change; the hurdle rate is a management choice, not a market price; and including a firm's cost of equity in the valuation of an instrument sits uneasily with the idea

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that value is a property of the instrument. Some firms treat KVA as a pricing hurdle (a floor below which they will not trade) rather than a valuation adjustment carried in the accounts, which is arguably the more honest treatment.

9.8 PUTTING THEM TOGETHER

Table 9.1 collects the adjustments for the trade used throughout this chapter, under two collateral arrangements. Two entries rest on inputs the examples did not state. DVA is computed at the dealer's own spread, taken as 49 basis points, on the mirror image of the client's exposure profile: it assumes the dealer's negative exposure has the same profile as its positive one, which holds for an at-market swap under the symmetric normal model of Example 9.2. And the cleared trade's capital is taken as averaging $28,000, set by the leverage ratio rather than by the risk-based rule: Example 27.3's trade exposure to the clearing house needs only about $1,950 at the 2% risk weight, but at a 3% leverage requirement it needs about $28,000 (Section 27.10). The totals are net of DVA.

Table 9.1: XVA on a five-year $100m swap under two collateral regimes, net of DVA.

Adjustment	Uncollateralized	Cleared	Driver
CVA	−94,000	≈ 0	counterparty default
DVA	+39,000	≈ 0	own default
FVA	−49,000	≈ 0	funding the exposure
MVA	—	−34,000	funding initial margin
KVA	−181,000	−13,000	regulatory capital
Total	**−285,000**	**−47,000**	

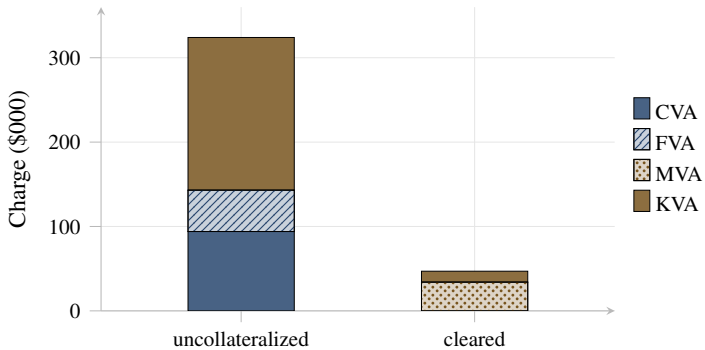


Figure 9.2: Composition of the charge in Table 9.1, gross of DVA.

Figure 9.2 plots the composition, and shows what the totals conceal: the two columns are not the same charge in different sizes but largely different charges. Three observations follow.

The adjustments are large. On the uncollateralized trade they total 28.5 basis points of notional net of DVA. DVA carries little weight in pricing (Section 9.3), so the charge a client is actually quoted is nearer the $324,000 before it: 32.4 basis points of notional, or about 7 basis points a year on the swap rate, against an interdealer bid–ask spread on a five-year swap of a fraction of a basis point. The XVA is not a correction to the price; on this trade it substantially *is* the price.

Clearing changes the composition of the charge rather than eliminating it. Credit and funding charges collapse toward zero, and are replaced by the cost of funding initial margin. The total falls (clearing is genuinely cheaper, as Chapter 8 explained it was designed to be),

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but not to zero. For this client the cleared column is a comparison rather than a choice: a nonfinancial corporate hedging its own business is usually exempt from mandatory clearing and from the uncleared margin rules (the end-user exception in the United States, the clearing threshold in the European Union), which is why uncollateralized corporate swaps still exist and why the left-hand column is the one a corporate treasurer is quoted.

Capital dominates. On the uncollateralized trade KVA is larger than everything else combined. That explains market structure changes that otherwise look puzzling: why dealers pushed clients into clearing, why long-dated uncollateralized corporate trades became expensive, and why balance-sheet-light business models proliferated.

9.9 WRONG-WAY RISK

Wrong-way risk arises when exposure increases in the same states in which the counterparty is more likely to default. The textbook example is buying credit protection on a sovereign from a bank in that country: the protection is worth most when the sovereign is in trouble, which is exactly when the bank is least able to pay. Chapter 25 shows this was not hypothetical.

Less dramatic cases are common. A dealer that has bought protection against falling oil prices from an oil producer has exposure that rises when oil prices fall, which is when the producer's credit is worst. A dealer receiving fixed from a highly levered borrower has exposure that rises as rates fall, which may correlate with the conditions that stress the borrower.

Right-way risk is the reverse and reduces CVA: a dealer that has bought protection against *rising* oil prices from the producer has exposure rising when oil rises, which is when the producer is strongest.

The insurance argument shows exactly where the first assumption of Section 9.2 did its work. What the dealer needs cover for is the exposure *at the moment the counterparty fails*, the expected exposure *given that* the default falls in each interval, and the standard calculation replaces it with the unconditional average. Under wrong-way risk the exposure in the scenarios where the counterparty actually defaults is systematically larger than the average across all scenarios, so the amount insured is too small, and (9.6) is a lower bound rather than an estimate.

Two modeling approaches are standard, and regulation avoids the question:

- *Intensity models* let the counterparty's default rate rise and fall with the driver of the exposure.
- *Copula approaches* join the default time and the exposure while leaving each one's own distribution as it was, the device Section 26.6 turns on portfolio credit.
- *Regulation* sidesteps both, covering general wrong-way risk with the multiplier of 1.4 (Section 27.5) and specific wrong-way risk with a rule that treats a legally connected reference entity as already defaulted (Section 27.9).

Example 27.4 puts a size on the effect: a dependence strong enough to double the exposure expected at default doubles the CVA. Where the dependence is definitional rather than statistical, no correlation parameter is adequate. Protection bought on a sovereign from a bank domiciled in it is not an exposure with a correlation adjustment; the exposure and the default are the *same event*, and the trade should be assessed as a jointly contingent claim.

9.10 THE XVA DESK

The organizational response to all this is a central **XVA desk**. When a trader books a client trade, the XVA desk charges the relevant adjustments and takes the corresponding risk onto its own book. The originating desk is left with a clean, hedgeable market risk position; the XVA desk holds the credit, funding, and capital exposures across the whole firm. The centralization exists because these risks are *portfolio* properties, not trade properties: exposure is a netting-set

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quantity (9.4), so a new trade's CVA depends on everything already facing that counterparty, and only a desk that sees the whole book can price the increment.

Hedging. The XVA desk hedges CVA by doing what Section 9.2 priced: buying credit protection on its counterparties, and hedging the sensitivity of the exposure itself to rates and FX. Because the exposure is an option, CVA has a delta and a vega: sensitivities to the underlying market variable and to its volatility. Hedging CVA is harder than it sounds: liquid CDS exists for a few hundred issuers, while a dealer faces thousands of counterparties, most with no traded credit, so much CVA is hedged by proxy, with index CDS and correlation assumptions, leaving basis risk whenever a specific issuer diverges from its proxy.

Other instruments fill parts of the gap. A **contingent CDS** pays the loss on a notional that is itself defined by a referenced derivative's value at default, so it insures the exposure exactly rather than its average, and is bought where one large exposure justifies the bespoke cost. Banks also transfer CVA risk in portfolio form, through credit-linked notes and guarantees sold to investors such as insurers and credit funds that want the spread and can carry the risk. And what a desk hedges follows the capital charge as well as the price: SA-CVA recognizes hedges of the exposure's rate and FX sensitivities and BA-CVA does not (Section 27.6), so a desk under SA-CVA hedges the market side of CVA as well as the credit side.

FVA and MVA are hedged, to the extent they are, by managing the firm's own funding profile; in most banks the treasury charges them to the desks through funds-transfer pricing, at an internal rate for the term and liquidity of the funding each trade consumes. KVA is generally not hedged at all; it is managed by not doing the trades.

Why the XVA desk loses money in a crisis

An XVA desk that has hedged carefully can still report large losses, for three recurring reasons.

Proxy basis. When a specific counterparty's spread widens while the index used to hedge it does not, the CVA rises and the hedge does not respond. In a broad credit sell-off the proxy works; in an idiosyncratic default it does not, and idiosyncratic defaults are exactly when CVA matters.

Cross-gamma. CVA depends on exposure *and* on credit, and the exposure depends on rates and FX, so a rate move changes CVA even with credit spreads unchanged, and a spread move changes how much a rate move matters. That interaction is CVA's **cross-gamma**. Hedging the credit sensitivity alone leaves that interaction unhedged, and the cross-terms are largest when both move together, which is the wrong-way risk of Section 9.9 as the hedging desk sees it.

Hedging costs. Buying CDS protection consumes capital and funding of its own, so a desk that hedges its CVA fully has converted a credit exposure into a capital and funding exposure, which is an improvement, not a cure.

The deeper issue is that CVA hedging is an attempt to hedge an option on a portfolio whose composition changes daily, written on a counterparty whose credit may not trade, in a market where correlations move when it matters. Doing it imperfectly is not a failure of execution; it is the nature of the problem.

9.11 THREE DIFFERENT NUMBERS

A recurring source of confusion inside banks is that the same trade carries three XVA figures that are all correct and do not agree.

The accounting number is what appears in published financial statements. It must satisfy fair value accounting standards (IFRS 13 internationally and ASC 820 in the United States), which require an exit price, what a market participant would pay, including the risk that either party does not perform, and allow a portfolio of trades with one counterparty to be valued

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net. That admits CVA, DVA, and now FVA, which any market participant would price, and generally excludes KVA, because a firm's own hurdle rate is not a market participant's.

The regulatory number is what drives capital. It includes the CVA capital charge of Section 9.7, applies prudential filters (removing DVA, as Section 9.3 described), and follows rules that may differ from the firm's own models.

The pricing number is what the desk charges the client. It includes everything the trade actually costs the firm, KVA included, because a trade that does not clear its capital hurdle destroys value regardless of how it is booked.

The client has an accounting number of its own, seen from the other side: its fair value carries CVA on the dealer and DVA on itself. Where it has designated the swap as a hedge of a loan, those adjustments have no counterpart in the hedged item, so they show up in its profit as hedge ineffectiveness.

The three diverge systematically, and the divergence is not an error to be reconciled away. A trade can be profitable on the accounting number, capital-destroying on the pricing number, and punitive on the regulatory one. Firms that manage to only one of the three make predictable mistakes: accounting alone produces trades that consume capital without earning it, regulatory alone an optimization of rules rather than of economics, and pricing alone published results that surprise the market. The practical resolution is that pricing decisions use the pricing number, published accounts use the accounting number, and the difference is explained rather than eliminated.

9.12 XVA AND THE MEANING OF VALUE

The chapter began with the claim that "the value of the swap" does not identify a number. What survives can be stated precisely. The cash flows on their own still have a well-defined value: the forward rates of Chapter 7, discounted on the collateral curve, a number two dealers will agree on to within a rounding error. What is not objective is the price, which depends on who the counterparty is, what the master agreement and CSA say, how the dealer funds itself, what capital the trade attracts, and what is already on the books between the two parties, so two dealers quoting the same client on the same trade may legitimately differ by tens of basis points, and neither is wrong. This is a genuine change in what a derivative *is*. Before the crisis, a swap was a bundle of cash flows and its value was a property of those cash flows. It is now more accurate to say that a swap is a bundle of cash flows plus a relationship, and that a substantial part of its price is the cost of maintaining the relationship. The framework that prices the relationship remains contested, for three reasons.

The components overlap and the overlaps are not agreed. Besides the FVA–DVA problem of Section 9.4, capital is held against counterparty risk that CVA has already priced, and MVA and FVA both price funding. Every firm decides for itself what to include, so the totals are not comparable across firms.

Some components are not market prices. CVA is disciplined by hedging: a dealer that charges too little can be arbitraged. KVA is not, for the reasons Section 9.7 gave, and calling it a valuation adjustment gives it a status it has not earned.

The precision is illusory. Simulations over years, credit spreads taken from thin markets, and unobservable correlations cannot support a total reported to the nearest thousand dollars, as Table 9.1 reports it.

None of this argues for ignoring the adjustments. The costs they represent are real, and a firm that does not charge them will discover them elsewhere. It argues for treating XVA as a disciplined framework for allocating known costs, rather than as a valuation in the sense that Chapters 13 and 15 use the word.

Counterparty credit adjustments predate the crisis: Sorensen and Bollier (1994) and Duffie

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and Huang (1996) already treated a swap's counterparty exposure as an option that is not symmetric between the two parties, and by the mid-2000s large dealers ran CVA desks. What came after 2008 was everything else, collateral-rate discounting, funding, margin and capital, and the works that introduced each are set out in the Further Reading.

SUMMARY

Exposure is $\max(V(t), 0)$: counterparty risk is an option on the portfolio's value struck at zero, so it needs a distribution rather than a point estimate, and it is measured across the netting set, where an offsetting trade attracts a lower charge. An interest rate swap's exposure humps early in its life, while a cross-currency swap's rises to maturity because the notional is exchanged then. Expected exposure serves pricing and potential future exposure serves credit limits, and much internal disagreement comes from not saying which is meant.

CVA is the market price of the counterparty's option to default, priced the way everything in this book is priced, by what the hedge costs: CDS protection on the counterparty, sized to the expected exposure. It therefore uses market-implied rather than historical default probabilities, because it is the cost of a hedge and not a forecast of a loss. DVA is its mirror image, required if two parties are to agree on a value, but it turns a deterioration in a firm's own credit into a reported gain, so regulators filtered it out of capital and it survives in accounts rather than in pricing.

Three further adjustments price funding and capital. FVA charges an uncollateralized position at the dealer's own funding spread; Hull and White argued that funding cost is not a property of the instrument, and practice settled the question through accounting recognition. MVA funds initial margin, so clearing and margin rules converted part of the charge for counterparty risk into a funding charge rather than removing it. KVA prices capital at the firm's hurdle rate and is the most strained of the set, since it embeds a management choice in an instrument's value.

Together the adjustments can come to many times the interdealer spread: the XVA is not a correction to the price but substantially is the price. Clearing reduces the total without removing it. ColVA prices how collateral is remunerated and chosen; thresholds and the margin period of risk appear as CVA and FVA on the exposure they leave.

CVA the price and the CVA capital charge are different objects, the second held against the volatility of the first, so reducing one need not reduce the other. Wrong-way risk makes the standard calculation a lower bound. And the same trade carries three defensible numbers, for accounting, for regulatory capital and for pricing, and they differ by design rather than by error. XVA is managed centrally because the quantities are portfolio properties, and hedged imperfectly because most counterparties have no traded credit. The framework remains contested, but the costs are real, and a firm that does not charge them meets them somewhere else.

FURTHER READING

Gregory's *The xVA Challenge* is the standard practitioner reference and Green's *XVA* treats the whole set of adjustments in one volume; Brigo, Morini, and Pallavicini's *Counterparty Credit Risk, Collateral and Funding* gives the derivations this chapter states, in full. Burgard and Kjaer (2011) is the source of the partial-differential-equation representation with bilateral credit and funding costs. The FVA argument is worth reading in the original: Hull and White's "The FVA Debate" and their later "Collateral and Credit Issues in Derivatives Pricing" set out one side, Laughton and Vaisbrot's 2012 reply in *Risk* the other, and Andersen, Duffie, and Song's "Funding Value Adjustments" the resolution in terms of who bears the cost. Piterbarg's "Funding Beyond Discounting" is the source of collateral-rate discounting, and Green, Kenyon,

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and Dennis's "KVA: Capital Valuation Adjustment by Replication" introduced the capital charge. Capriotti (2011) explains the adjoint algorithmic differentiation that makes a nightly exposure run affordable at all.

PRACTICE QUESTIONS

9.1 Explain why counterparty exposure is $\max(V(t), 0)$ rather than $|V(t)|$, and why this makes counterparty risk an option.

9.2 Explain why an interest rate swap's expected exposure peaks partway through its life while a cross-currency swap's rises to maturity, and state which carries more counterparty risk and why.

9.3 Explain why exposure is computed on the netting set rather than trade by trade, and why a dealer might quote a better price for a trade that offsets its existing position with a client.

9.4 A three-year uncollateralized swap has expected exposures of \$1.2m, \$1.5m, and \$0.9m at mid-year points. The counterparty's CDS trades at 180 basis points, recovery is 40%, and discount factors are 0.97, 0.93, and 0.89. Compute the CVA, first as the cost of insuring the exposure and then as loss given default times the default rate the spread implies, and confirm the two agree.

9.5 Explain why CVA is computed from the counterparty's CDS spread rather than from its historical default rate, and what would happen to a dealer that used the historical rate.

9.6 State the argument for DVA and the objection to it, explain why regulators removed DVA from capital, and state what that leaves banks managing.

9.7 Set out the Hull–White argument against FVA and the practitioner response. In what sense was the debate settled and in what sense was it not? Identify where FVA and DVA arguably double-count, and state what a firm would have to believe to include both.

9.8 Using the trade of Example 9.4, recompute MVA at a funding spread of 90 basis points and average initial margin of \$2.5 million, compare it with the CVA of Example 9.2, and explain why KVA is nonetheless often the largest adjustment on a long-dated uncollateralized trade.

9.9 A portfolio's value at the peak of its exposure profile is normally distributed about zero with a standard deviation of \$9 million. Compute the expected exposure and the 95% potential future exposure, and explain which is used for pricing, which for credit limits, and why.

9.10 A client requests a \$3 million threshold in its CSA. Using a CDS spread of 150 basis points, a funding spread of 70 basis points, an average uncollateralized exposure of \$1 million, and a discount-weighted duration of 4.0, estimate the cost of the concession. How would the funding cost change if the client were instead a sovereign that insisted on a one-way CSA, under which only the dealer posts collateral?

9.11 Distinguish CVA the valuation adjustment from the CVA capital charge, explain why the latter exists, and give one action that reduces both and one that hedges the price while leaving most of the charge in place.

9.12 An XVA desk reports a large loss in a quarter during which it was fully hedged on credit spreads. Give three distinct explanations consistent with the hedging having been done competently.

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FURTHER QUESTIONS

9.13 A dealer prices a ten-year uncollateralized swap for a corporate client at theoretical value plus CVA only, ignoring FVA, MVA, and KVA. Describe what happens to the firm over the following decade, identifying which cost appears where in its accounts.

9.14 Give three examples of wrong-way risk, one specific and two general, and explain in each case why insuring the average exposure fails. How would you adjust the calculation in each case?

9.15 Argue that KVA does not belong in a valuation at all, and then argue that omitting it guarantees that a dealer's derivatives business destroys shareholder value. Which treatment would you adopt, and how would you present it to a trading desk?

9.16 Post-crisis reform moved derivatives into clearing to reduce counterparty risk. Using Table 9.1, assess how far it succeeded, and explain precisely what was reduced and what was substituted for it.

9.17 Most of a dealer's counterparties have no traded CDS. Explain how CVA is hedged in practice, what basis risk this leaves, and when that basis risk would be most damaging. Relate your answer to the default correlation of Chapter 24.

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